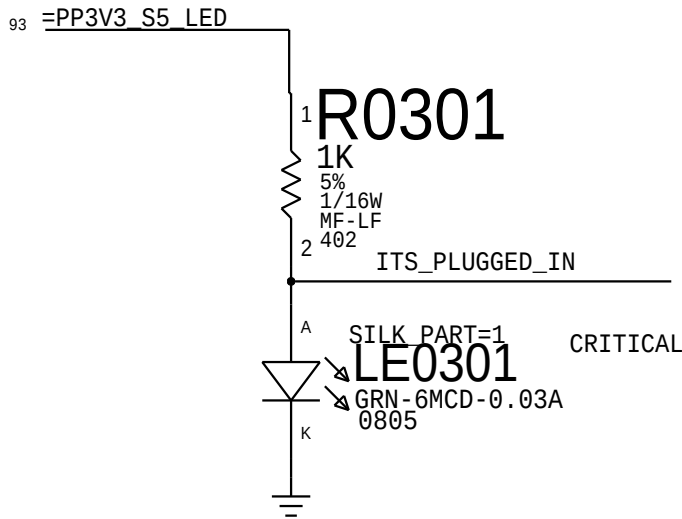


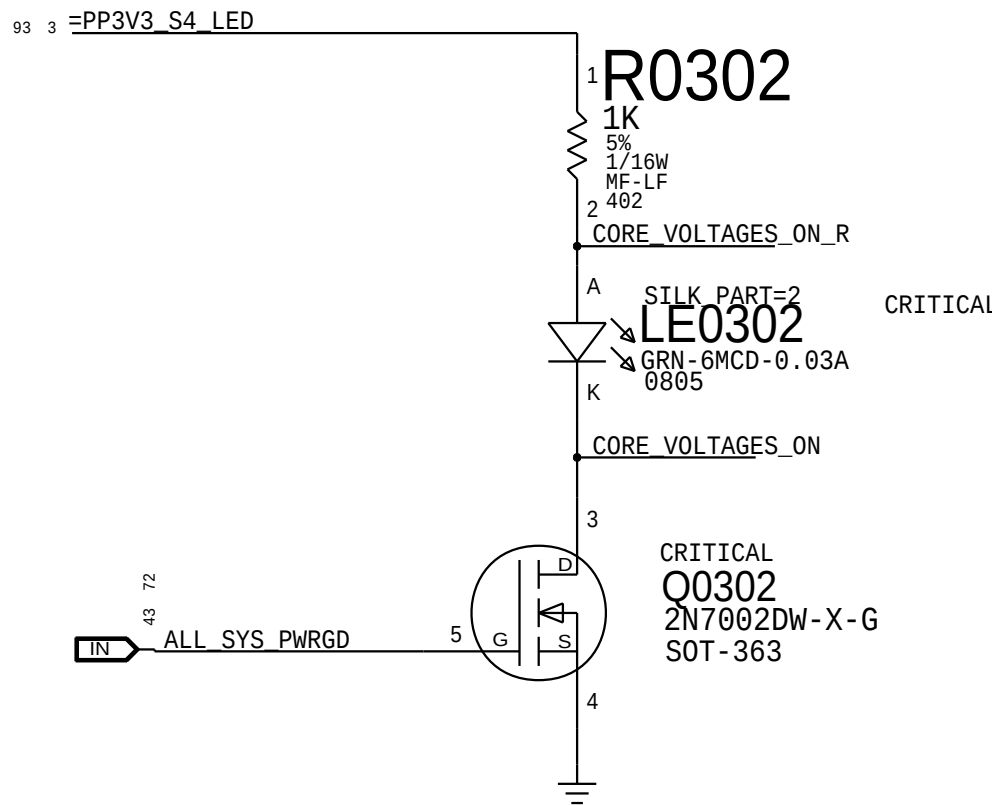
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BOM Variants																	
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985-00041		PCBA, DEV, MLB_SKL, J95				DEVELOPMENT, J95_DEVEL											
685-00059		PCBA, MLB_SKL, COMMON PARTS, J95				J95_COMMON											
639-00847		PCBA, MLB SKL, AMETHYST PROA, VRAM_HYNIX, 2GB, J95				J95, MLB_CMNPTS, CPU:ULT, SSD:Y, GPU:AM_PROA, FB:2G_HYNIX, EEEE:G99J											
639-00846		PCBA, MLB SKL, AMETHYST PROA, VRAM_ELPIDA, 2GB, J95				J95, MLB_CMNPTS, CPU:ULT, SSD:Y, GPU:AM_PROA, FB:2G_ELPIDA, EEEE:GCP9											
639-00849		PCBA, MLB SKL, AMETHYST XTA, VRAM_HYNIX, 4GB, J95				J95, MLB_CMNPTS, CPU:ULT, SSD:Y, GPU:AM_XTA, FB:4G_HYNIX, EEEE:GCR6											
639-00848		PCBA, MLB SKL, AMETHYST XTA, VRAM_ELPIDA, 4GB, J95				J95, MLB_CMNPTS, CPU:ULT, SSD:Y, GPU:AM_XTA, FB:4G_ELPIDA, EEEE:GCR7											
639-01033		PCBA, MLB SKL, CTO, AMETHYST PROA, VRAM_HYNIX, 2GB, J95				J95, MLB_CMNPTS, CPU:CTO, SSD:Y, GPU:AM_PROA, FB:2G_HYNIX, EEEE:G99J											
639-01032		PCBA, MLB SKL, CTO, AMETHYST PROA, VRAM_ELPIDA, 2GB, J95				J95, MLB_CMNPTS, CPU:CTO, SSD:Y, GPU:AM_PROA, FB:2G_ELPIDA, EEEE:GCP9											
639-01035		PCBA, MLB SKL, CTO, AMETHYST XTA, VRAM_HYNIX, 4GB, J95				J95, MLB_CMNPTS, CPU:CTO, SSD:Y, GPU:AM_XTA, FB:4G_HYNIX, EEEE:GCR6											
639-01034		PCBA, MLB SKL, CTO, AMETHYST XTA, VRAM_ELPIDA, 4GB, J95				J95, MLB_CMNPTS, CPU:CTO, SSD:Y, GPU:AM_XTA, FB:4G_ELPIDA, EEEE:GCR7											
BOM Groups																	
BOM GROUP		BOM OPTIONS															
J95_COMMON		COMMON, ALTERNATE, J95_PROGPARTS, GPU_AM_EN, XDP, SPEAKERID, FBVDDQ_DFLT:1V5, SMCRES: SUP_VR_BULKCAP: CURRENT, GPU_AME_NEP															
J95_PROGPARTS		SMC:PROG, BOOTROM:PROG, TBTRON:PROG, CIVROM:PROG, CAMROM:PROG															
J95_DEVEL		XDP_CONN, T112, DEVEL_AUDIO															
CPUS																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
337S00122	1	CPU, SKL, QHQF, ES2, Q0, 2.6, 95W, LGA1151	CPU	CRITICAL	CPU:CTO												
337S00114	1	CPU, SKL, QHBE, ES, P0, 4/2, 2.2, 95W, LGA1151	CPU	CRITICAL	CPU:ULT												
337S00121	1	CPU, SKL, QHQ0, ES2, Q0, 2.2, 65W, LGA1151	CPU	CRITICAL	CPU: BEST												
CPU SOCKET																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
511S00002	1	SOCKET_MOLEX, LGA1151, H4	U0500	CRITICAL													
ASICS																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
337S00124	1	IC, PCH-H, SKL, SUPRSKU_C0, QHPT, ES2, BGA1151	U1100	CRITICAL													
338S1247	1	IC, TBT, FR-4C, A0, PRQ_C10, SR13C, FCBGA288	U2000	CRITICAL													
343S00616	1	IC, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL													
Programmable Parts																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
341S00050	1	IC, EFI, V0174, J78	U5210	CRITICAL	BOOTROM:PROG												
335S00006	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK												
341S00236	1	IC, SMC-B1, EXTERNAL, VXXXX, POC-S, J78AM	U5000	CRITICAL	SMC:PROG												
338S1214	1	IC, SMC12-B1, 40MHZ/500MIPS, MCU, 1578GA	U5000	CRITICAL	SMC:BLANK												
341S00152	1	IC, T29, EPROM, FR, VTBD, POC, J95	U2000	CRITICAL	TBTRON:PROG												
335S0915	1	IC, SRL SPI FLASH ROM, 4MBIT, 50MHZ, USON8	U2000	CRITICAL	TBTRON:BLANK												
341S0912	1	IC, ENET ROM, NUMONYX, V1.15, J16/J166/J17	U3900	CRITICAL	CIVROM:PROG												
335S0054	1	1-MBIT, SPI FLASH ROM 501C, HF	U3900	CRITICAL	CIVROM:BLANK												
341S0378	1	IC, CAMERA FLASH, V7230, J16/J17	U4202	CRITICAL	CAMROM:PROG												
335S0052	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK												
Bar Code Labels / EEEE #'s																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
825-7896	1	LABEL, MLB, 2D	EEEE_G99J	CRITICAL	EEEE:G99J												
825-7896	1	LABEL, MLB, 2D	EEEE_GCP9	CRITICAL	EEEE:GCP9												
825-7896	1	LABEL, MLB, 2D	EEEE_GCR6	CRITICAL	EEEE:GCR6												
825-7896	1	LABEL, MLB, 2D	EEEE_GCR7	CRITICAL	EEEE:GCR7												
J95 SCHEMATIC / PCB #'S																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
051-00321	1	SCH, MLB SKL, J95	SCH1	CRITICAL	J95												
020-00134	1	PCBF, MLB SKL, J95	PCB1	CRITICAL	J95												
685-00059	1	PCBA, MLB SKL COMMON PARTS, J95	CMNPTS	CRITICAL	MLB_CMNPTS												
J95 ALTERNATES																	
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:													
377S0147	377S0126		ALL	USB Diode Array													
197S0481	197S0480		ALL	25MHX PCH XTAL													
197S0479	197S0478		ALL	12MHZ CAM/NXP XTAL													
372S0186	372S0185		ALL	Alternate Temp Diode													
107S00030	107S00029		ALL	Alt 5mOhm sense													
107S0255	107S0240		ALL	Alt 1mOhm sense													
341S00016	341S3912		U3990	IC, ENETROM, ADE3D0, V1, 15, 178													
138S00681	138S0638		ALL	10uF Caps													
138S0060	138S0775		ALL	Single-source 1uF 402													
128S0398	128S0220		ALL	3.3V INPUT CAP													
378S0391	378S0390		ALL	DEGUG LEDS													
311S00008	311S0271		ALL	AND GATE													
155S0030	155S0316		ALL	FROM 80.000 OHM, 75.0-0A													
376S00016	376S0644		ALL	MOSFET, P-CH, 20V, 1A													
376S00001	376S0659		ALL	MOSFET, P-CH, 20V													
376S00015	376S0610		ALL	XSTR, FET, DL NCH, 60V													
138S1103	138S0719		ALL	4.7uF 0402													
311S00004	311S0370		ALL	Config. Logic Gate													
311S00075	311S0377		ALL	AND GATE													
132S0447	132S0448		ALL	0.1uF 0201													
138S0055	138S0750		ALL	4.7uF 0402													
353S00095	353S3328		ALL	2:4 Diff MUX													
311S00072	311S0657		ALL	Dual 2 INP AND GATE													
GPU and VRAM																	
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION												
337S00001	1	IC, GPU, AMD, PS, AMST, XTA, ES, 250B, BGA1737	U8700	CRITICAL	GPU:AM_XTA												
337S00002	1	IC, GPU, AMD, PS, AMST, PROA, ES, 250B, BGA1737	U8700	CRITICAL	GPU:AM_PROA												
333S00027	4	IC, GDDR5, 4GBIT, 128MX32, 56BPS, GEMMA	U9200, U9250, U9300, U9350	CRITICAL	FB:4G_HYNIX												
333S00027	4	IC, GDDR5, 4GBIT, 128MX32, 56BPS, GEMMA	U9400, U9450, U9500, U9550	CRITICAL	FB:4G_HYNIX												
333S0766	4	IC, GDDR5, 4GBIT, 128MX32, 56BPS, GEMMA	U9200, U9250, U9300, U9350	CRITICAL	FB:4G_ELPIDA												
333S0766	4	IC, GDDR5, 4GBIT, 128MX32, 56BPS, GEMMA	U9400, U9450, U9500, U9550	CRITICAL	FB:4G_ELPIDA												
333S0734	4	IC, GDDR5, 2GBIT, 64MX32, 66BPS, 29NM	U9200, U9250, U9300, U9350	CRITICAL	FB:2G_HYNIX												
333S0734	4	IC, GDDR5, 2GBIT, 64MX32, 66BPS, 29NM	U9400, U9450, U9500, U9550	CRITICAL	FB:2G_HYNIX												
333S0701	4	IC, GDDR5, 2GBIT, 64MX32, 66BPS, 30NM	U9200, U9250, U9300, U9350	CRITICAL	FB:2G_ELPIDA												
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2 OF 96			

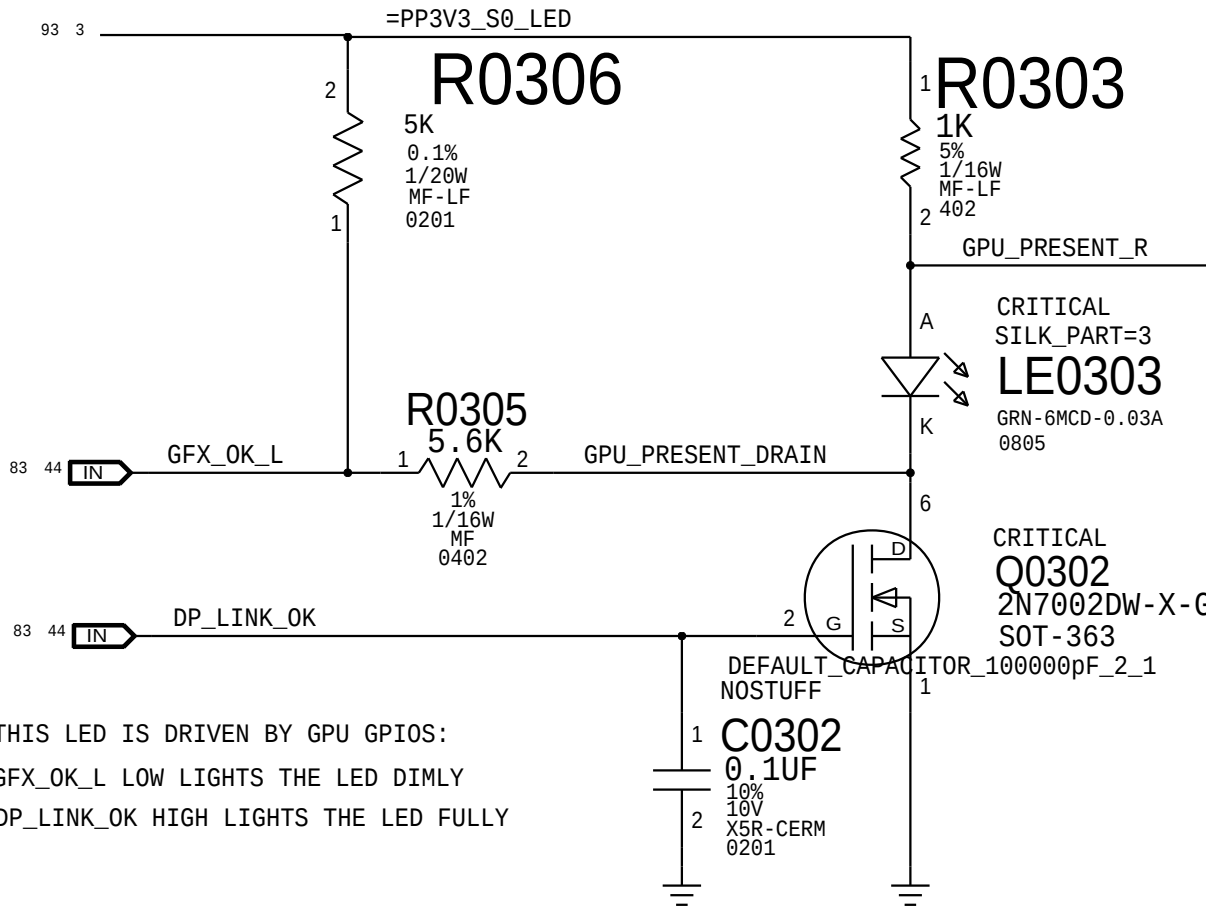
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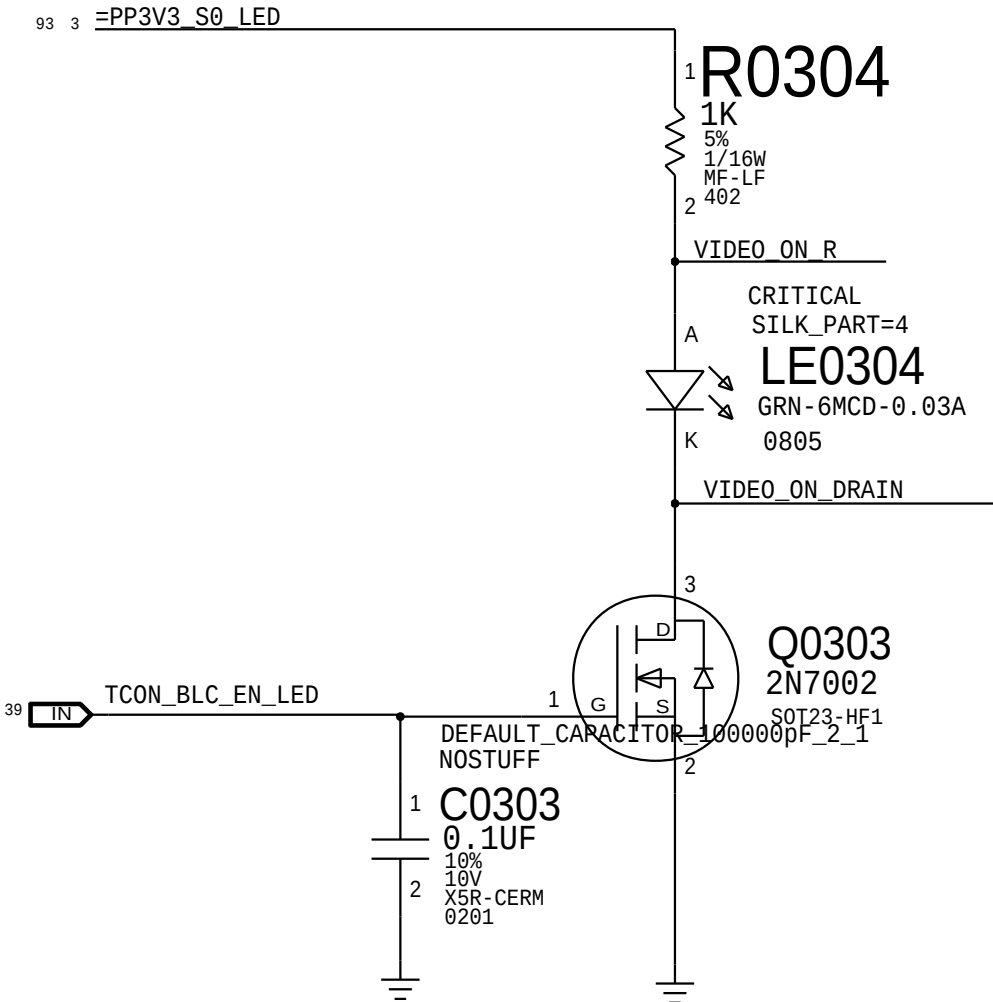
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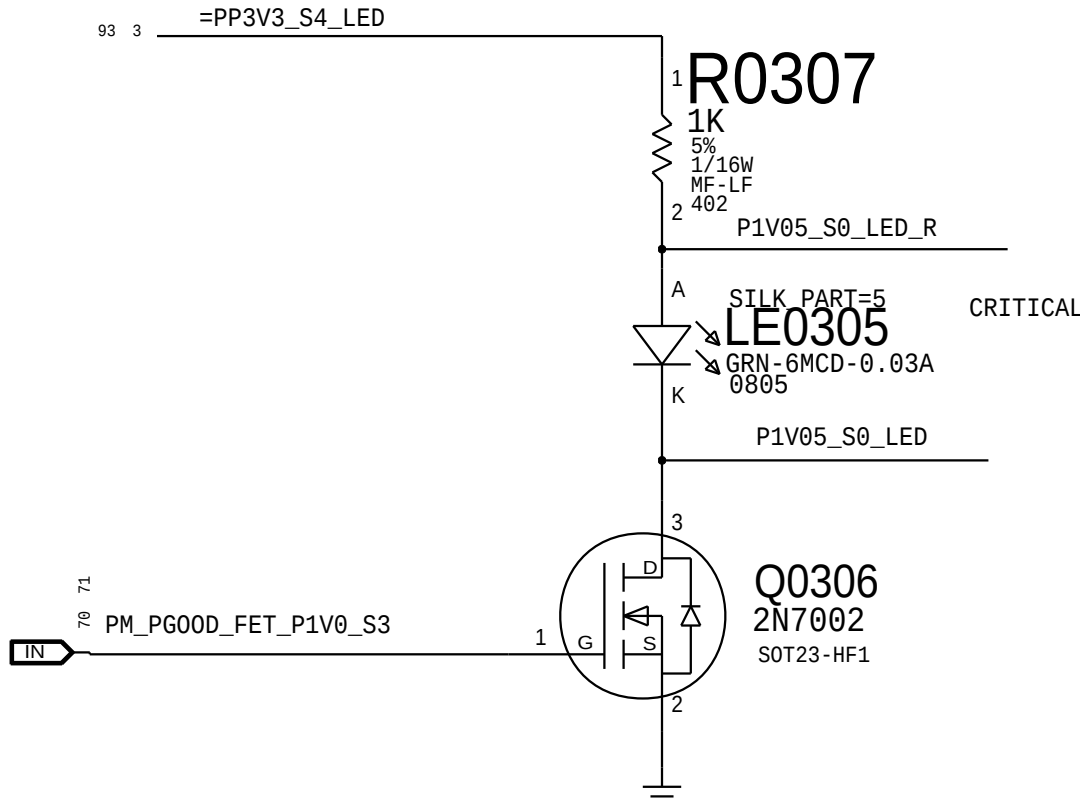
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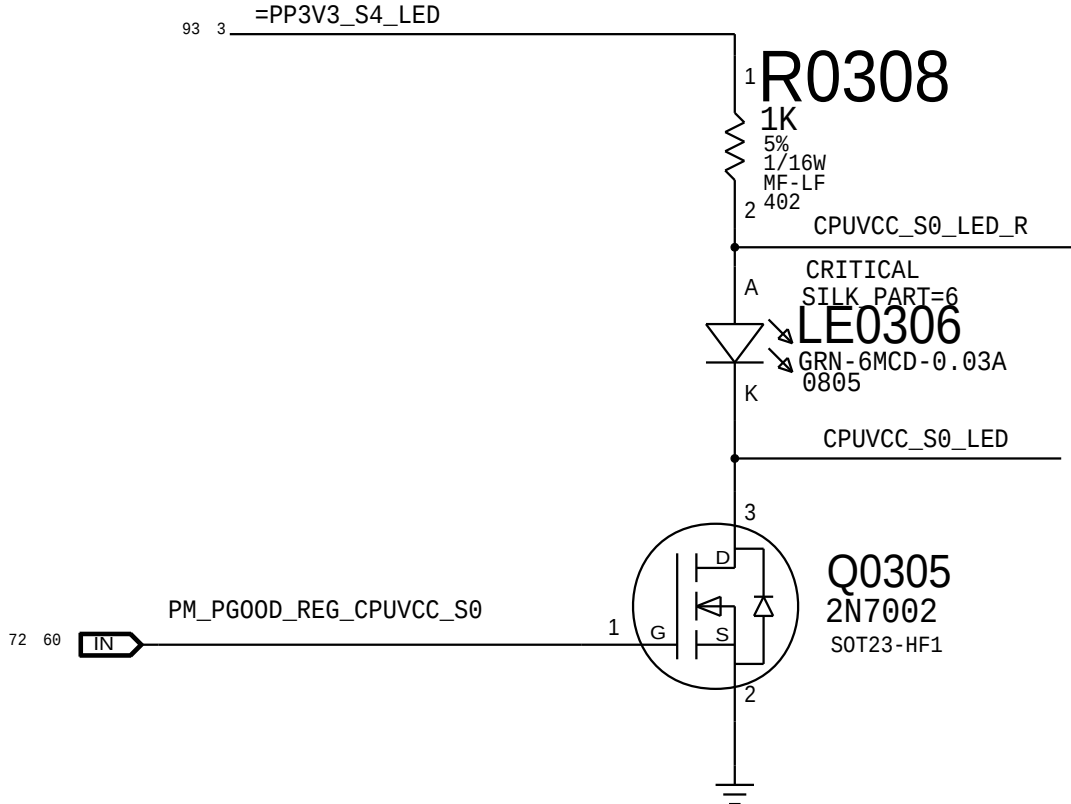
VIDEO ON Led



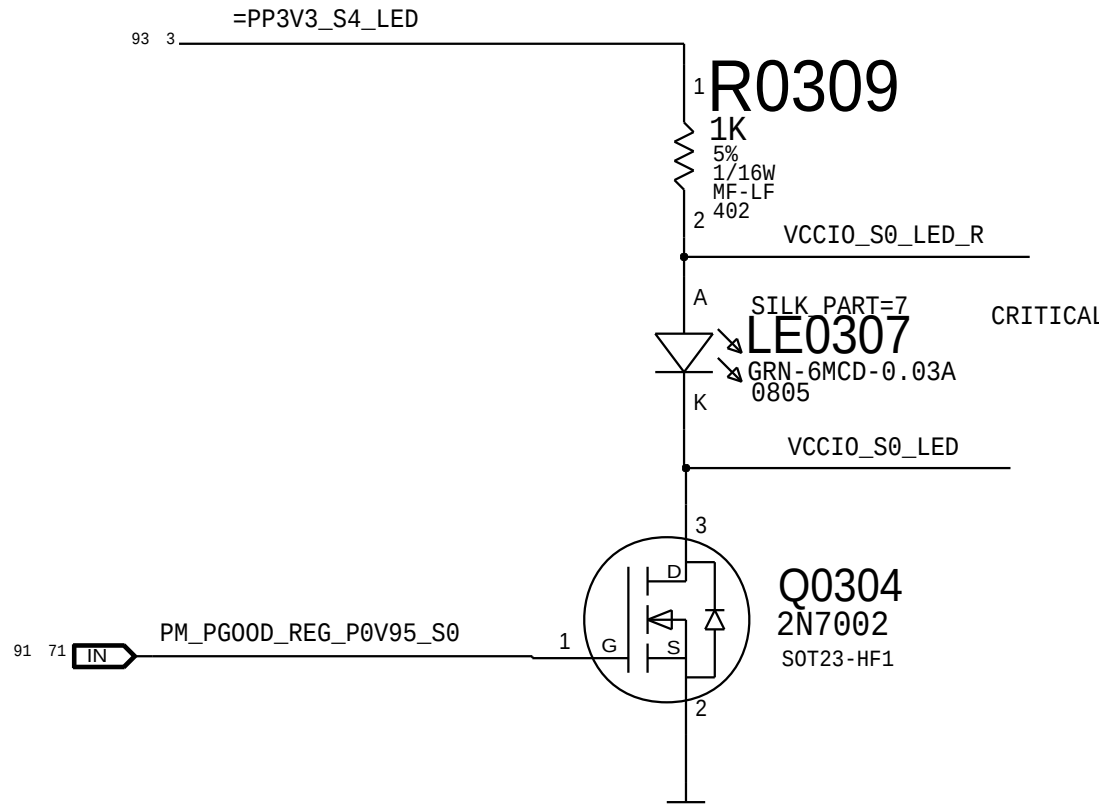
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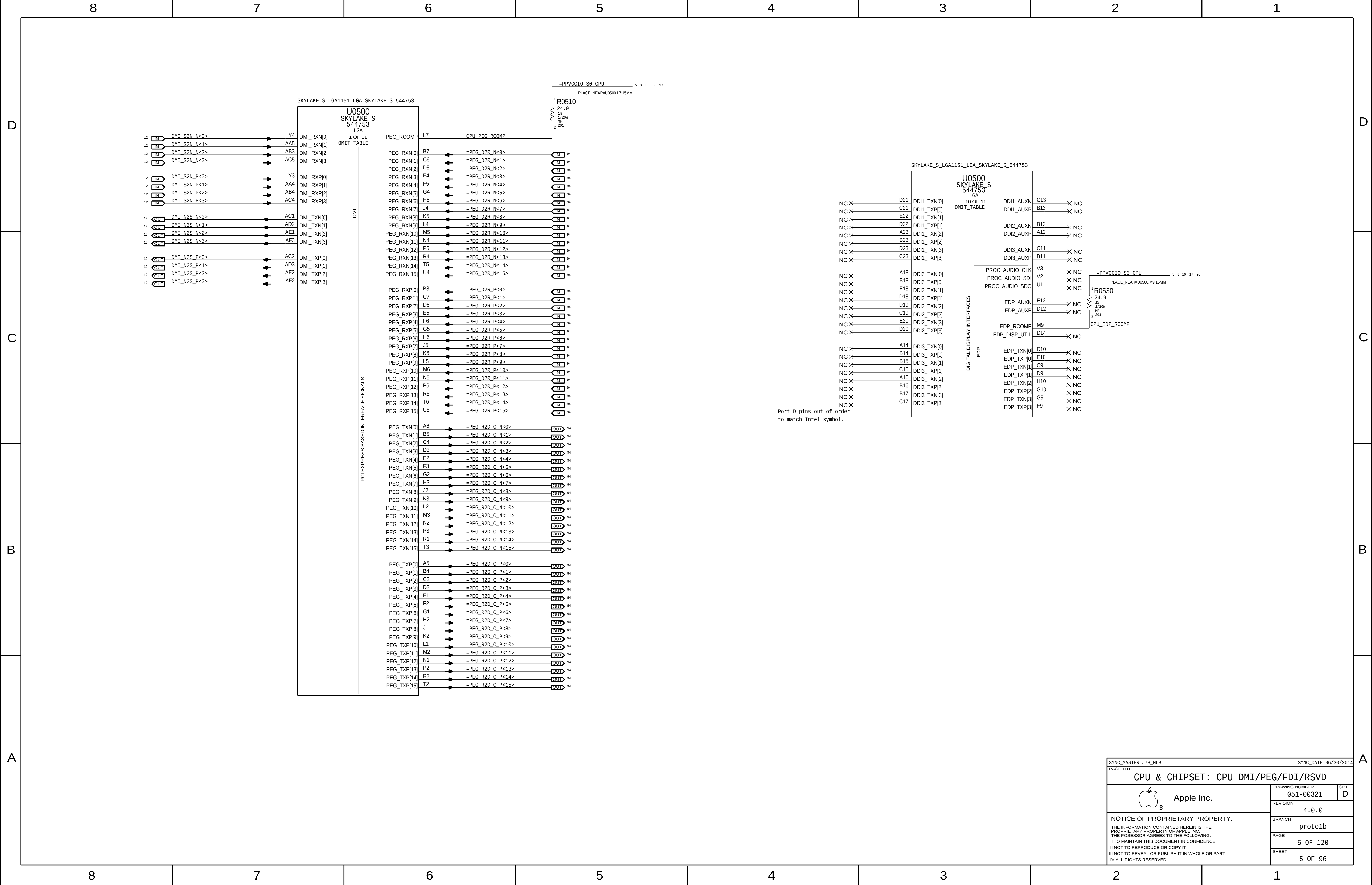


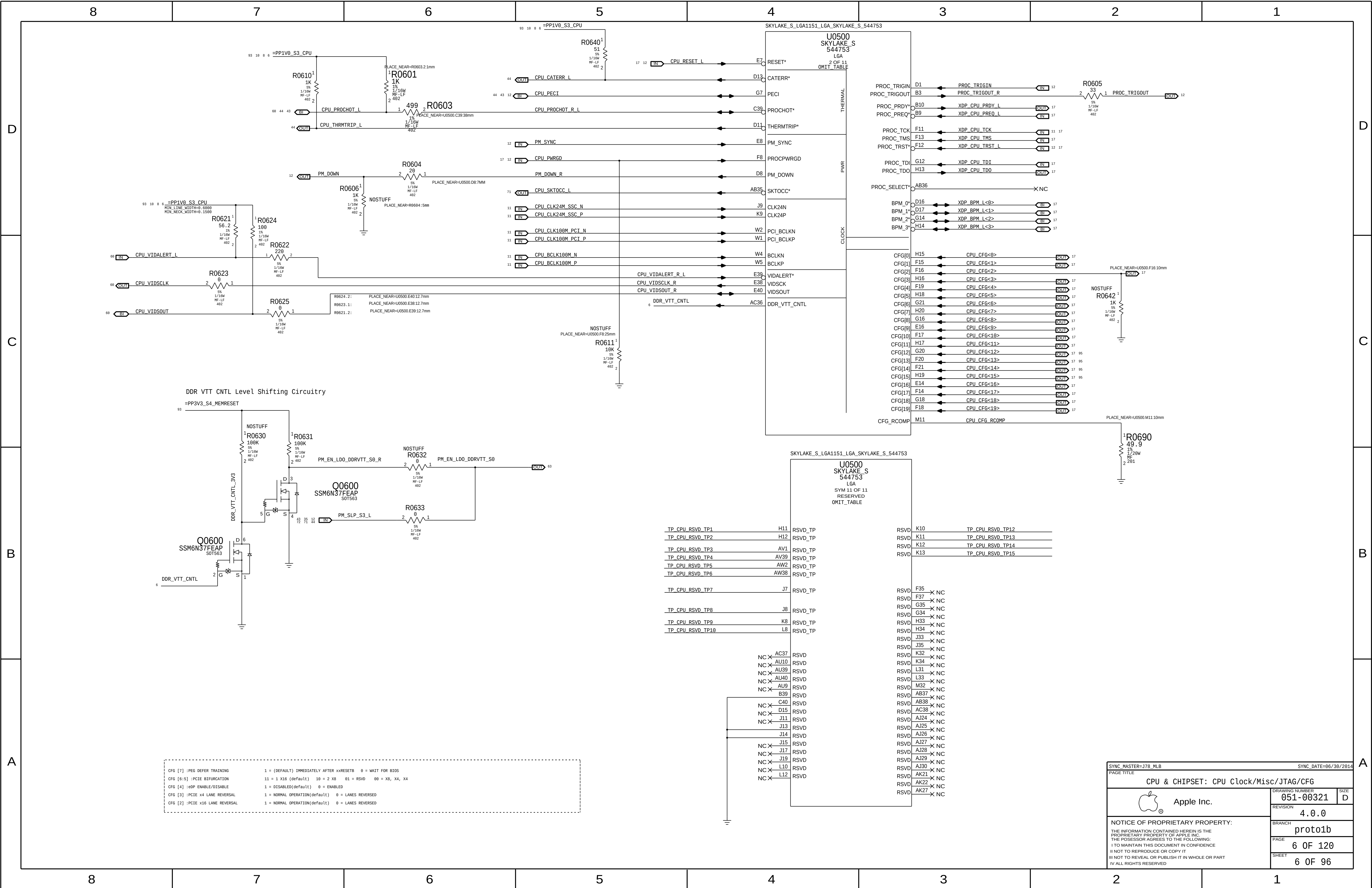
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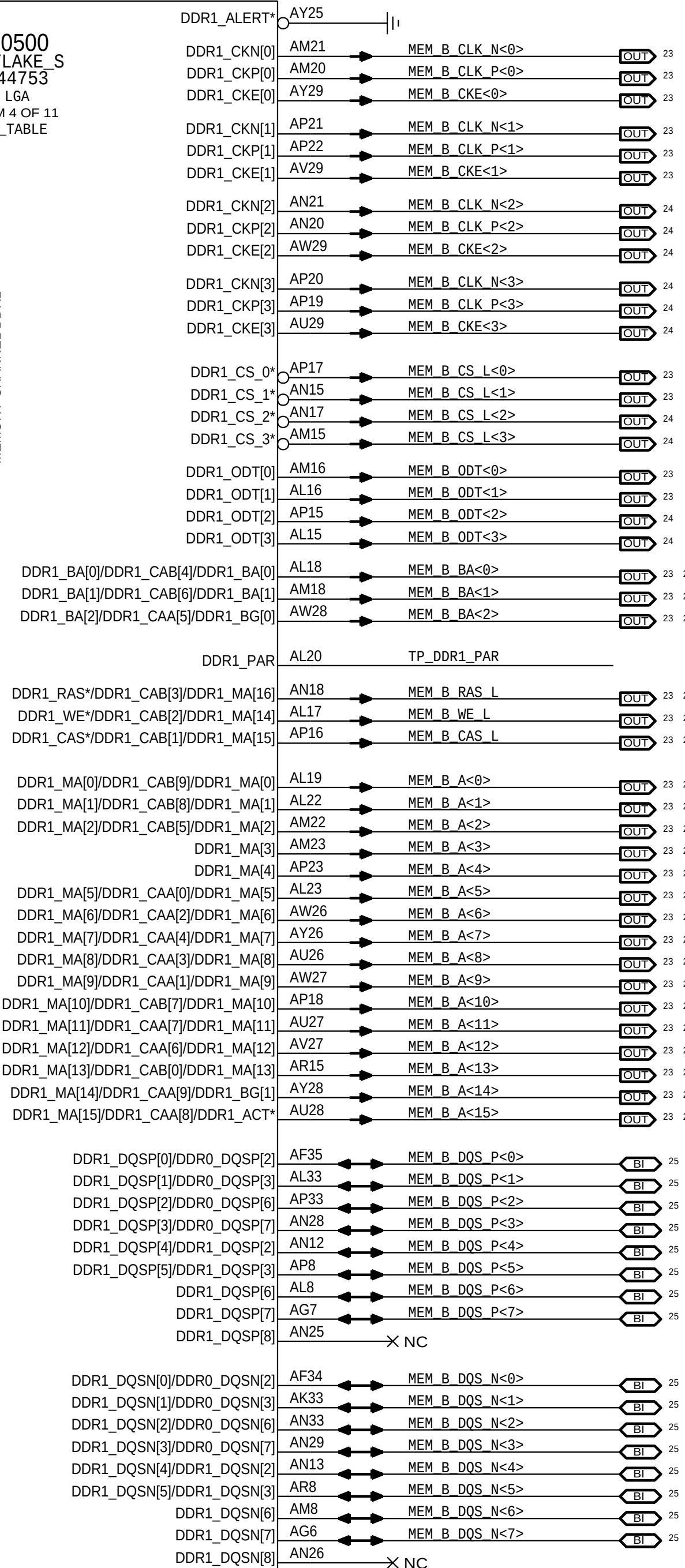
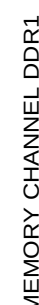
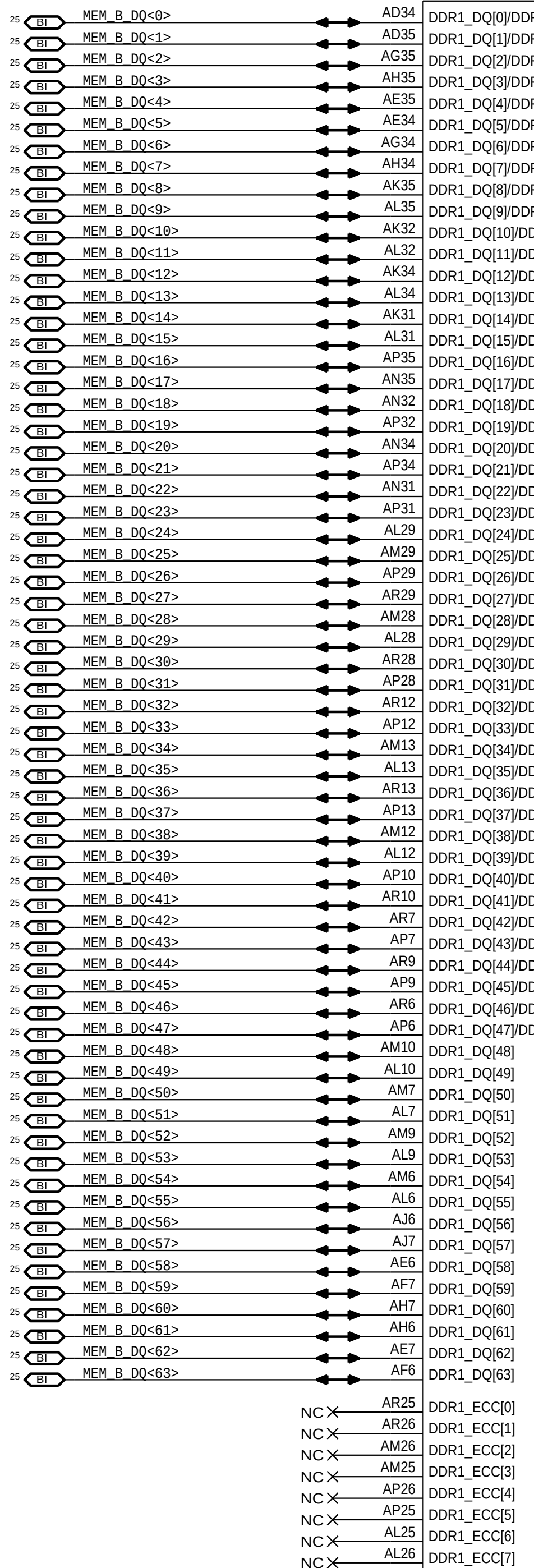
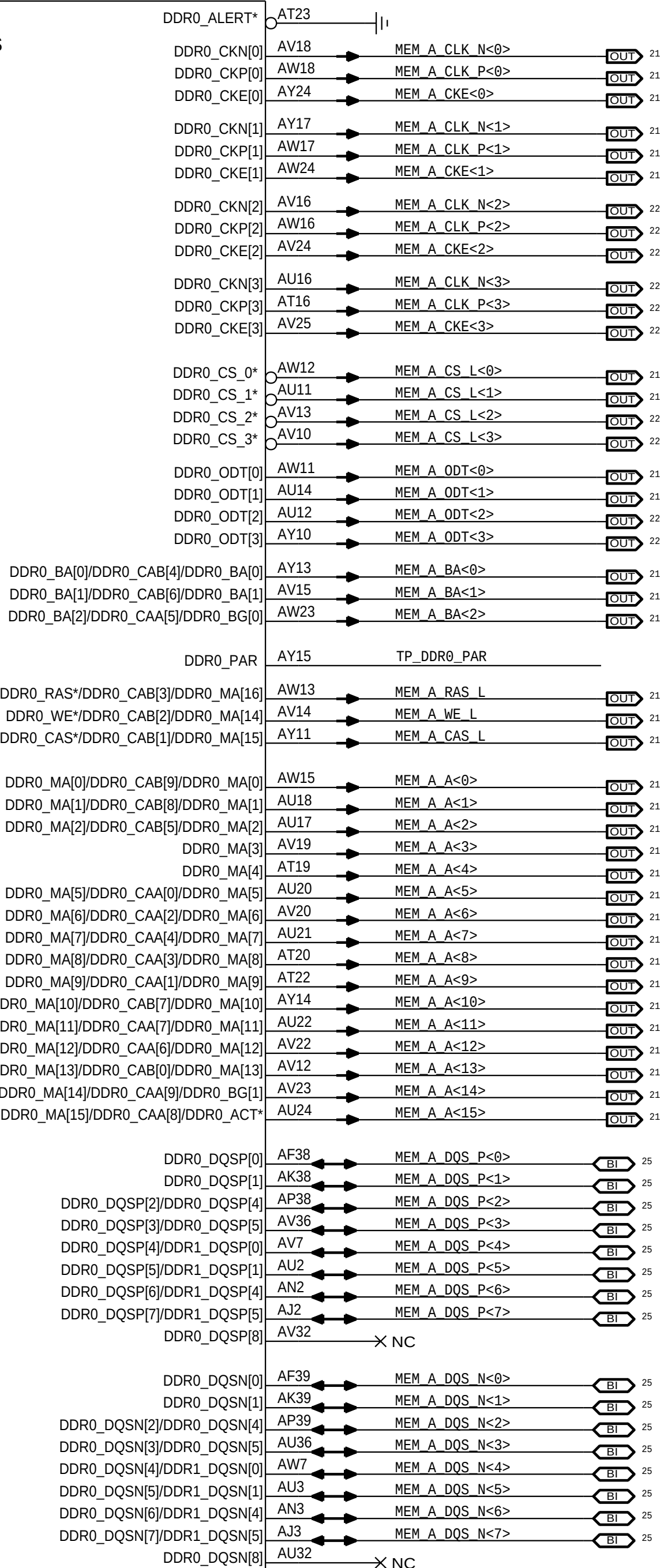
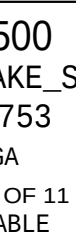
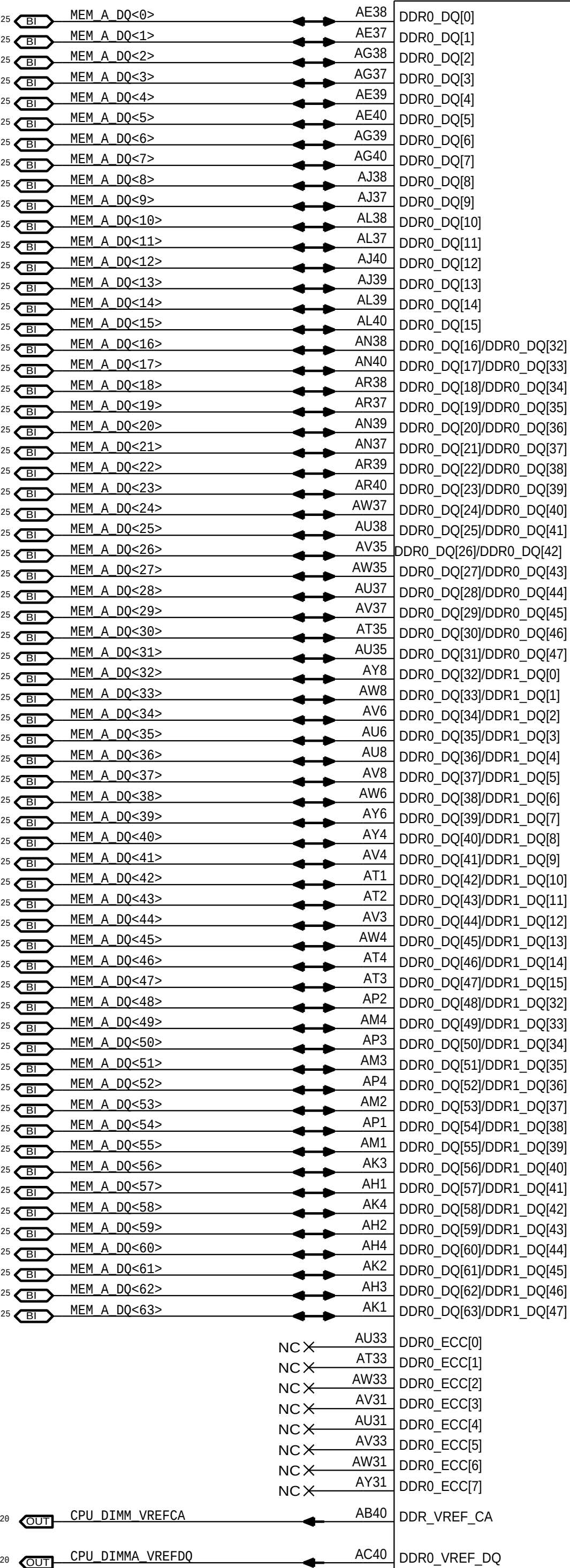


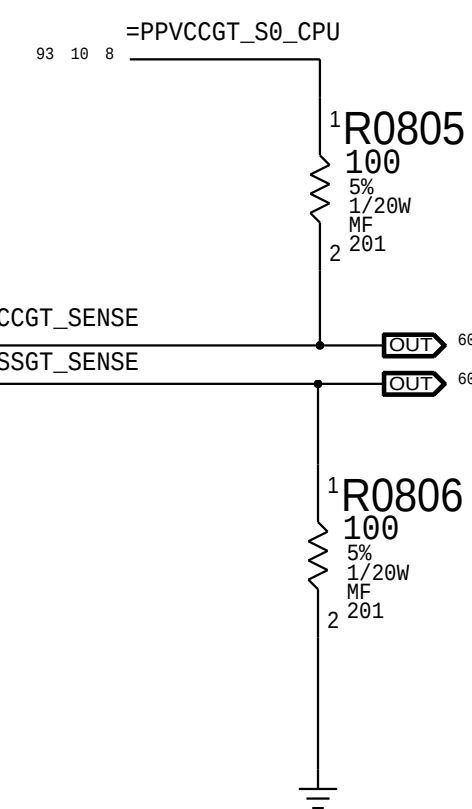
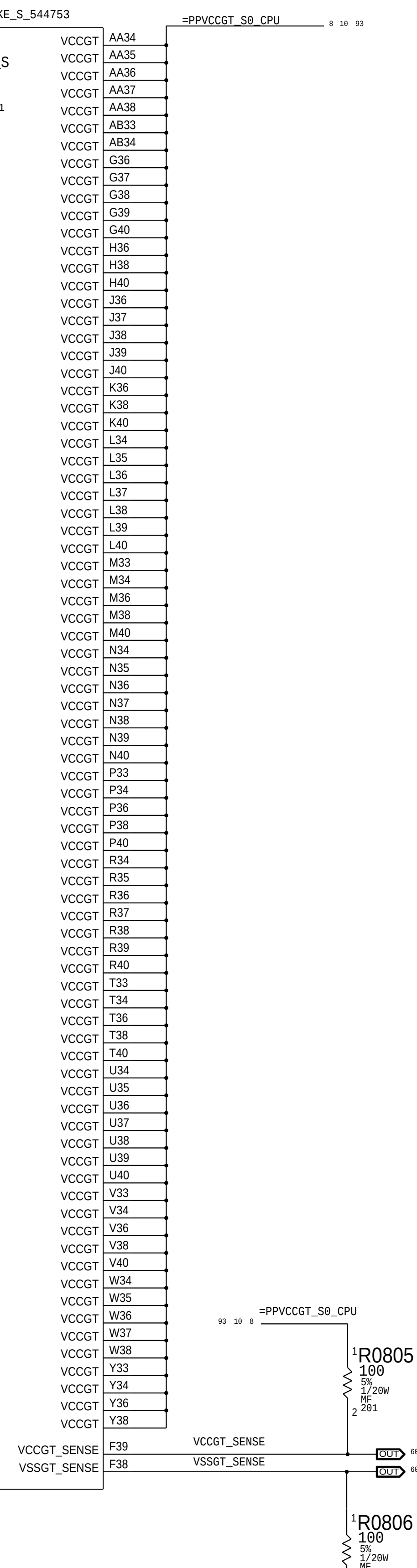
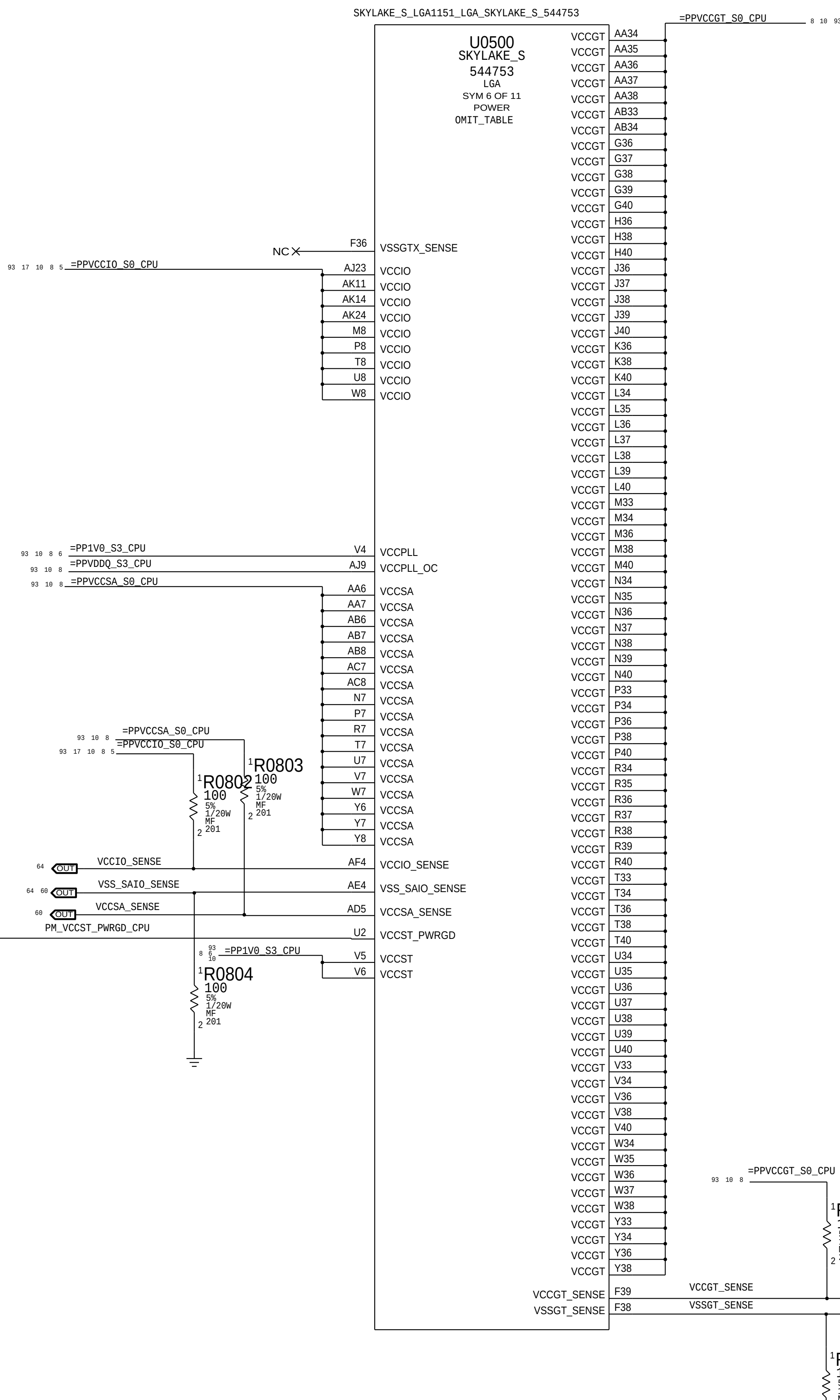
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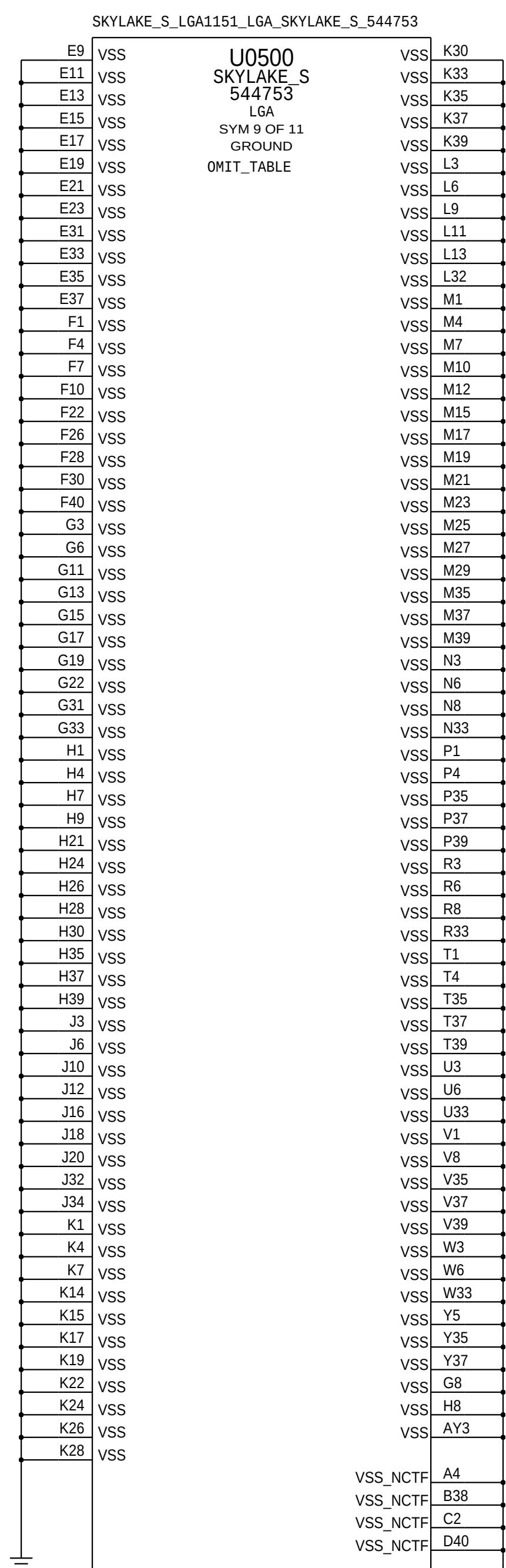
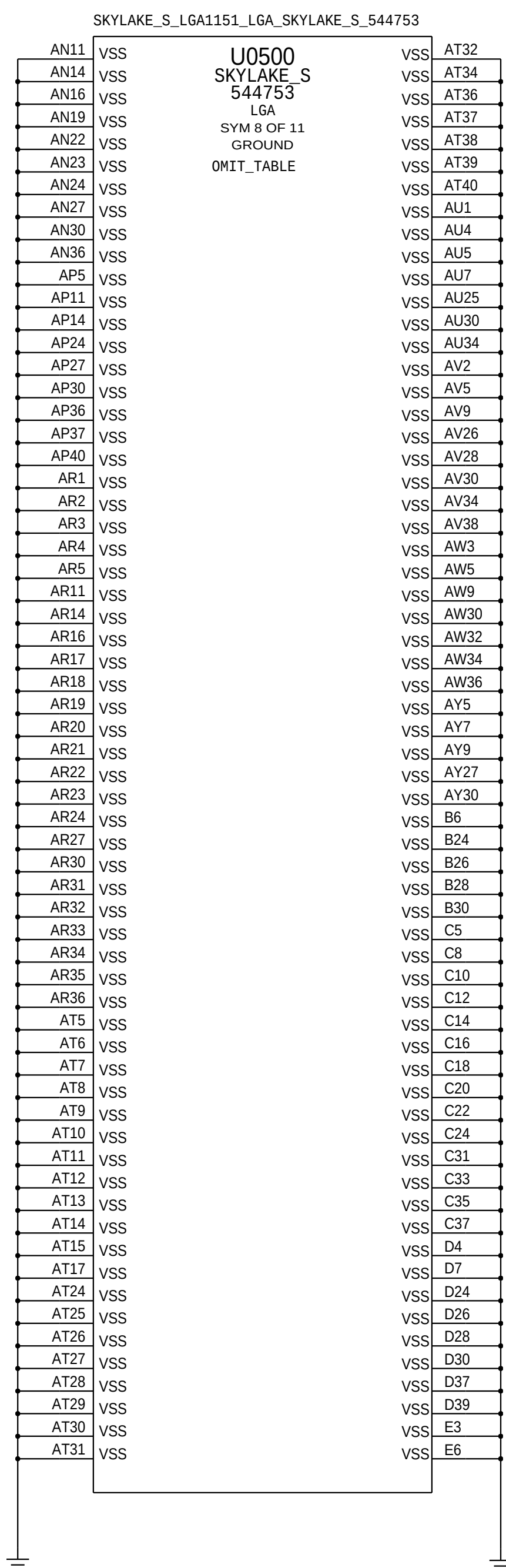
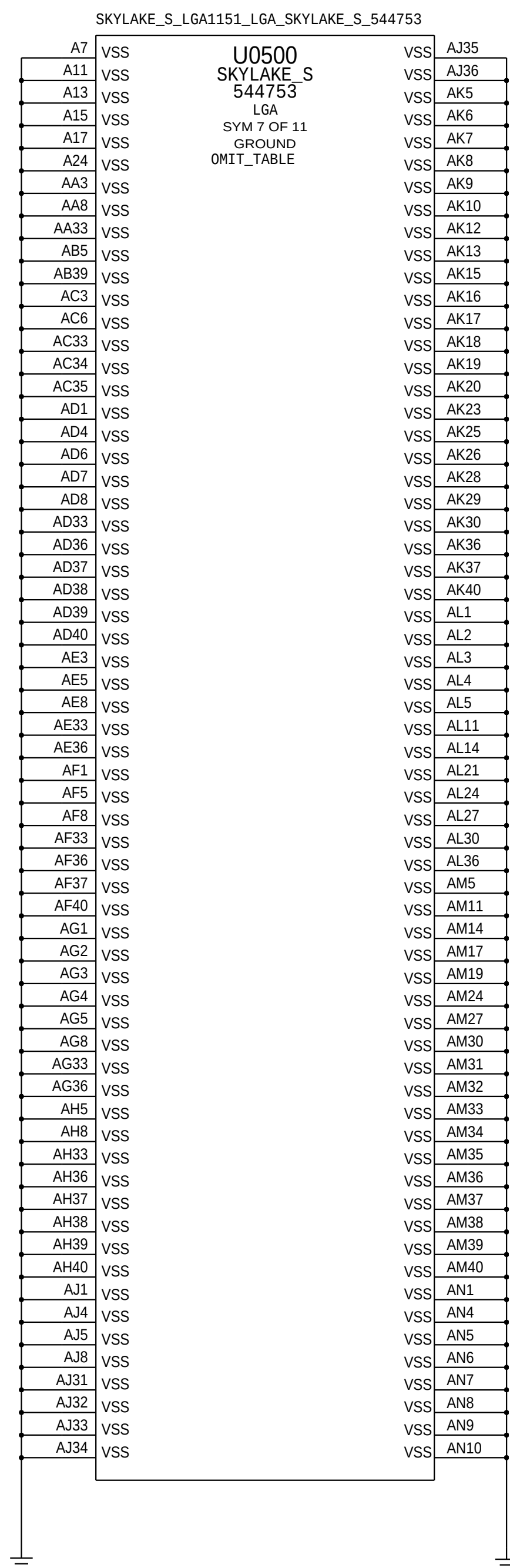








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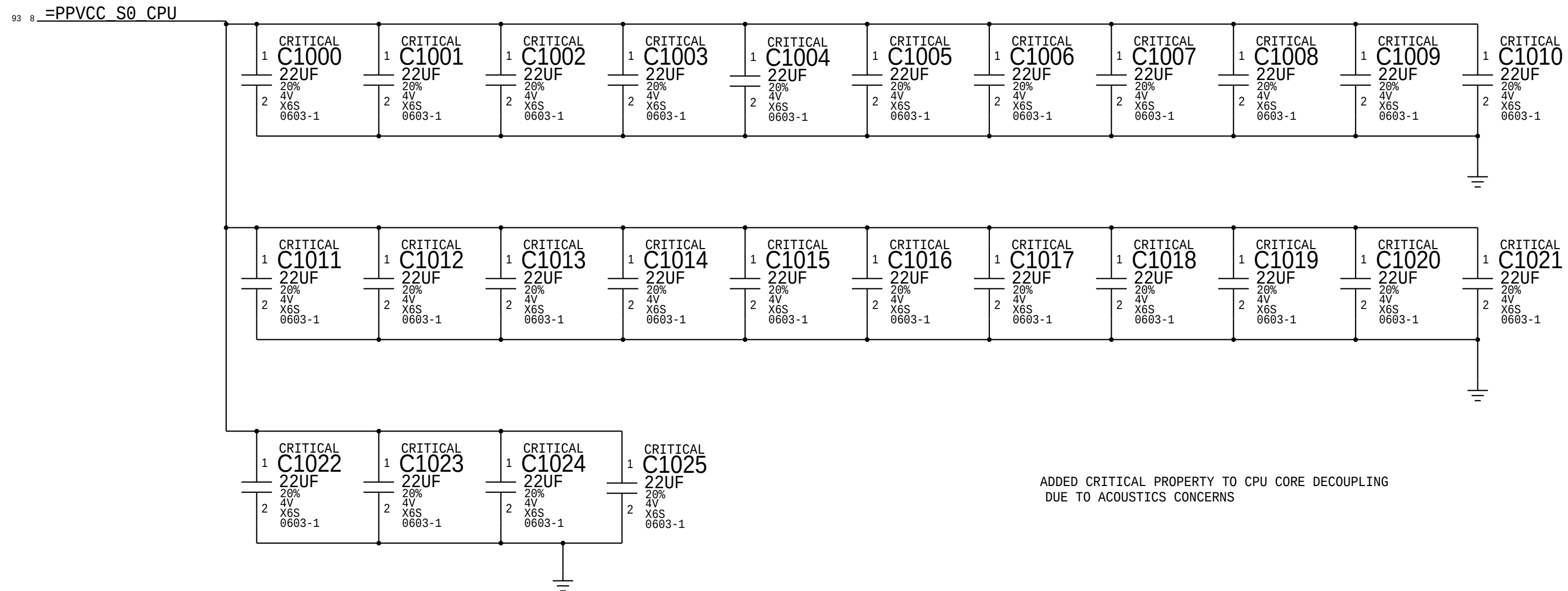


CPU VCORE DECOUPLING

```
Intel Recommendation:12x 22UF 0805 (top side cavity)
                      6x 22UF 0603 (top side cavity)
                      5x 22UF 0805 (top side outside cavity)
```

Apple Implementation: 26x 22UF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



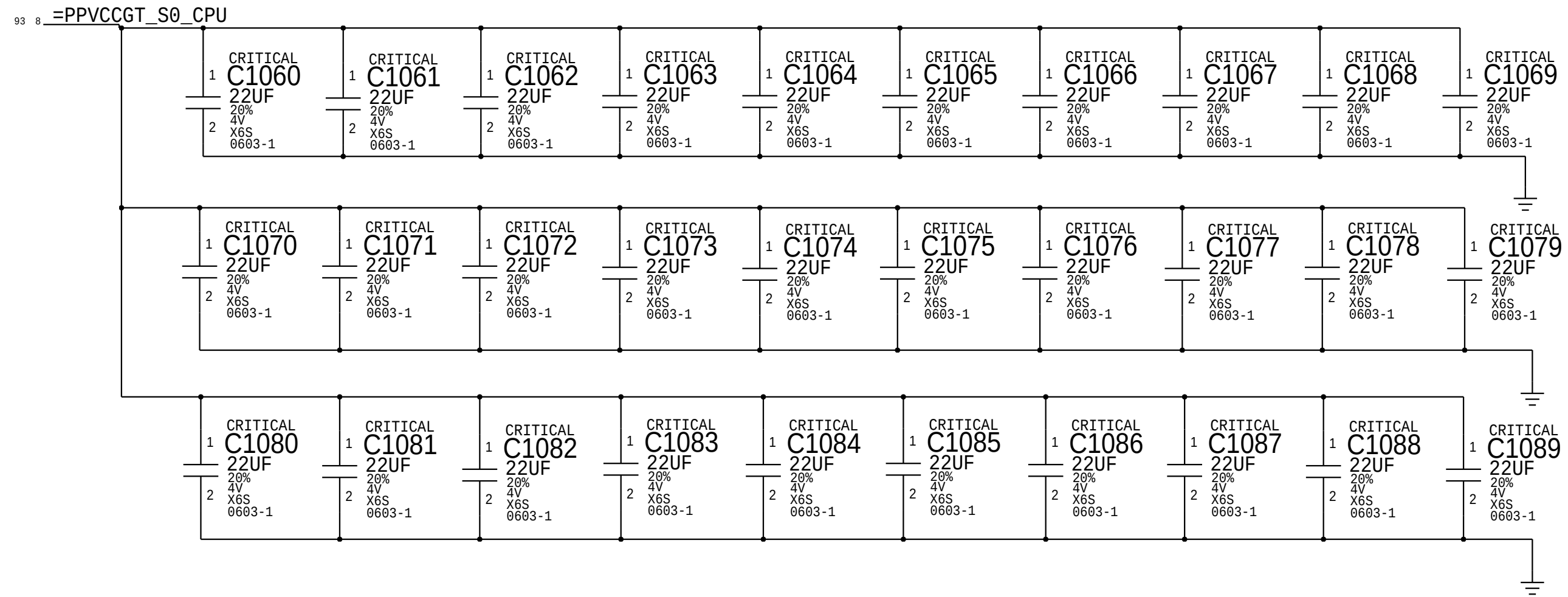
ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

CPU GT DECOUPLING

Intel Recommendation: 9x 47UF 0805 (top side cavity)
4x 47UF 0805 (top side outside cavity)

Apple Implementation: 30x 22uF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

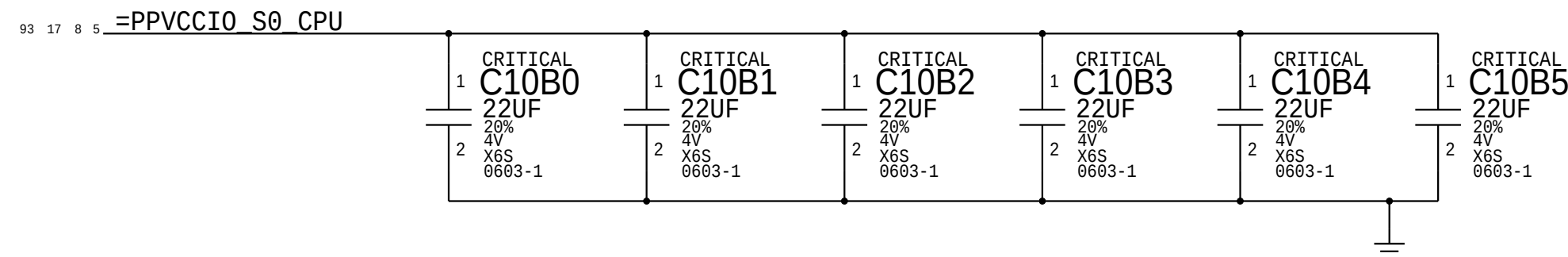


CPU VCCIO DECOUPLING

```
Intel Recommendation:5x 22UF 0603 (top side cavity)
                    1x 22UF 0805 (top side cavity)
```

Apple Implementation:(following Intel recommendation w/ 0603)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

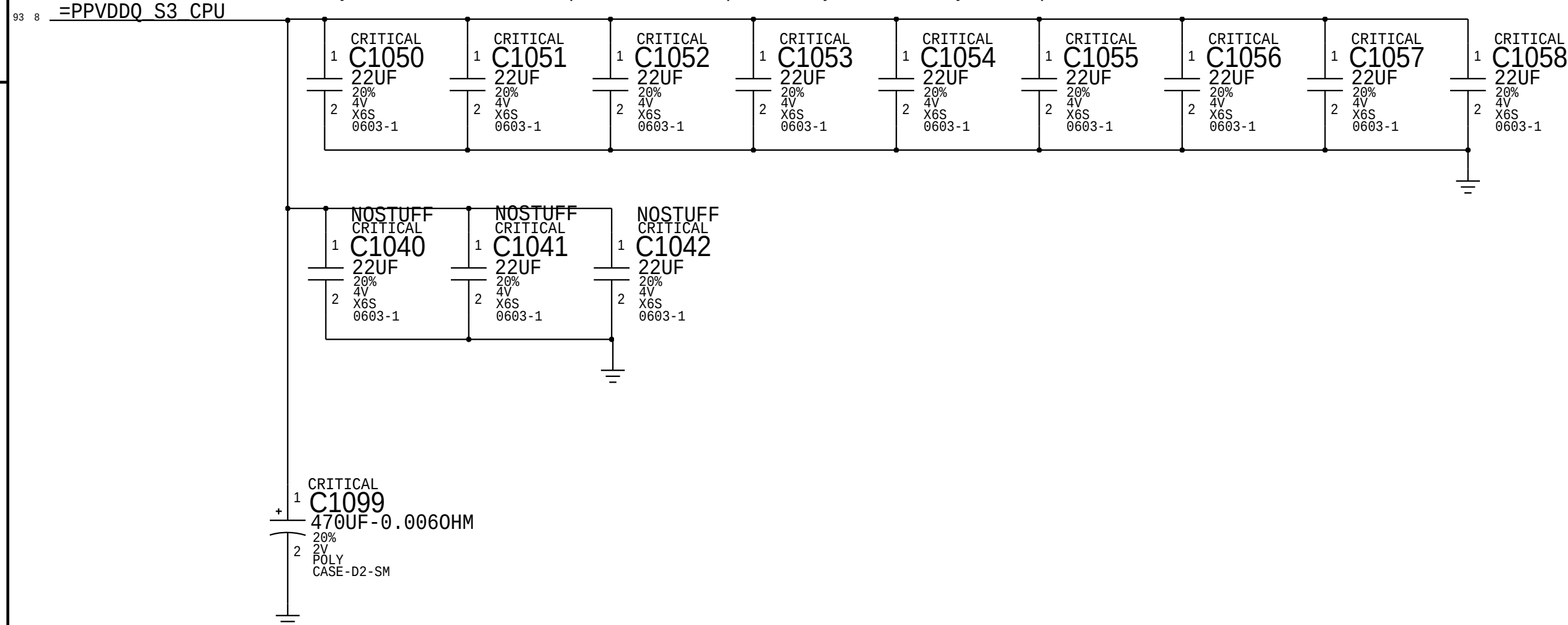


Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation: 4x 22UF 0603 (top side outside cavity)

Apple Implementation: 9x 22UF 0603 (J78 carry over)

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

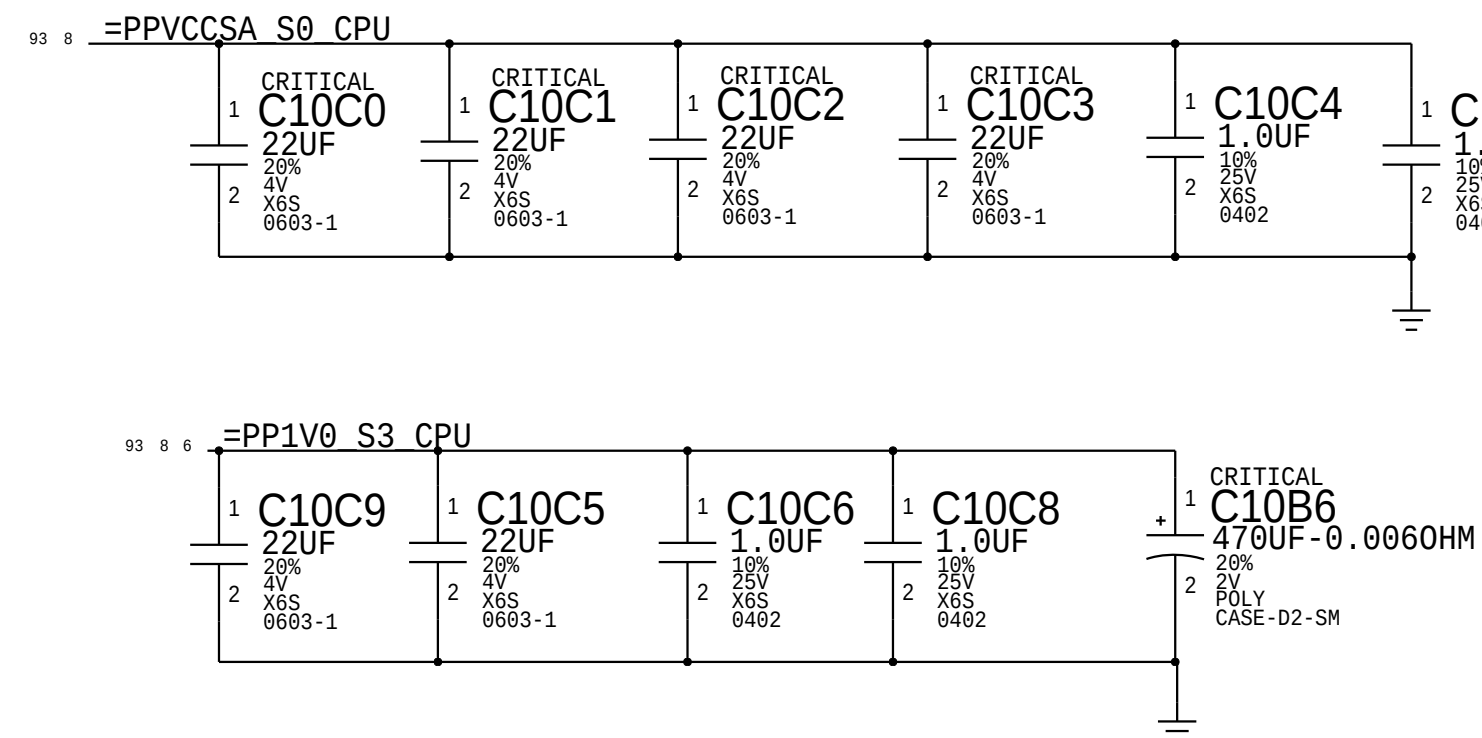


CPU VCCSA / VCCST+VCCPLL DECOUPLING

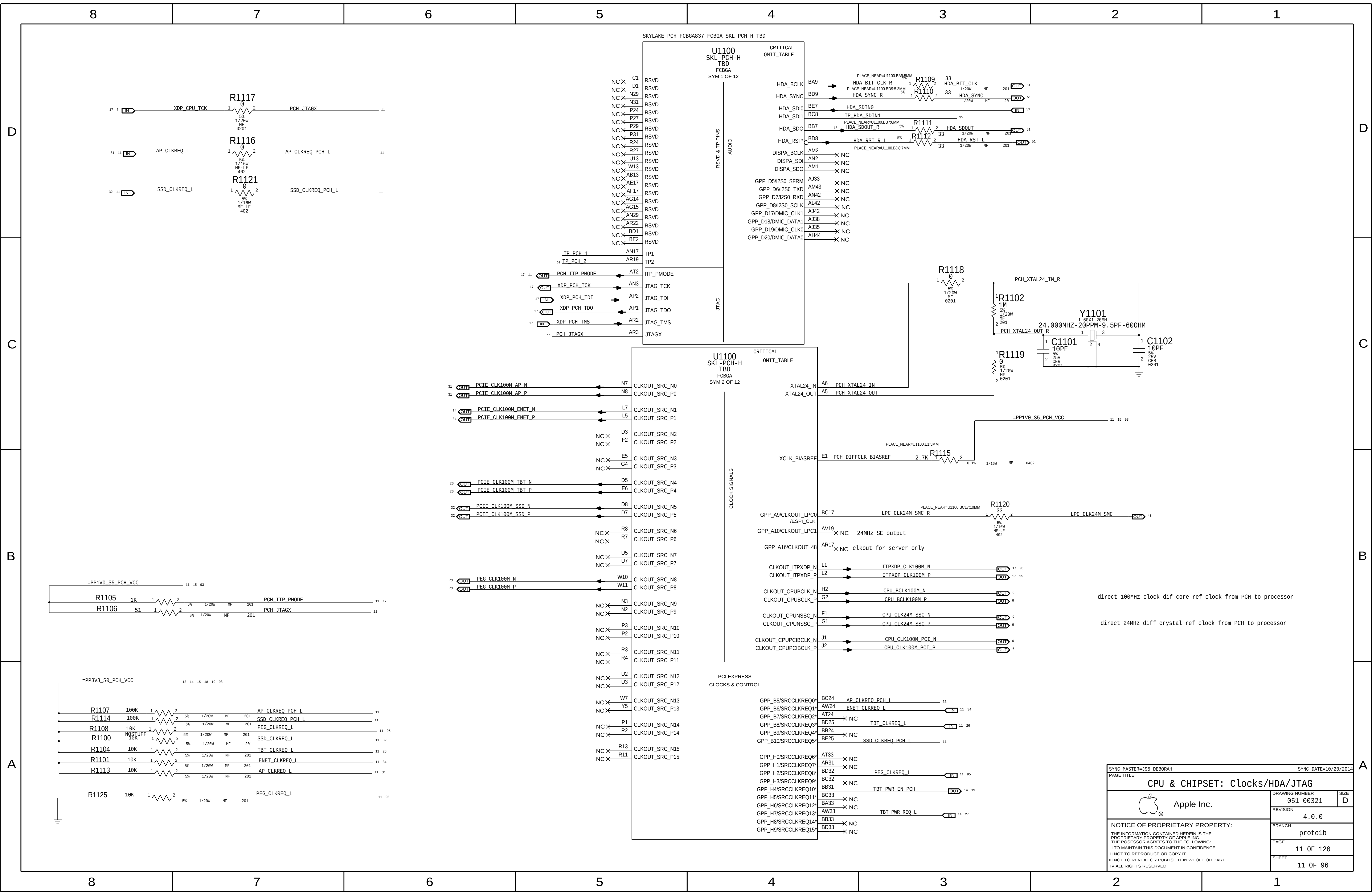
Intel Recommendation: 2x 22UF 0603 near top side cavity

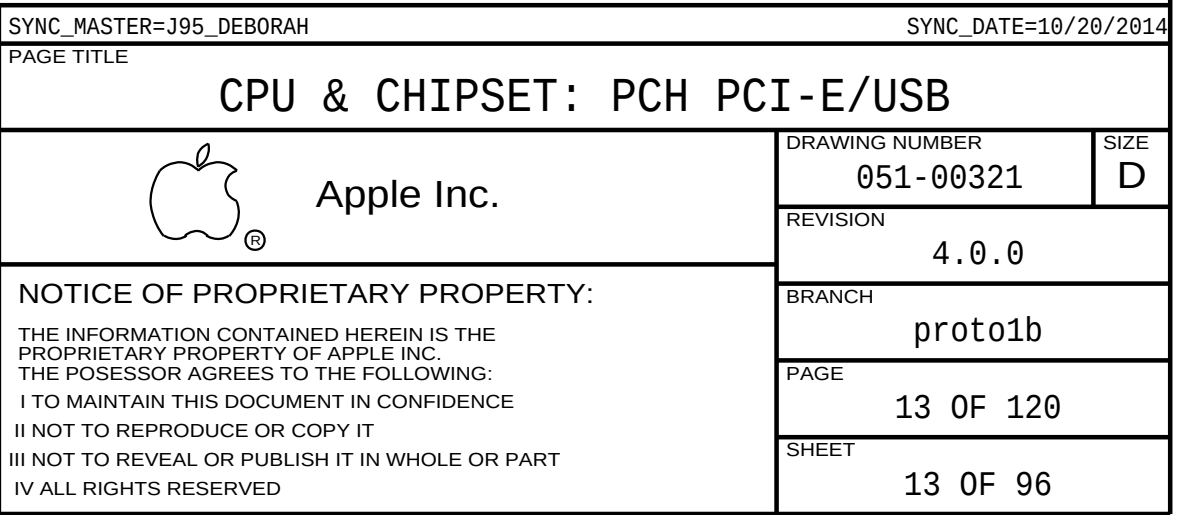
Apple Implementation: VCCST/VCCPLL: 1X 22UF 0603/2X 1UF 0402

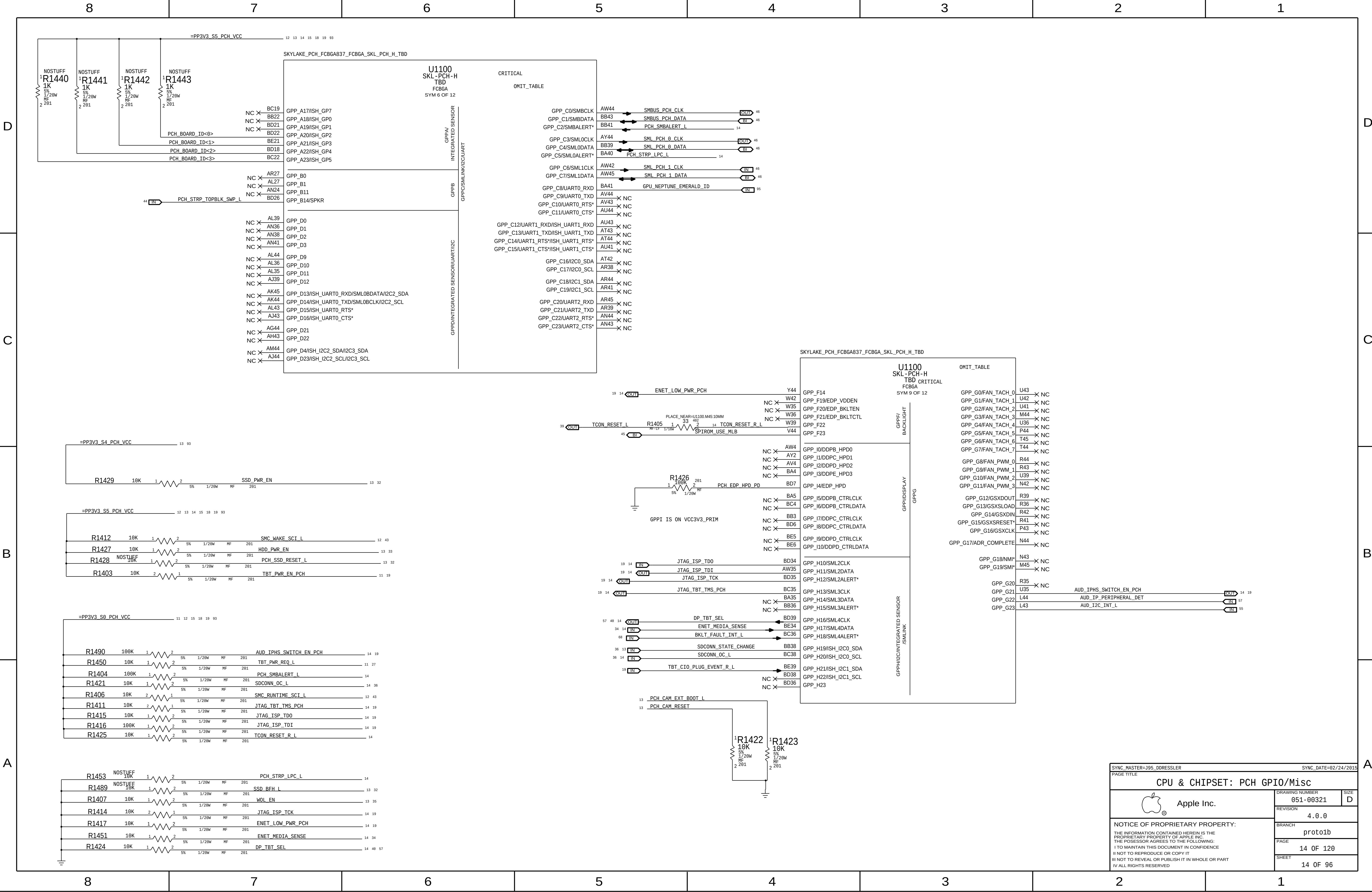
Layout Note: These caps should be placed on top side cavity.



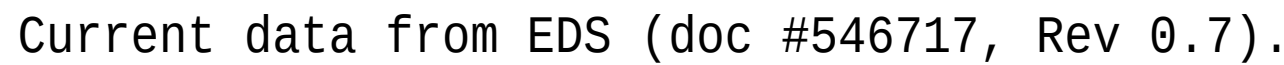
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		PAGE	14 OF 120
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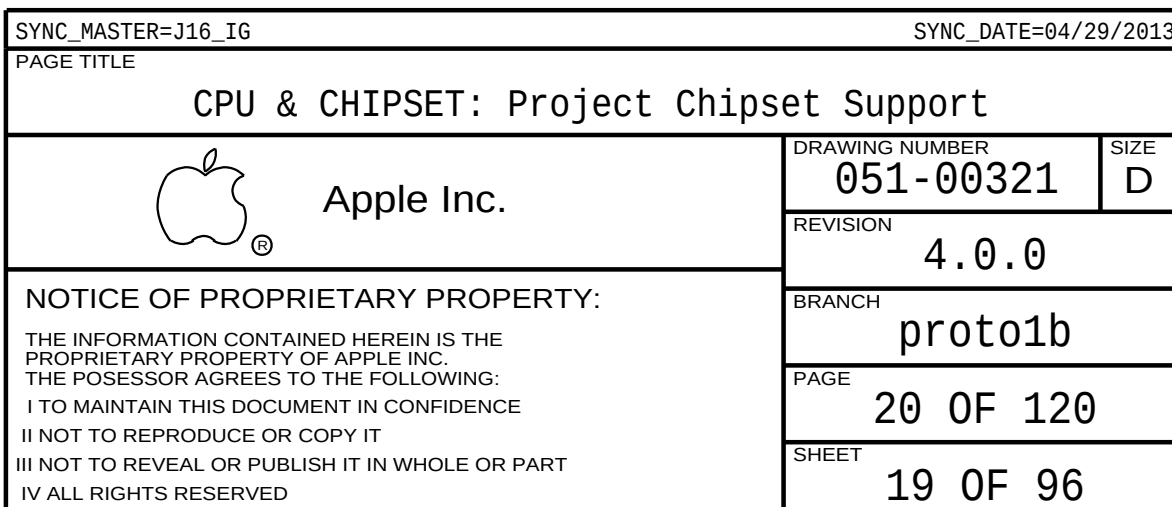
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D

C

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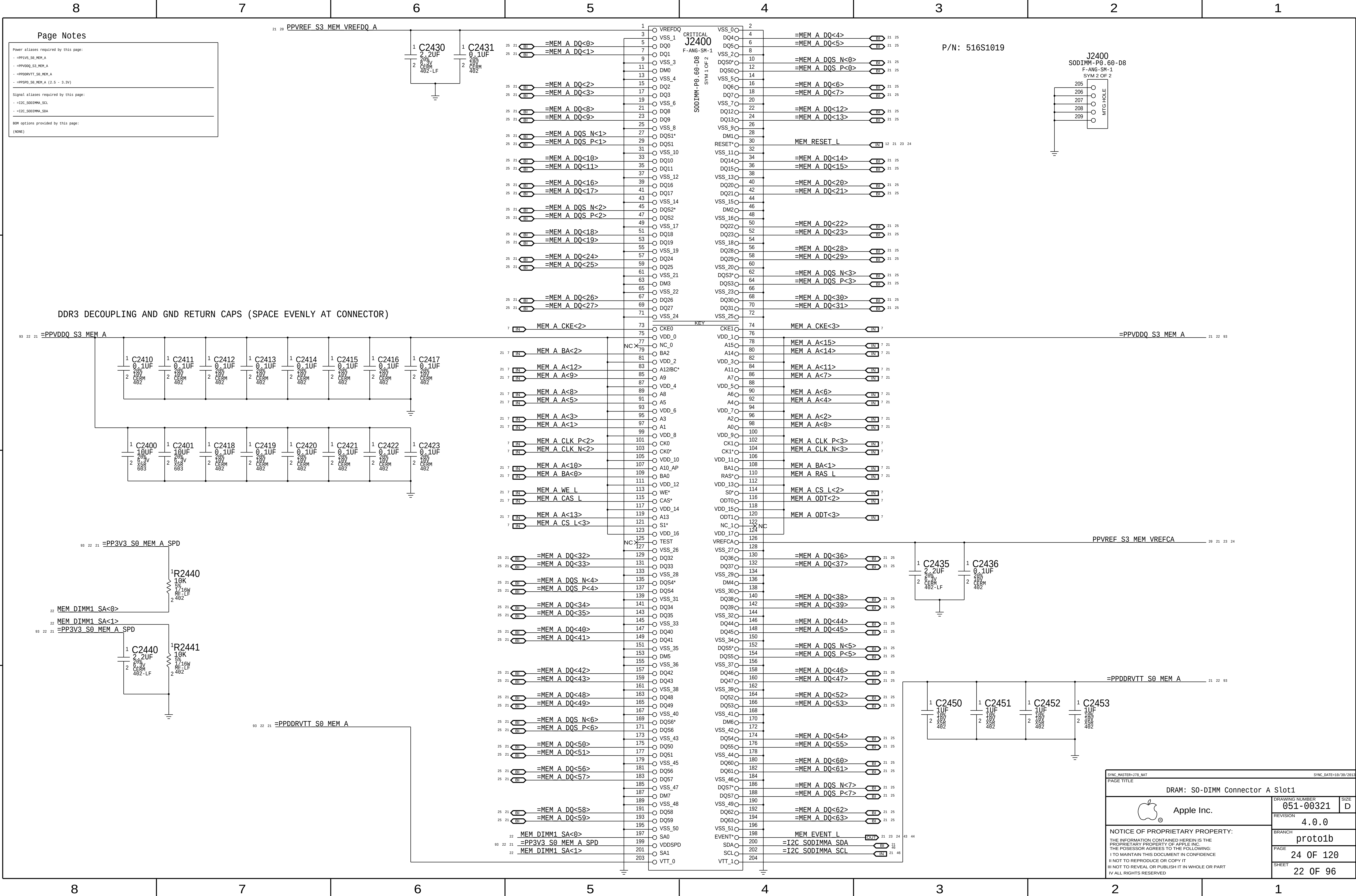
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B

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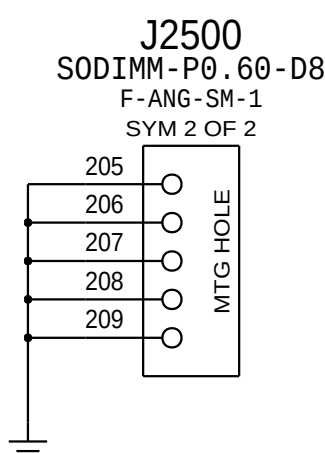
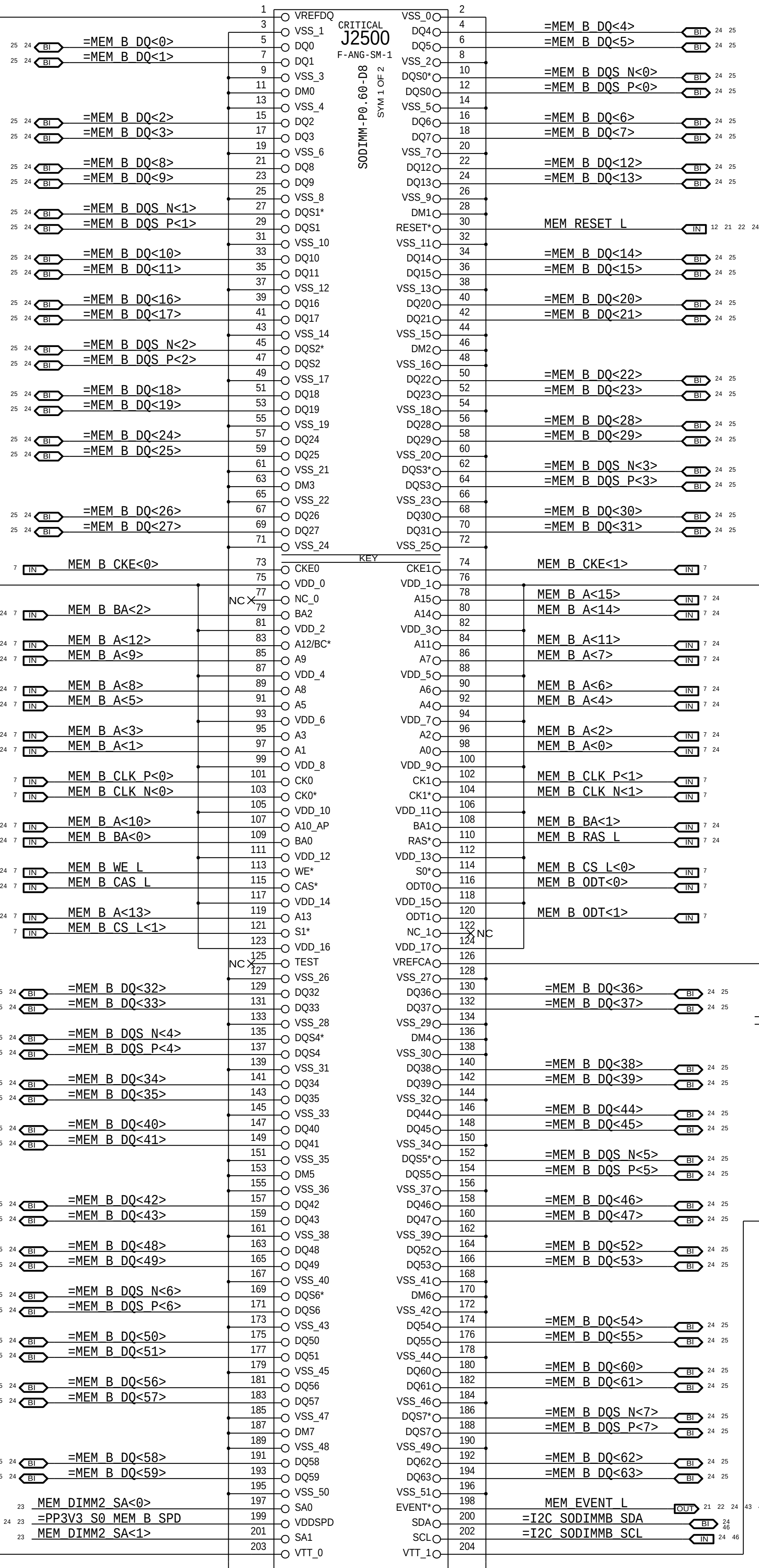
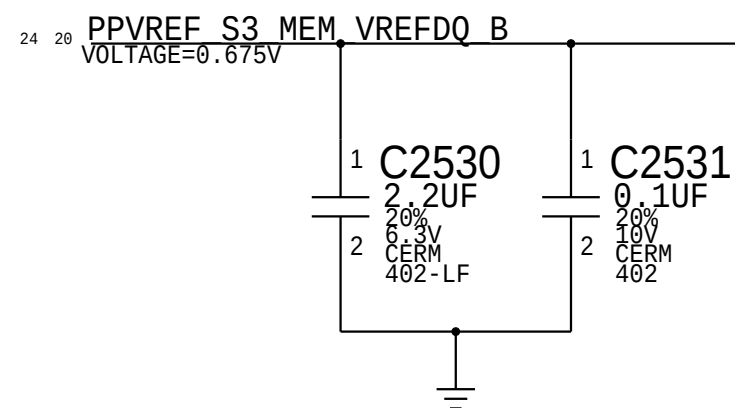
- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2

Signal aliases required by this page:

- #I2C_SODIMMA_SCL
- #I2C_SODIMMA_SDA

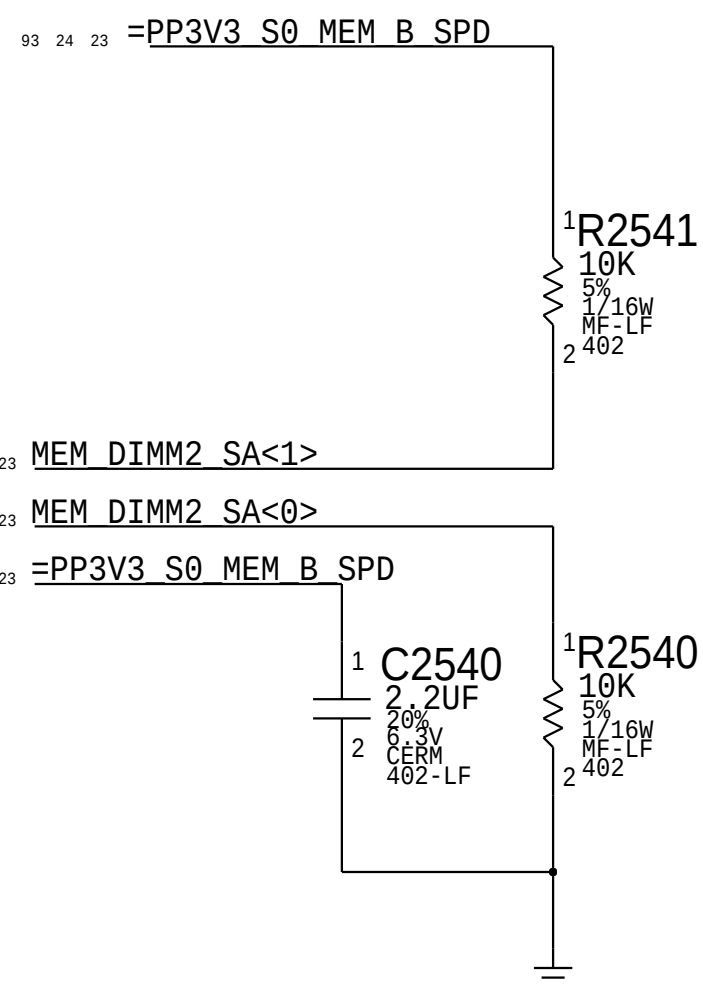
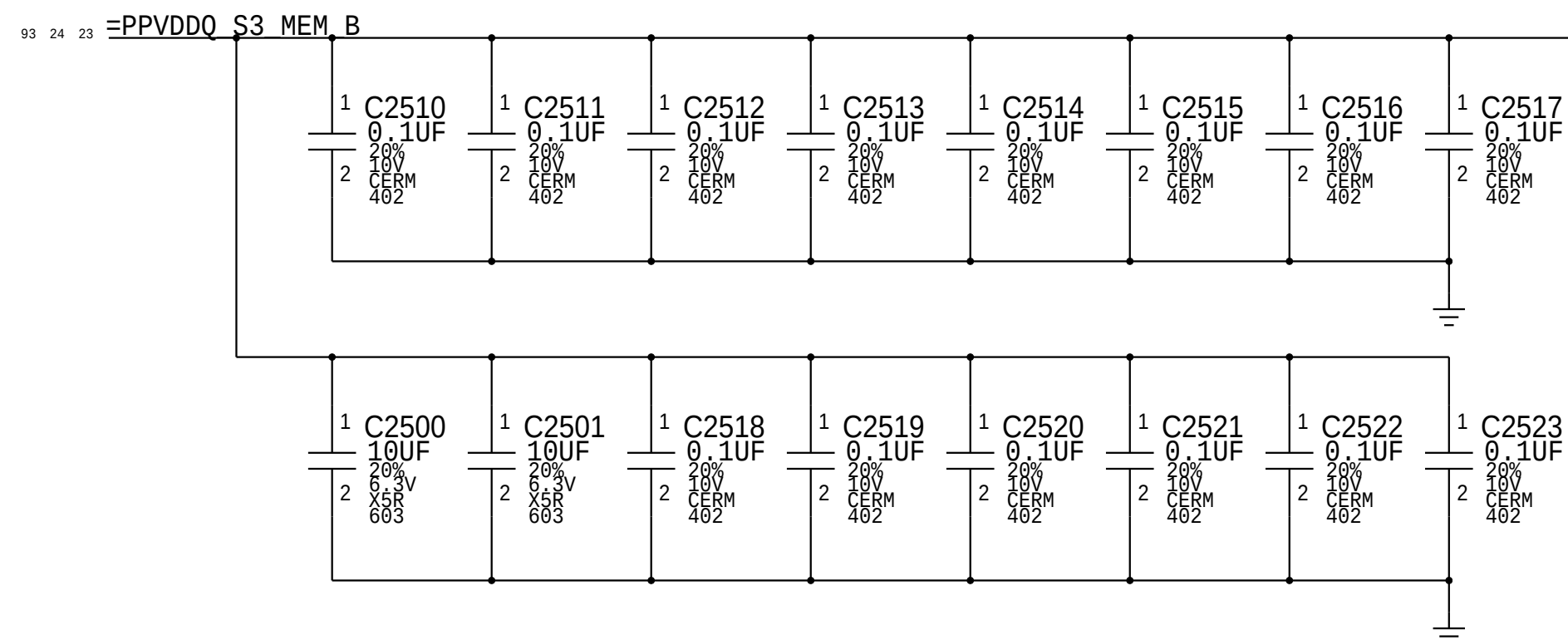
BOM options provided by this page:

(NONE)



P/N: 516S1019

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



SYNC_MASTER=J78_NAT		SYNC_DATE=10/30/2013	
PAGE TITLE			
DRAM: APPLE-DIMM CONNECTOR B SLOT0			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
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	4.0.0		
	BRANCH		proto1b
	PAGE		25 OF 120
		SHEET	23 OF 96

Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2

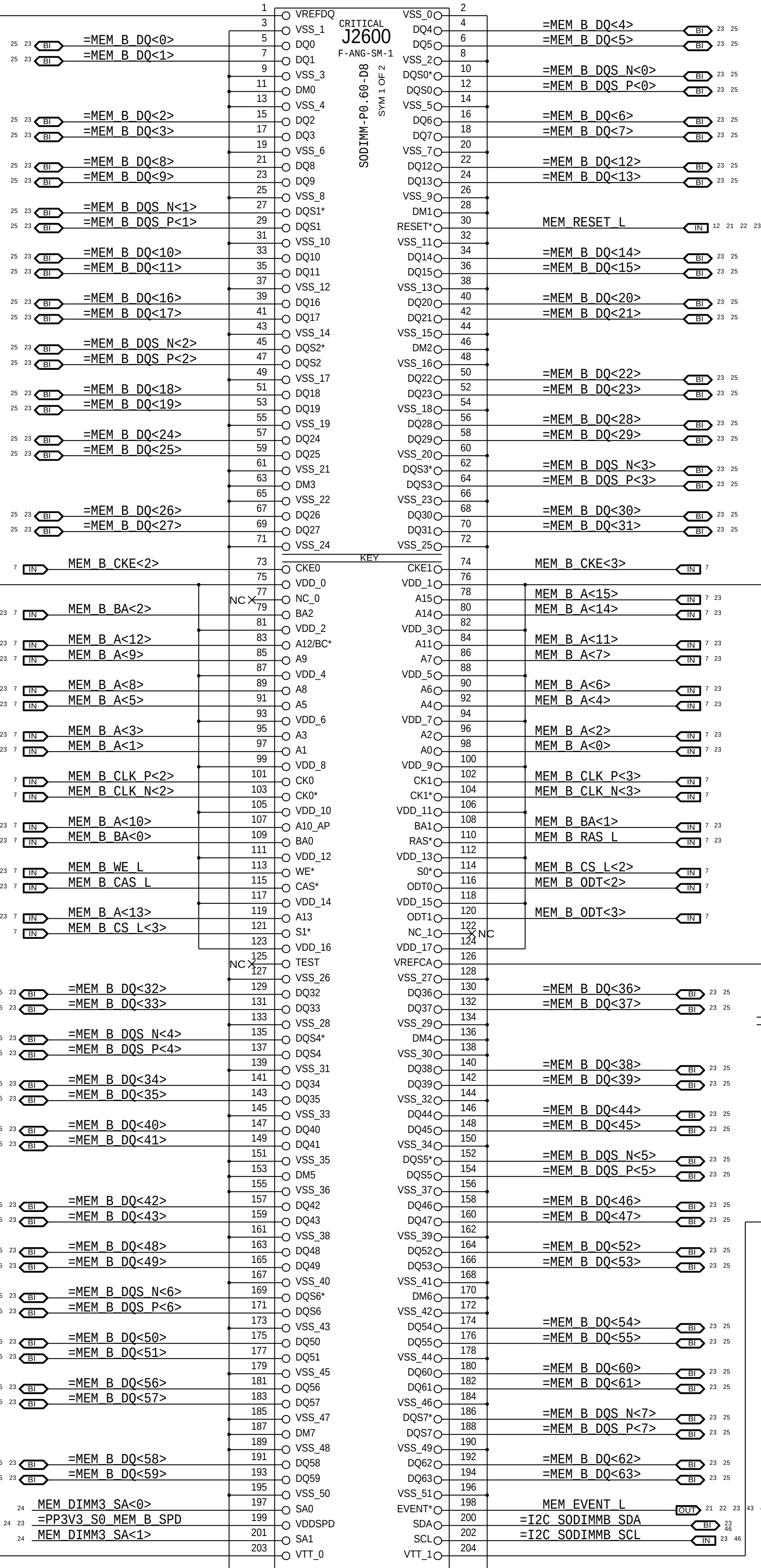
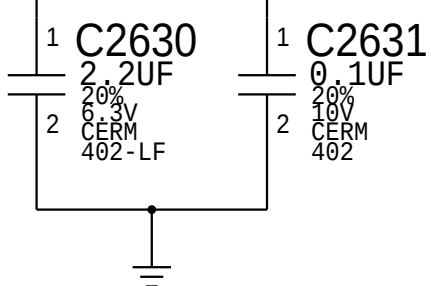
Signal aliases required by this page:

- #I2C_SODINMA_SCL
- #I2C_SODINMA_SDA

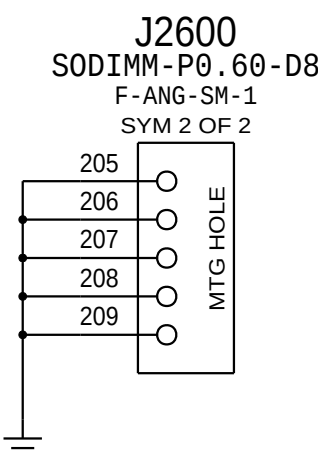
BOM options provided by this page:

(NONE)

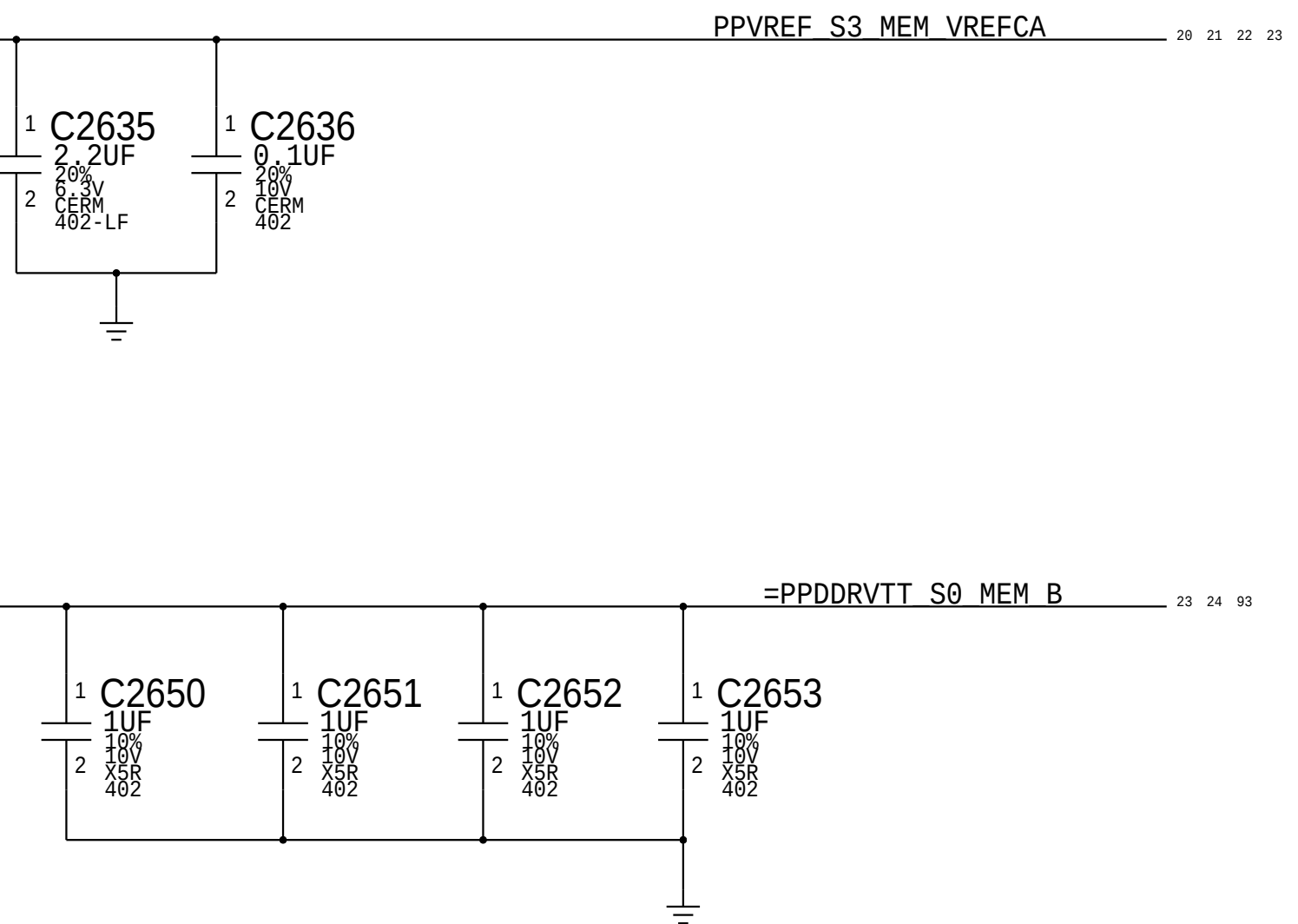
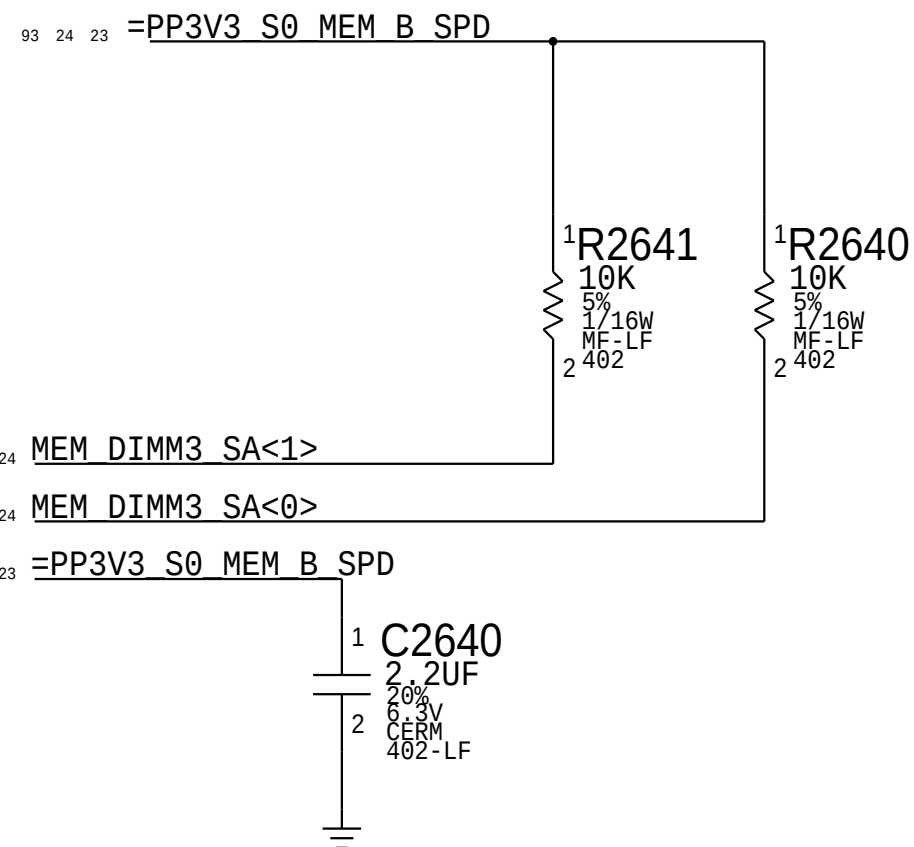
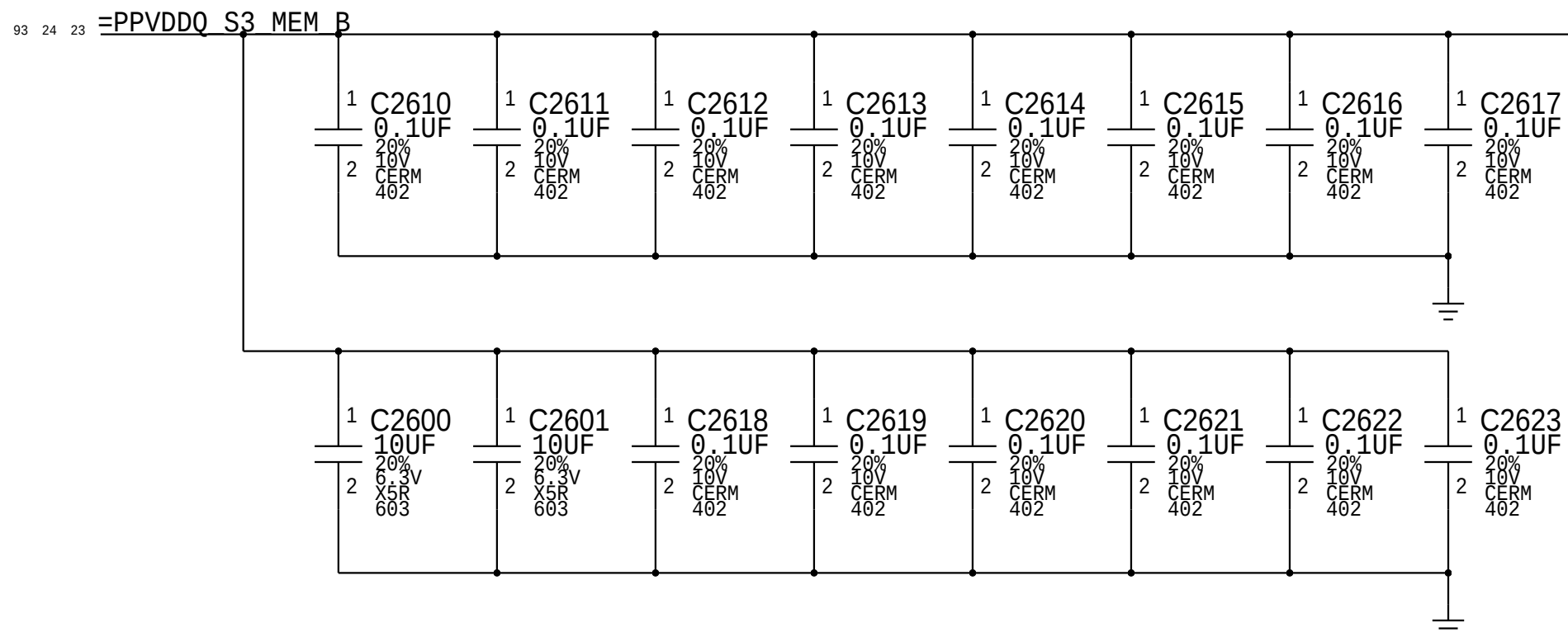
23 28 PPVREF S3 MEM VREFDQ B
VOLTAGE=0.675V



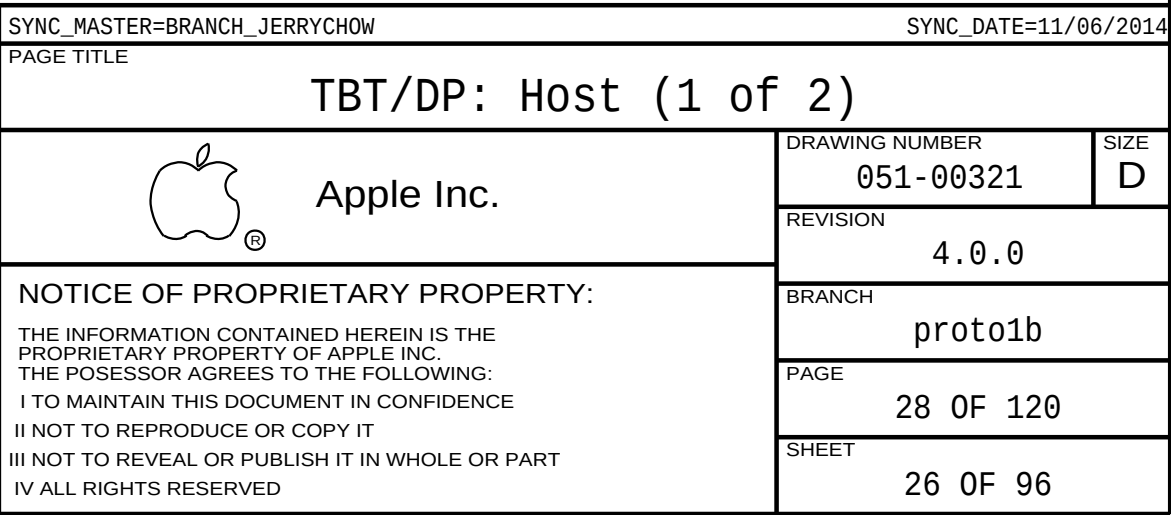
P/N: 516S1019

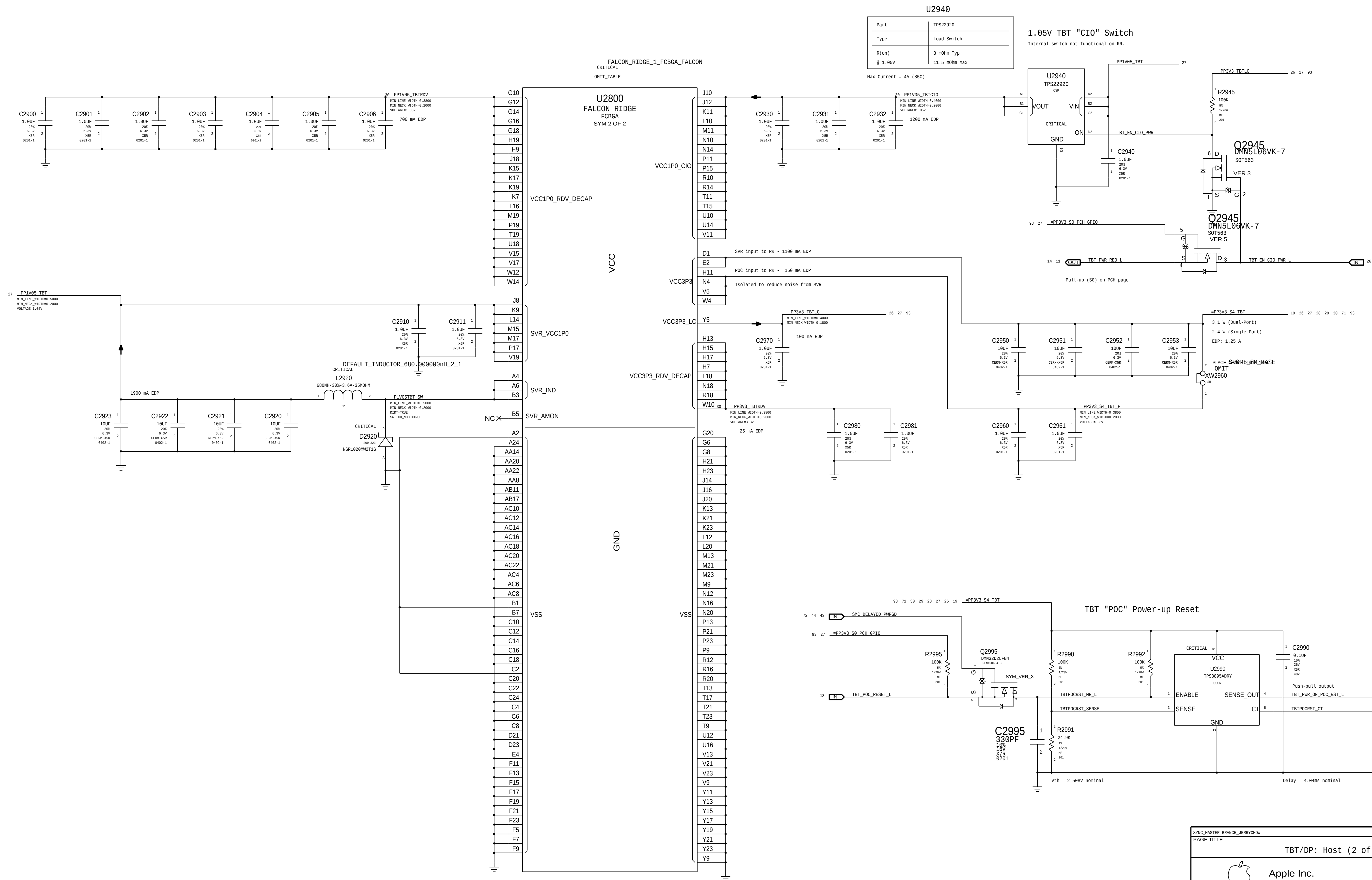


DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



SYNC_MASTER=J78_NAT		SYNC_DATE=10/30/2013	
PAGE TITLE			
DRAM: SO-DIMM CONNECTOR B SLOT1			
 Apple Inc.		DRAWING NUMBER	
		051-00321	
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		4.0.0	
		BRANCH	
		proto1b	
		PAGE	
		26 OF 120	
		SHEET	
		28 OF 96	





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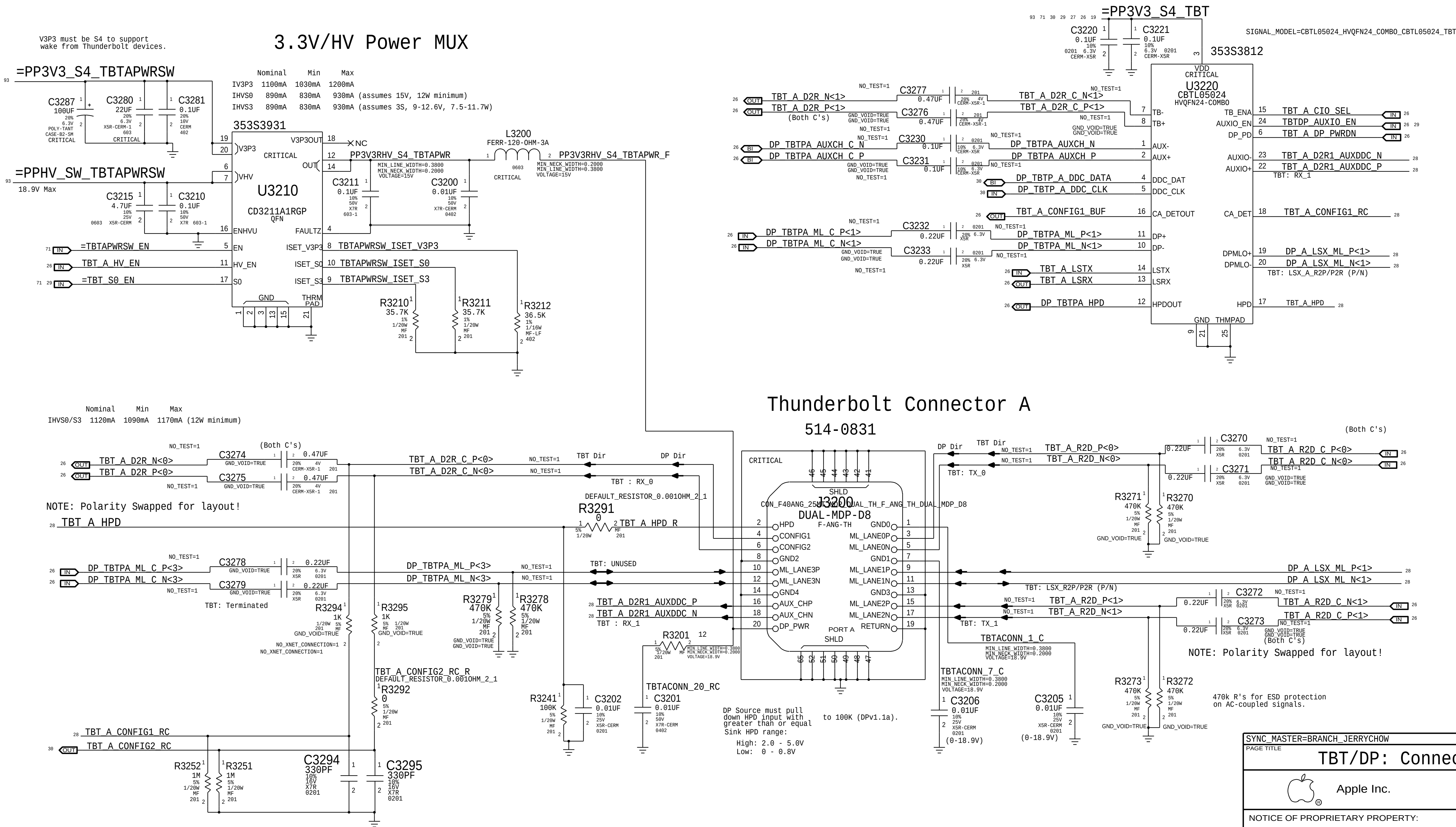
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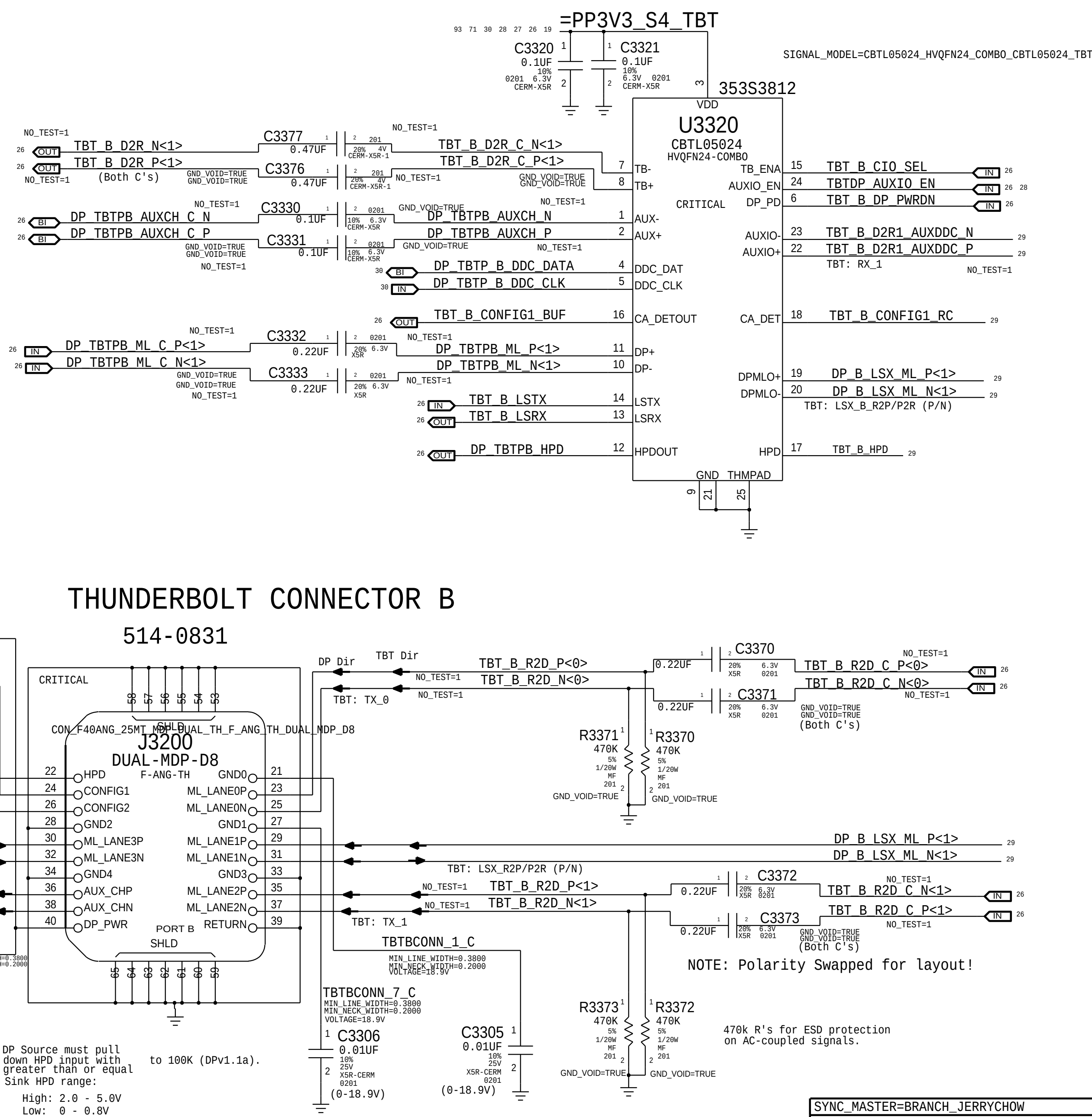
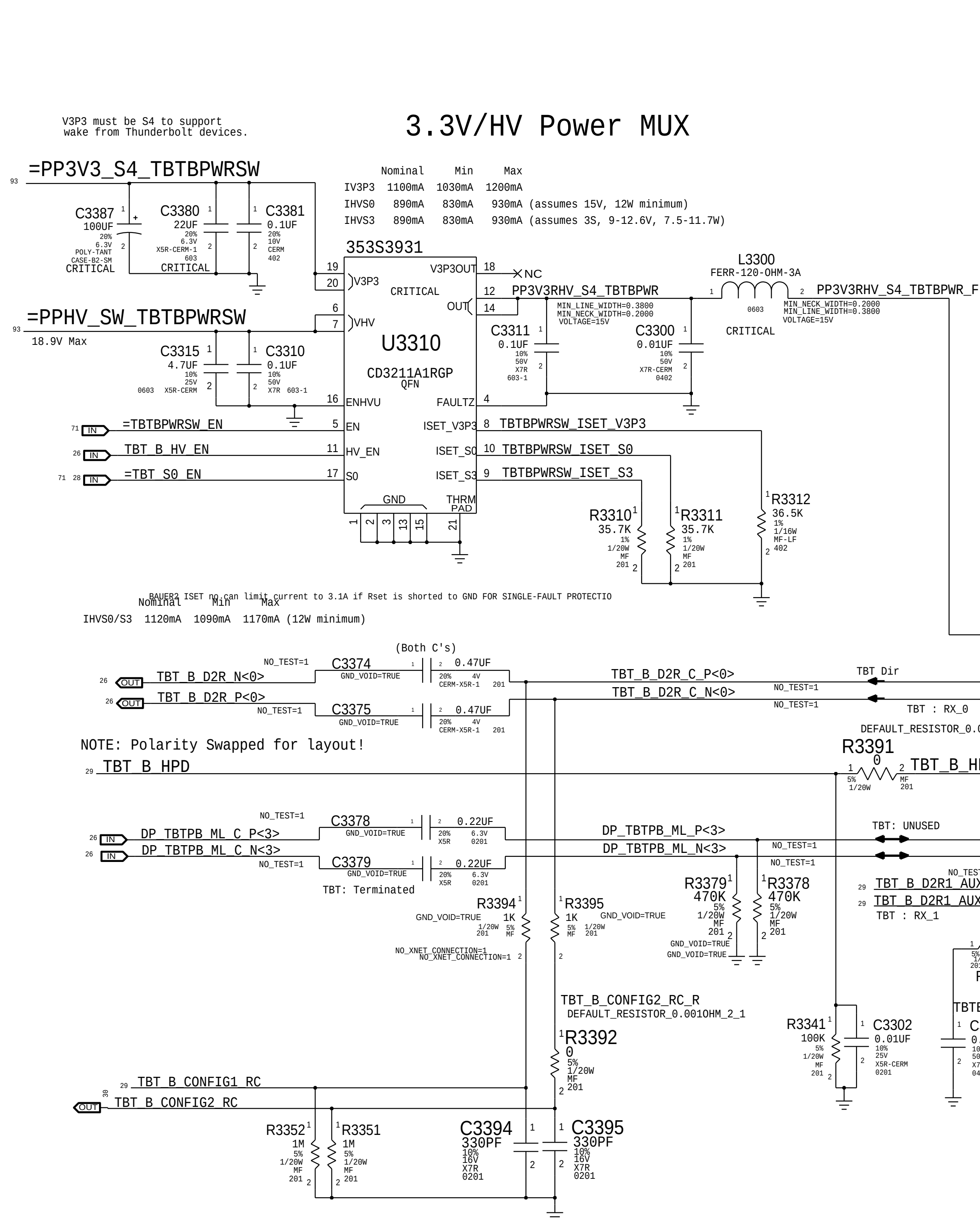
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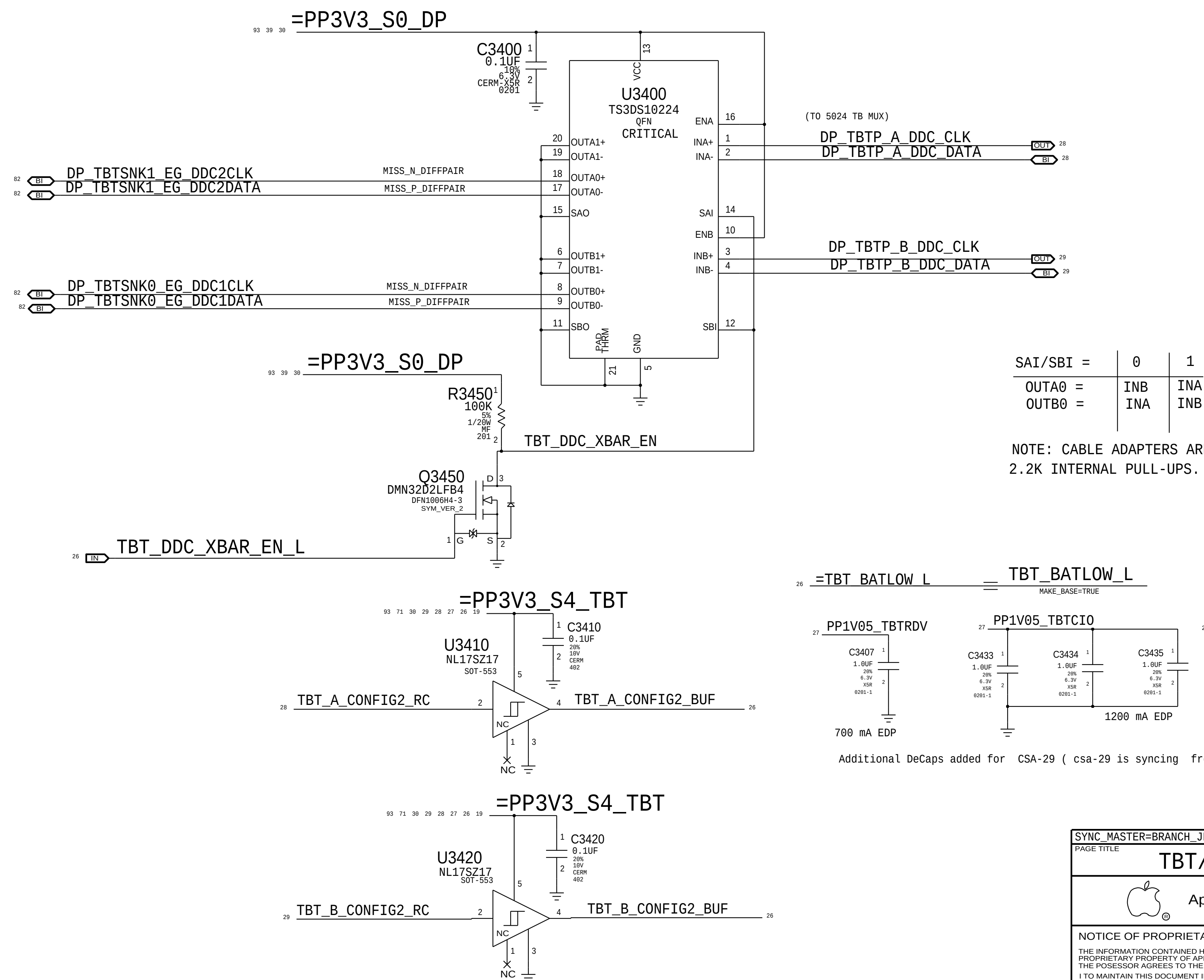
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SYNC_MASTER=BRANCH_JERRYCHOW		SYNC_DATE=09/10/2014	
PAGE TITLE			
TBT/DP: Connector B			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
	REVISION		
NOTICE OF PROPRIETARY PROPERTY:		4.0.0	
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		proto1b	
		PAGE	33 OF 120
		SHEET	29 OF 96

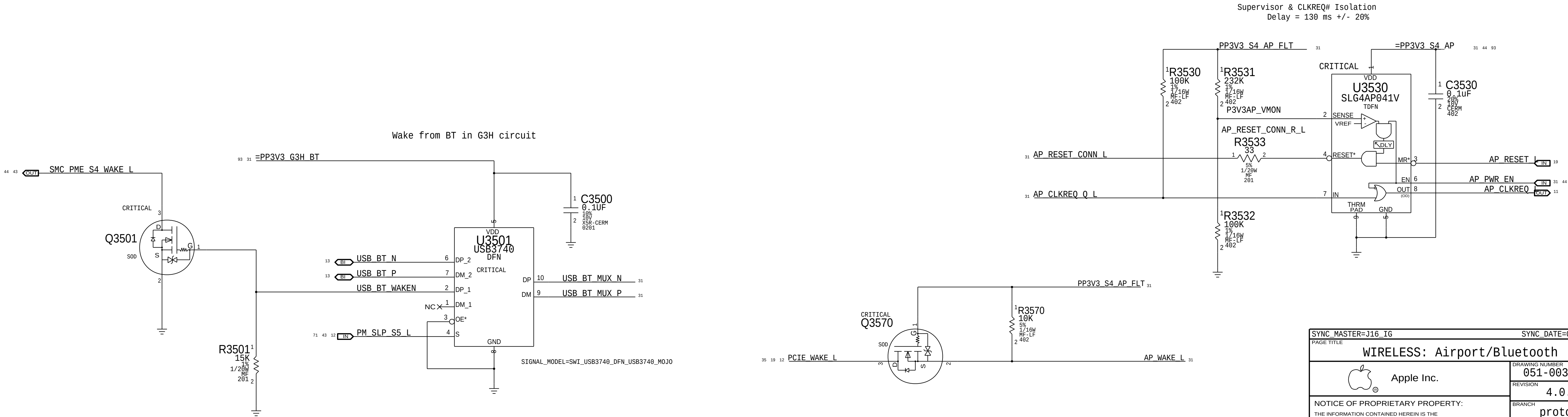
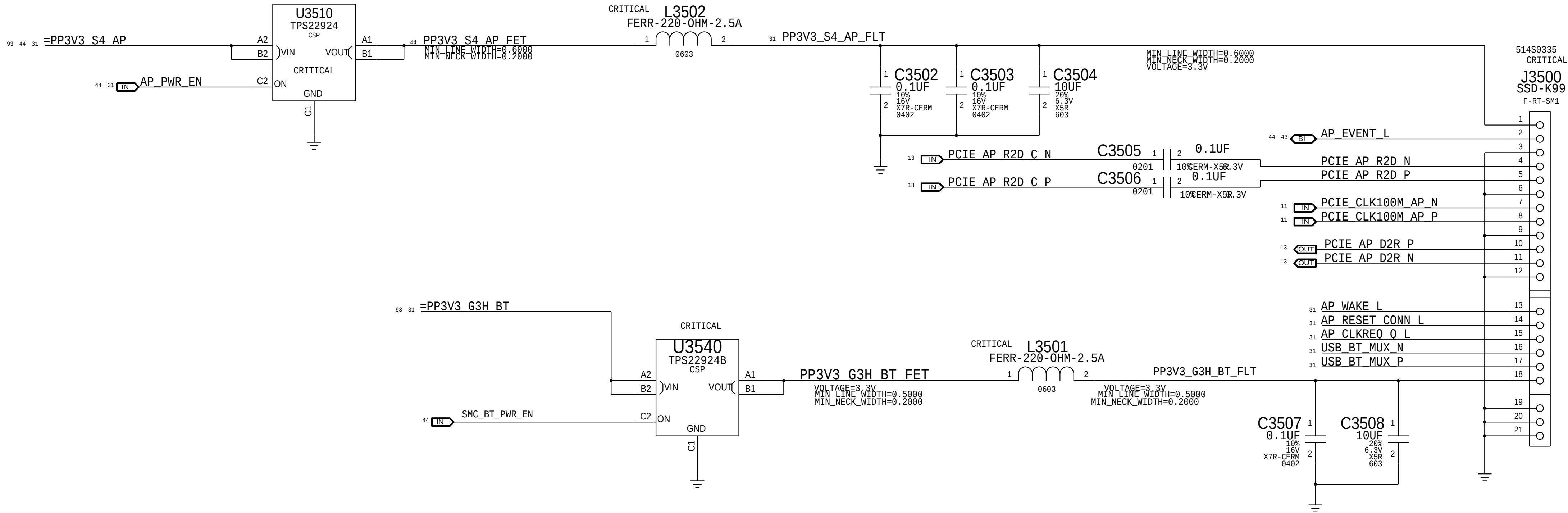
Dual-Port Host DDC Crossbar



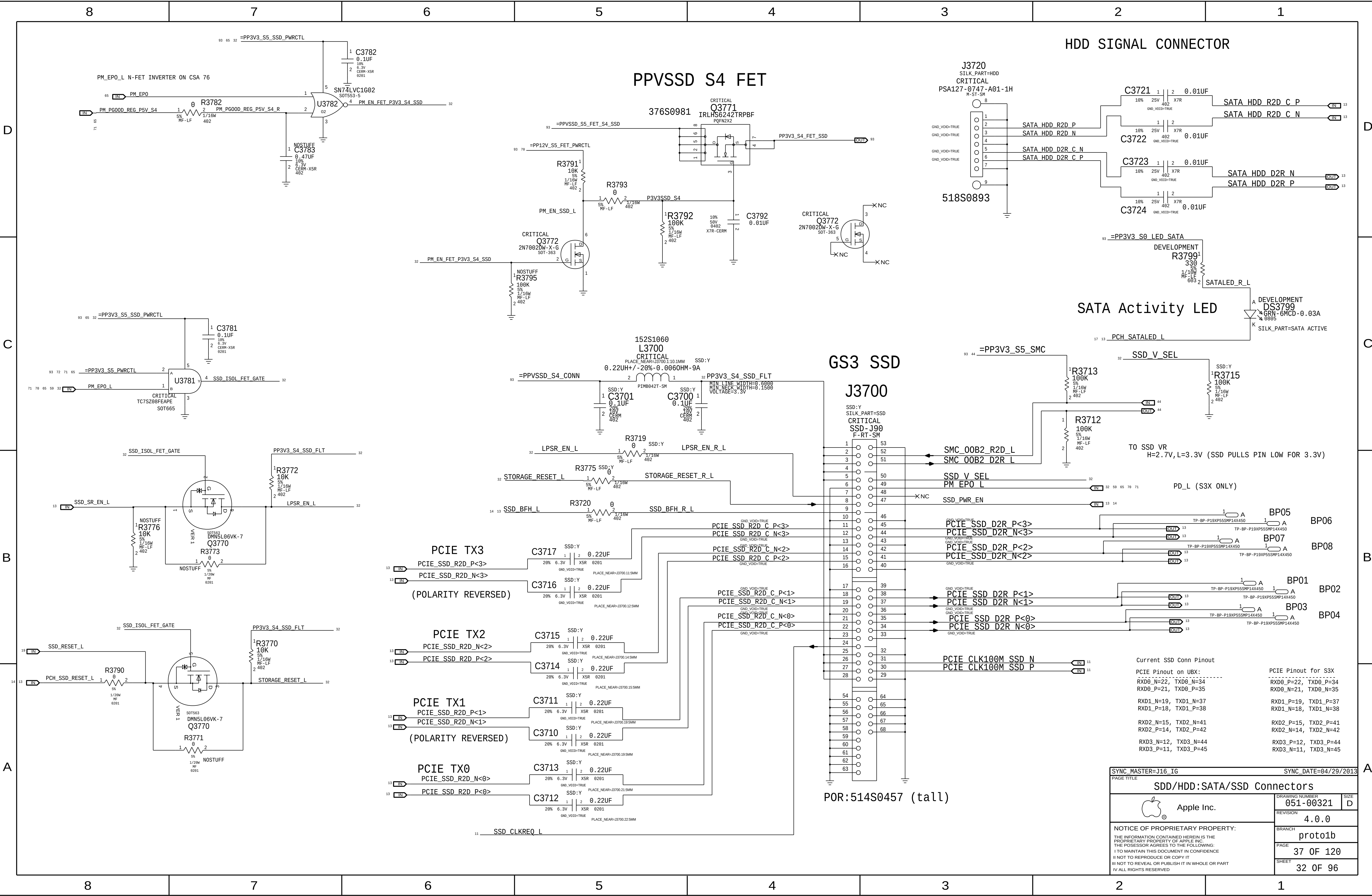
SYNC_MASTER=BRANCH_JERRYCHOW		SYNC_DATE=11/05/2014	
PAGE TITLE			
TBT/DP: DDC Crossbar			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
	REVISION		
	4.0.0		
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BRANCH		proto1b	
PAGE		34 OF 120	
SHEET		30 OF 96	

AP & BT Load Switch	
SWITCH	TPS22924B
CHANNEL	N-TYPE
RDS(ON)	18.4 MOHM @3.3V
LOADING	2 A (EDP)

AIRPORT
BLUETOOTH

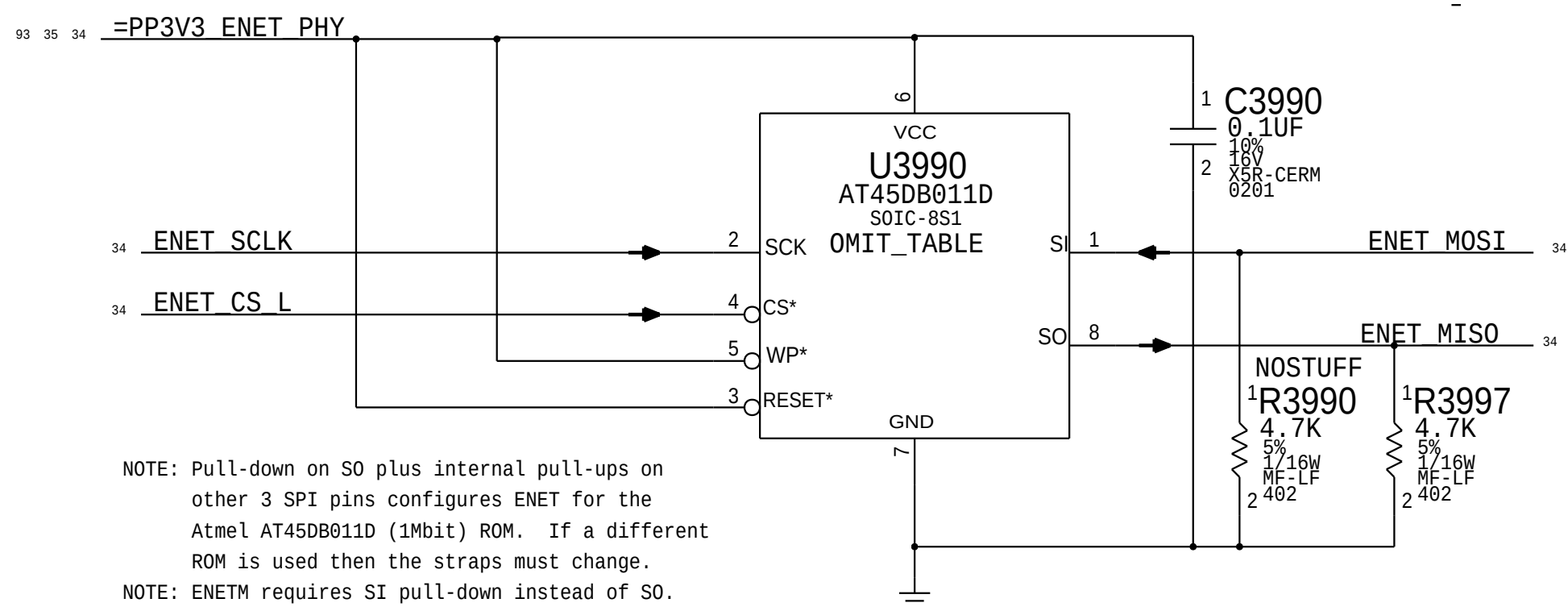
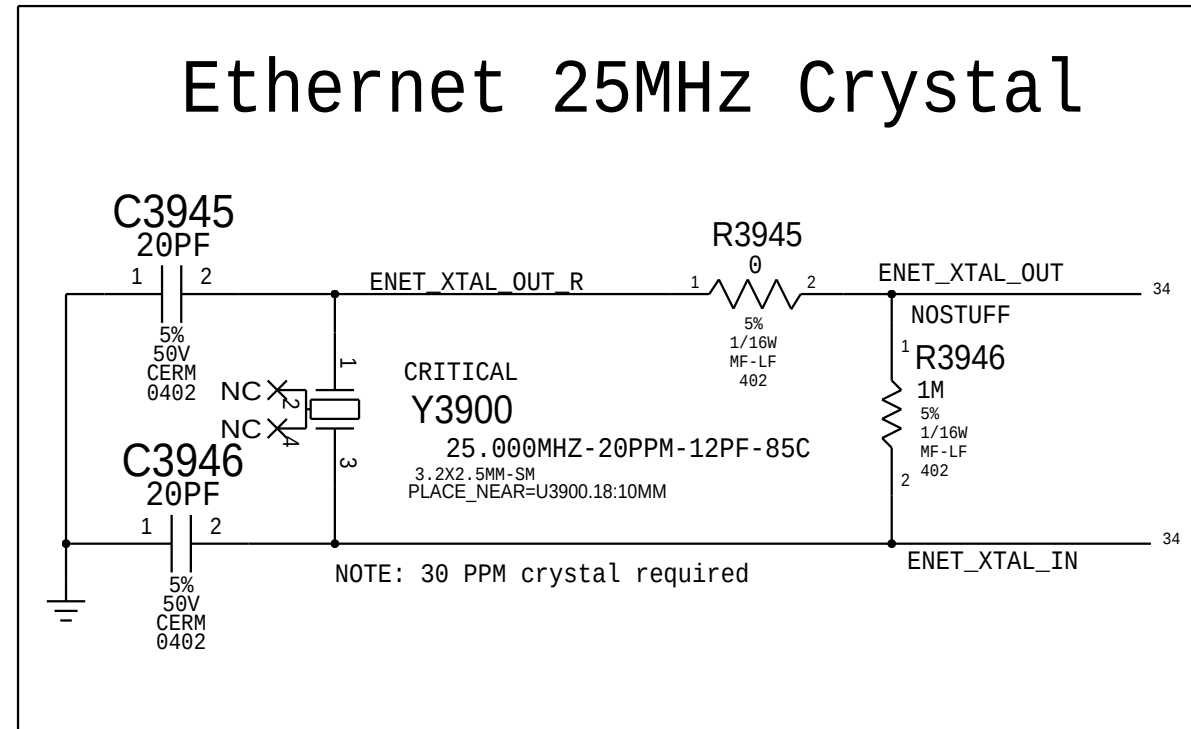
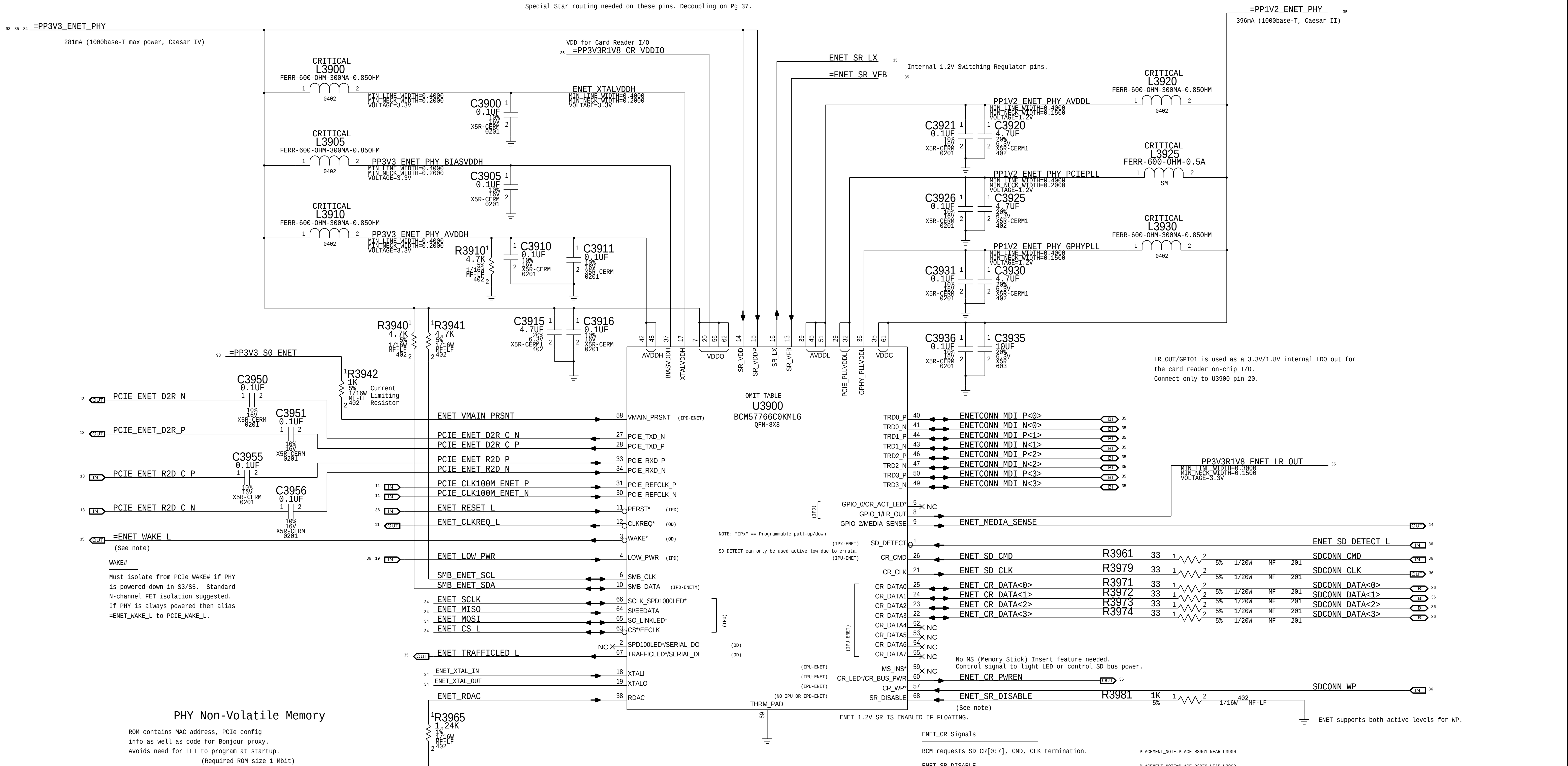
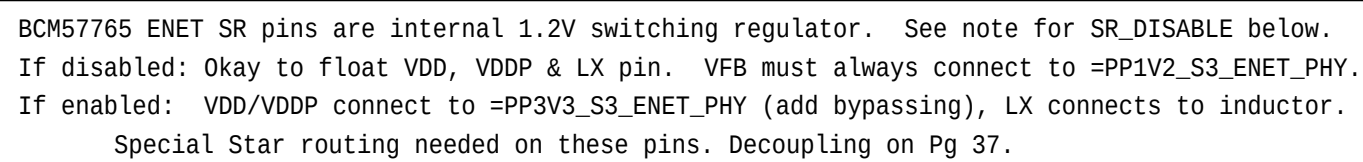


SYNC_MASTER=J16_I6		SYNC_DATE=04/29/2013	
PAGE TITLE			
WIRELESS: Airport/Bluetooth			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
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		PAGE	35 OF 120
		SHEET	31 OF 96



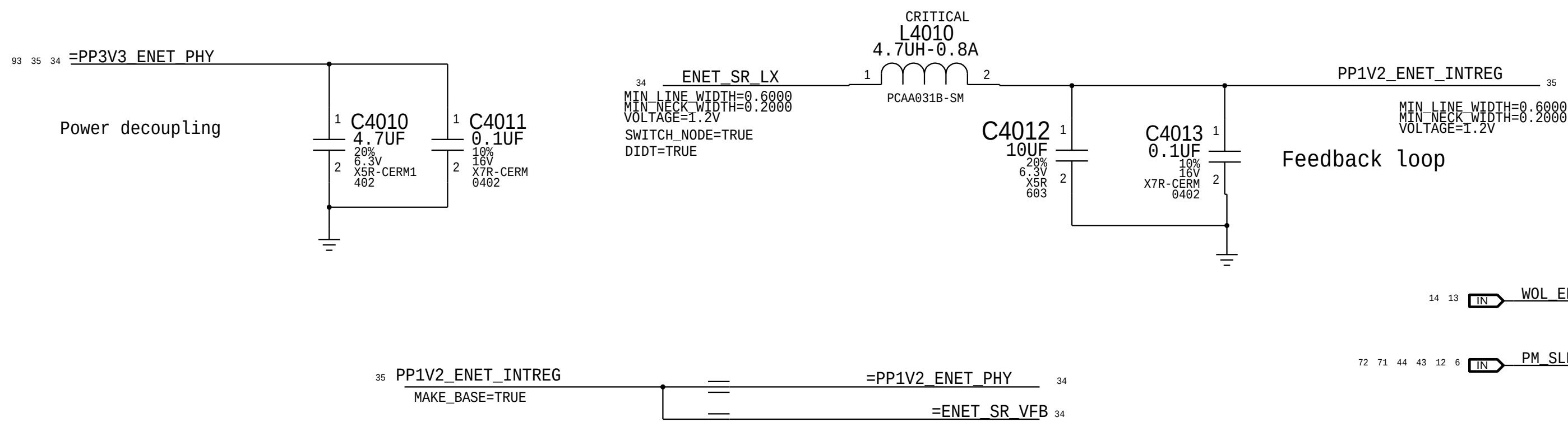
Current SSD Conn Pinout		PCIE Pinout on USB:		PCIE Pinout for S3X	
RXD0_N=22,	TXD0_N=34	RXD0_P=21,	TXD0_P=35	RXD0_P=22,	TXD0_P=34
RXD0_P=21,	TXD0_P=35	RXD0_N=21,	TXD0_N=35	RXD0_N=21,	TXD0_N=35
RXD1_N=19,	TXD1_N=37	RXD1_P=18,	TXD1_P=38	RXD1_P=19,	TXD1_P=38
RXD1_P=18,	TXD1_P=38	RXD1_N=18,	TXD1_N=38	RXD1_N=18,	TXD1_N=38
RXD2_N=15,	TXD2_N=41	RXD2_P=14,	TXD2_P=42	RXD2_P=15,	TXD2_P=41
RXD2_P=14,	TXD2_P=42	RXD2_N=14,	TXD2_N=42	RXD2_N=14,	TXD2_N=42
RXD3_N=12,	TXD3_N=44	RXD3_P=11,	TXD3_P=45	RXD3_P=12,	TXD3_P=44
RXD3_P=11,	TXD3_P=45	RXD3_N=11,	TXD3_N=45	RXD3_N=11,	TXD3_N=45

SYNC_MASTER=J16_I6		SYNC_DATE=04/29/2013	
PAGE TITLE			
SDD/HDD:SATA/SSD Connectors			
	Apple Inc.	DRAWING NUMBER	051-00321
		SIZE	D
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		BRANCH	proto1b
		PAGE	37 OF 120
		SHEET	32 OF 96



SYNC_MASTER=J16_IG		SYNC_DATE=04/29/2013	
PAGE TITLE			
ETHERNET: PHY (CAESAR IV)			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00321	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		4.0.0	
		BRANCH	proto1b
		PAGE	39 OF 120
		SHEET	34 OF 96

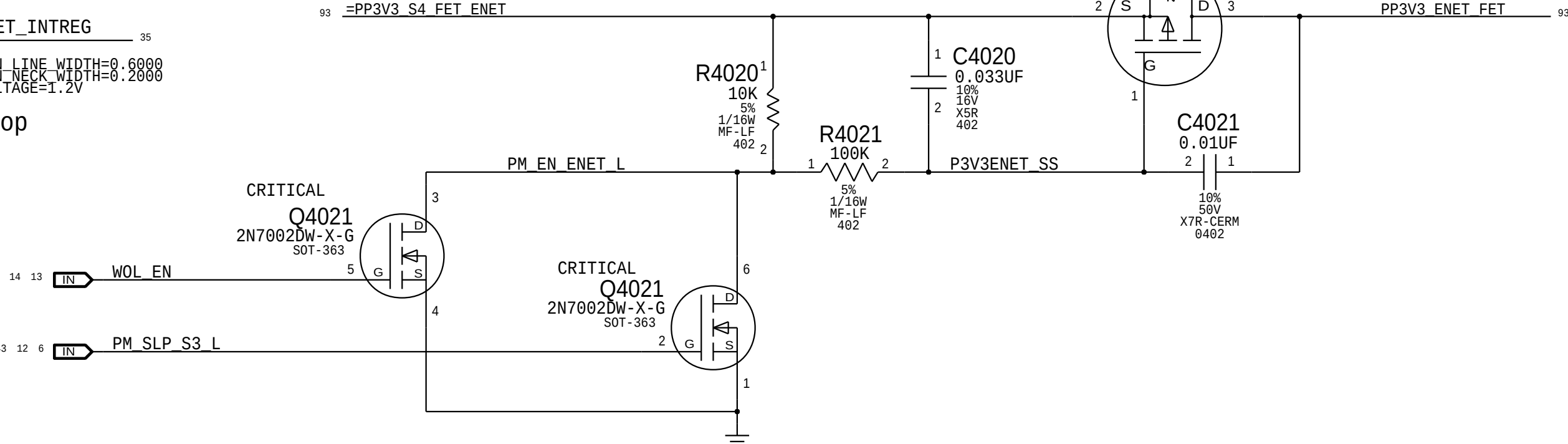
CAESAR IV 1.2V INT.VR CMPTS



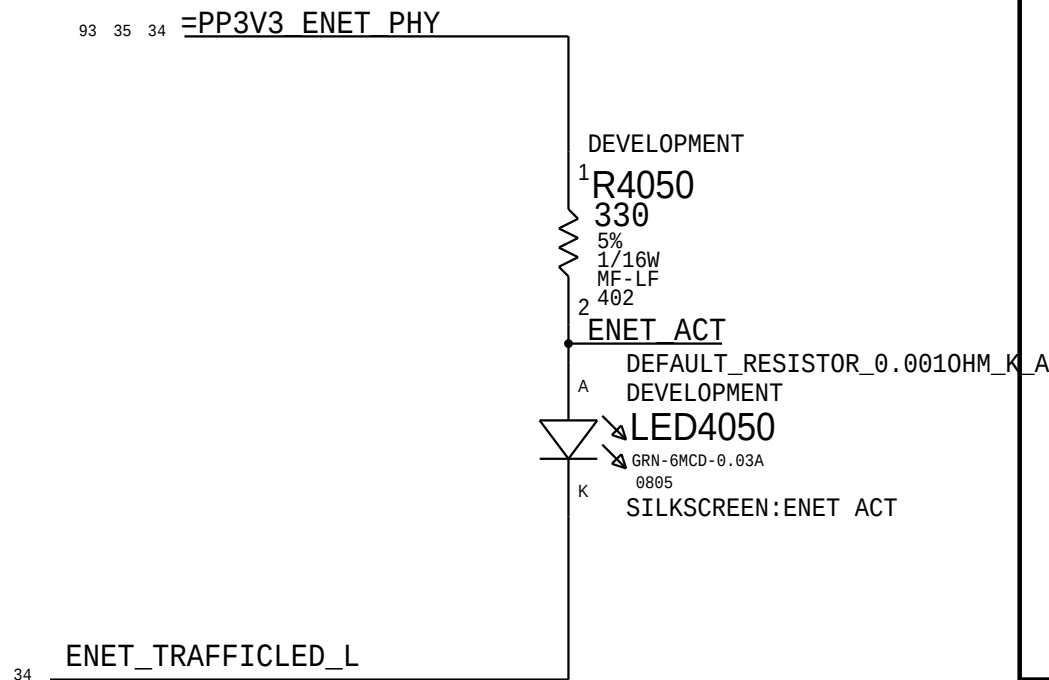
ENET Enable Generation

"ENET" = "S8" || ("S4" && "WOL_EN")

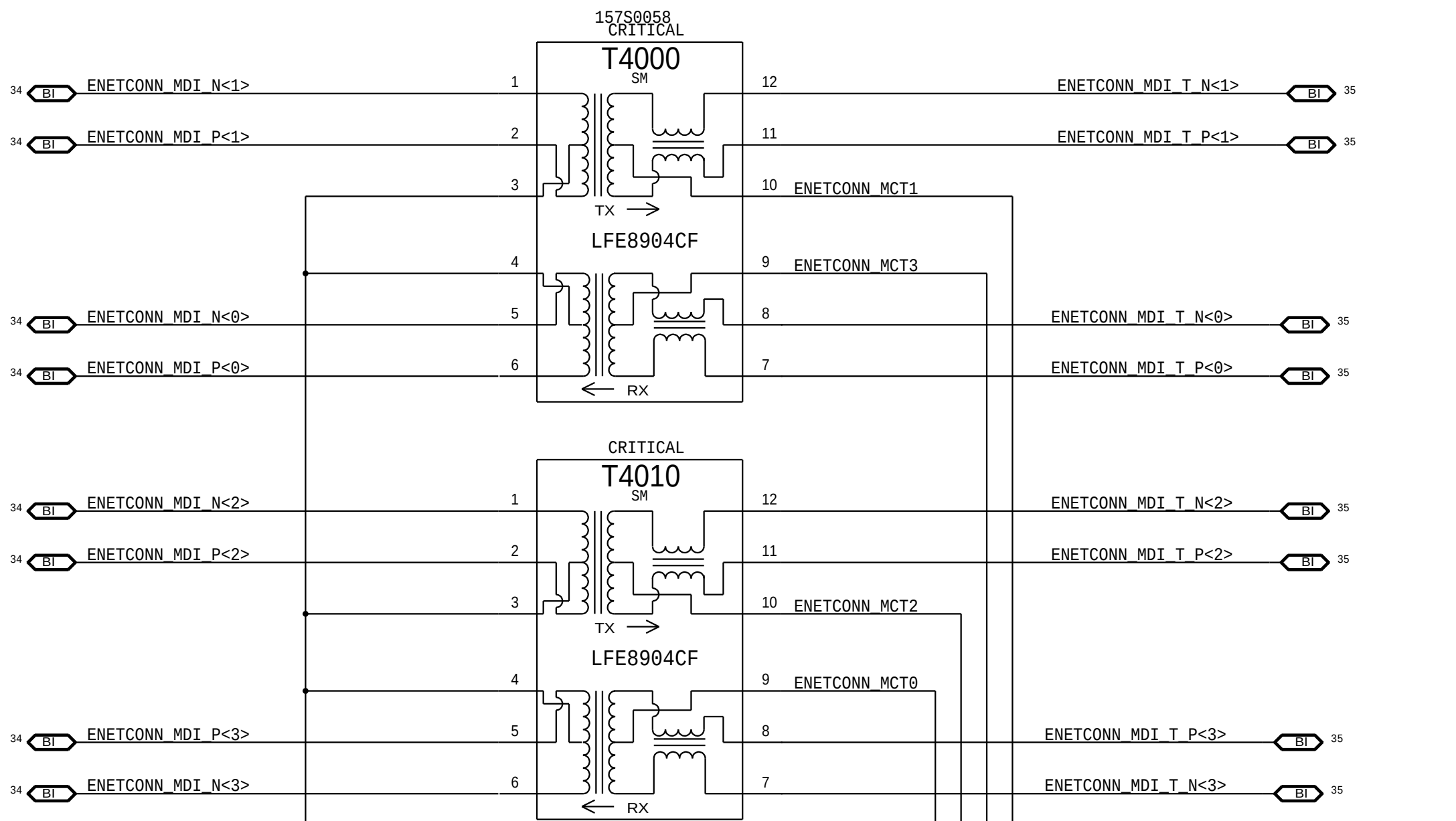
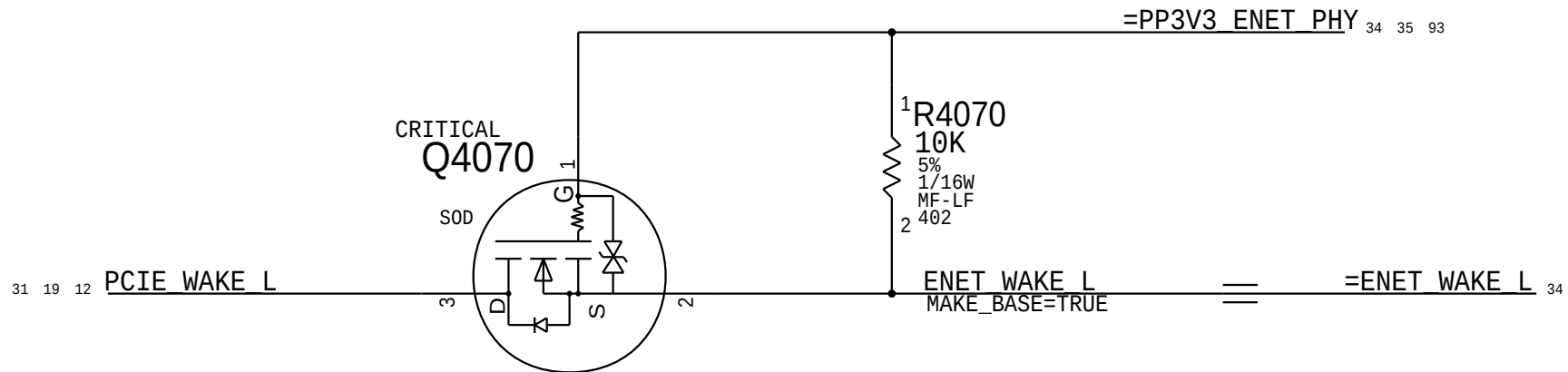
3.3V ENET FET



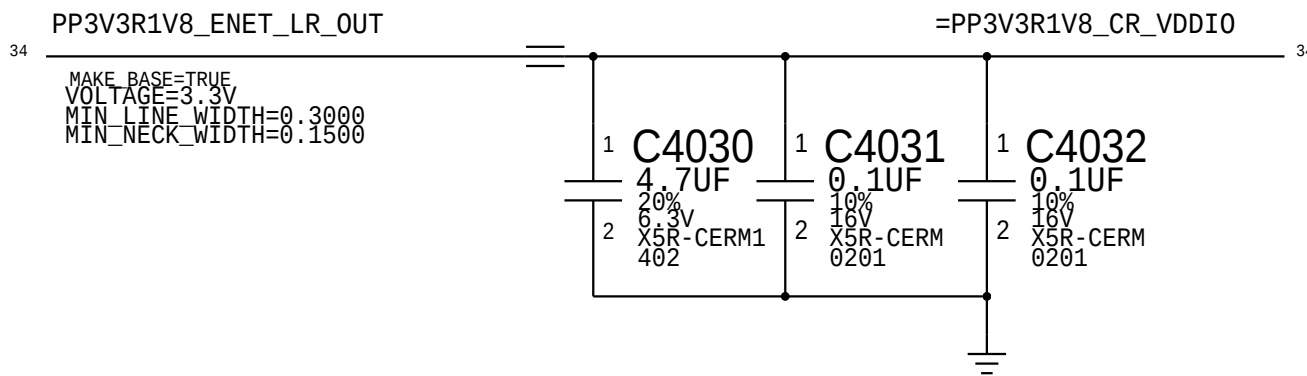
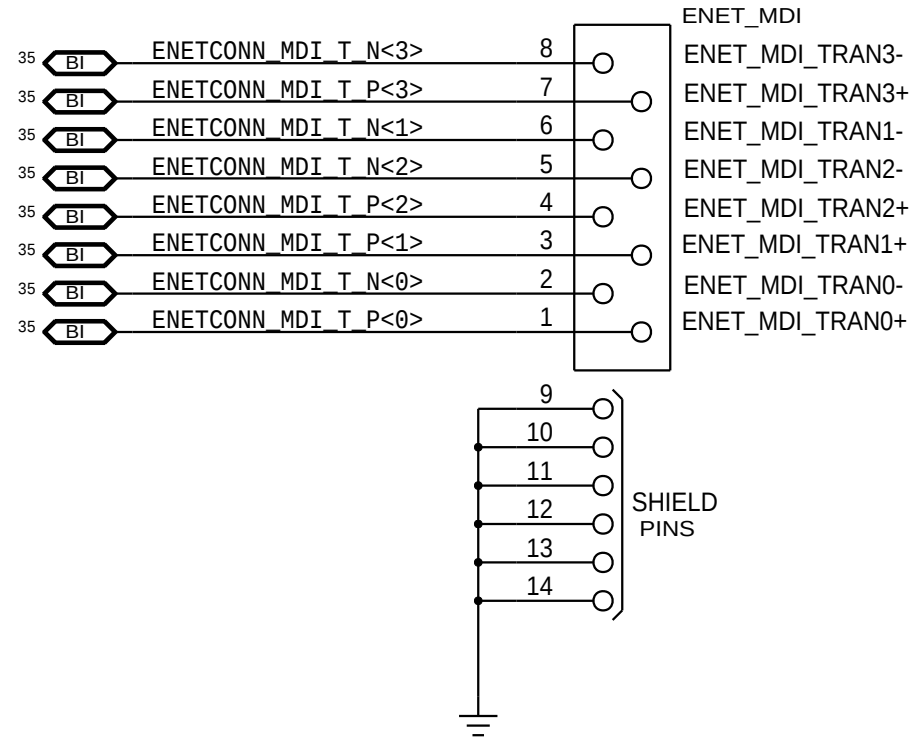
CAESAR IV ACTIVITY LED



CAESAR IV WAKE# ISOLATION



RCPT-RJ45-D8
F-ANG-TH



SYNC_MASTER=J16_IG		SYNC_DATE=12/07/2012	
PAGE TITLE			
ETHERNET: Support & Connector			
 Apple Inc.	DRAWING NUMBER		051-00321
	SIZE		D
	REVISION		4.0.0
	BRANCH		proto1b
	PAGE		40 OF 120
	SHEET		35 OF 96
	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		

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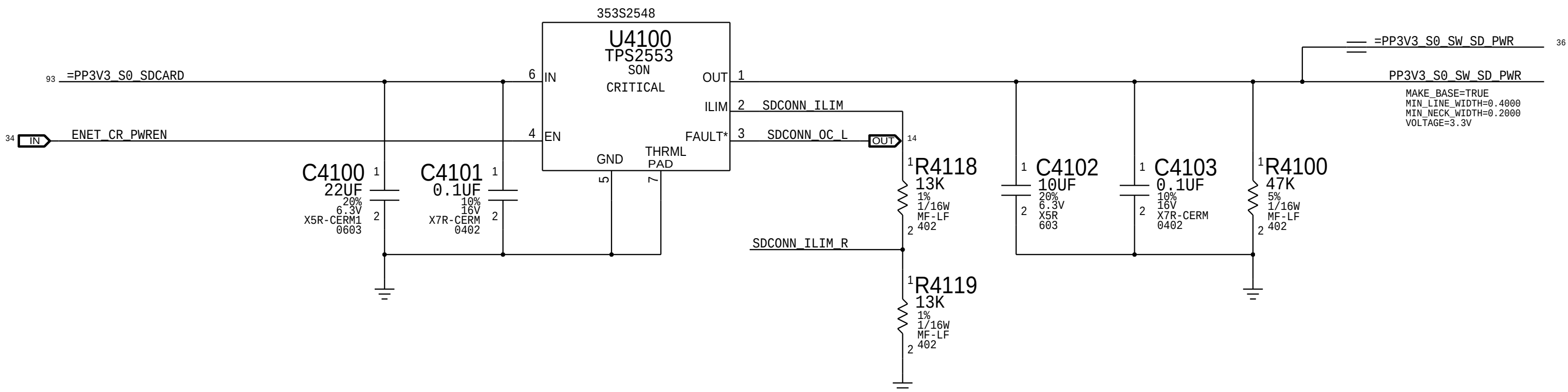
D

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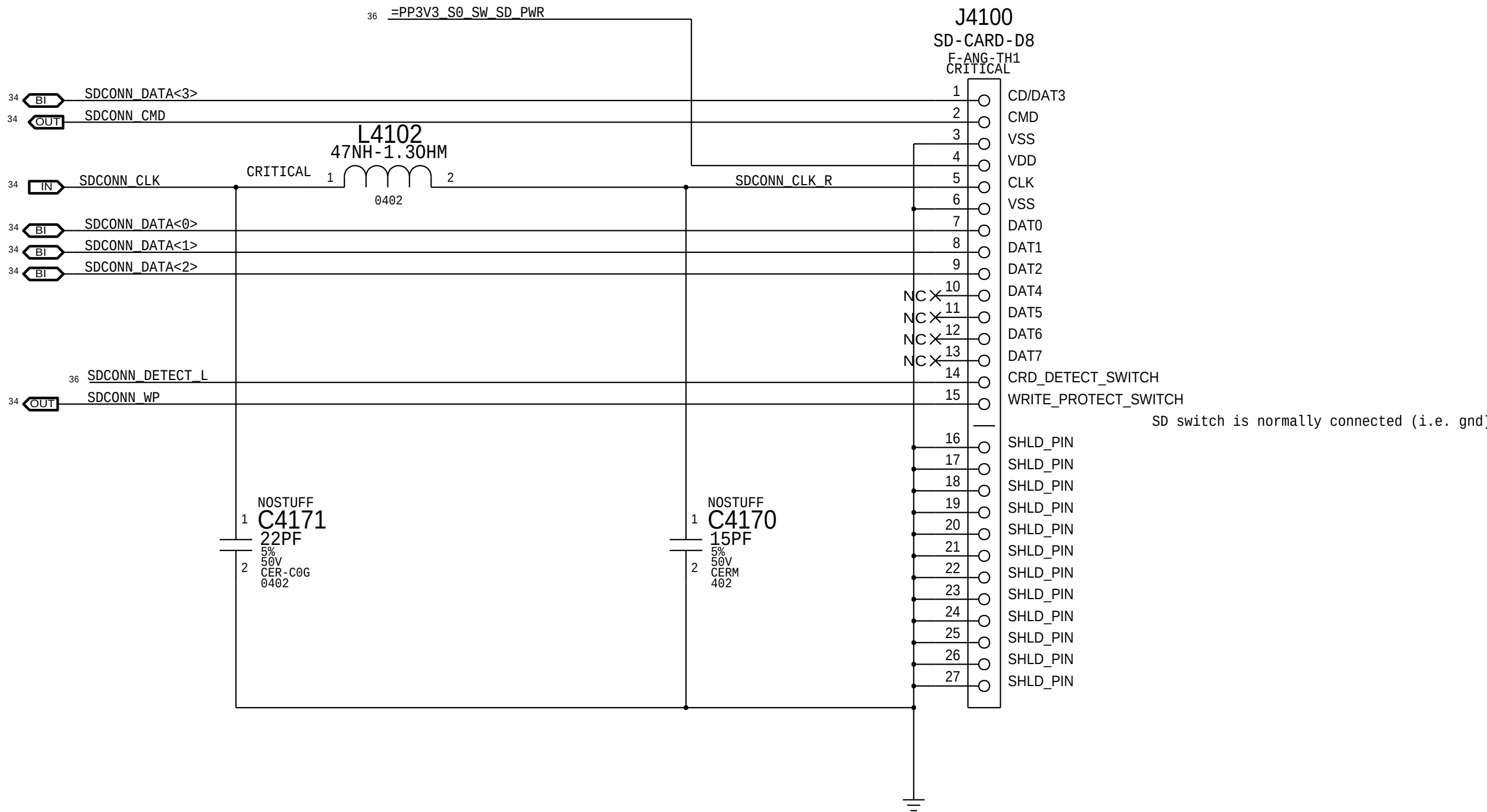
A

SD CARD 3.3V OVERCURRENT PROTECTION CHIP

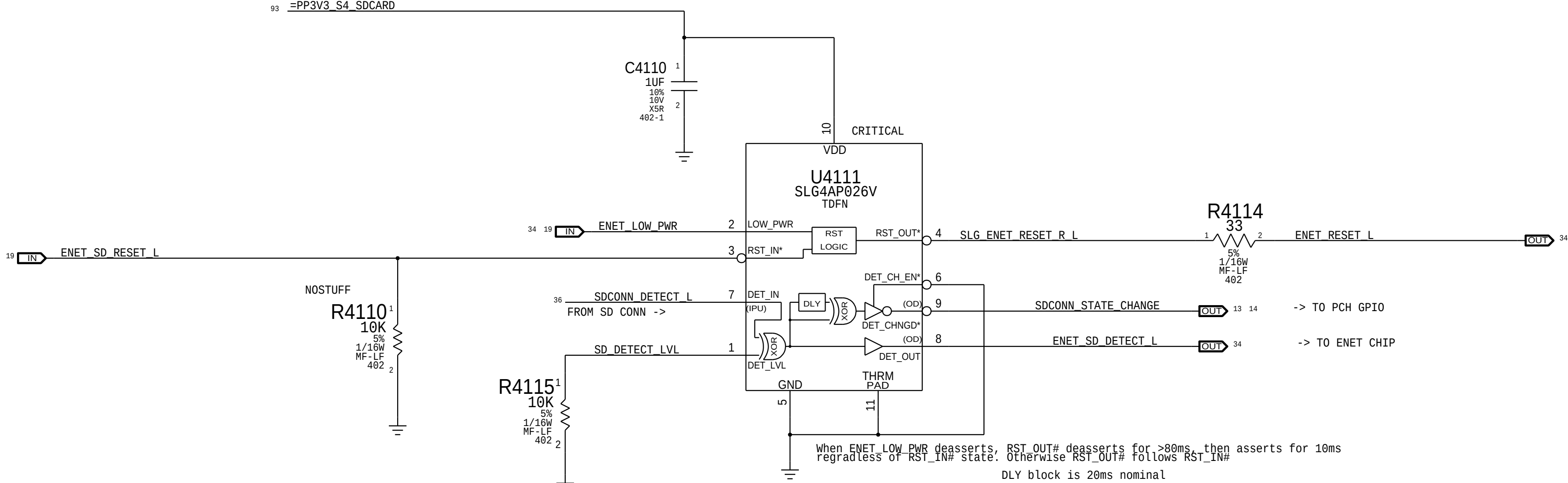


J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



SYNC_MASTER=J16_IG		SYNC_DATE=04/29/2013	
PAGE TITLE			
SD CARD: Connector			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION		
	4.0.0		
	BRANCH		
	proto1b		
	PAGE		
41 OF 120			
SHEET			
36 OF 96			

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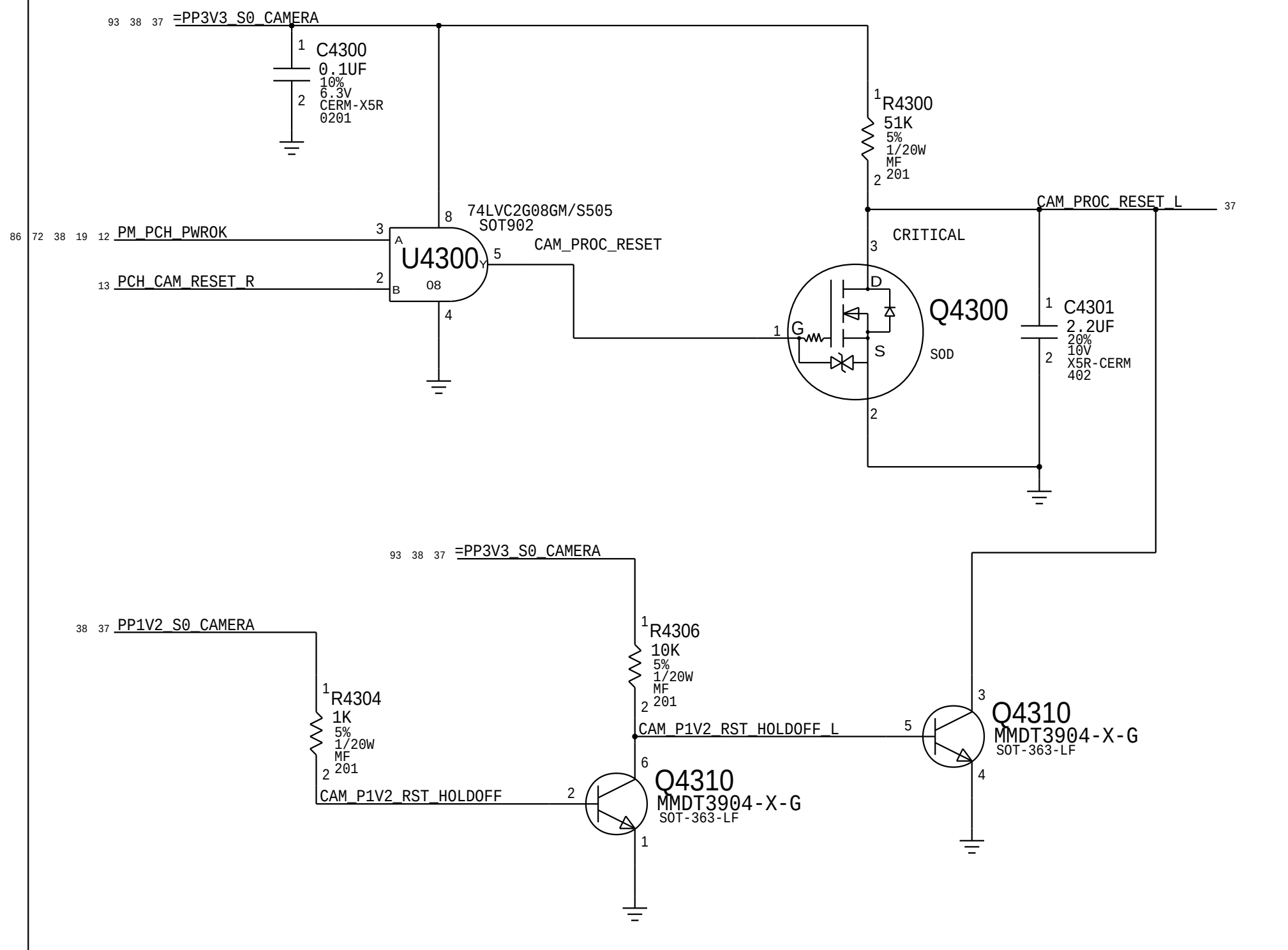
D

C

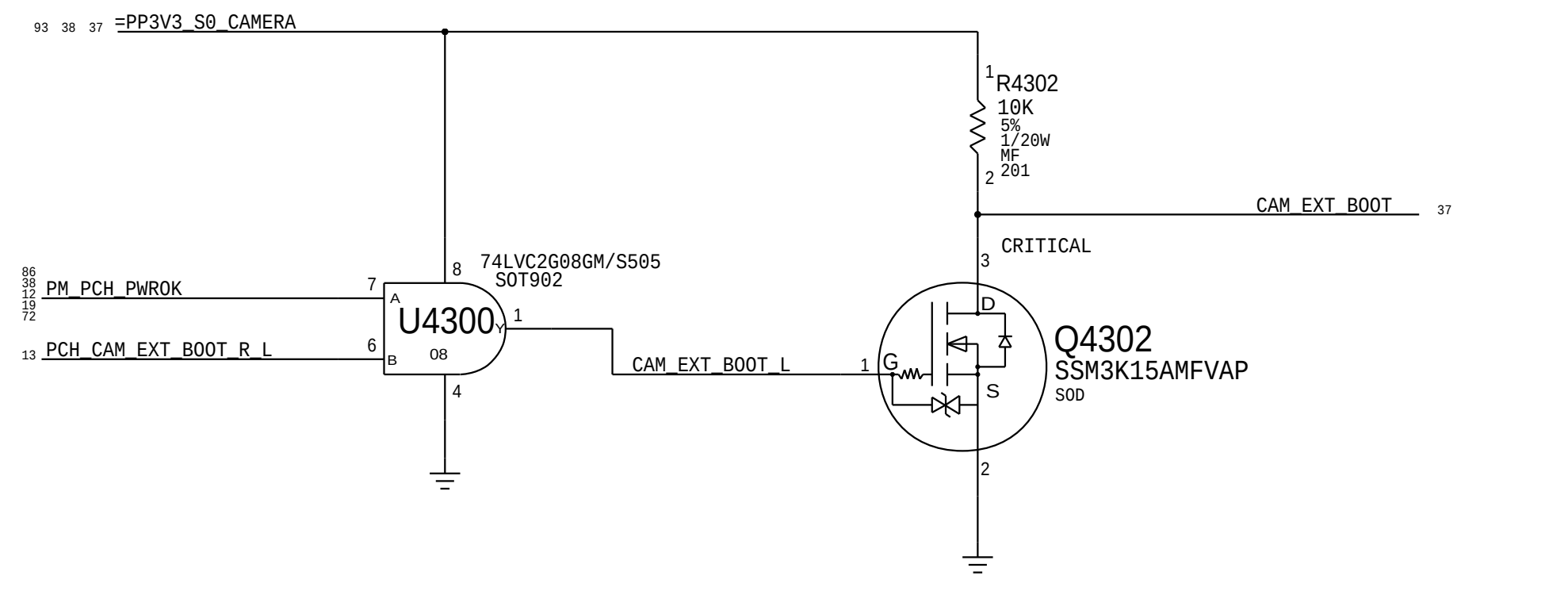
B

A

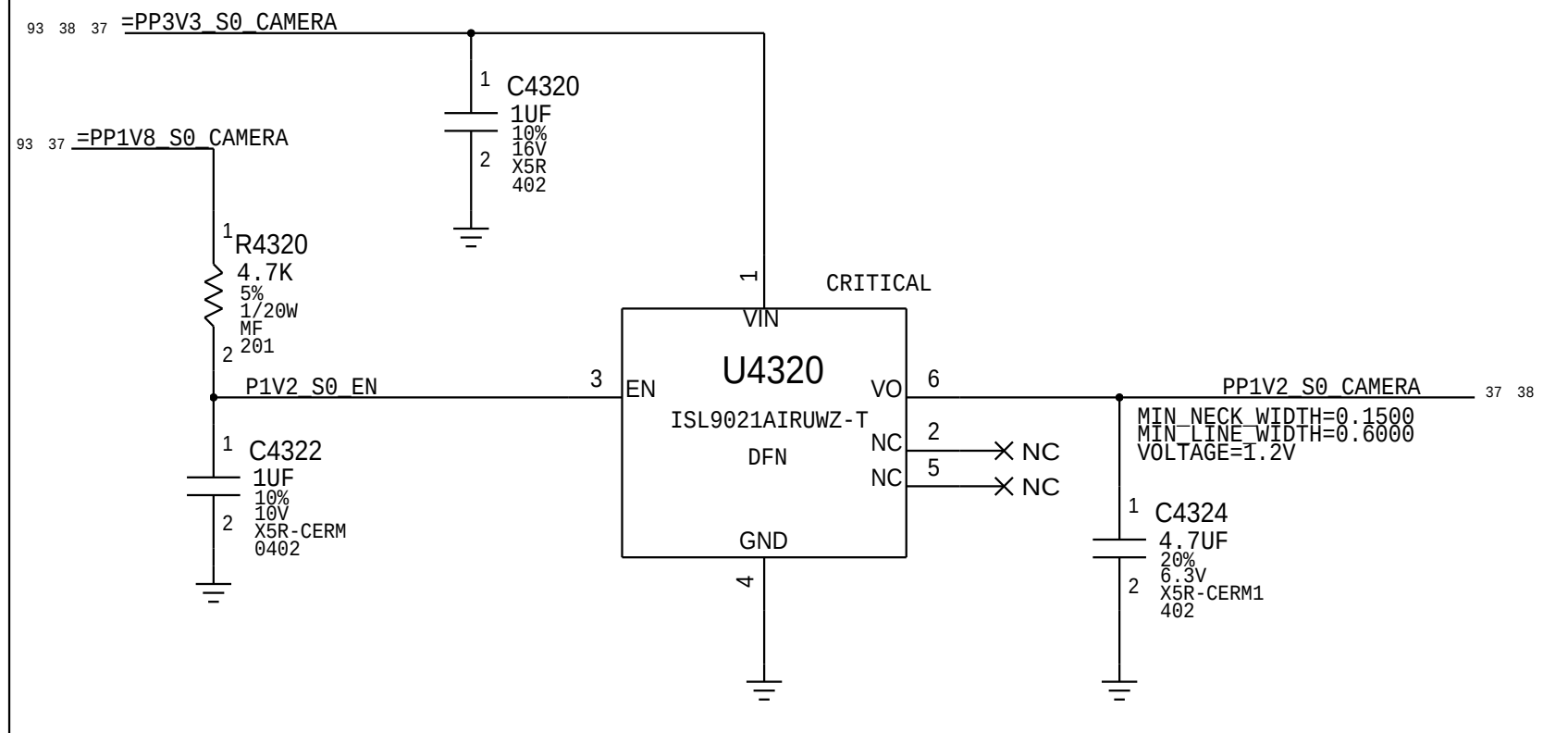
Camera Processor Reset



Camera Processor ExtBoot Cntl



PP1V2_S0_CAMERA VREG



SYNC_MASTER=J78_NAT		SYNC_DATE=11/05/2013	
PAGE TITLE			
CAMERA: Controller Support			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
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	BRANCH		
	proto1b		
	PAGE		43 OF 120
	SHEET		38 OF 96

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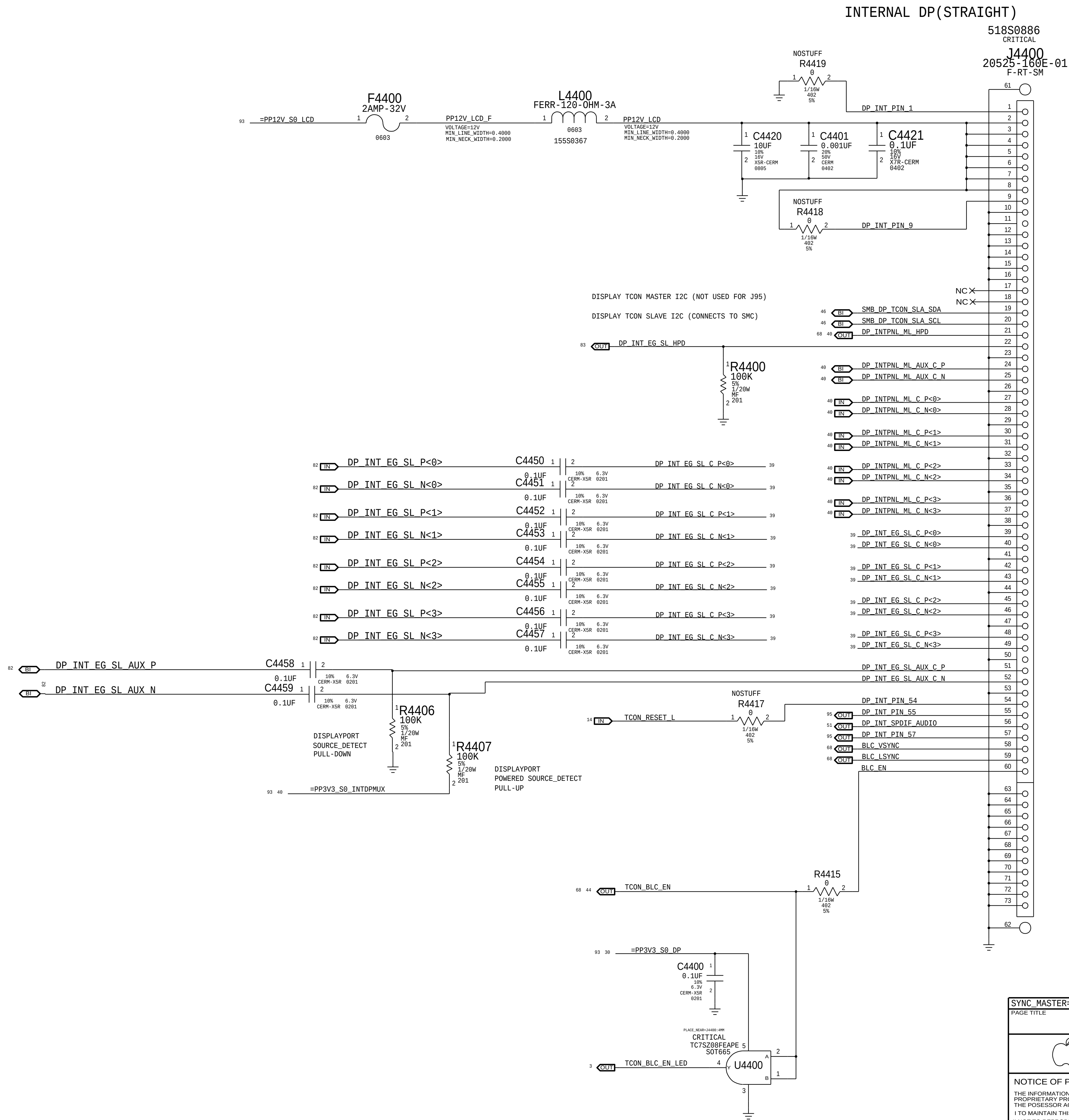
A

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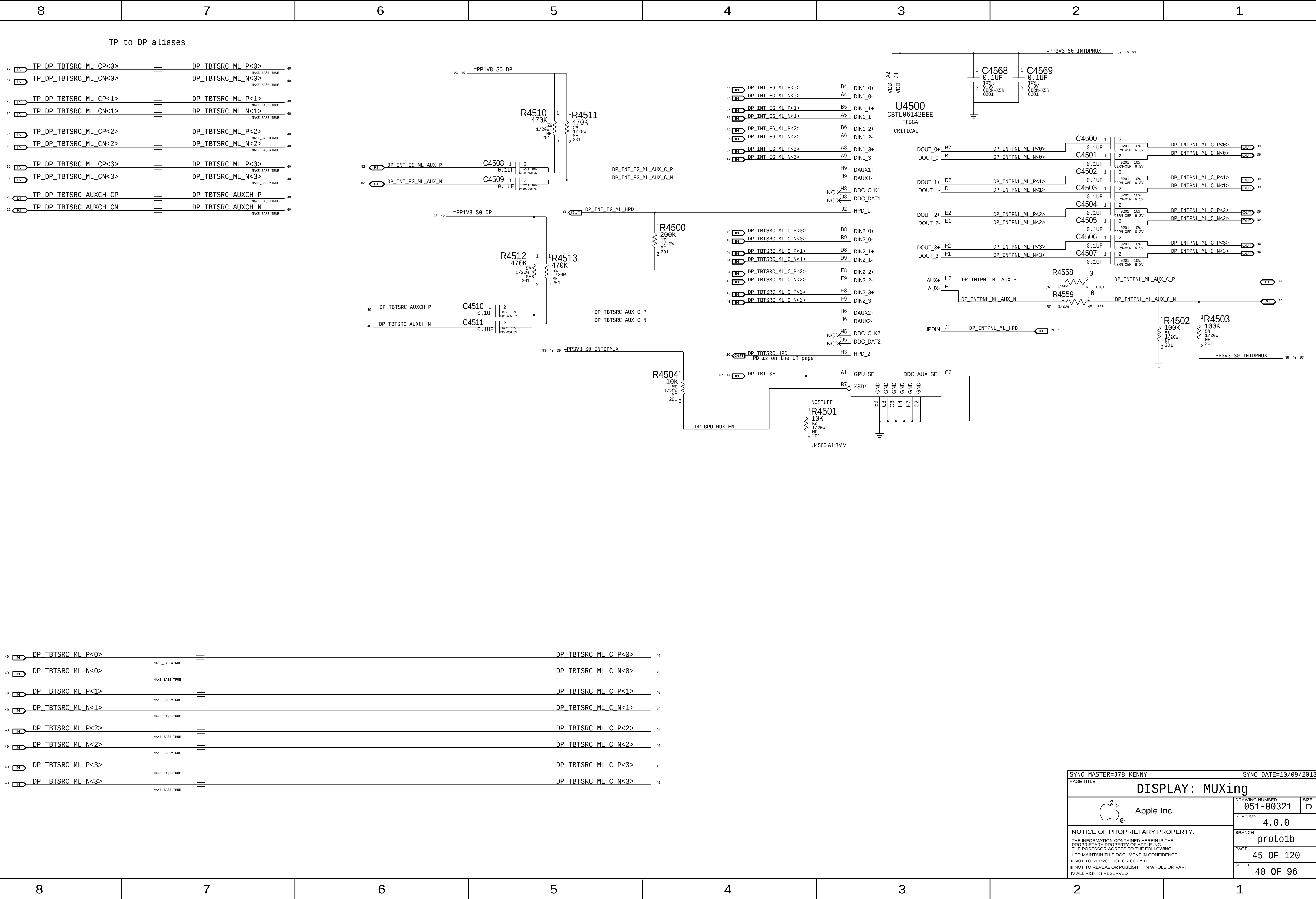
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SYNC_MASTER=J95_ANDREW		SYNC_DATE=1/27/2015	
PAGE TITLE			
DISPLAY: Support			
 Apple Inc.	DRAWING NUMBER	051-00321	SIZE D
	REVISION	4.0.0	
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	PAGE	44 OF 120	
	SHEET	39 OF 96	



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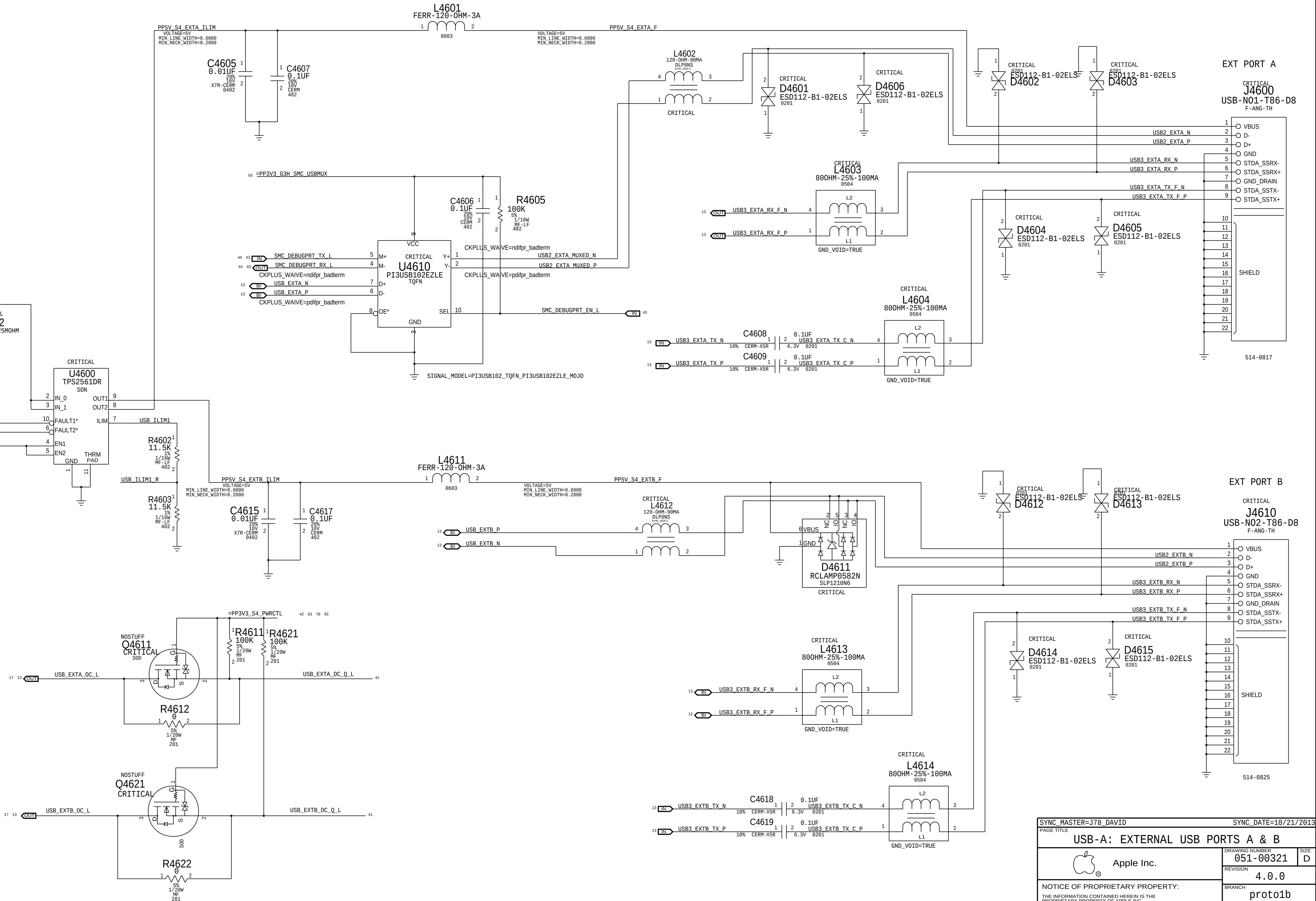
A

D

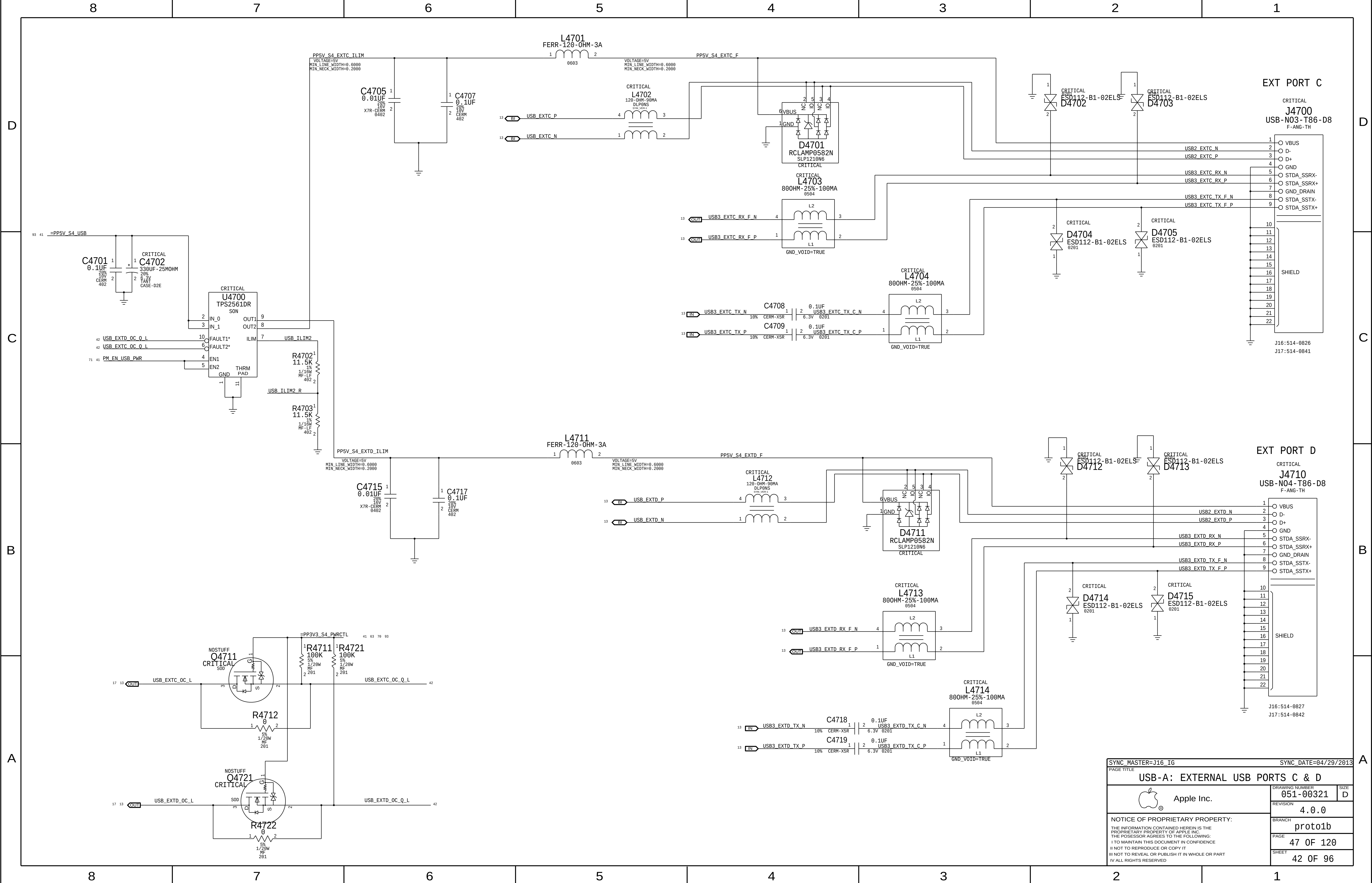
C

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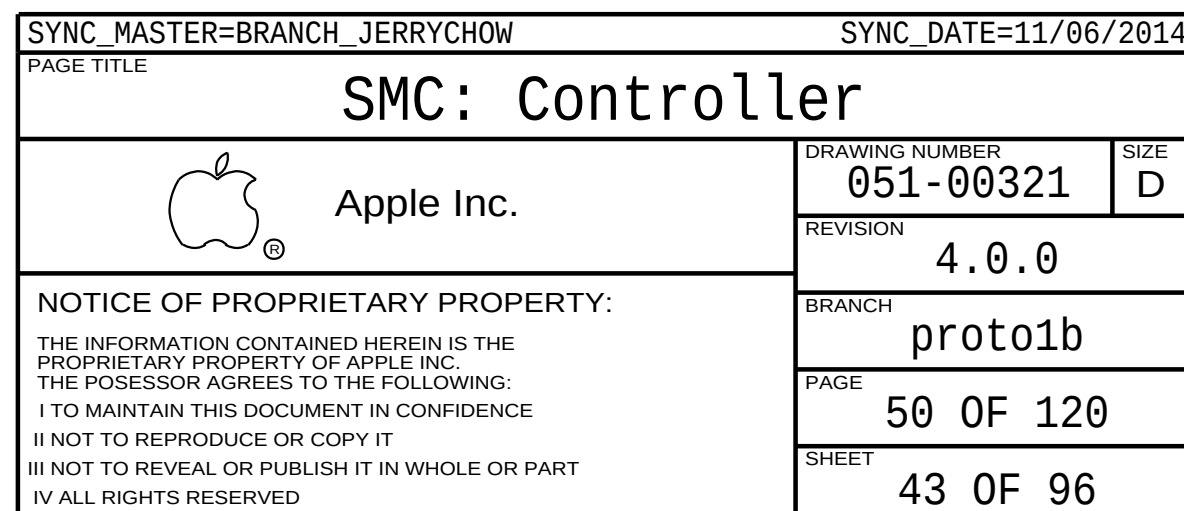


SYNC_MASTER=J78_DAVID		SYNC_DATE=10/21/2013	
PAGE TITLE			
USB-A: EXTERNAL USB PORTS A & B			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	proto1b
		PAGE	46 OF 120
		SHEET	41 OF 96



SYNC_MASTER=J16_I6		SYNC_DATE=04/29/2013	
PAGE TITLE			
USB-A: EXTERNAL USB PORTS C & D			
	Apple Inc.	DRAWING NUMBER	051-00321
		REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	proto1b
		PAGE	47 OF 120
		SHEET	42 OF 96

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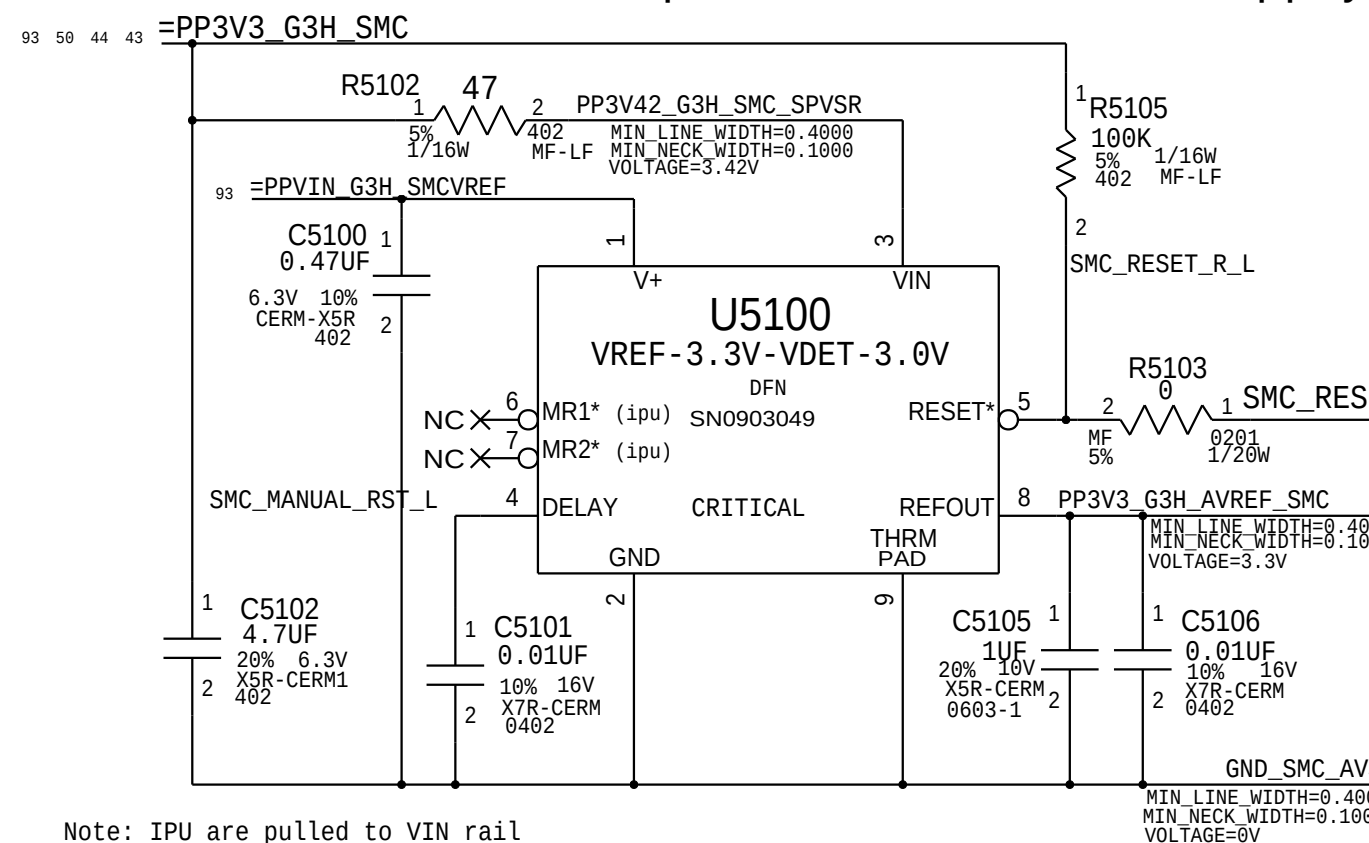
D

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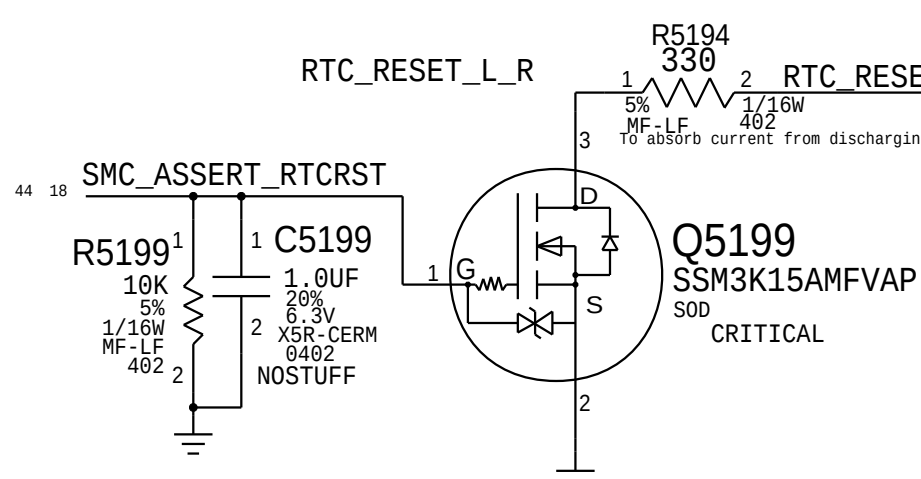
A

SMC Supervisor and AVREF Supply



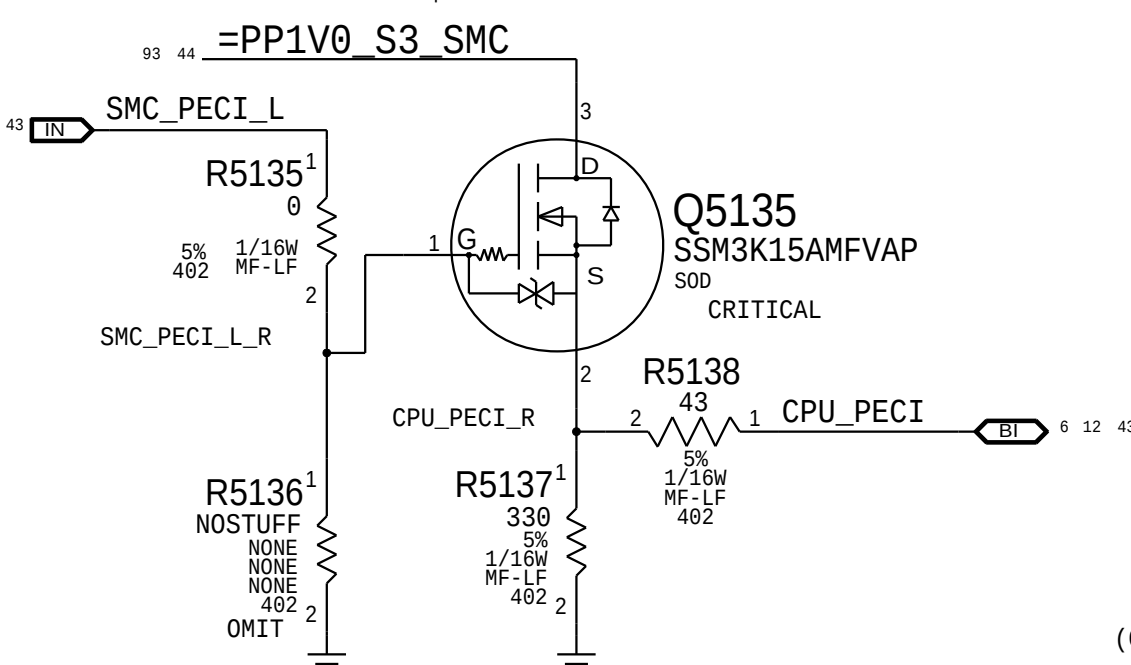
Note: IPU are pulled to VIN rail

SMC Controlled RTC Reset

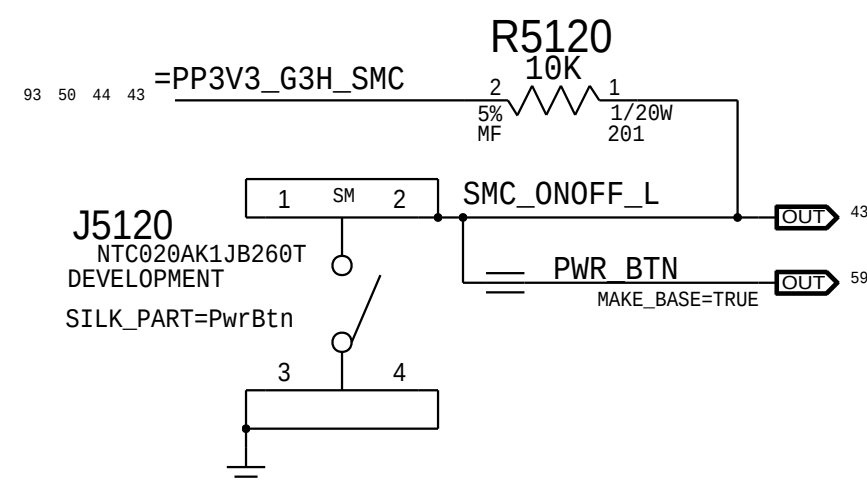


PECI Support

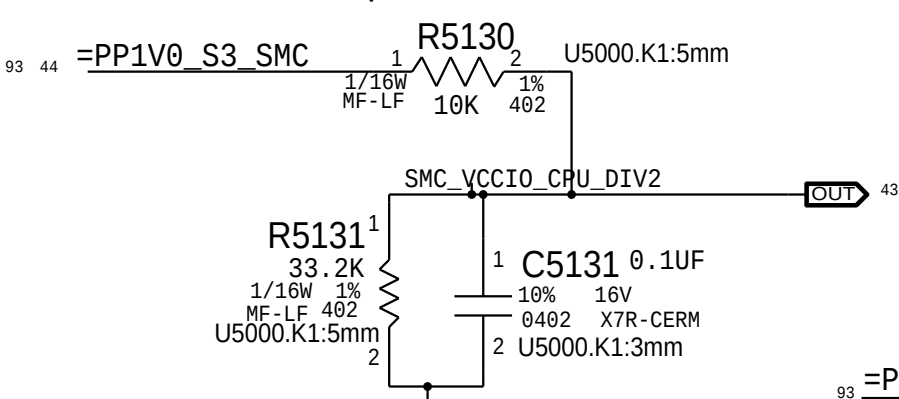
Level-shifter that allows SMC to drive Peci
Place this circuit near the Tee point to minimize reflections



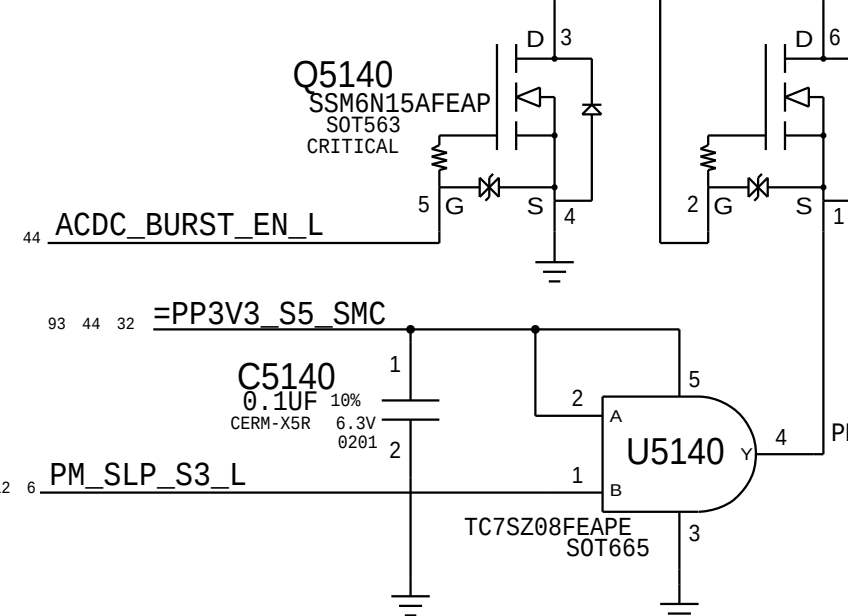
Power Button



Comparator Vref



AC/DC Burst Mode Enable



ADC Channel Aliases

SMC_ADC0	VSNS_P12VG3H	VD2R
SMC_ADC1	VSNS_P12VG3H	ID2R
SMC_ADC2	VSNS_P12VS0_CPUCORE	VD20
SMC_ADC3	VSNS_P12VS0_CPUCORE	ID20
SMC_ADC4	VSNS_CPUVCC	VC0C
SMC_ADC5	VSNS_CPUVCC	IC0C
SMC_ADC6	VSNS_CPUVCC_GT	VC0G
SMC_ADC7	VSNS_CPUVCC_GT	IC0G
SMC_ADC8	VSNS_CPUVCC_I0	IC0I
SMC_ADC9	VSNS_P1V35S0	IC0M
SMC_ADC10	VSNS_CPUVCC_SA	IC0S
SMC_ADC11	VSNS_P12VS0_FBVDDQ	IG1F
SMC_ADC12	VSNS_GPUCORE_ALT	VG0C
SMC_ADC13	VSNS_GPUCORE_ALT	IG0C
SMC_ADC14	VSNS_GPU_VDDCI	VG0I
SMC_ADC15	VSNS_GPU_VDDCI	IG0I
SMC_ADC16	VSNS_P12VS0_GPU_AUX	IG1A
SMC_ADC17	VSNS_P12VS0_GPUCORE	IG1C
SMC_ADC18	VSNS_P12VS0_HDD	IH02
SMC_ADC19	VSNS_HDD0S	IH05
SMC_ADC20	VSNS_SSD_S4	VH1R
SMC_ADC21	VSNS_SSD_S4	IH1R
SMC_ADC22	VSNS_VDDQ3_DDR	VM0R
SMC_ADC23	VSNS_VDDQ3_DDR	IM0R

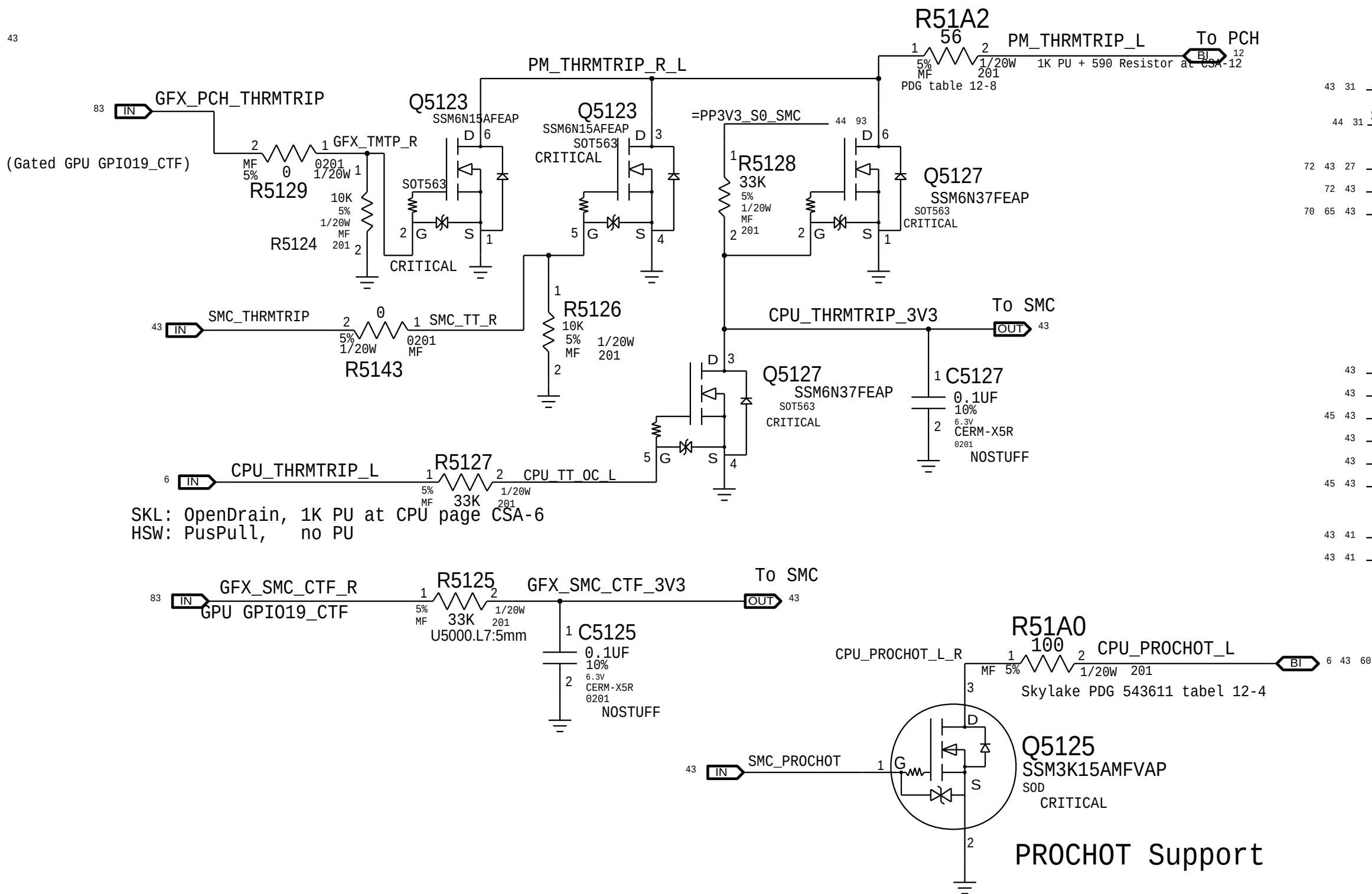
Project-specific Aliases

SMC_PN5	ACDC_BURST_EN_L	
SMC_PJ3	SMC_00B2_R2D_L	
SMC_PJ2	SMC_00B2_D2R_L	
SMC_PP8	SMC_ACDC_I0	
SMC_PH2	SMC_ASSERT_RTCRST	
SMC_PL6	SMC_WIFI_PWR_EN	
SMC_PN3	TCON_BLC_EN	(LED-4)
SMC_PH7	GFX_OK_L	(LED-3)
SMC_PN7	DP_LINK_OK	(LED-3)

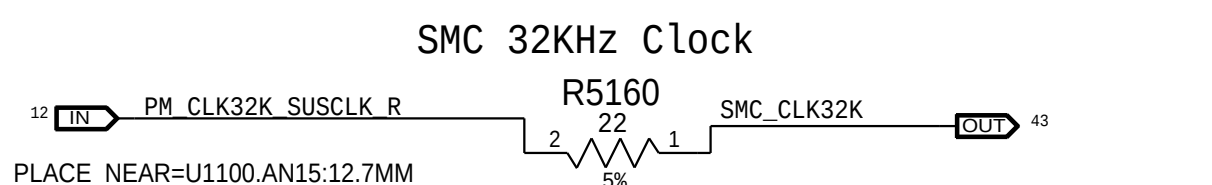
Unused Project-specific

SMC_PF0	NC_SMC_PF0	
SMC_PF1	NC_SMC_PF1	
SMC_PF2	NC_SMC_PF2	
SMC_PF3	NC_SMC_PF3	
SMC_PL7	SMC_BT_PWR_EN	(J95 new)
SMC_PP4	NC_SMC_S4_WAKESRC_EN	
SMC_PP5	NC_SMC_PP5	
SMC_PP6	NC_SMC_PP6	NO_TEST=1
SMC_PP7	NC_SMC_PP7	NO_TEST=1
SMC_DP_HPD_L	NC_SMC_DP_HPD_L	NO_TEST=1
SMC_PME_S4_DARK_L	NC_SMC_PME_S4_DARK_L	NO_TEST=1
SMC_PC4	NC_SMC_SS_PWRGD_VIN	NO_TEST=1
SMC_PQ3	NC_G3_POWERON_L	NO_TEST=1
SMC_PN6	NC_SMC_G3_WAKESRC_EN	
SMBUS_SMC_4_ASF_SCL	NC_SMBUS_SMC_4_ASF_SCL	
SMBUS_SMC_4_ASF_SDA	NC_SMBUS_SMC_4_ASF_SDA	NO_TEST=1
SMBUS_SMC_5_G3H_SCL	NC_SMBUS_SMC_5_G3H_SCL	NO_TEST=1
SMBUS_SMC_5_G3H_SDA	NC_SMBUS_SMC_5_G3H_SDA	NO_TEST=1
SMC_00B1_R2D_L	NC_SMC_00B1_R2D_L	NO_TEST=1
SMC_BATLOW_L	NC_SMC_SMC_BATLOW_L	NO_TEST=1
SMC_PH3	NC_SMC_PH3	NO_TEST=1

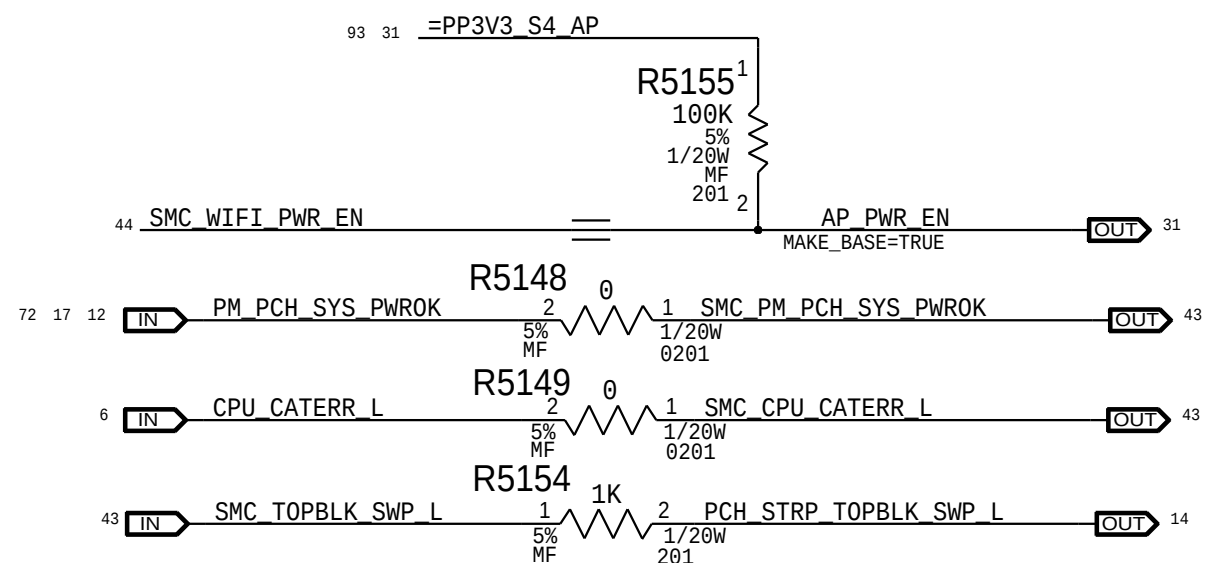
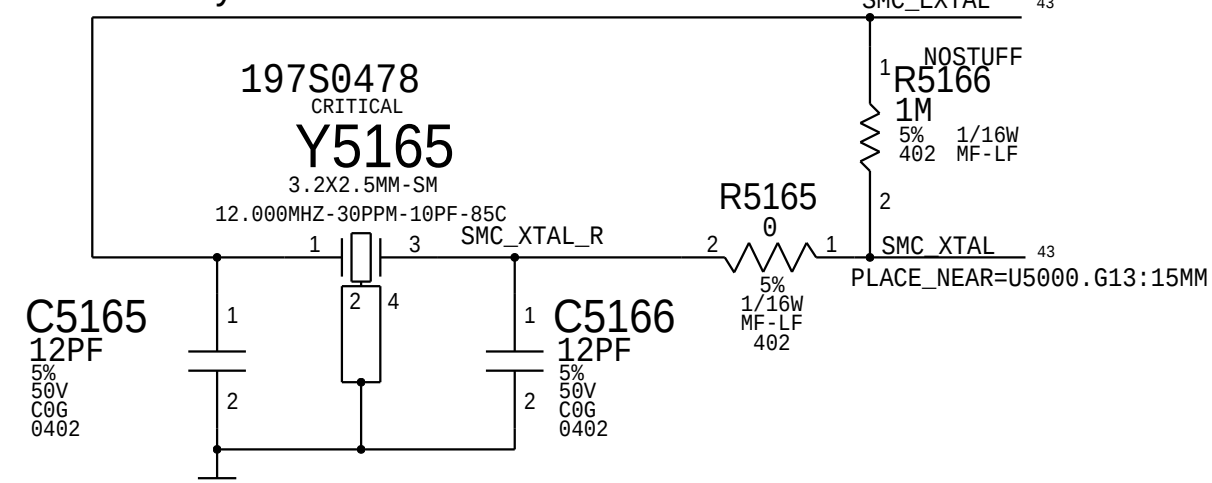
Platform Thermal Control



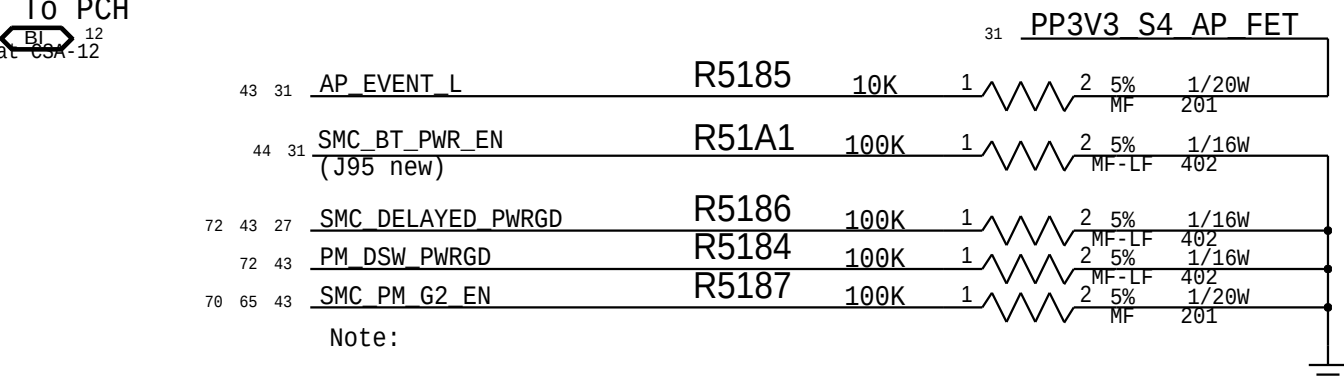
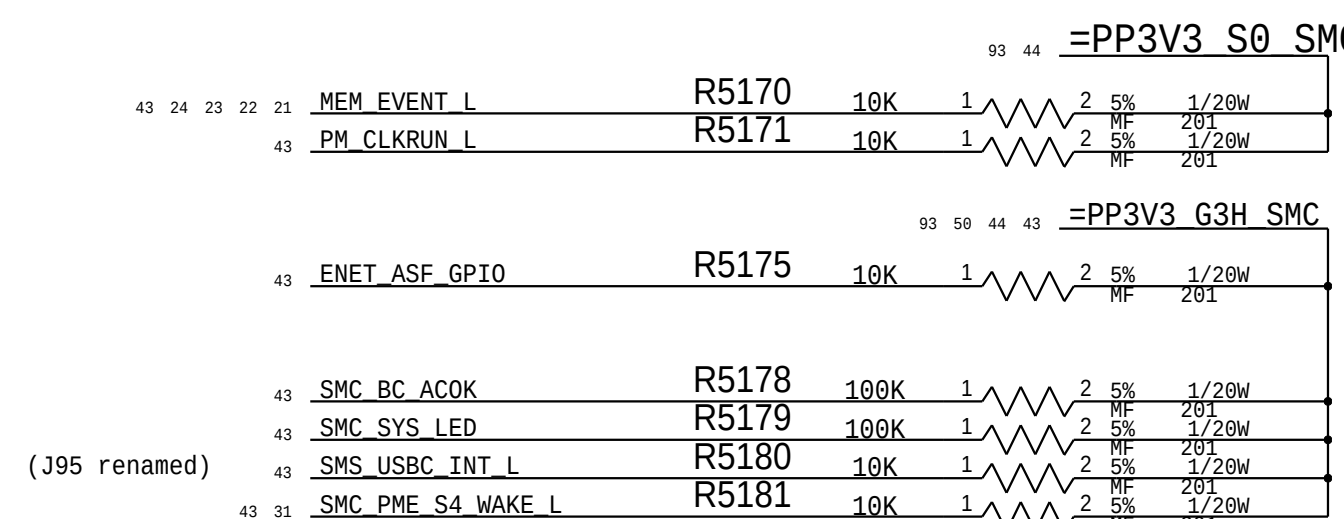
PROCHOT Support



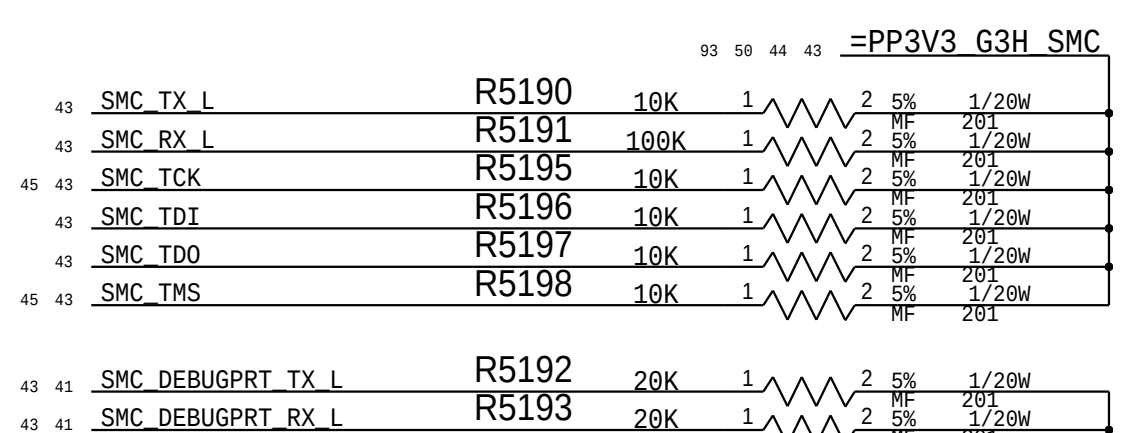
SMC Crystal



Arch Pull Up/Down



Serial/JTAG Interface Pull-ups



SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=11/06/2014

PAGE TITLE

SMC: Controller Support

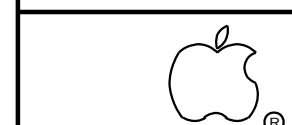
DRAWING NUMBER 051-00321 SIZE D

REVISION 4.0.0

BRANCH proto1b

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SHEET 44 OF 96



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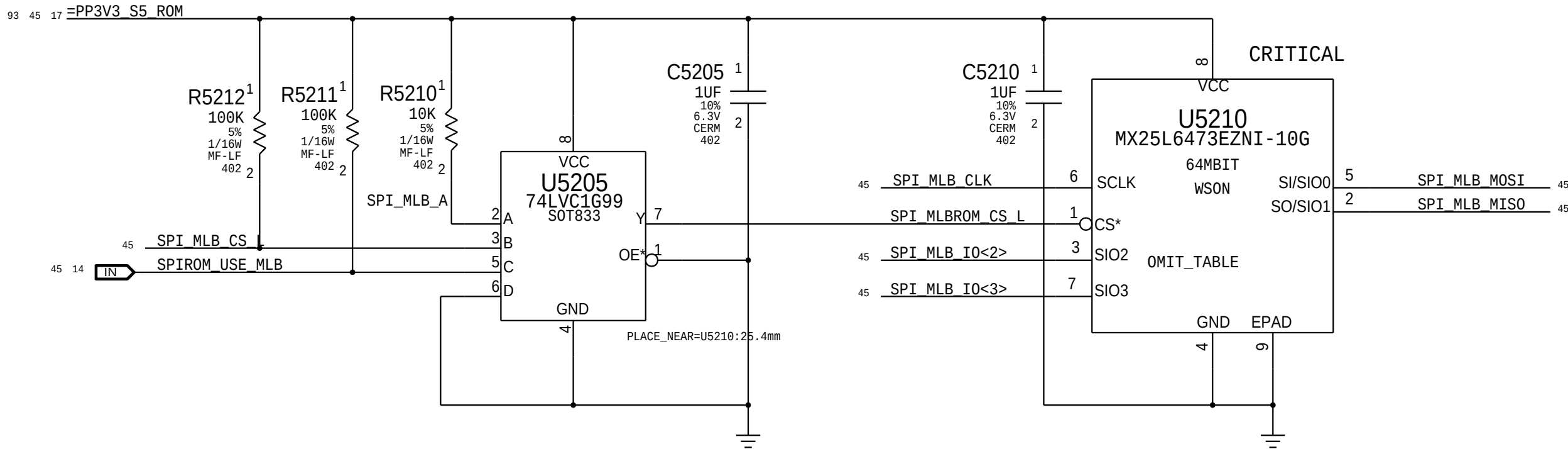
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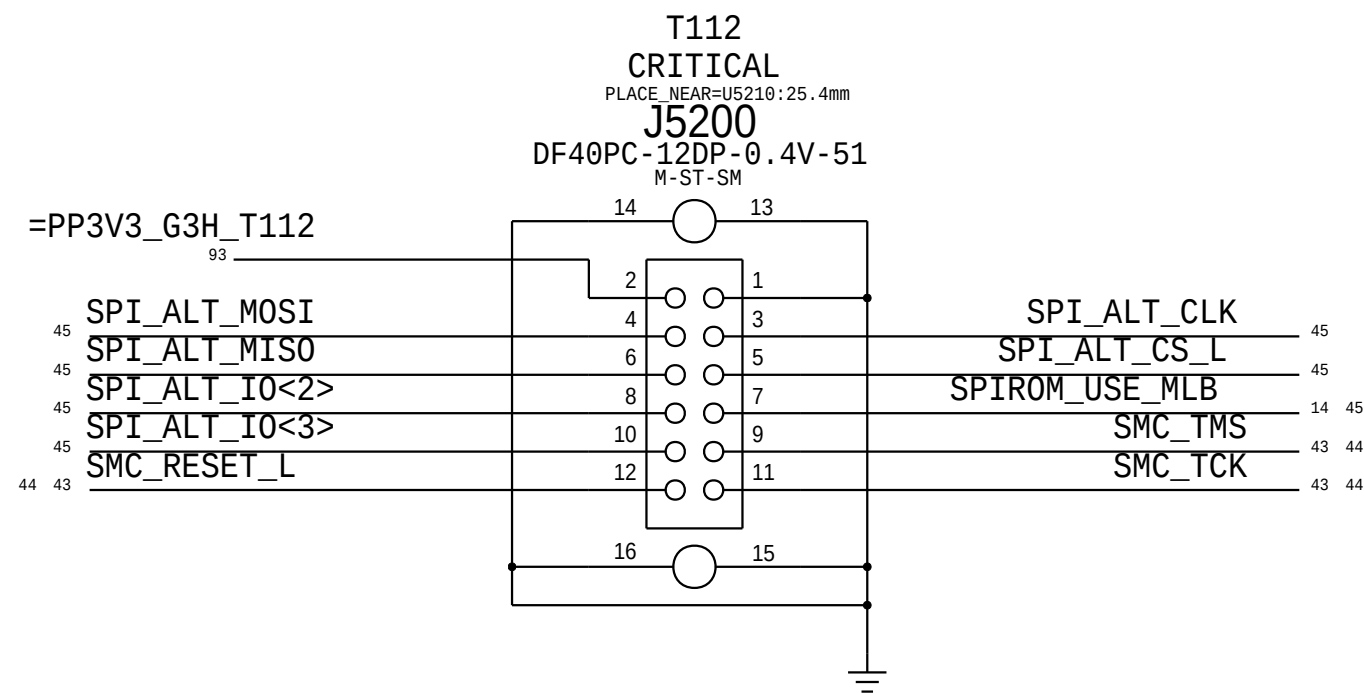
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SPI BootROM

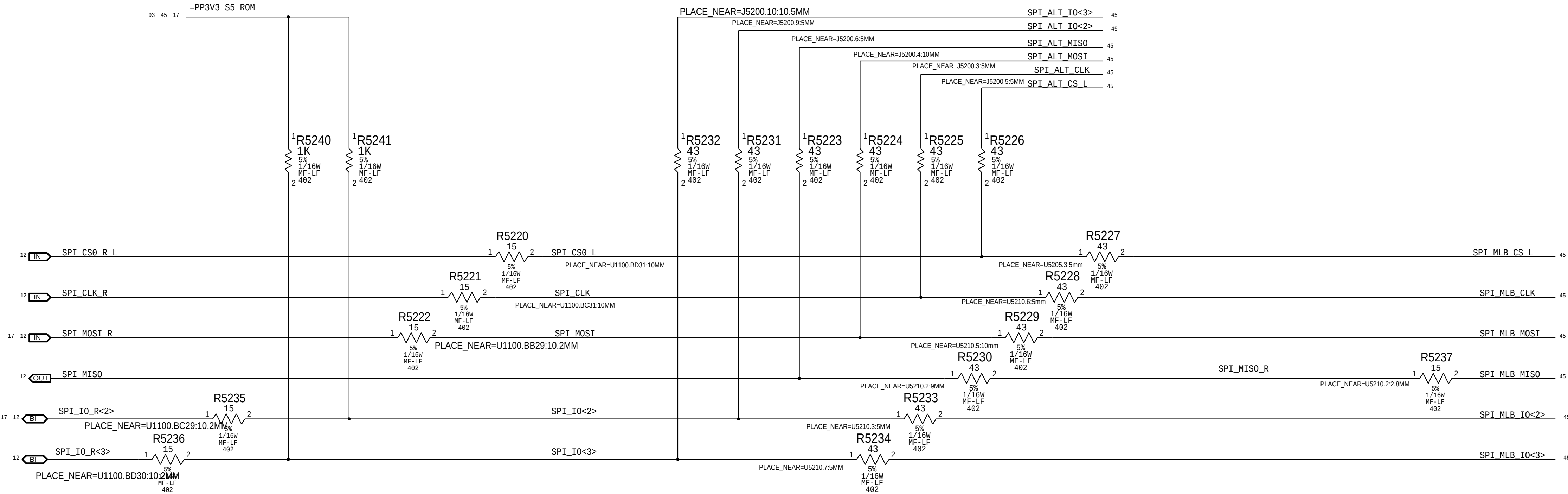
QUAD_IO MODE (MODE 0 & 3) SUPPORTED
SPI FREQ: 50 MHZ FOR PCH



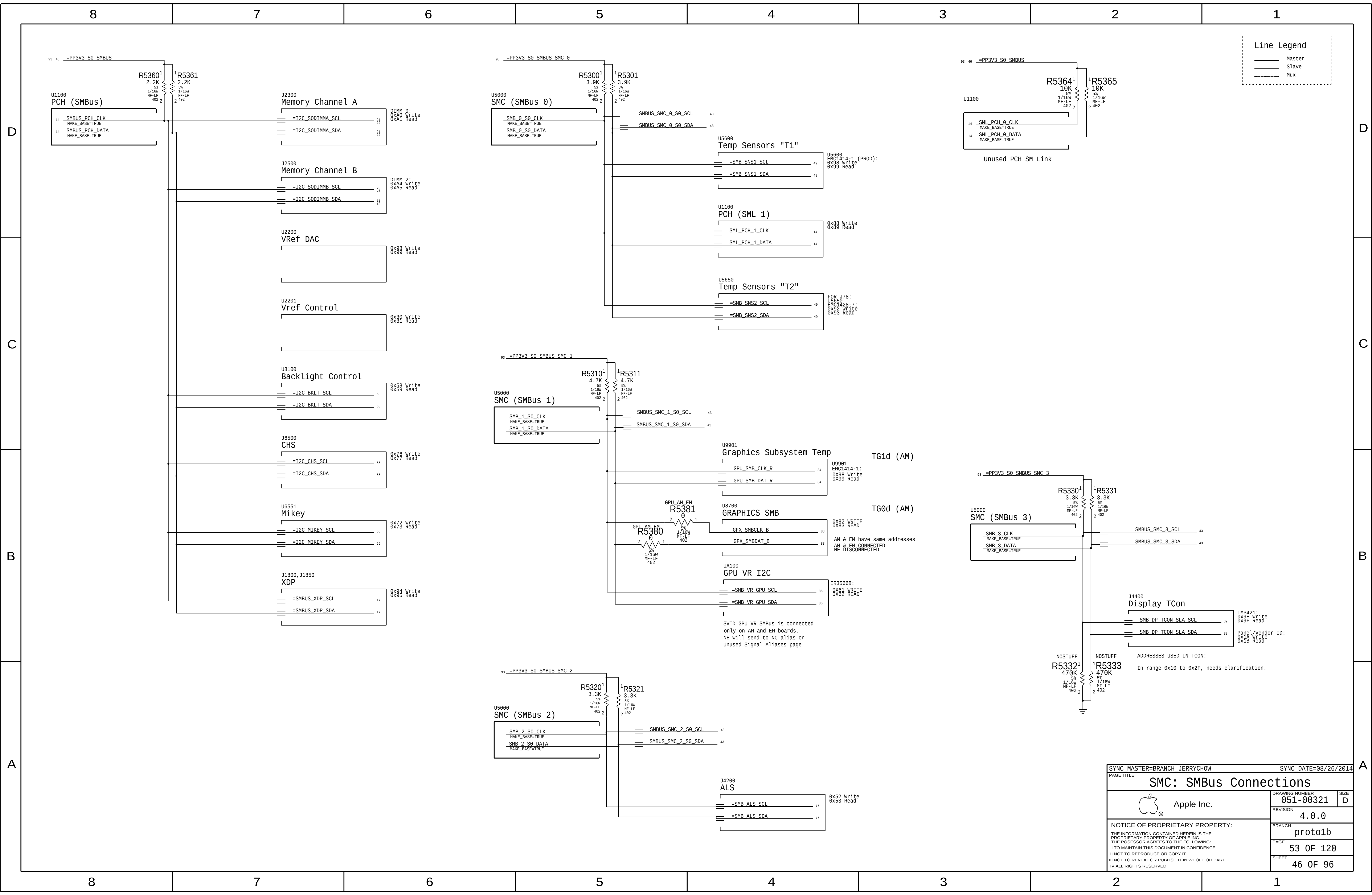
T112 Connector



SPI Series Termination



SYNC_MASTER=J17_TONY		SYNC_DATE=03/13/2013	
PAGE TITLE			
CPU & CHIPSET: SPI and Debug Connector			
 Apple Inc.	DRAWING NUMBER	051-00321	SIZE D
	REVISION	4.0.0	
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		PAGE	52 OF 120
		SHEET	45 OF 96



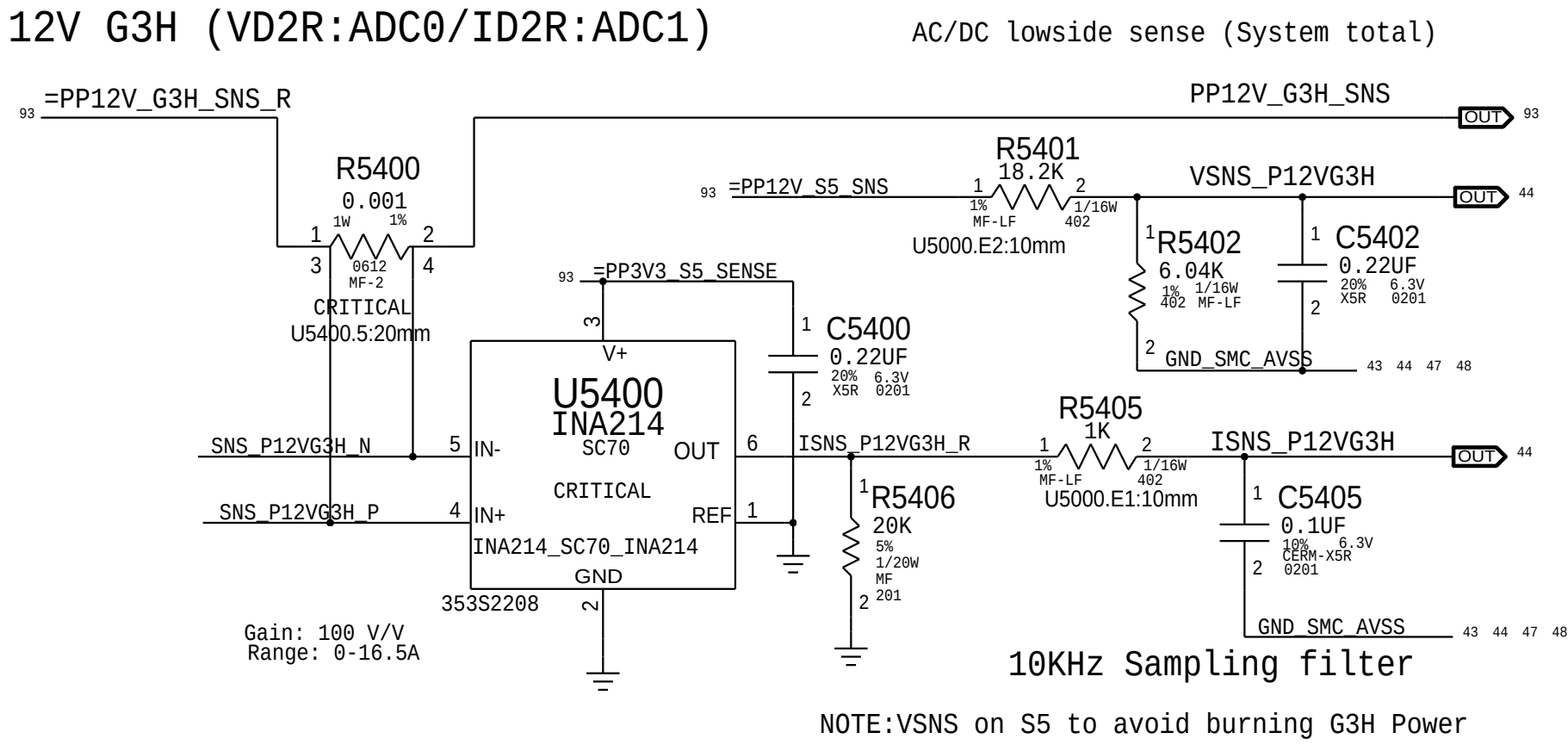
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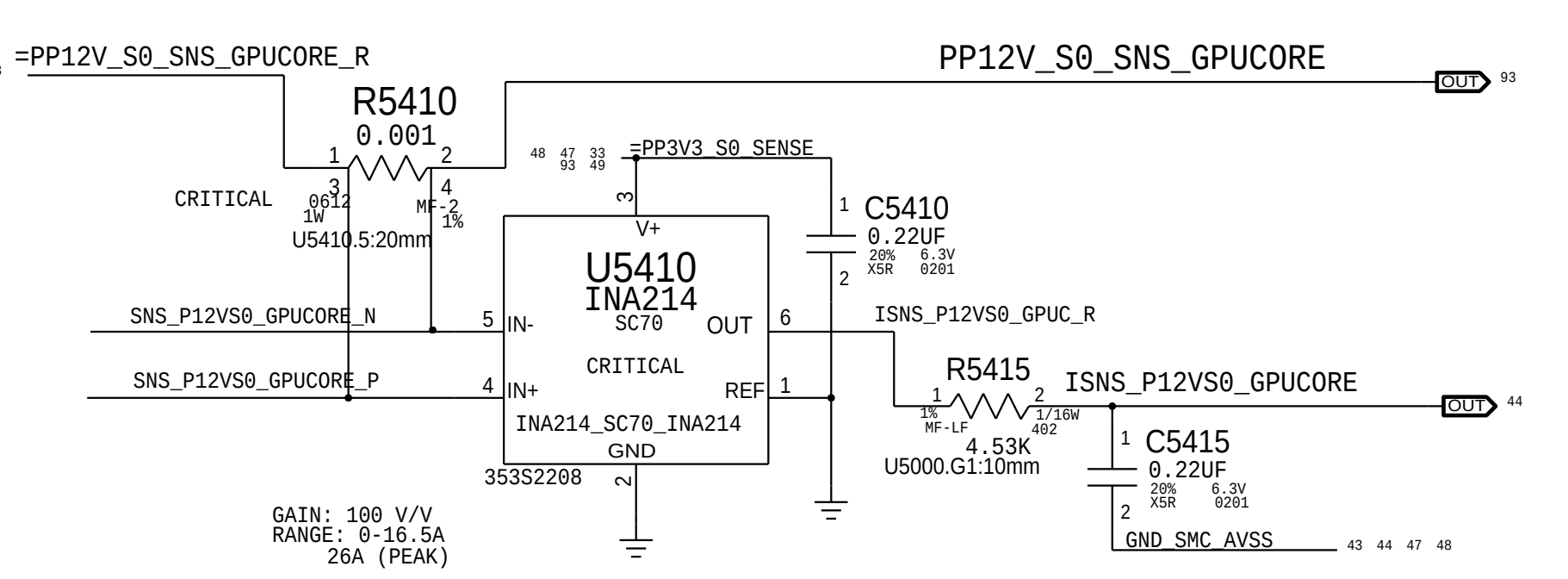
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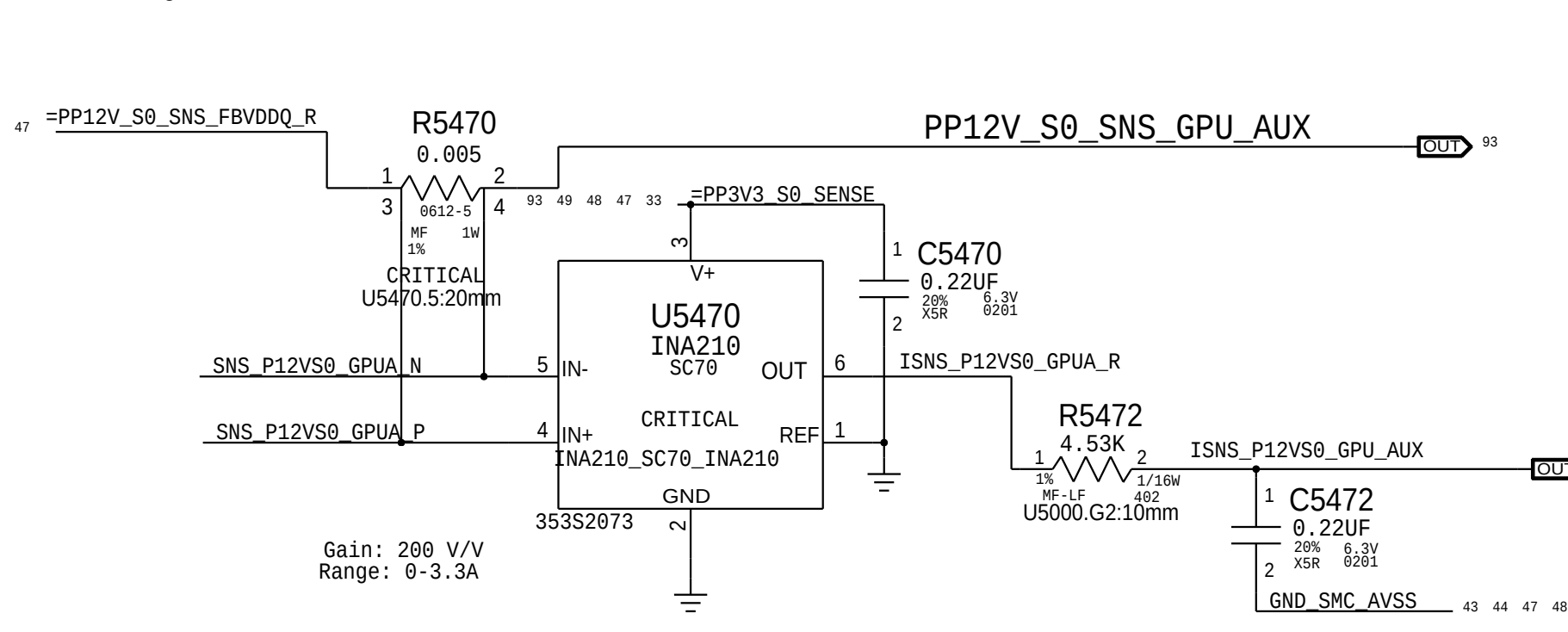
12V G3H (VD2R:ADC0/ID2R:ADC1)



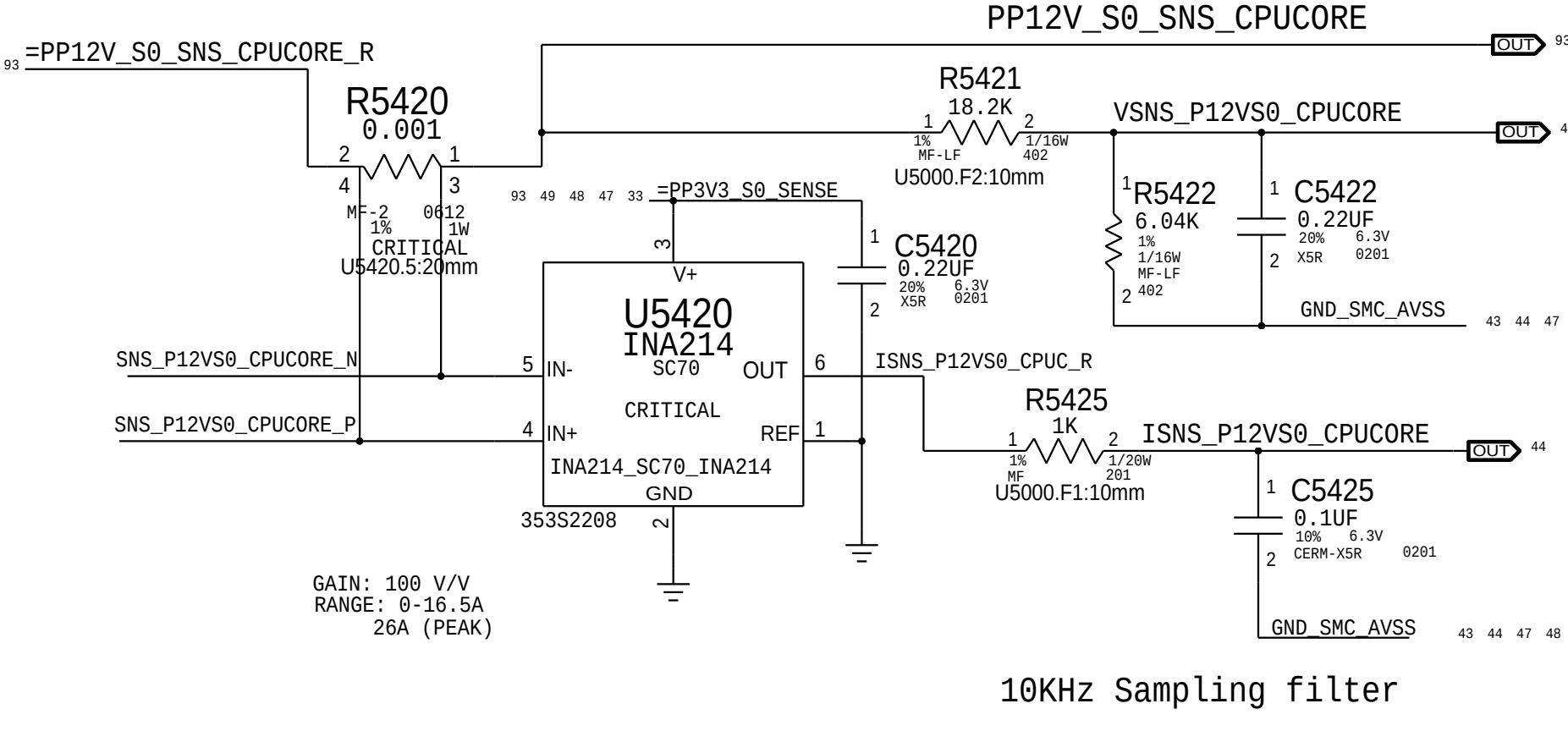
PP12V_S0_GPU (VG1C=VD20, IG1C:ADC17)



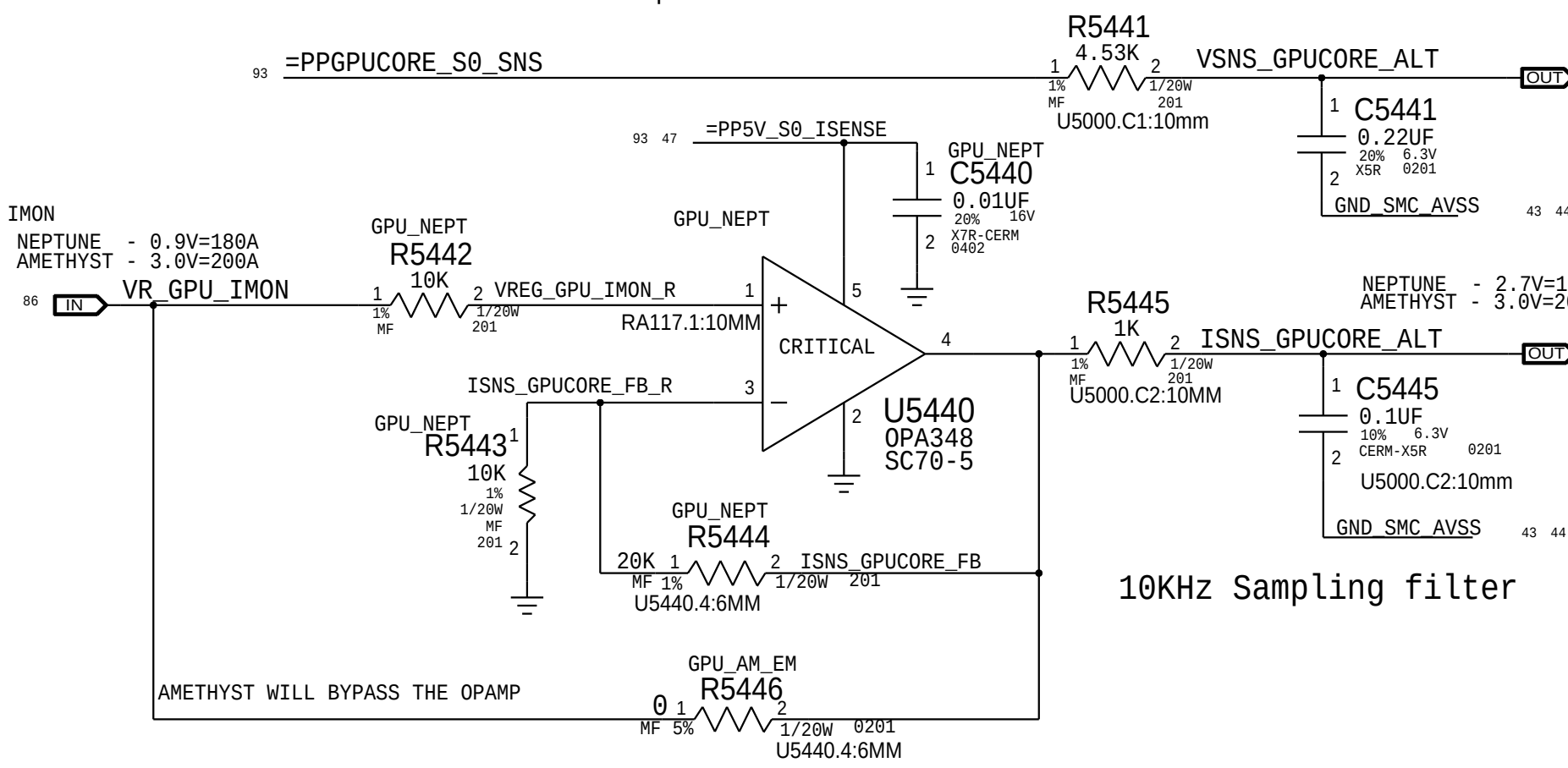
GPU AUX RAILS (VG1A=VD20, IG1A:ADC16)



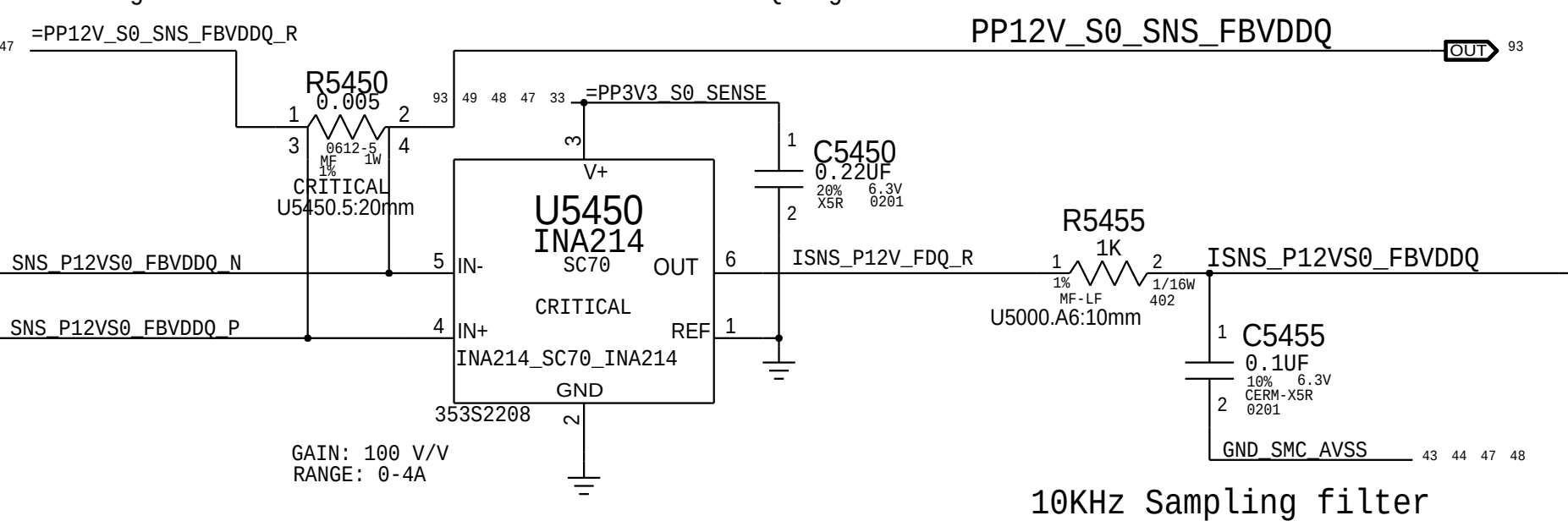
PP12V_S0_CPU (VD20:ADC2 /ID20:ADC3)



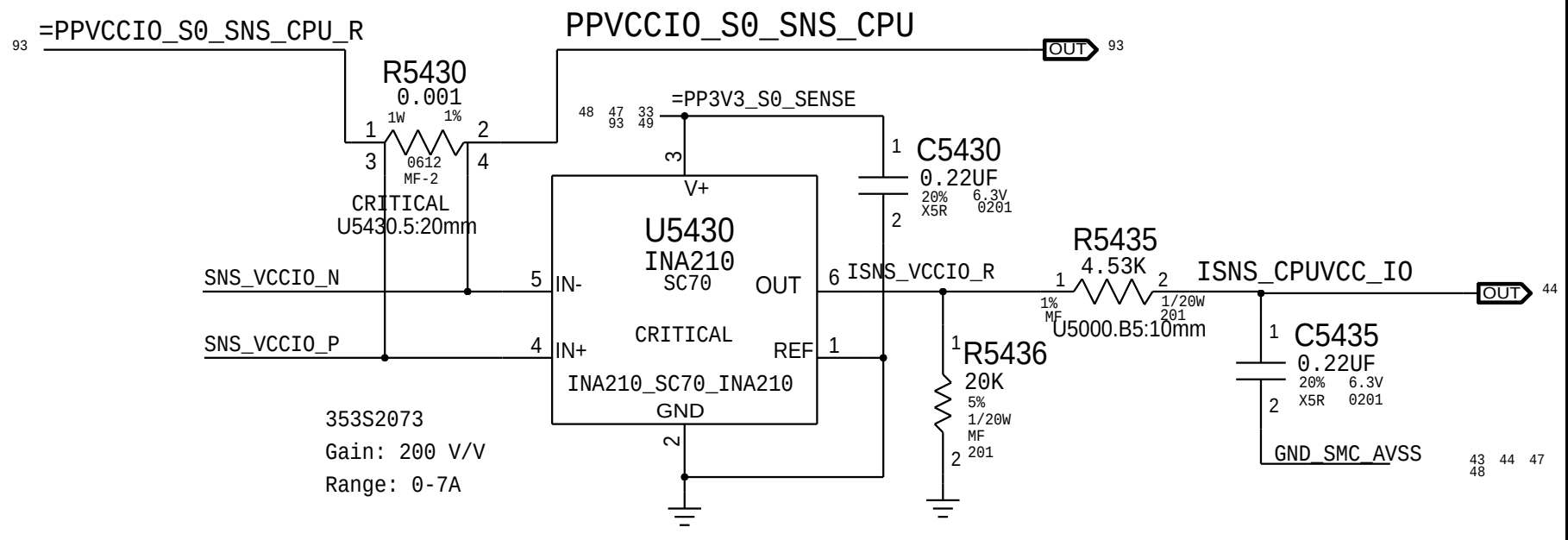
GPU Core - Alt (VG0C:ADC12/IG0C:ADC13)



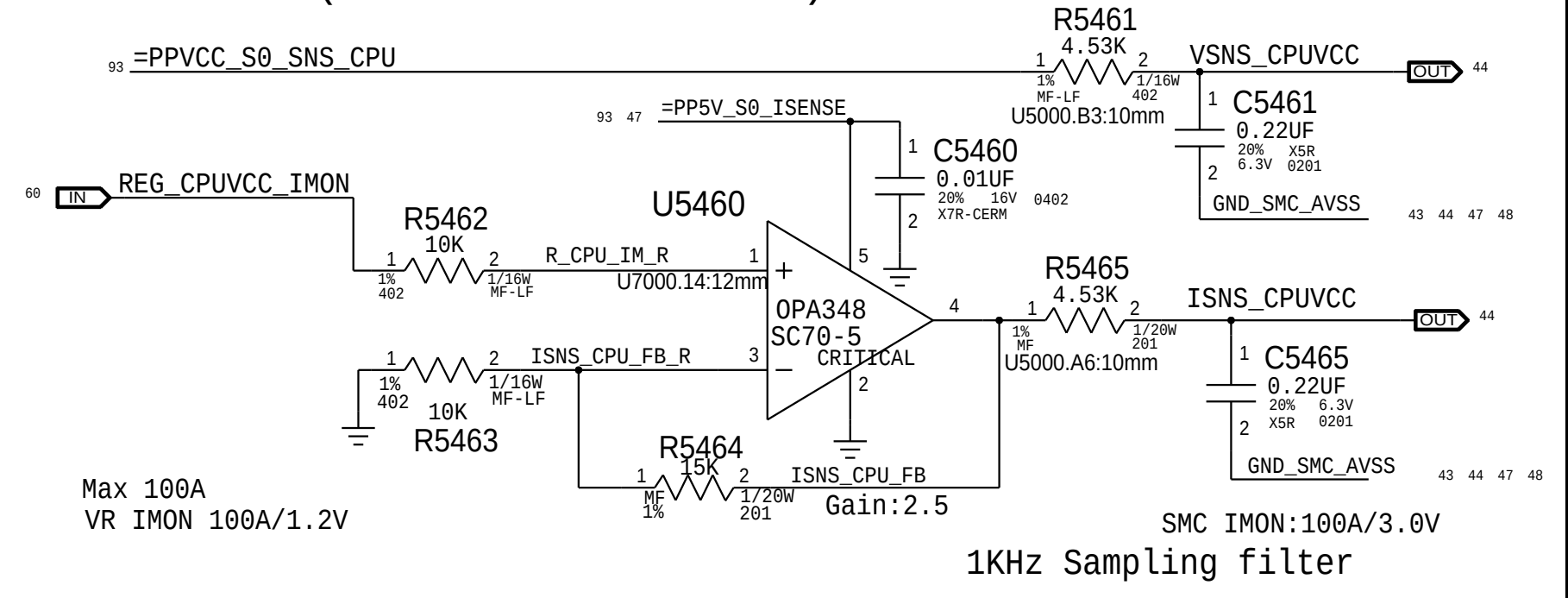
GPU FB (VG1F=VD20, IG1F:ADC11)



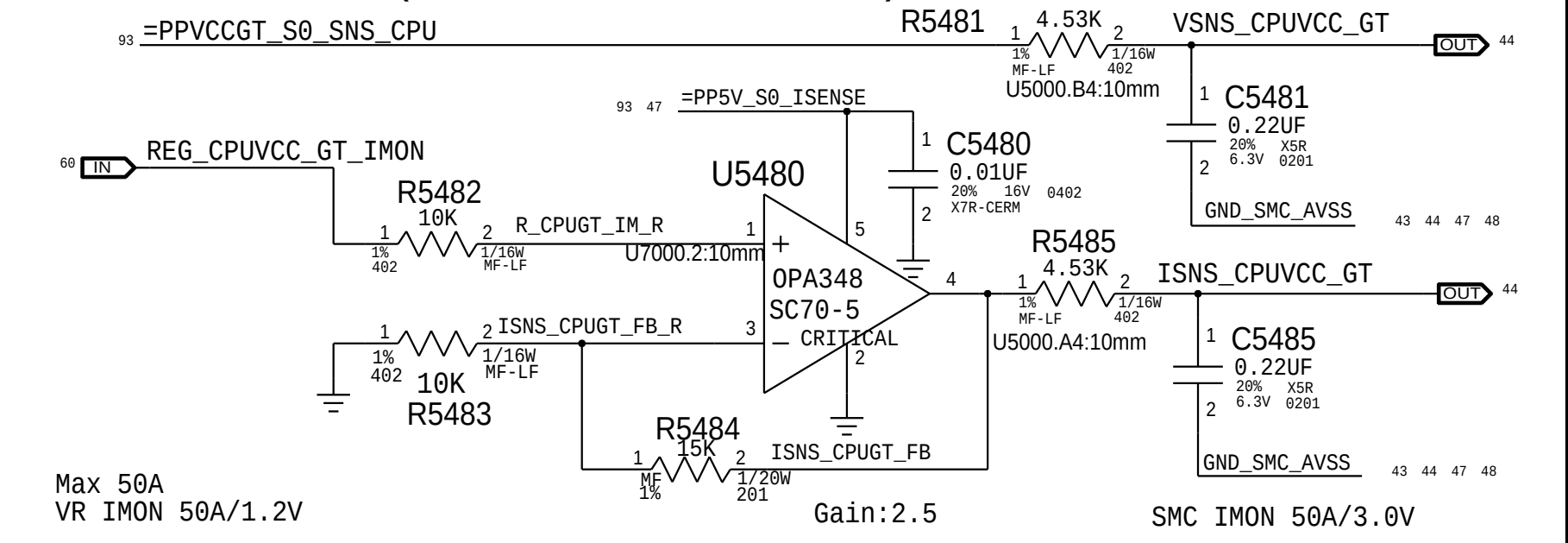
CPU VCCIO (VC0I=tbd, IC0I:ADC8)



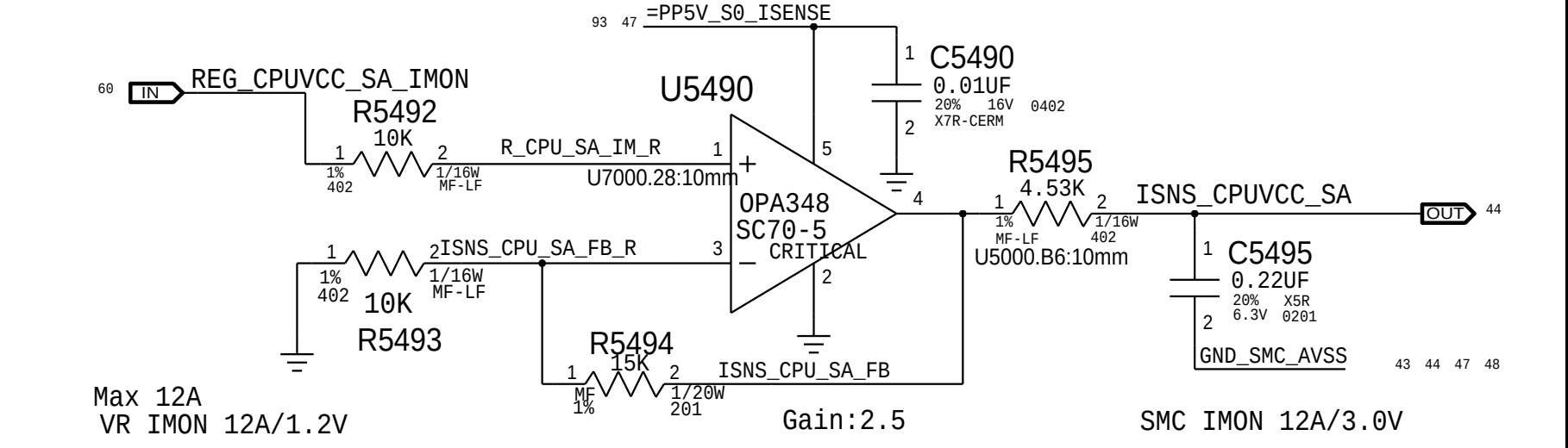
CPU Core (VC0C:ADC4/IC0C:ADC5)



CPU Core GT (VC0G:ADC6/IC0G:ADC7)



CPU Core VCC_SA (VC0S=1.05V, IC0S:ADC10)



SYNC_MASTER=BRANCH_JERRYCHOW		SYNC_DATE=09/10/2014	
PAGE TITLE			
SENSORS: I and V Sense			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
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		PAGE	54 OF 120
		SHEET	47 OF 96

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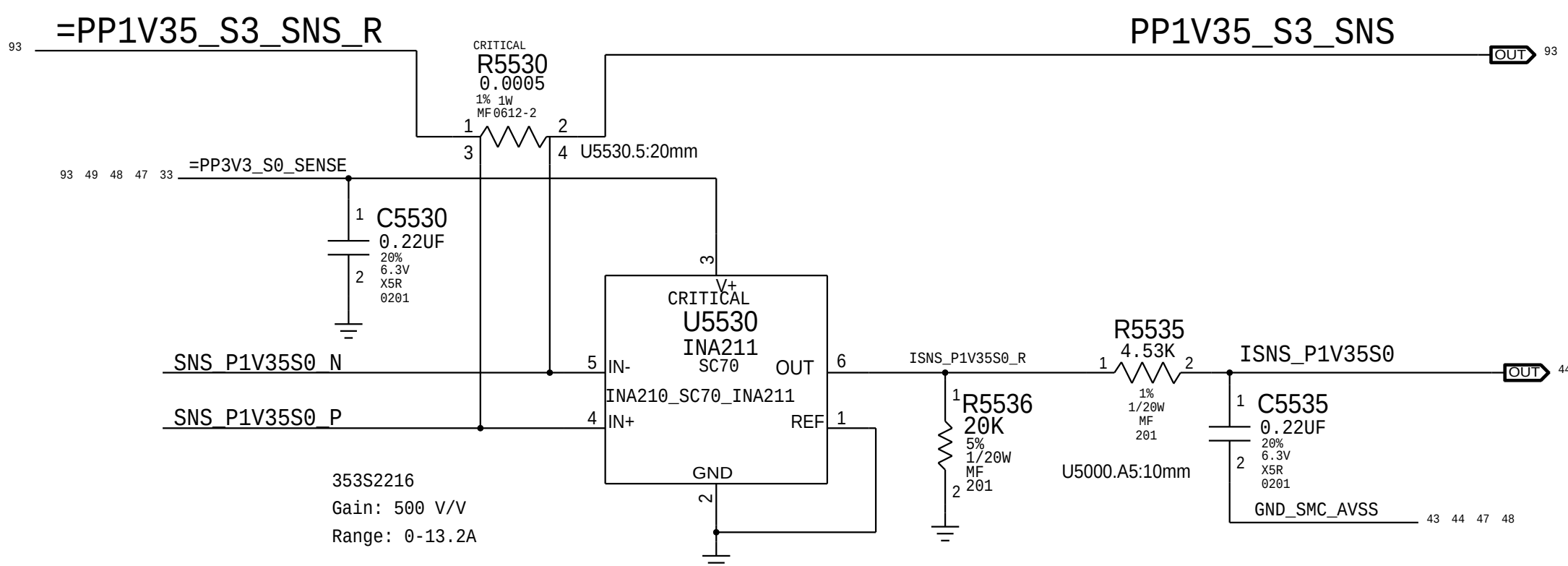
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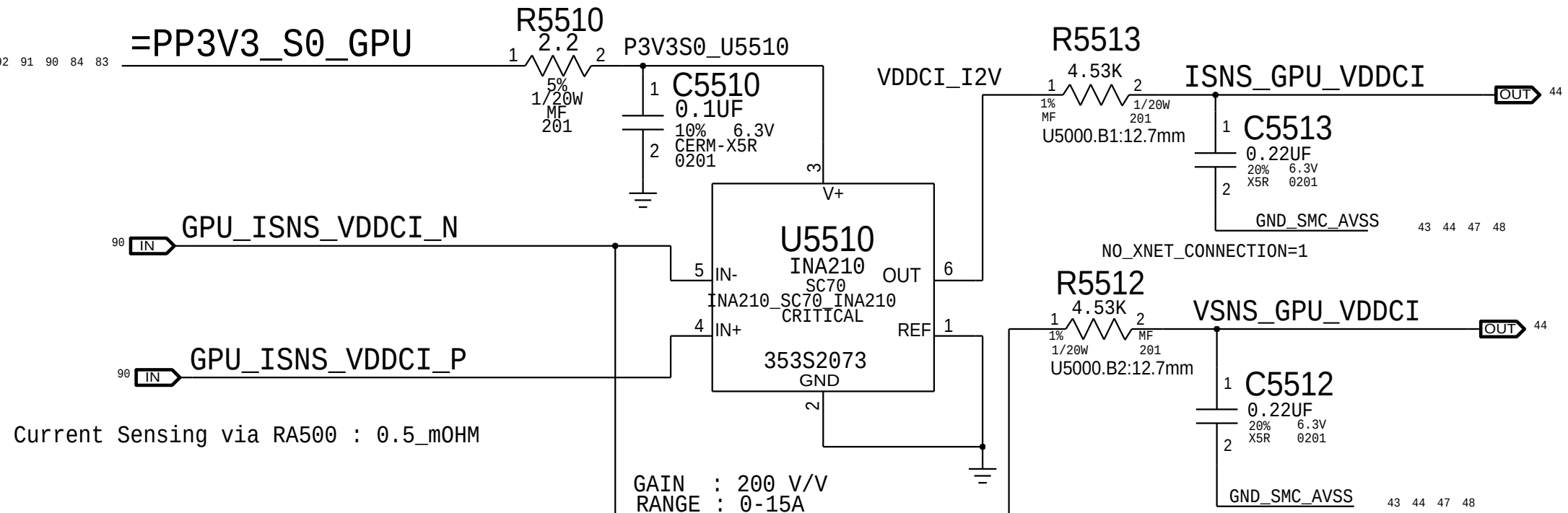
A

VDDQ S3 (VC0M=VM0R, IC0M:ADC9)

VDDQ lowside sense for S0-DIMM modules

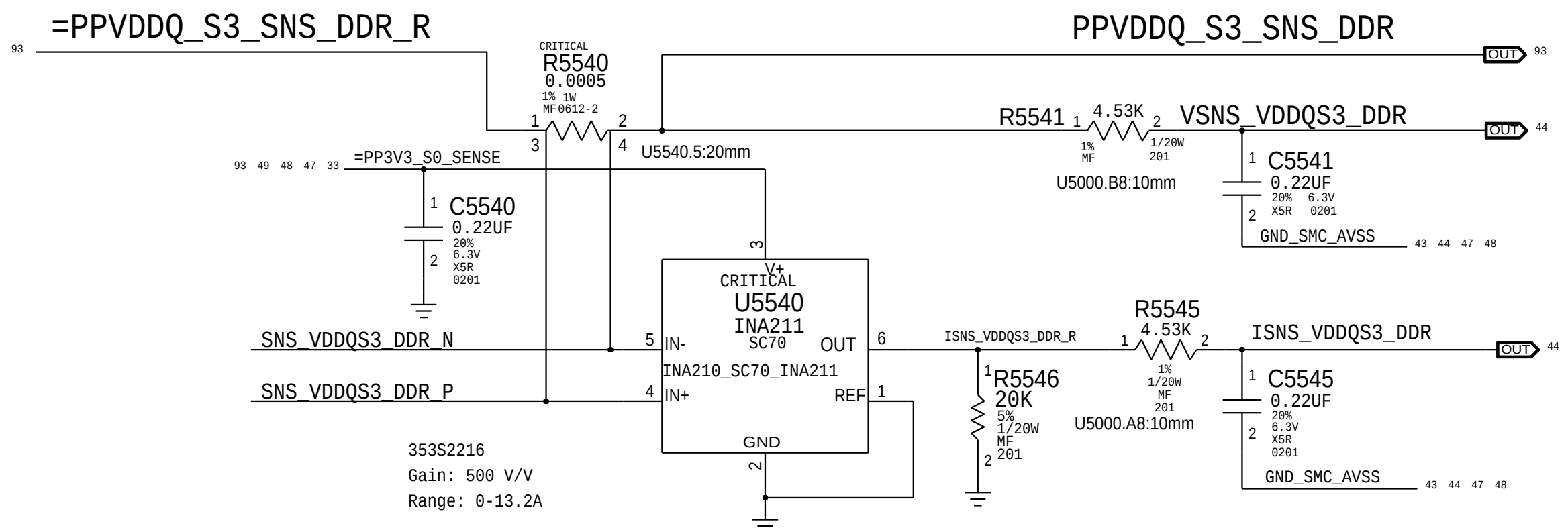


GPU_VDDCI S0 (VG0I:ADC14 /IG0I: ADC15)



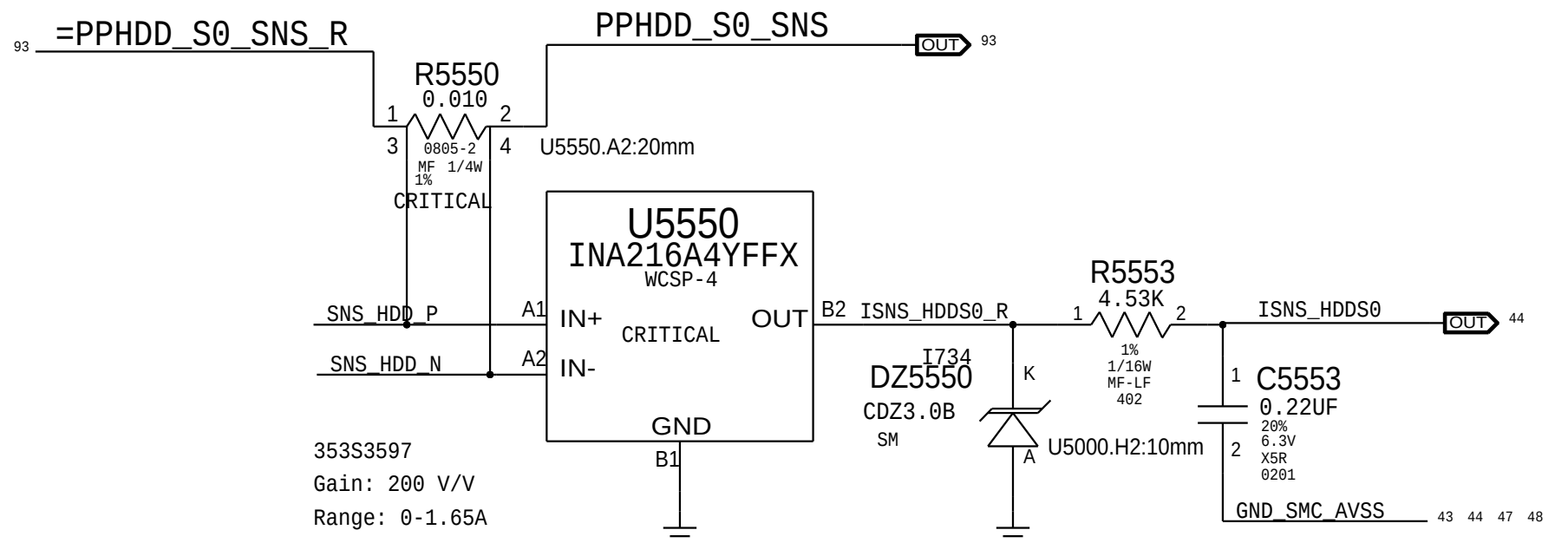
VDDQ S3 (VM0R:ADC22/ IM0R:ADC23)

VDDQ lowside sense for S0-DIMM modules



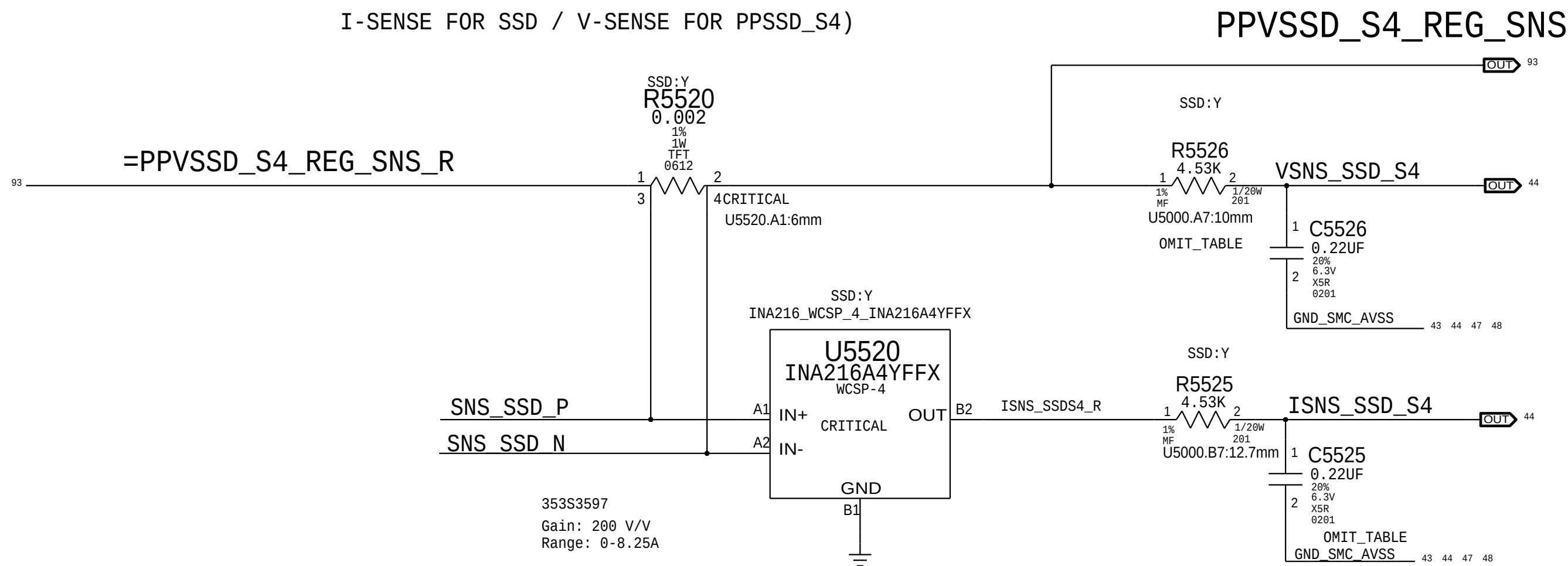
HDD S0 (VH05=5V, IH05:ADC19)

I/V-sense for HDD 5V



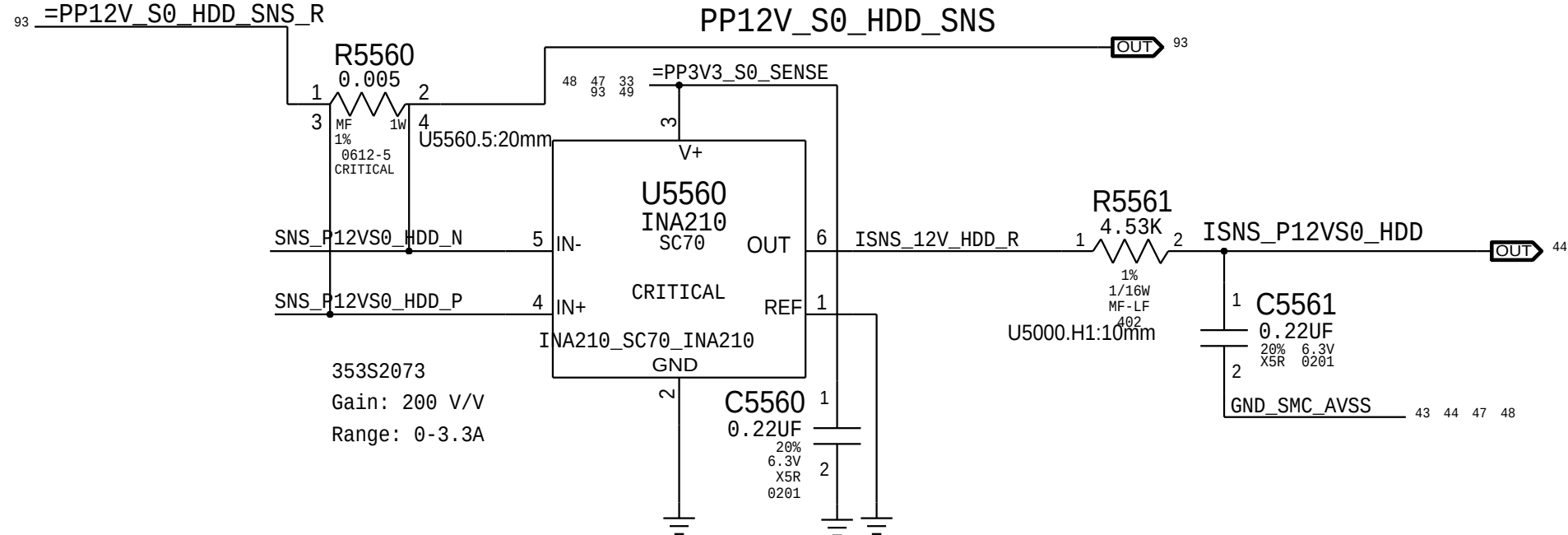
SSD S4 (VR1R:ADC20 / IH1R:ADC21)

I-SENSE FOR SSD / V-SENSE FOR PPSSD_S4)



PP12V_S0_HDD (VH02=VD20, IH02:ADC18)

HDD 12V CURRENT SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S8394	2	CAP, 0.22UF, 201	C5525, C5526	SSD:Y
117S8281	2	RES, 0 OHM, 201	C5525, C5526	SSD:N

SYNC_MASTER=BRANCH_JERRYCHOW		SYNC_DATE=09/10/2014	
PAGE TITLE			
SENSORS: I and V Sense(Continued)			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
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		PAGE	55 OF 120
		SHEET	48 OF 96

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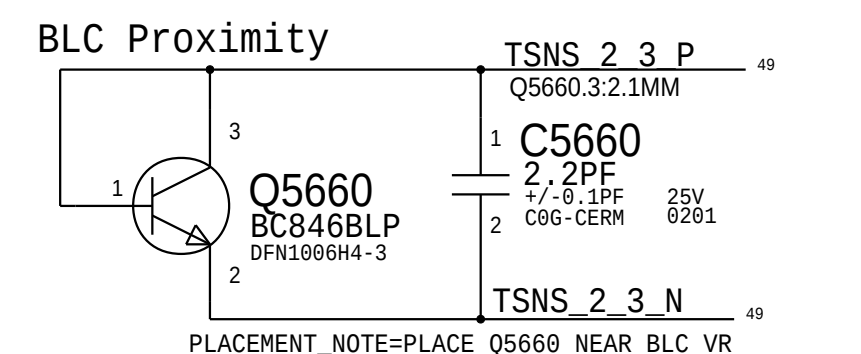
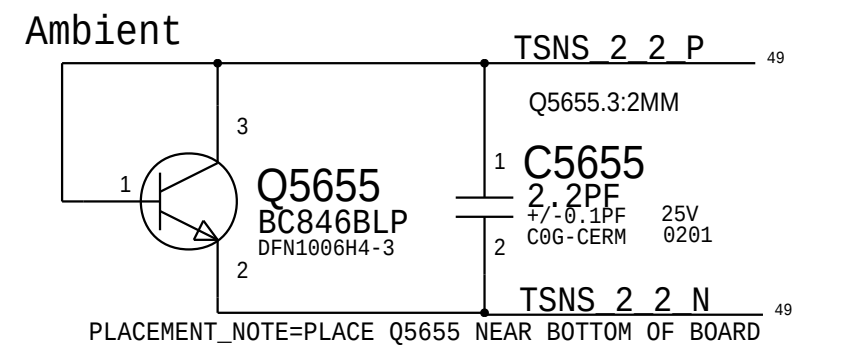
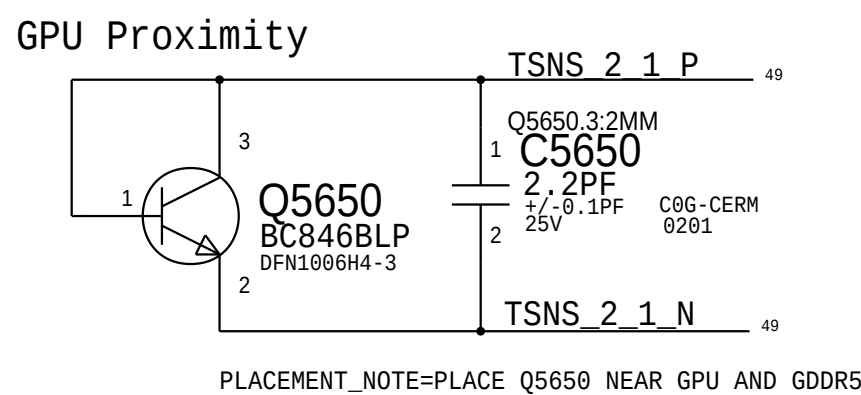
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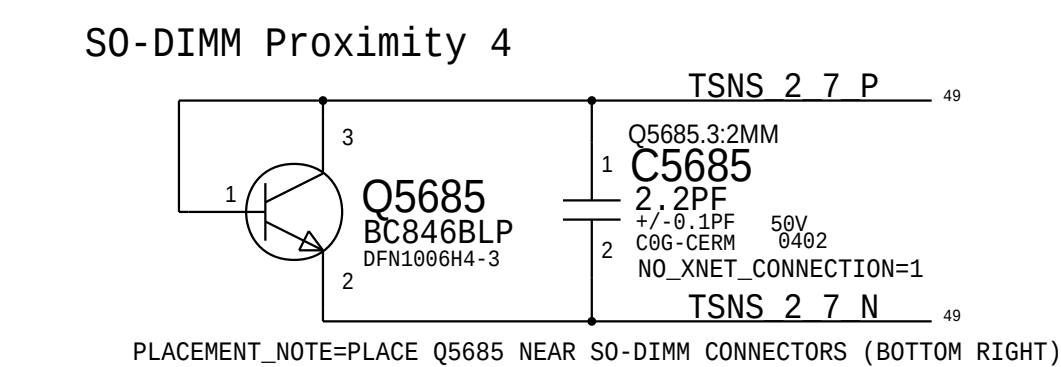
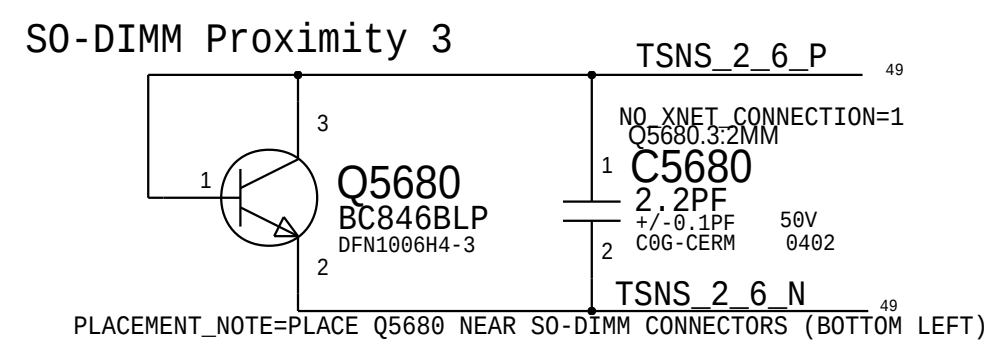
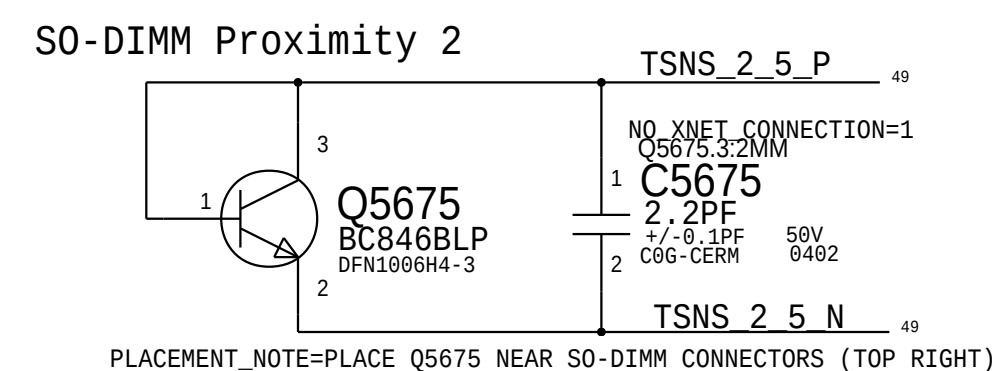
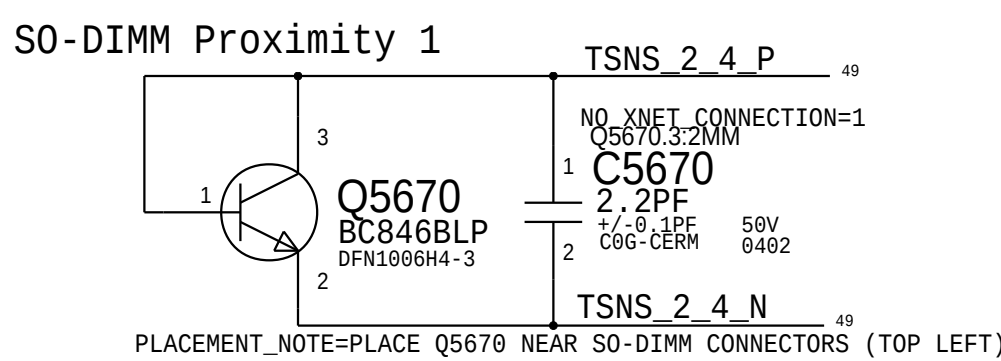
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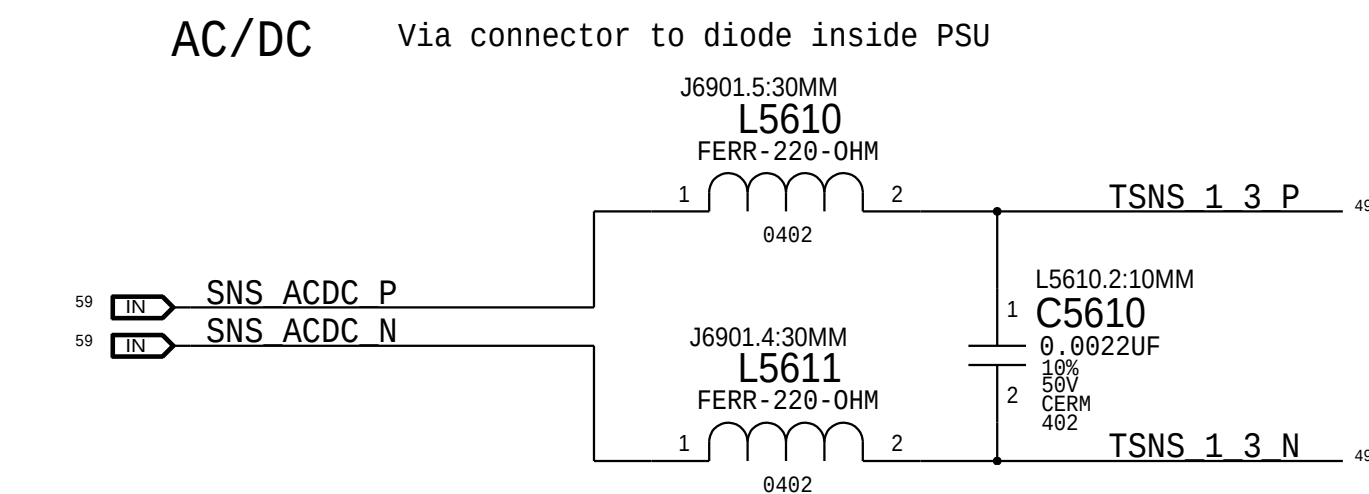
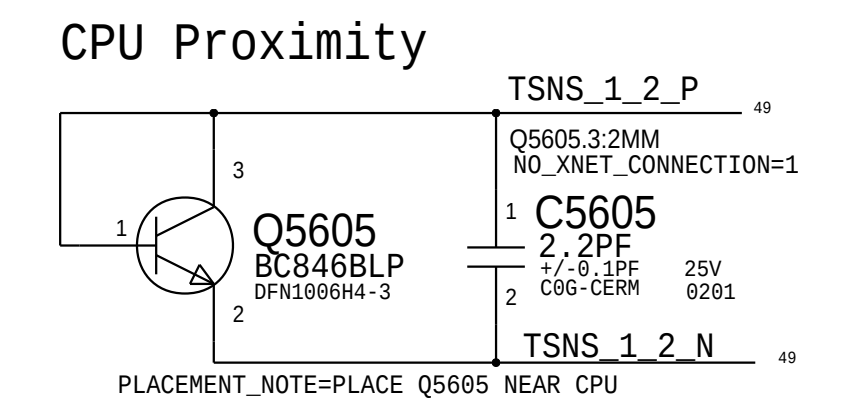
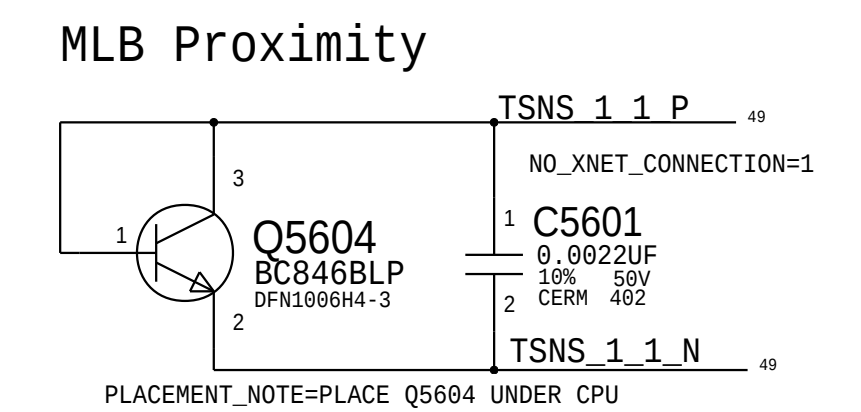
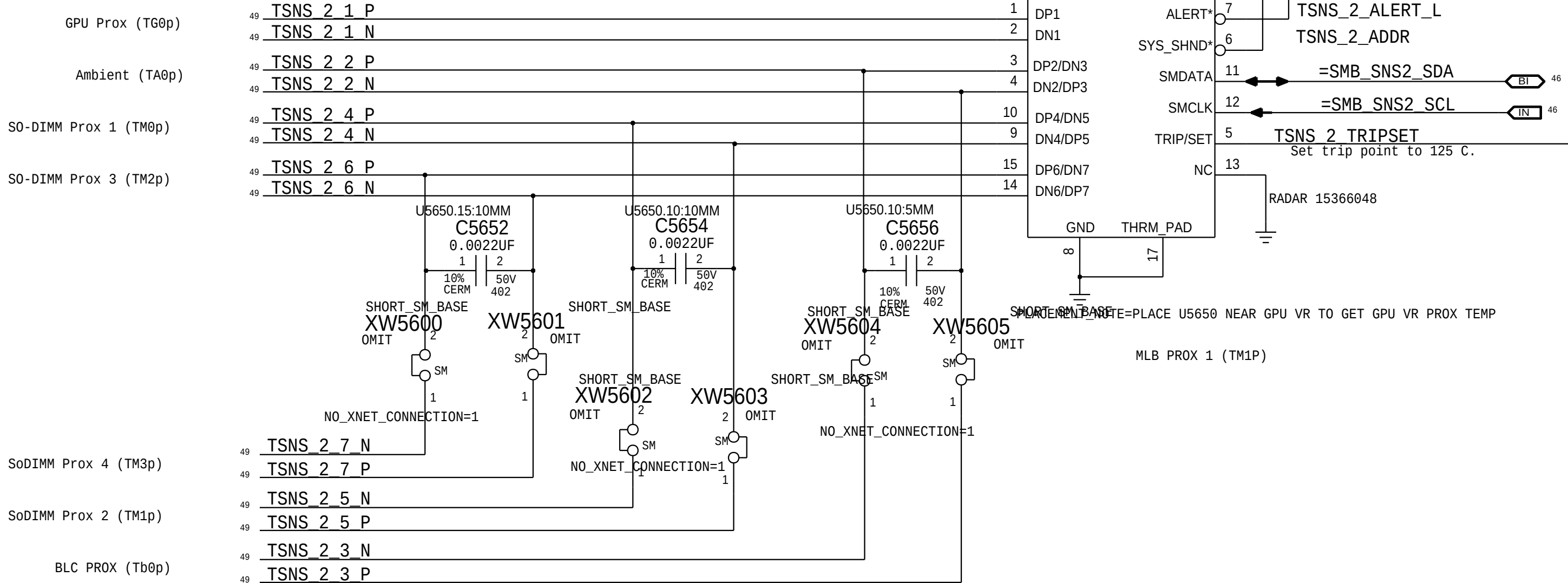


NEED TO FIND LOCATION

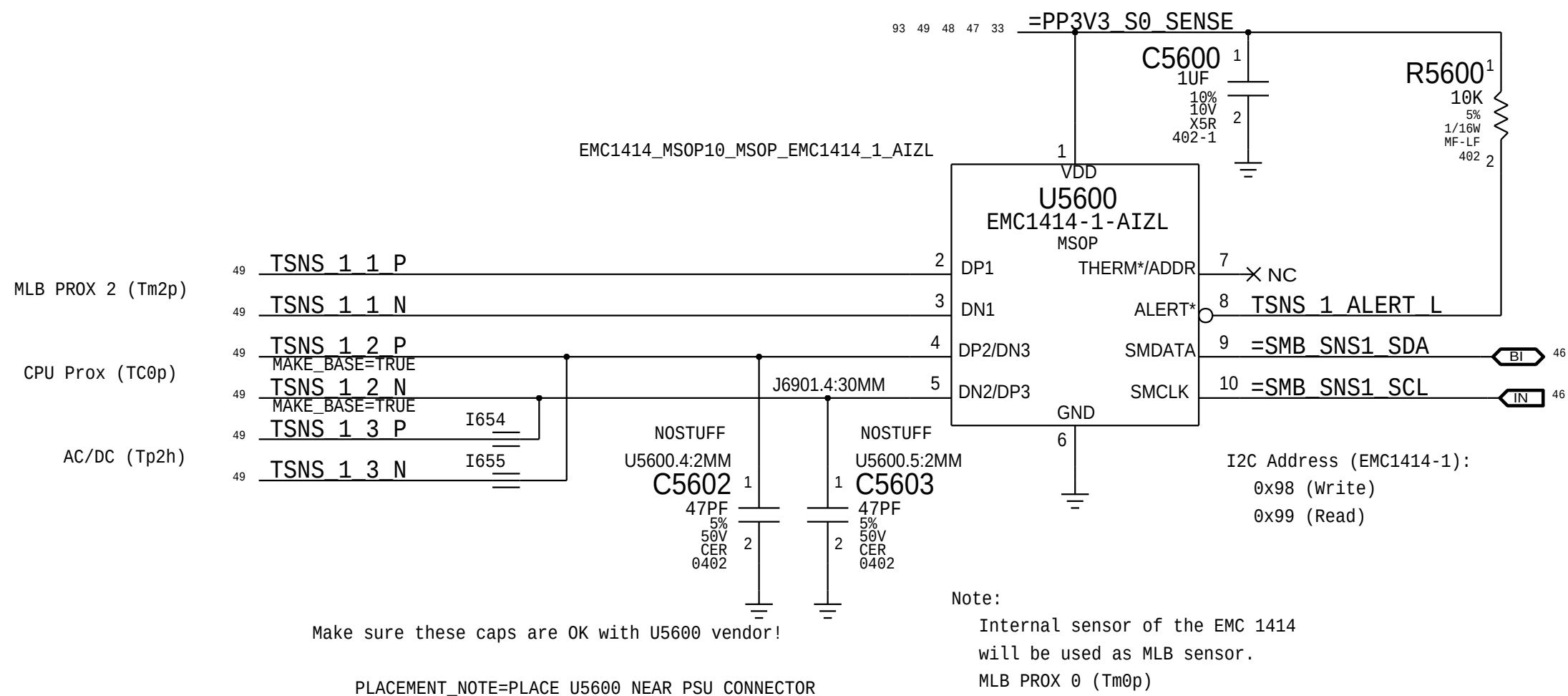


TEMP SENSOR T2 EMC1428: NEAR GPU VR

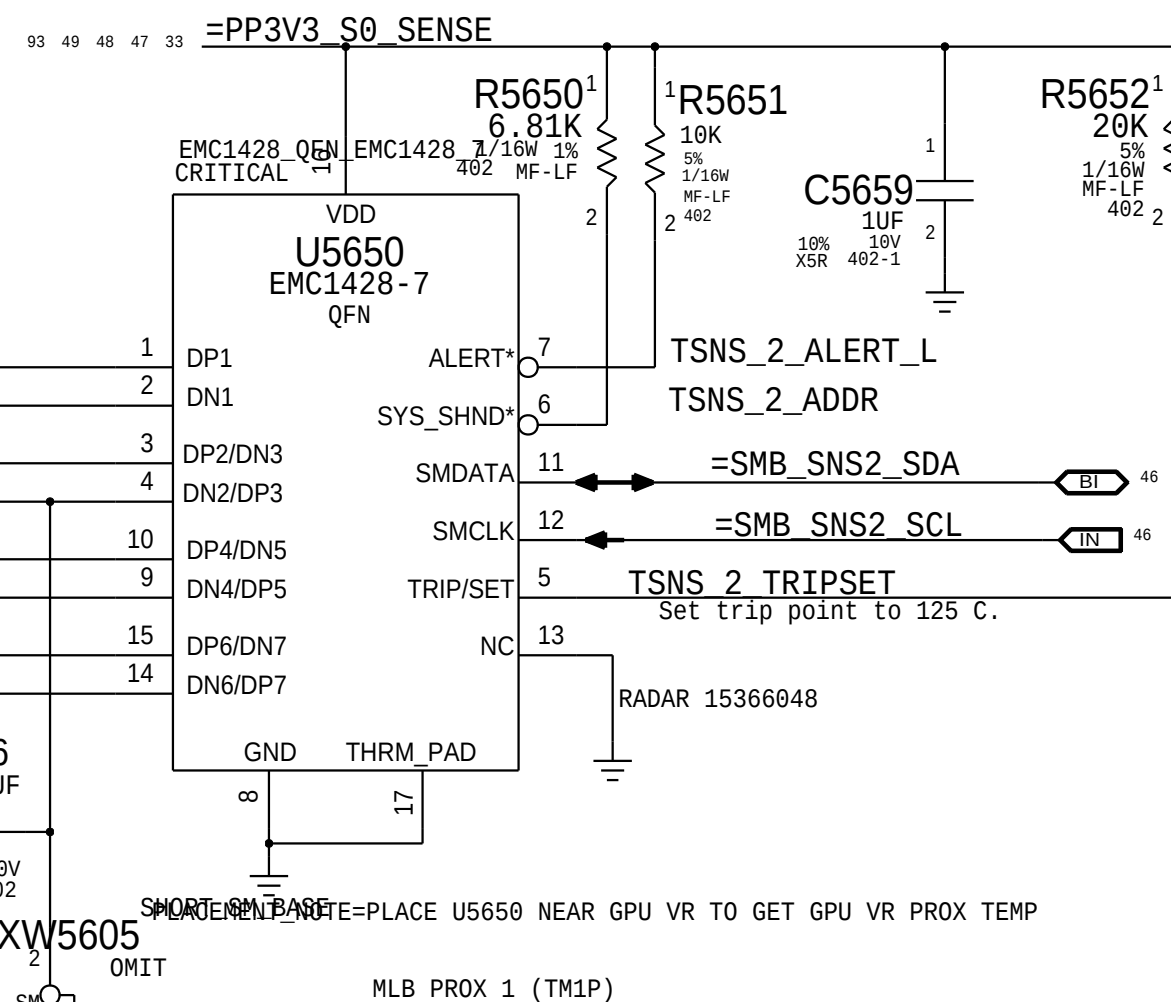
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode



Temperature Sensor T1 EMC1414: Near PSU Conn



SNS T2: TEMP SENSOR IC



SYNC_MASTER=BRANCH_JERRYCHOW		SYNC_DATE=09/10/2014	
PAGE TITLE			
SENSORS: Temperature Sensors			
 Apple Inc.		DRAWING NUMBER	051-00321
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		PAGE	56 OF 120
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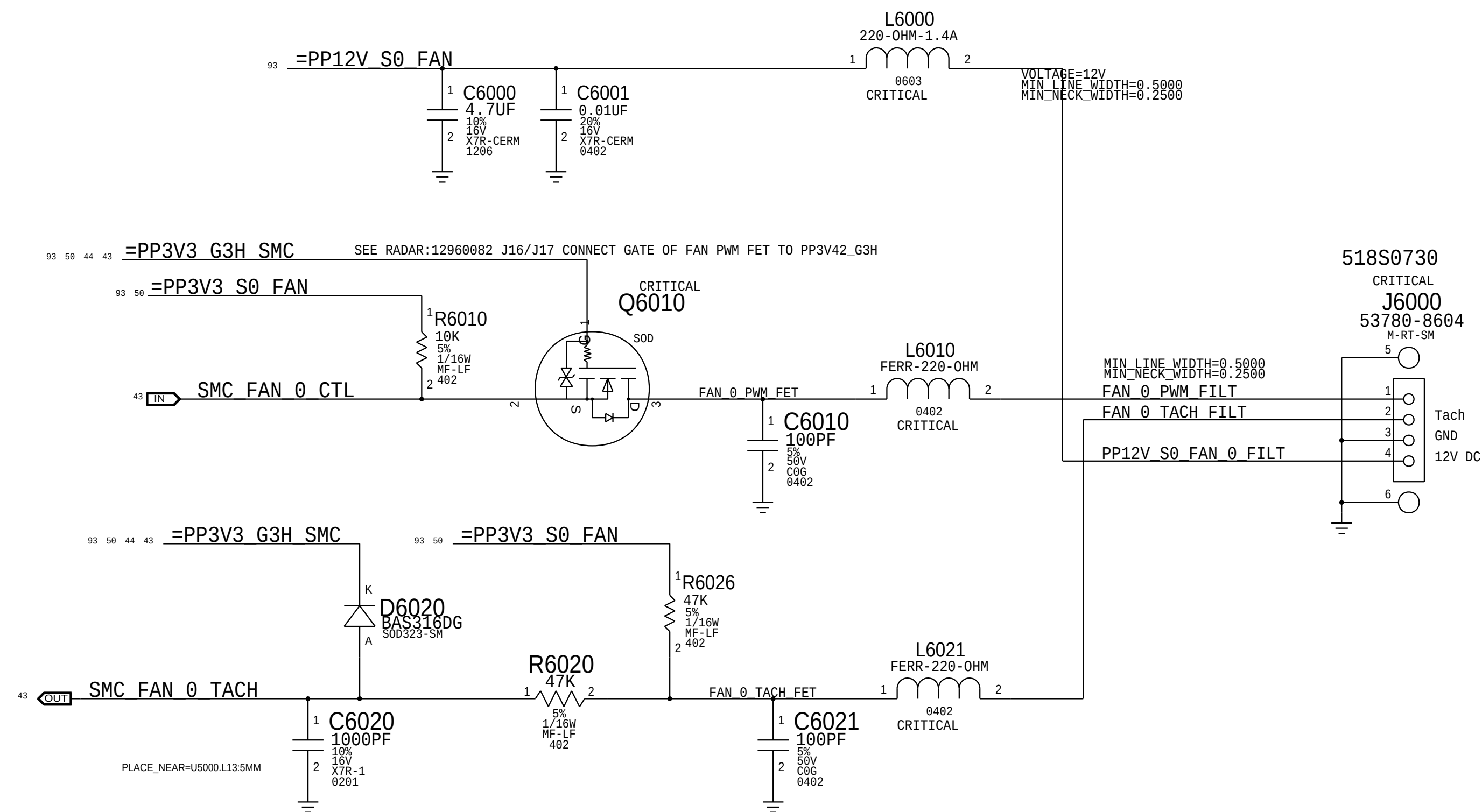
SMC Fan 0 (System)

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

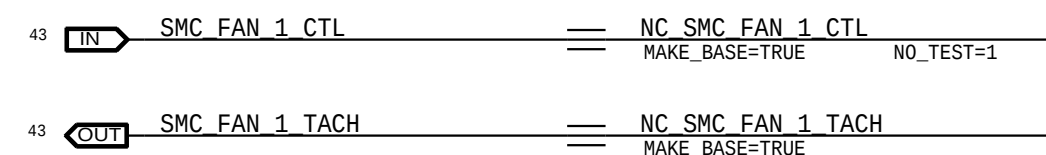
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC_MASTER=J16_IG		SYNC_DATE=04/29/2013	
PAGE TITLE			
FAN: System Fan			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
	REVISION		
	4.0.0		
	BRANCH		
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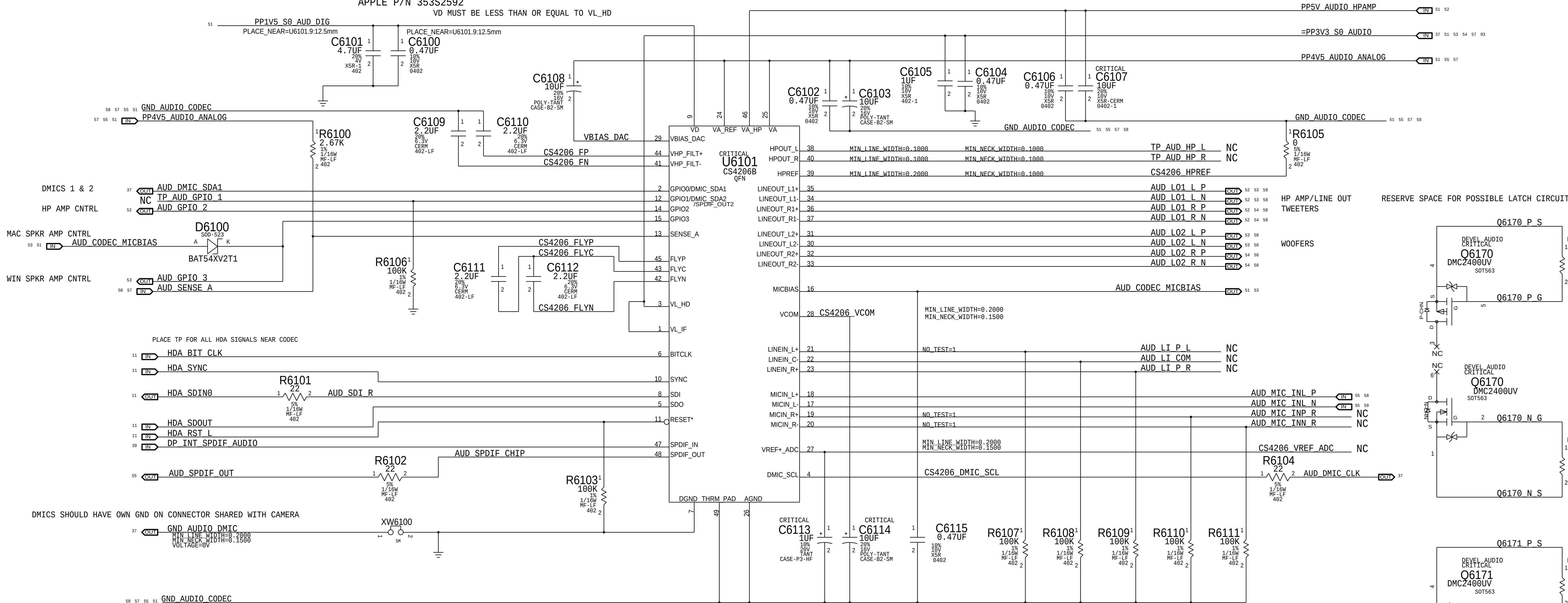
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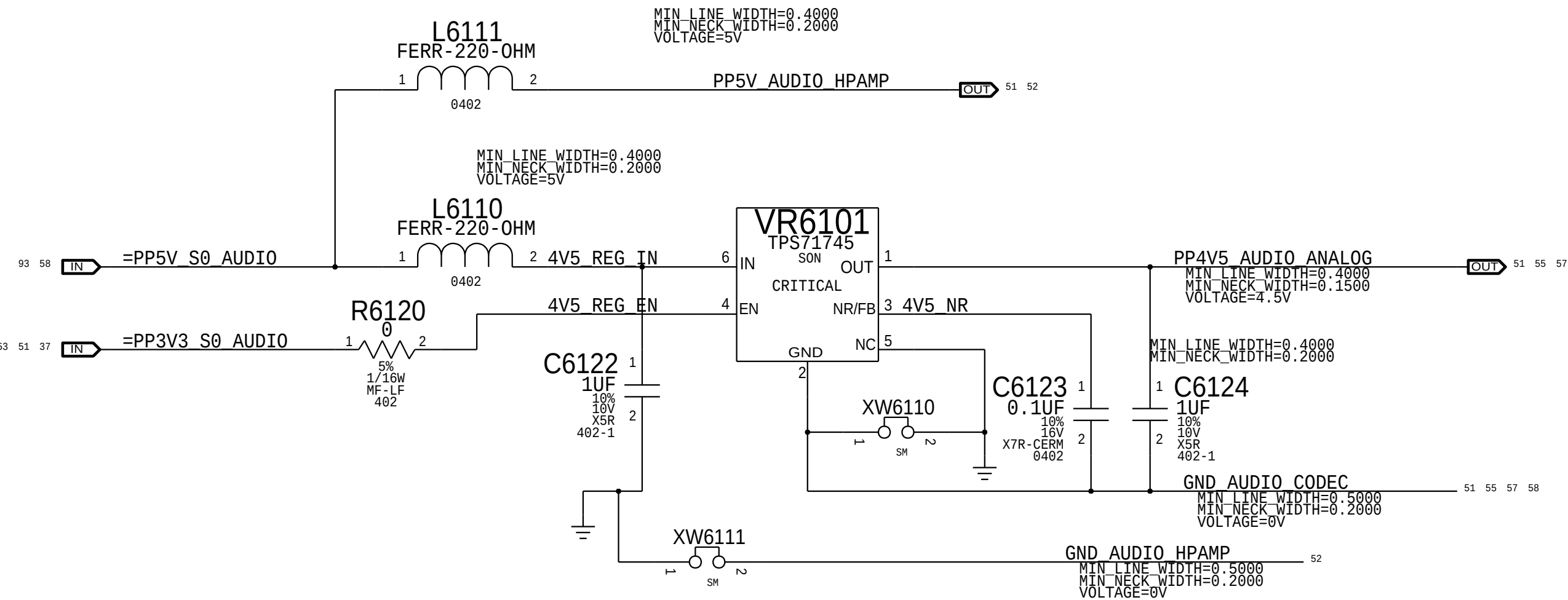
AUDIO CODEC
APPLE P/N 353S2592

PLACE C6100 AS CLOSE TO PIN 9 AS POSSIBLE

VD MUST BE LESS THAN OR EQUAL TO VL_HD



APPLE P/N 353S2456
4.5V POWER SUPPLY FOR CODEC



PLACE XW6110 BENEATH U6101, BETWEEN PINS 2 & 5

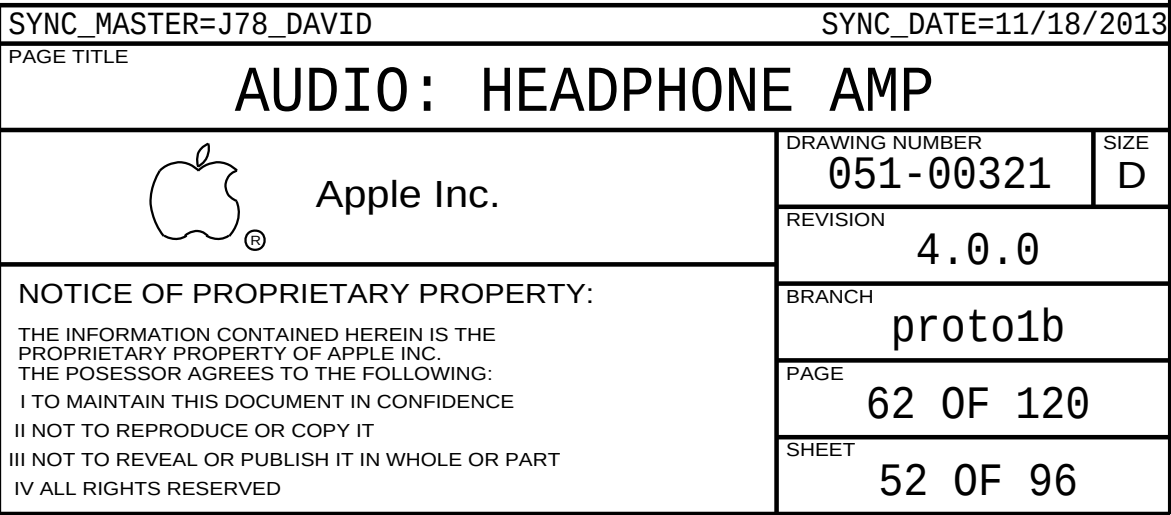
DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

HP AMP/LINE OUT
TWEETERS

WOOFERS

DEVEL AUDIO
CRITICAL
Q6171
DMC2400UV
SOT563

SYNC_MASTER=J17_DIRK		SYNC_DATE=03/07/2013		
PAGE TITLE				
AUDIO: CODEC/REGULATORS				
 Apple Inc.		DRAWING NUMBER	SIZE	
		051-00321	D	
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		4.0.0		
		BRANCH	proto1b	
		PAGE	61 OF 120	
		SHEET	51 OF 96	



LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6304
0 OHM
NOSTUFF

R6305
0 OHM
NOSTUFF

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

R6307
0 OHM
47 KOHM
NOSTUFF
NOSTUFF
NOSTUFF

SYNC_MASTER=J78_DAVID		SYNC_DATE=11/18/2013	
PAGE TITLE			
AUDIO: LEFT SPKR AMP			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
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		PAGE	63 OF 120
		SHEET	53 OF 96

RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

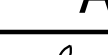
PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON R6404
OFF NOSTUFF

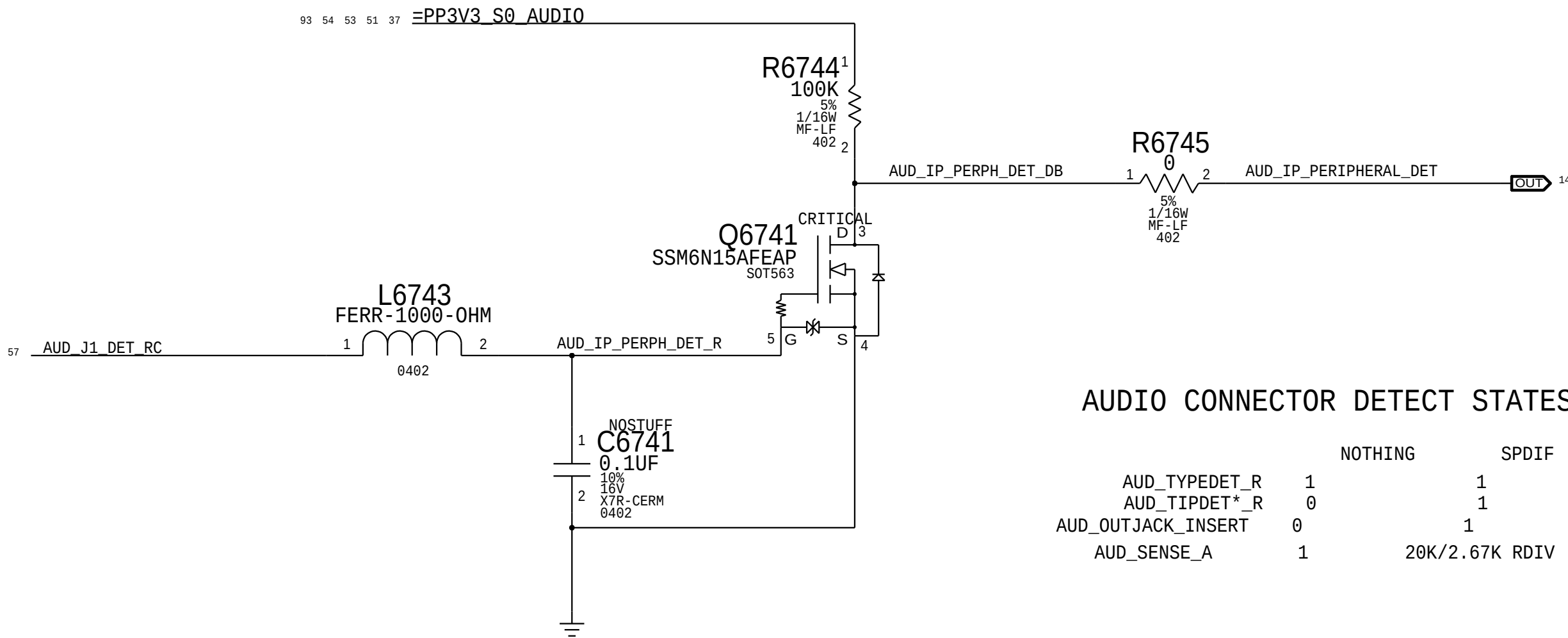
R6405
NOSTUFF
0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

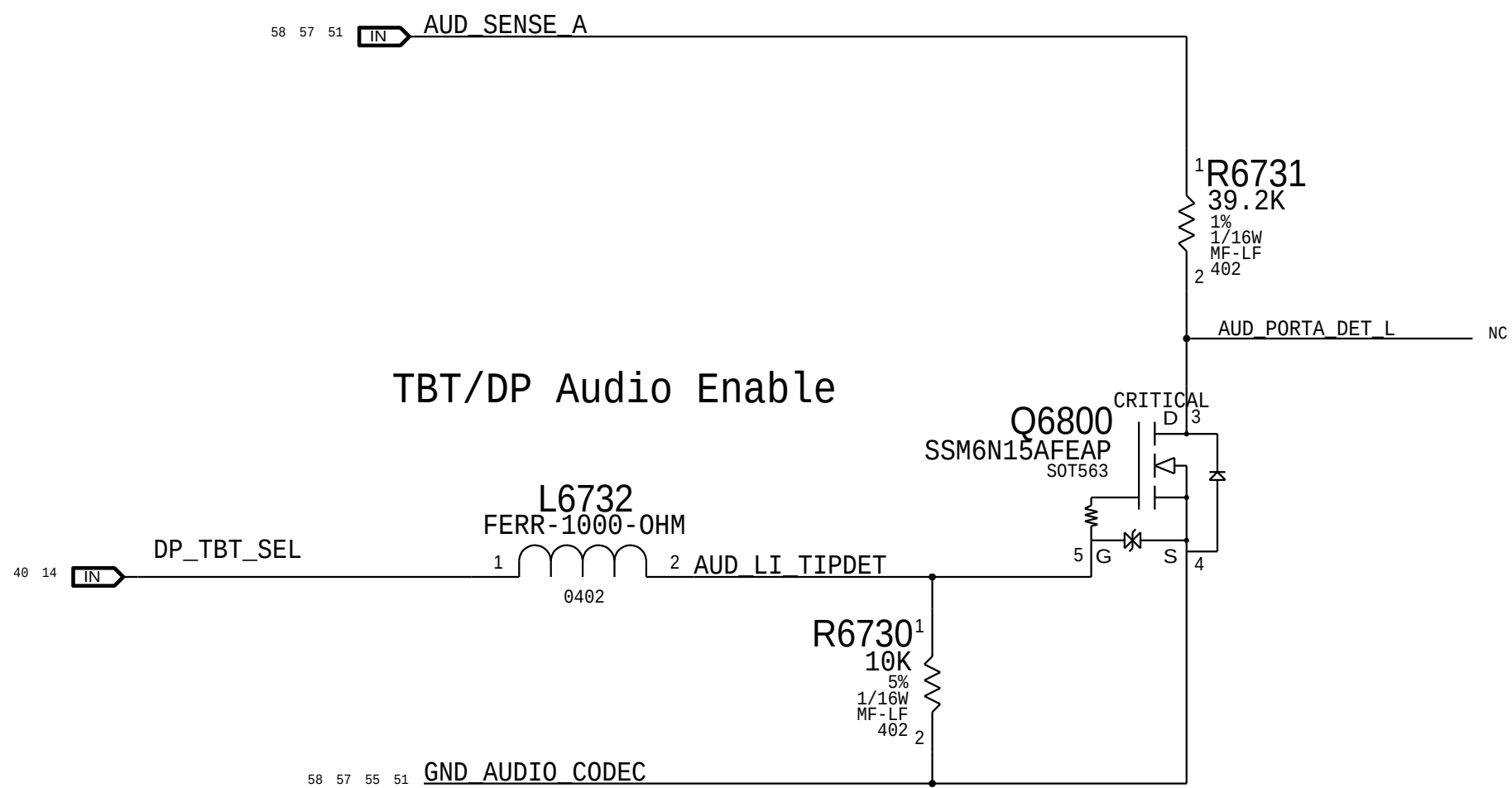
GAIN R6406 R6407
+9 DB NOSTUFF 0 OHM
+12 DB NOSTUFF 47 KOHM
+15 DB NOSTUFF NOSTUFF
+18 DB 47 KOHM NOSTUFF
+24 DB 0 OHM NOSTUFF

SYNC_MASTER=J78_DAVID		SYNC_DATE=11/18/2013	
PAGE TITLE			
AUDIO: RIGHT SPKR AMP			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00321	D
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		BRANCH	proto1b
		PAGE	64 OF 120
		SHEET	54 OF 96

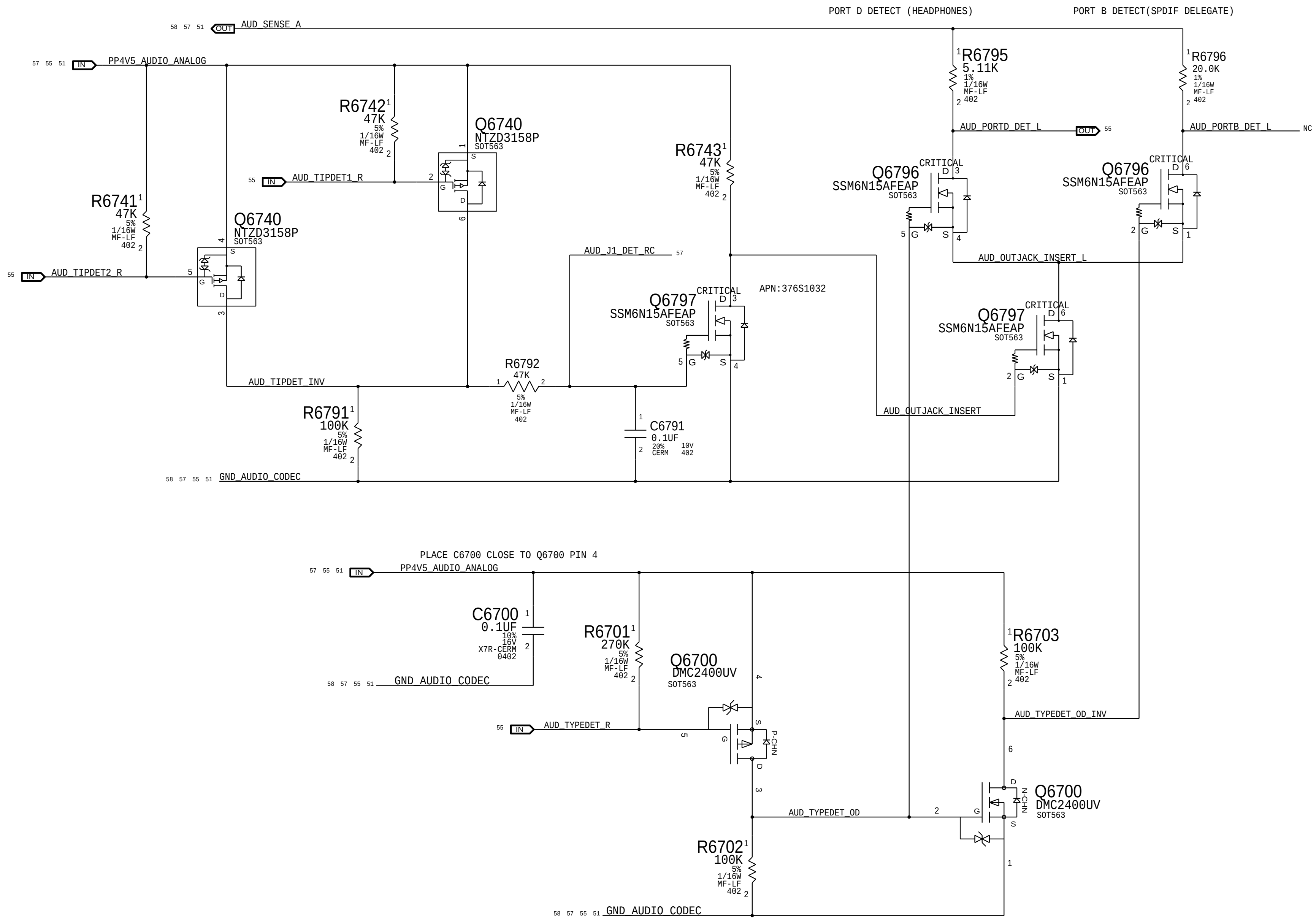
IPHS HS Detect Debounce CKT

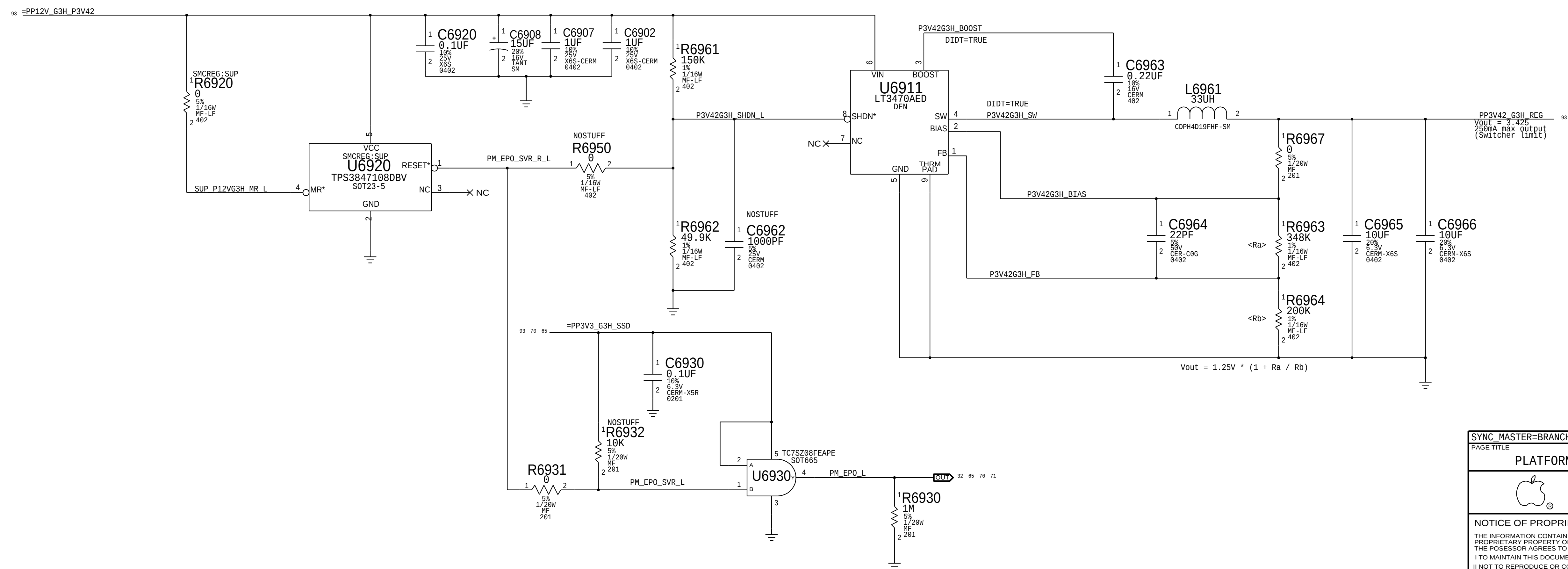
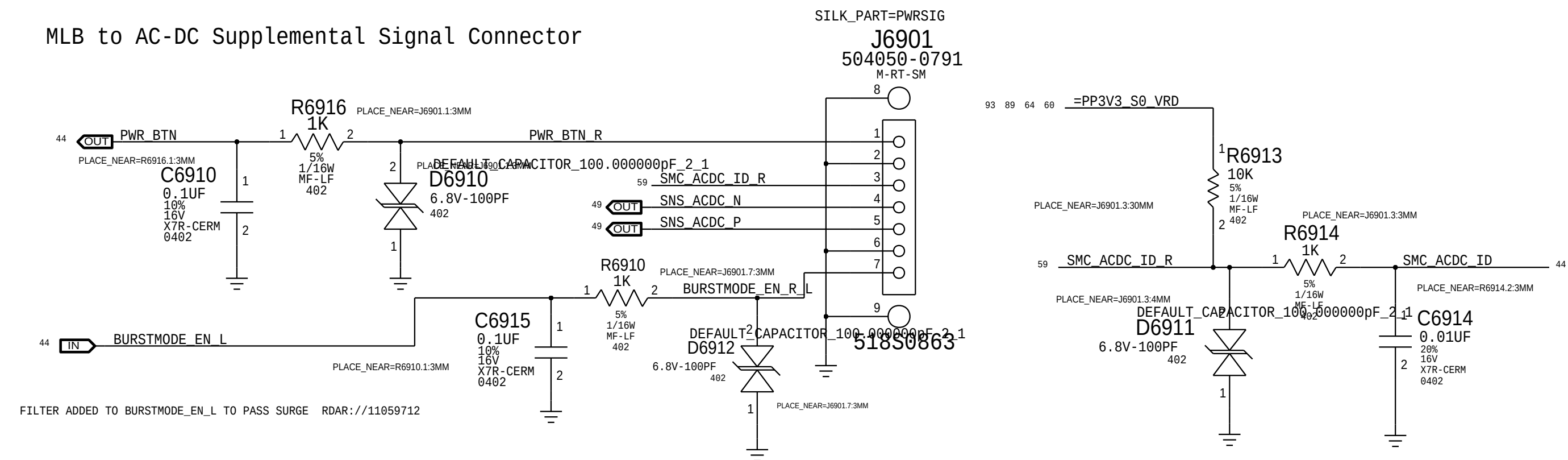
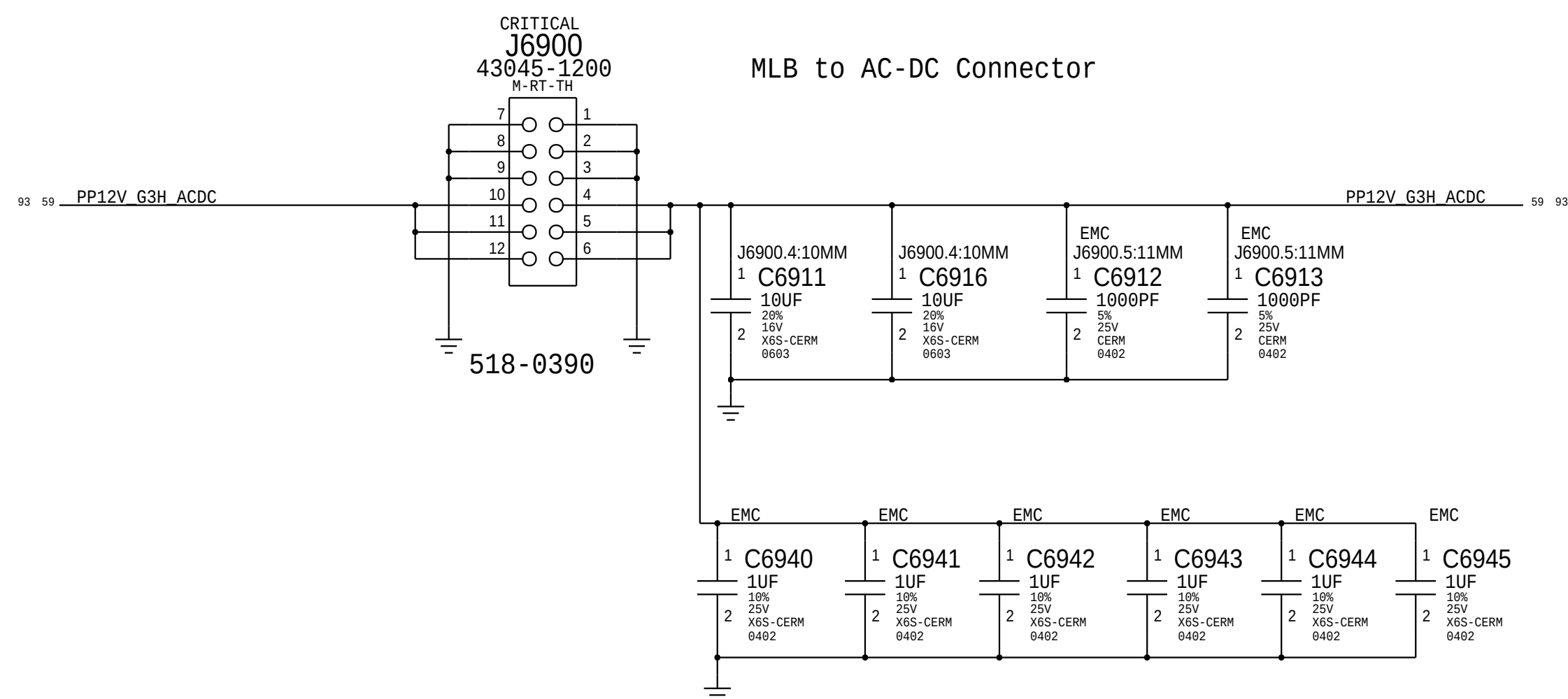


Target Display Mode Detect

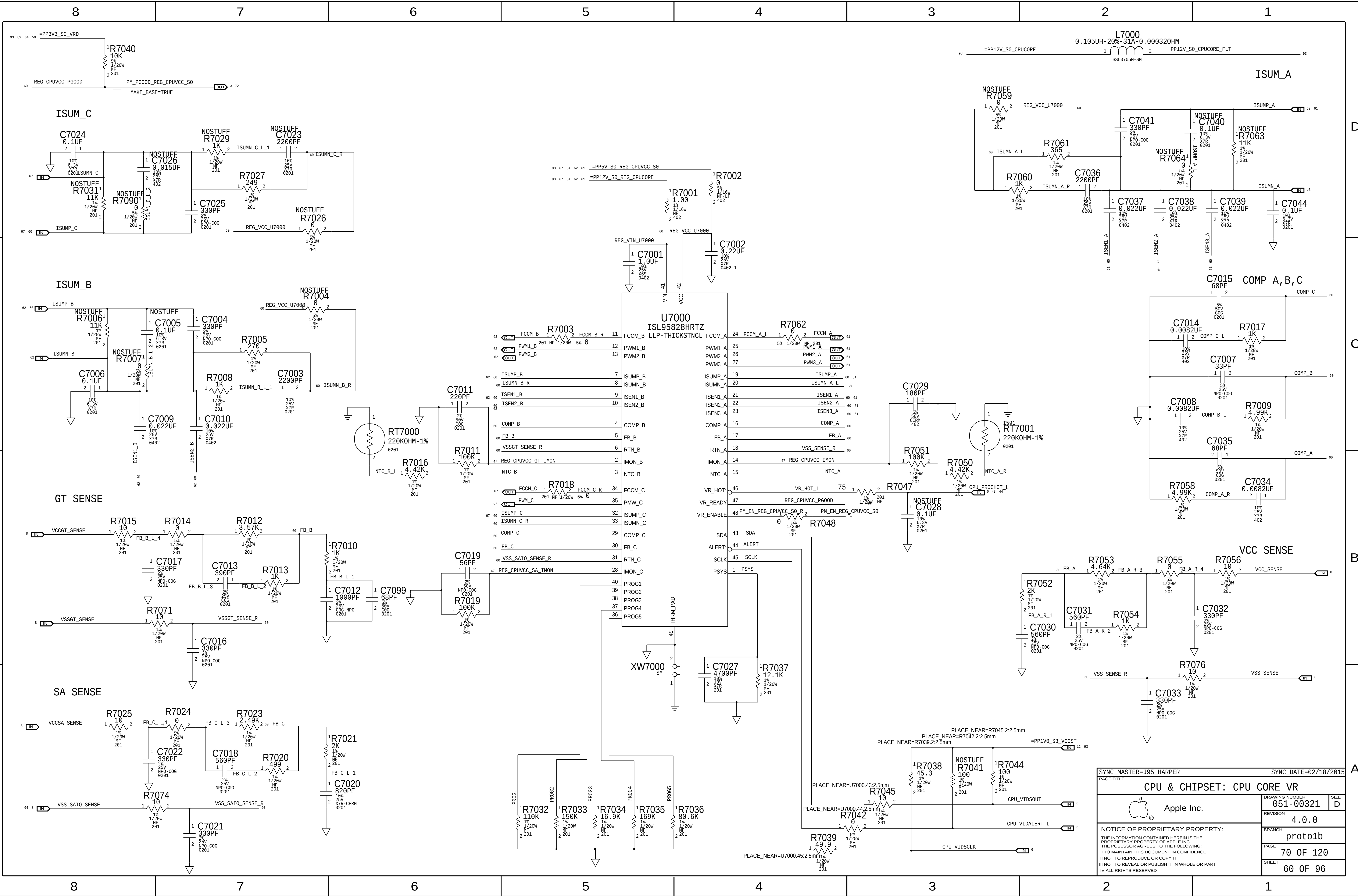


TBT/DP Audio Enable





SYNC_MASTER=BRANCH_HCHENG		SYNC_DATE=10/29/2014	
PAGE TITLE			
PLATFORM POWER: Connectors / VReg G3Hot			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
	REVISION		
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		proto1b	
		PAGE	69 OF 120
		SHEET	59 OF 96



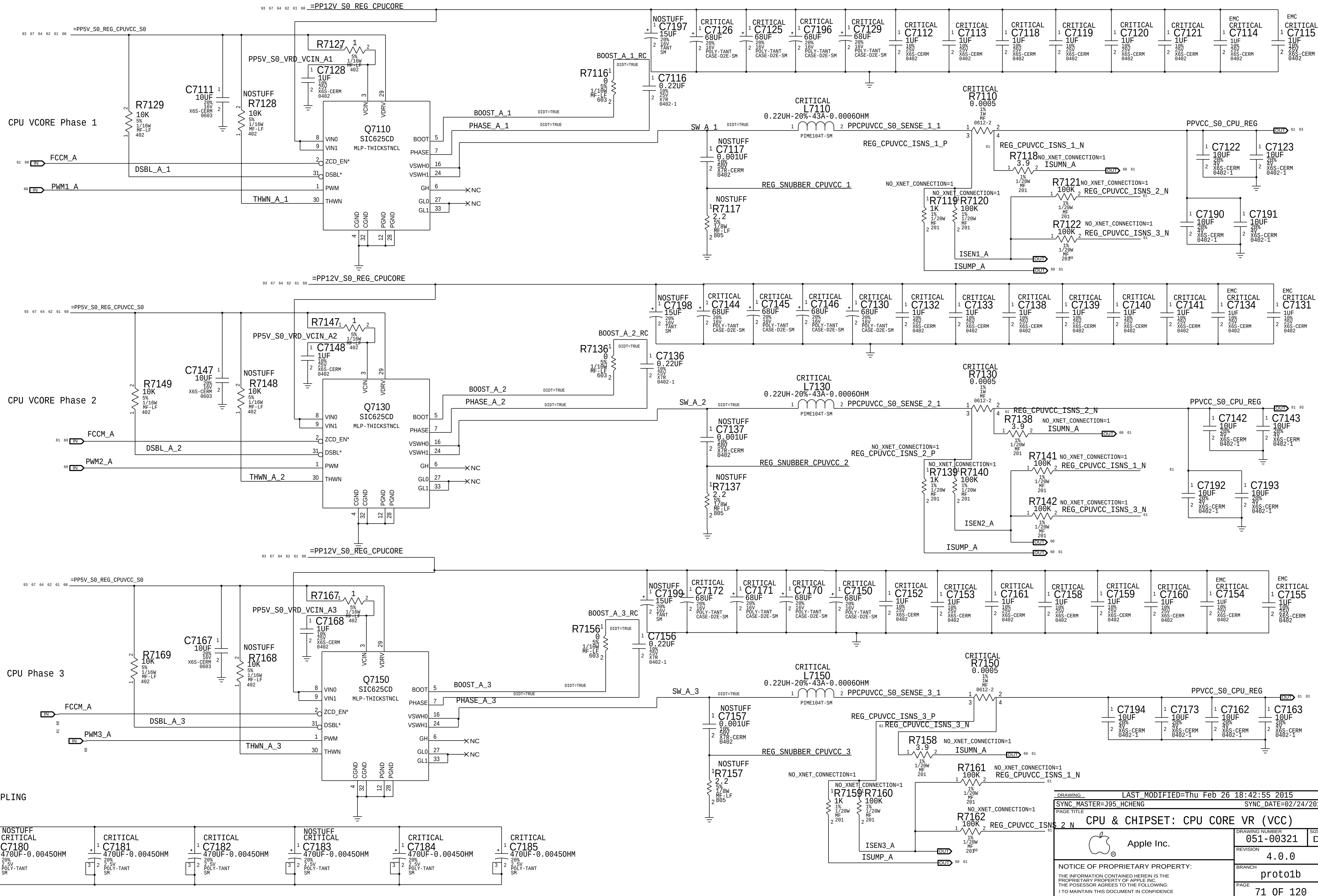
D

C

B

A

CPU VCC Regulator
EDC = 100A
TDC = 70A



D

C

B

A

DRAWING		LAST_MODIFIED=Thu Feb 26 18:42:55 2015	
SYNC_MASTER=J95_HCHENG		SYNC_DATE=02/24/2015	
PAGE TITLE			
CPU & CHIPSET: CPU CORE VR (VCC)			
DRAWING NUMBER		SIZE	
051-00321		D	
REVISION		4.0.0	
BRANCH		proto1b	
PAGE		71 OF 120	
SHEET		61 OF 96	
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8

7

6

5

4

3

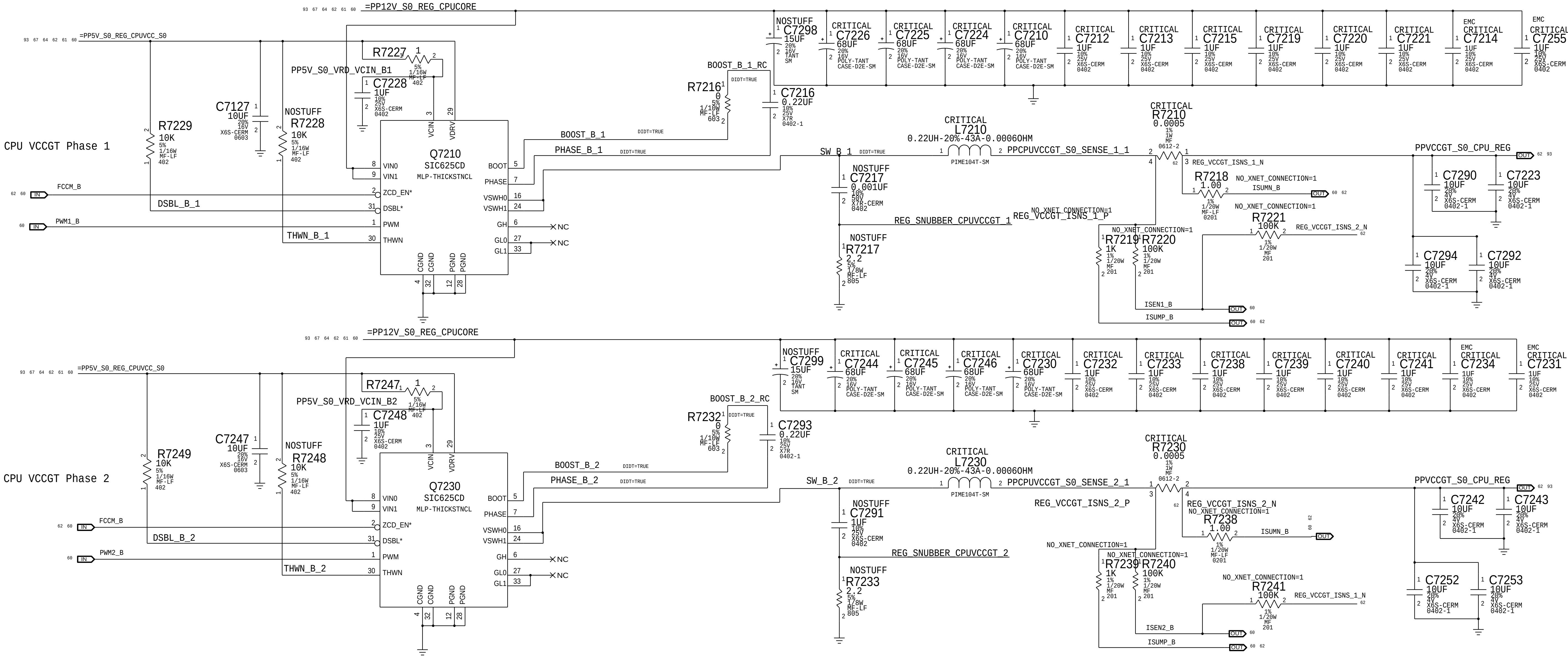
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1

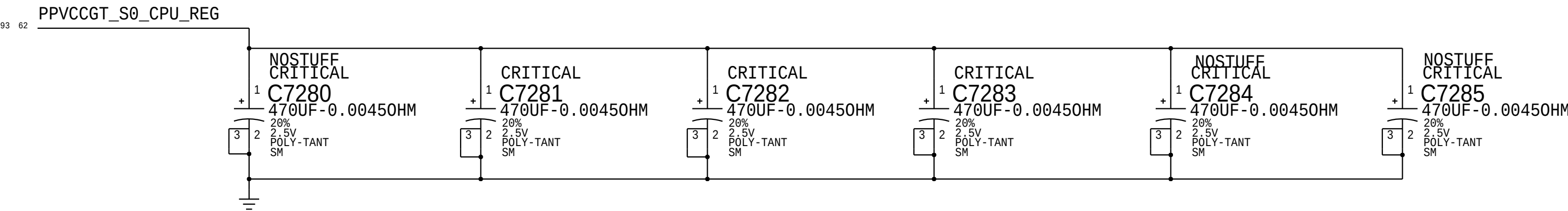
CPU VCCGT Regulator

EDC = 51A

TDC = 37A



CPU VCCGT OUTPUT DECOUPLING

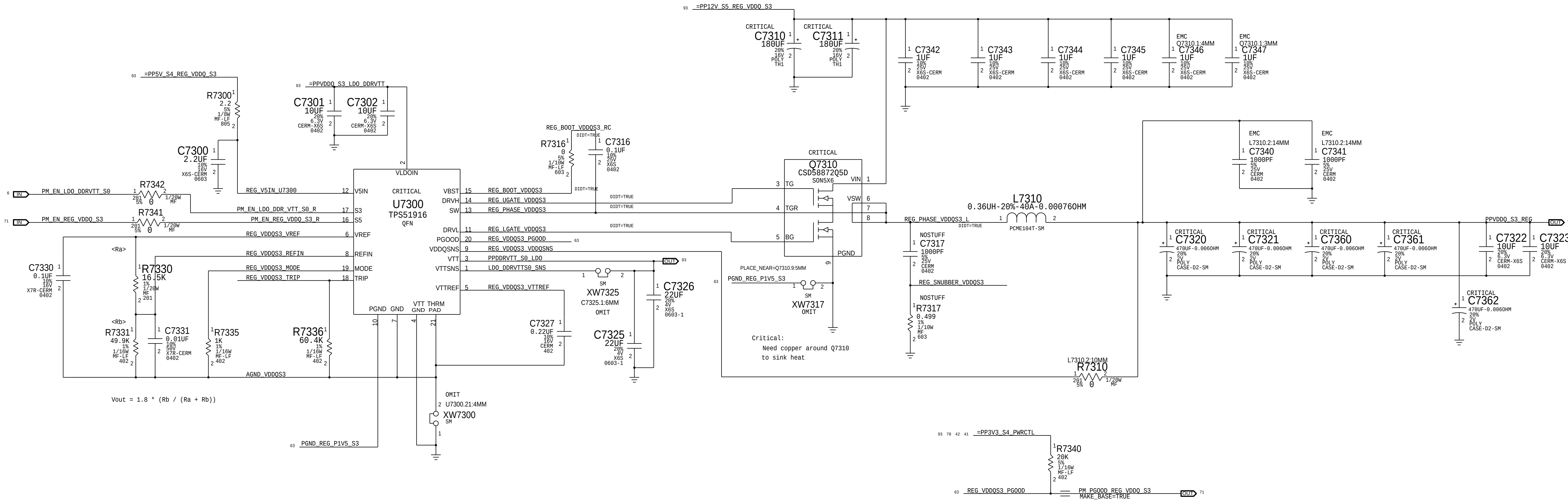


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PAGE TITLE			
CPU & CHIPSET: CPU CORE VR (VCCGT)			
 Apple Inc.	DRAWING NUMBER		051-00321
	REVISION		4.0.0
	BRANCH		proto1b
	PAGE		72 OF 120
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VDDQ (1.35V) S3 REGULATOR

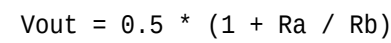
EDC = 20A

TDC = 14A



SYNC_MASTER=BRANCH_HCHENG		SYNC_DATE=10/29/2014			
PAGE TITLE					
CPU & CHIPSET: CPU VDDQ VR					
 Apple Inc.		DRAWING NUMBER	051-00321	SIZE	D
		REVISION	4.0.0		
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		PAGE	73 OF 120		
		SHEET	63 OF 96		

EDC = 5.5A
TDC = 5.5A



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S8358	2	CAP, 470uF, 0.0060HM, 2V, D2	C7540, C7541	VR_BULKCAP:CURRENT
128S8381	2	CAP, 470uF, 0.00450HM, 2.5V, SM	C7540, C7541	VR_BULKCAP:FUTURE

DRAWING		LAST_MODIFIED=Thu Feb 26 18:42:55 2015	
SYNC_MASTER=J95_HARPER		SYNC_DATE=02/13/2015	
PAGE TITLE			
CPU & CHIPSET: CPU VCCIO VR			
 Apple Inc.	DRAWING NUMBER		SIZE
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	REVISION		
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		PAGE	
		75 OF 120	
		SHEET	
		64 OF 96	

3.3V S5 Regulator

EDC = 12.46A

TDC = 9.4A

5V S4 Regulator

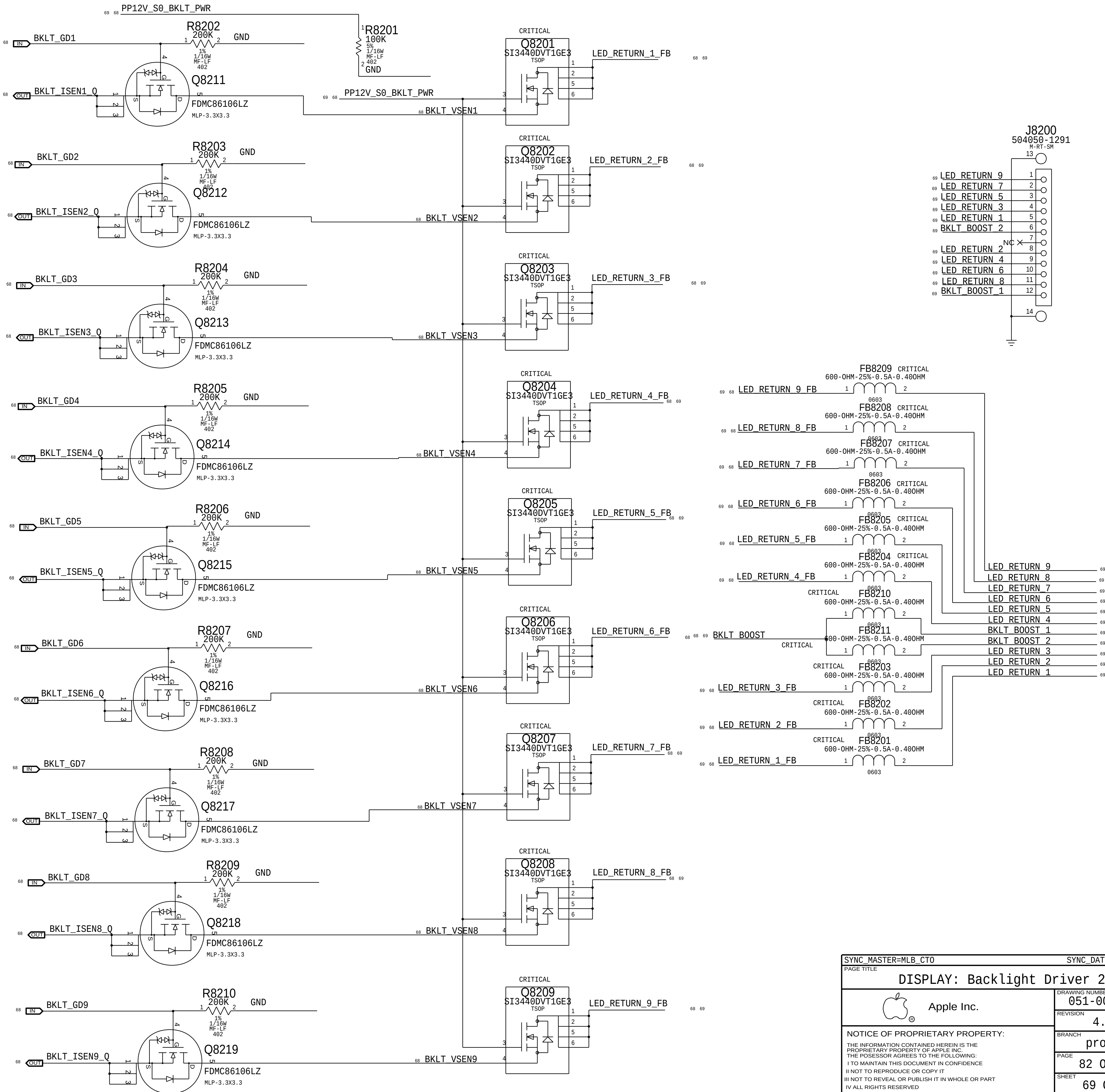
EDC = 6.5A

TDC = 5.9A

3V3 S5 FET NON SSD

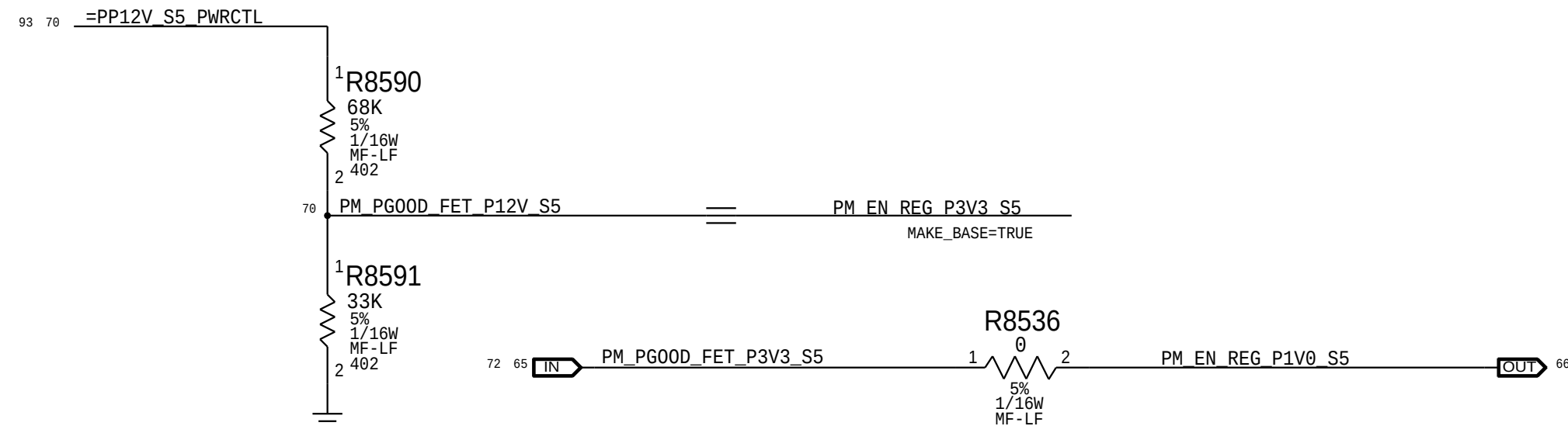
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PAGE TITLE			
PLATFORM POWER: 3.3V S5/5V S4 VR			
	DRAWING NUMBER		SIZE
	051-00321		D
Apple Inc.	REVISION		
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	BRANCH		
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	PAGE		
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IV ALL RIGHTS RESERVED			
SHEET		65 OF 96	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1256	376S1973		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8201 TO FB8211

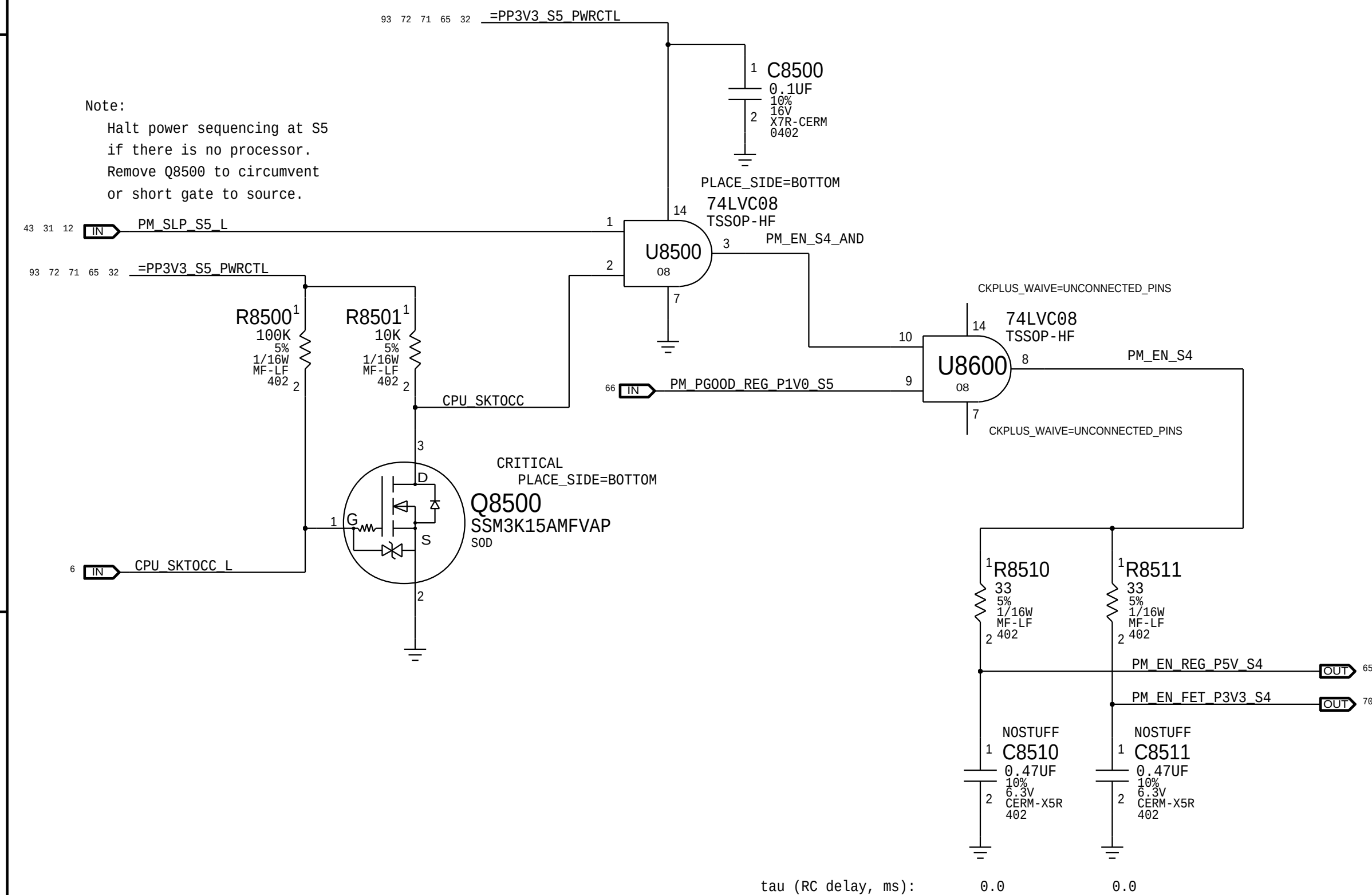


SYNC_MASTER=MLB_CTO		SYNC_DATE=05/14/2014	
PAGE TITLE			
DISPLAY: Backlight Driver 2			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00321	D
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		BRANCH	proto1b
		PAGE	82 OF 120
		SHEET	69 OF 96

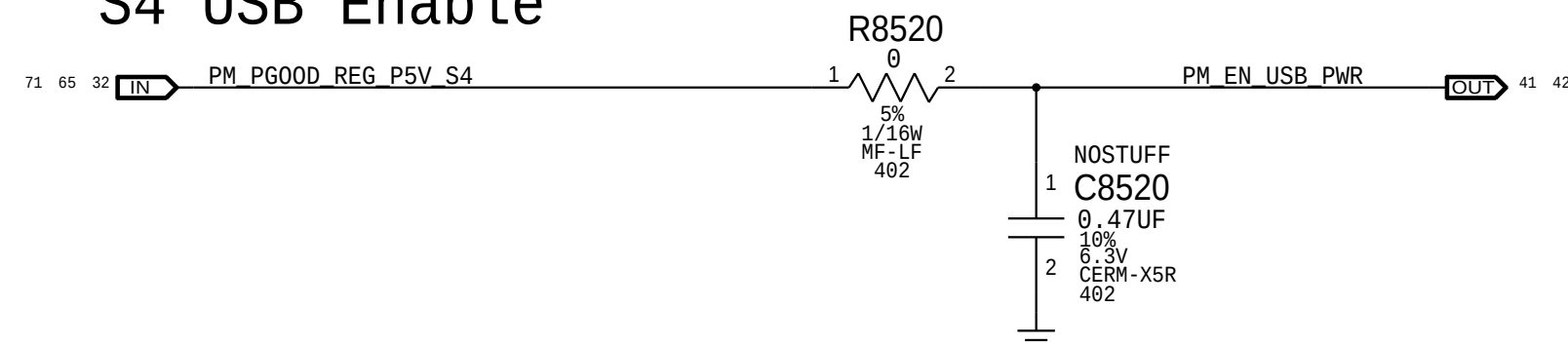
S5 Enable



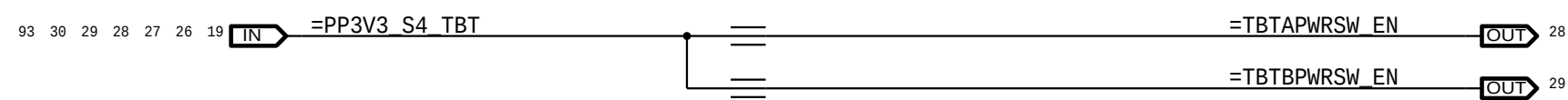
S4 Enables



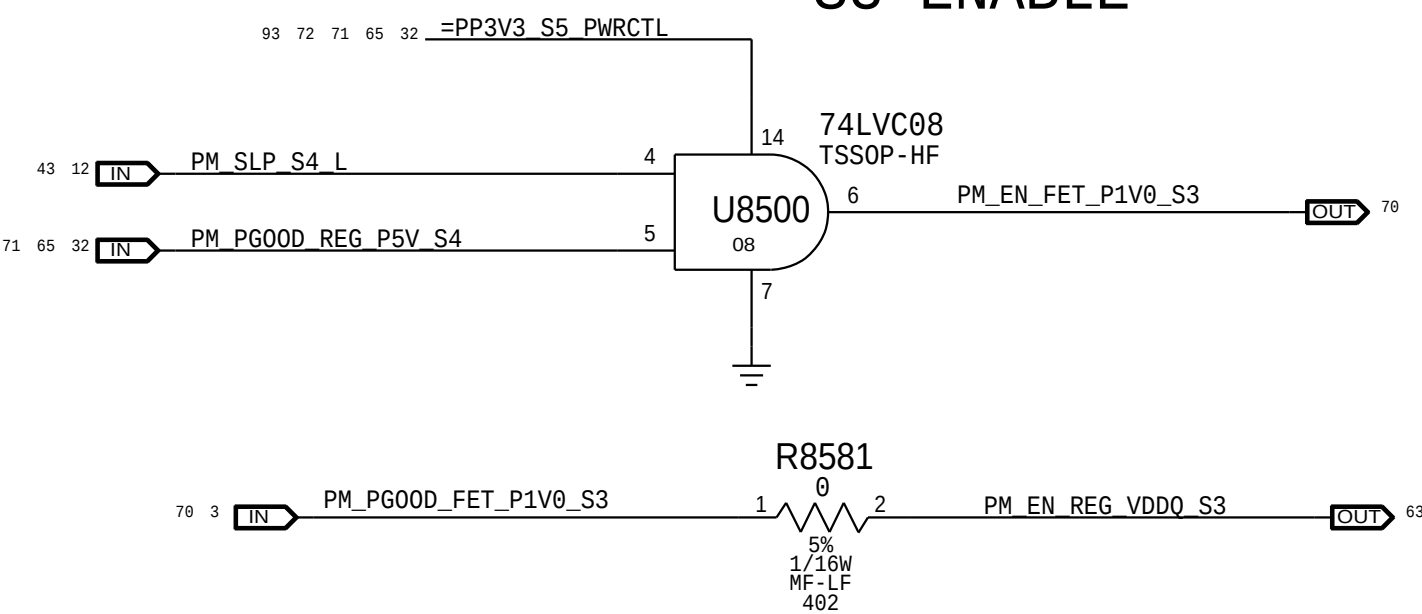
S4 USB Enable



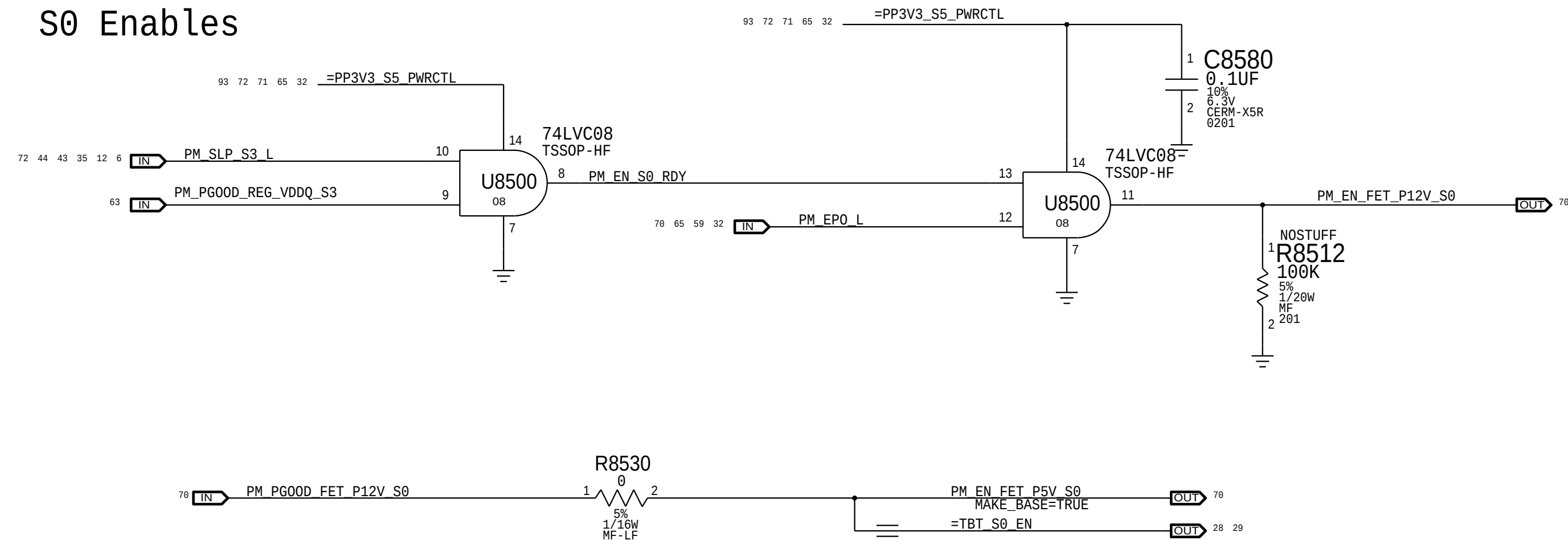
S4 TBT S4 Port Enable



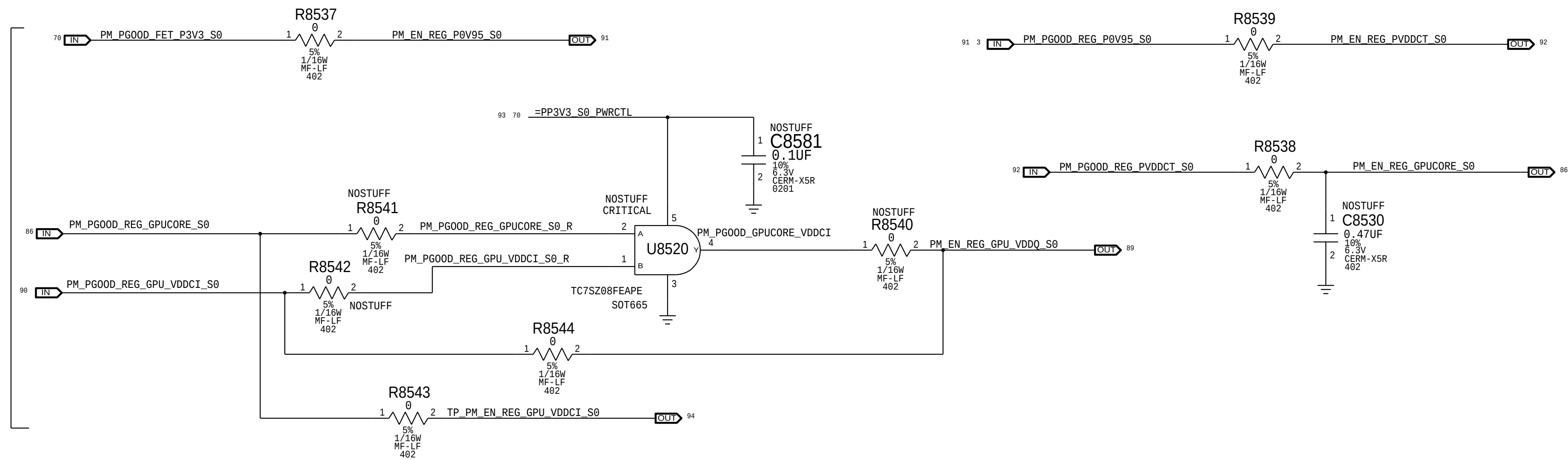
S3 ENABLE



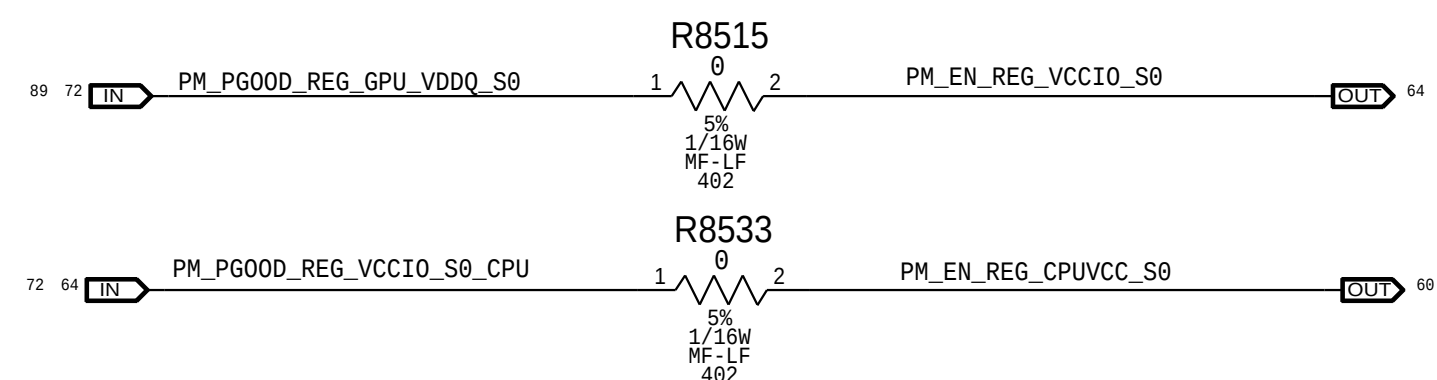
S0 Enables



S0 GPU SEQUENCING



S0 CPU Sequencing



Power Sequencing requirements

System:

1. 12V_ACDC -> 3V42G3H -> 12V_S5/12V_S5_SSD -> 3V3_S5 -> 1V0_S5 -> 5V_S4/3V3_S4 -> 1V0_S3

Intel CPU:

2. 1V0_S3 -> VDDQ_S3 -> 12V_S0 -> 5V_S0 -> 3V3_S0 -> GPU Rails -> VCCIO_S0-> VCC_CPU_S0 (VCCGT & VCCSA)

AMD GPU:

3. 3V3_S0 -> 0V95 -> 1V8 (VDD_CT) -> GPUCORE (VDDC) -> VDDCI -> FBVDDQ

SYNC_MASTER=J78_KENNY		SYNC_DATE=12/18/2013	
PAGE TITLE			
PLATFORM POWER: Regulator Enables			
 Apple Inc.	DRAWING NUMBER		SIZE
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		PAGE	
		85 OF 120	
		SHEET	
		71 OF 96	

D

C



D

C

B

A

[illegible]

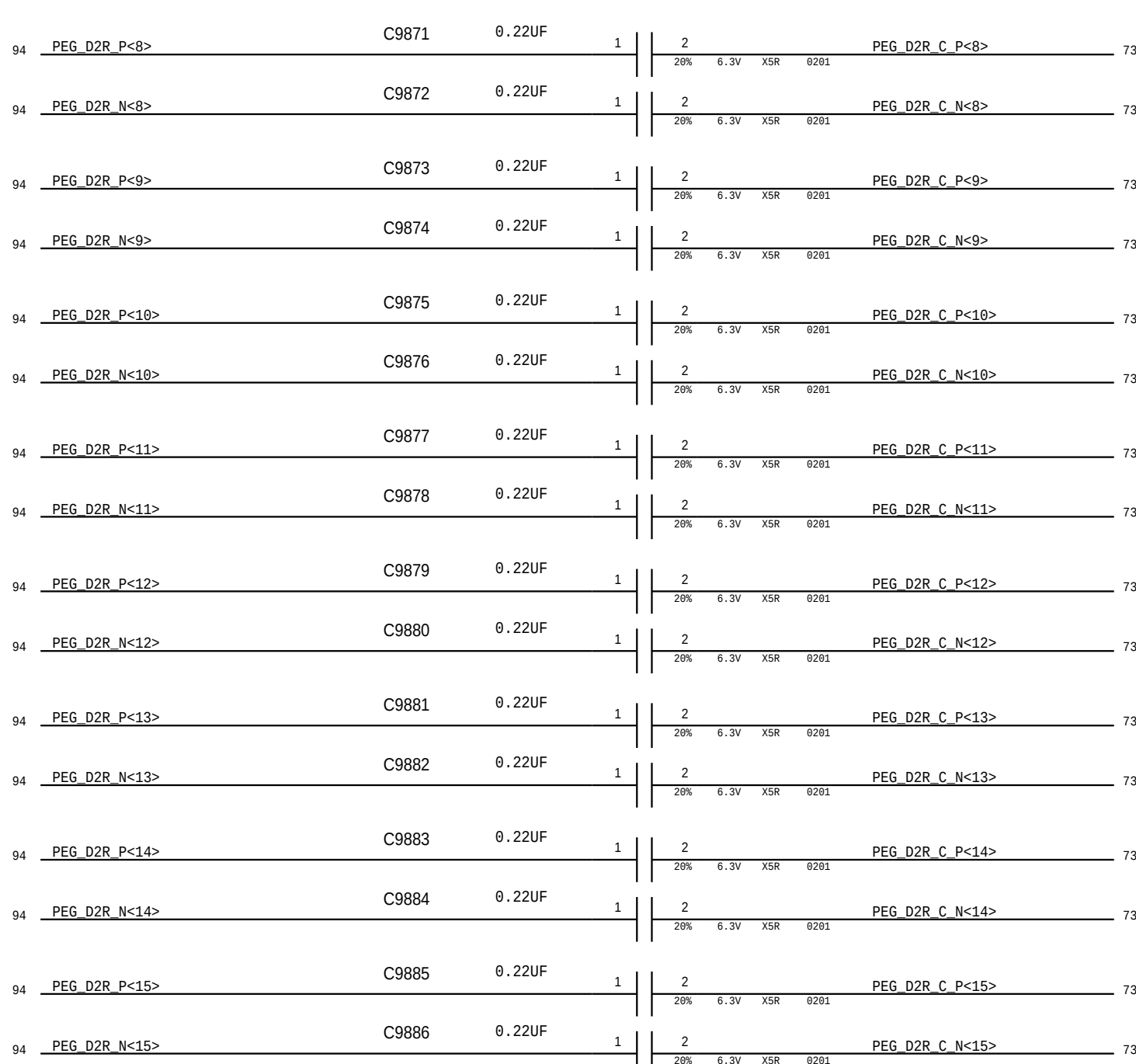
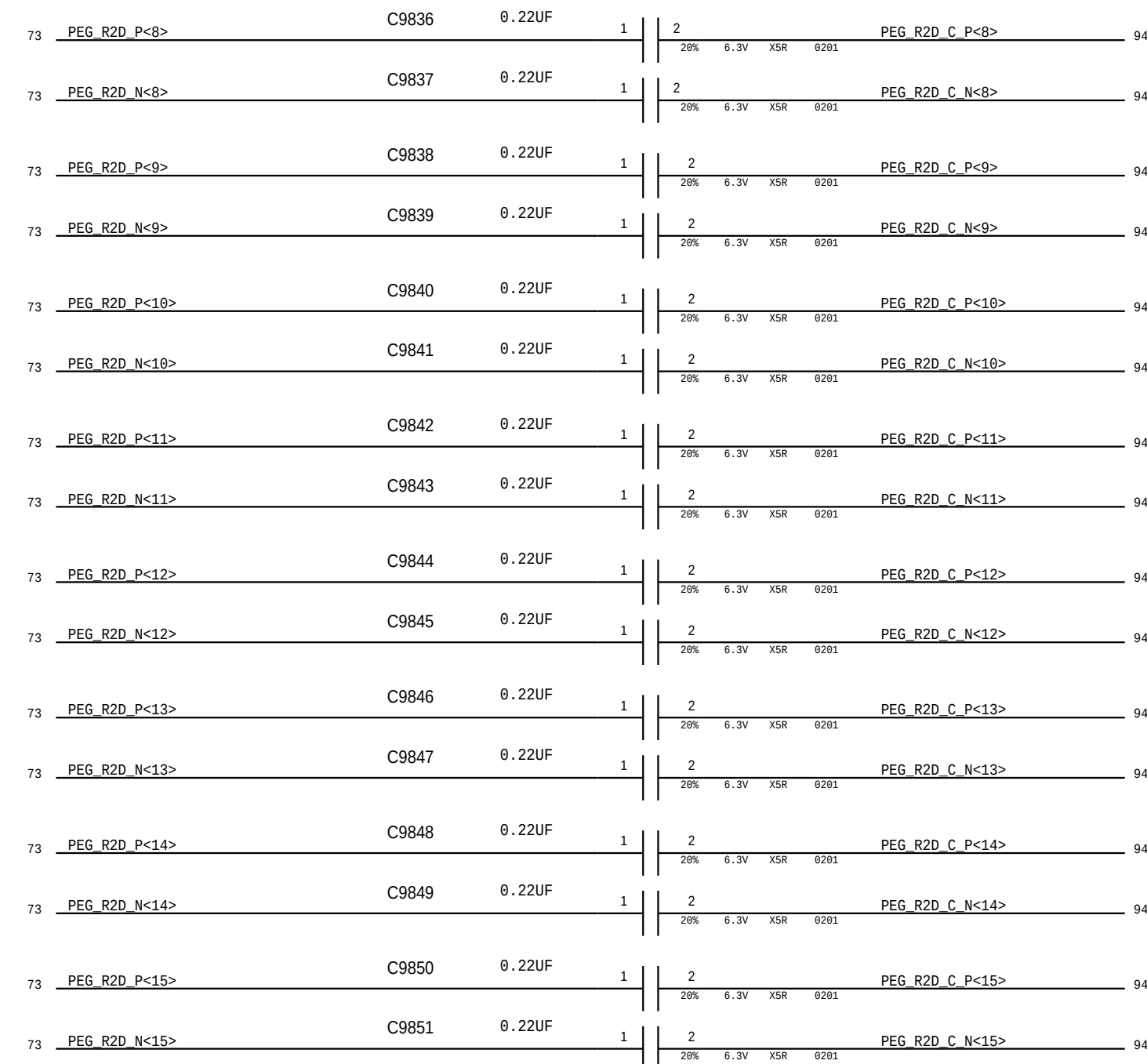
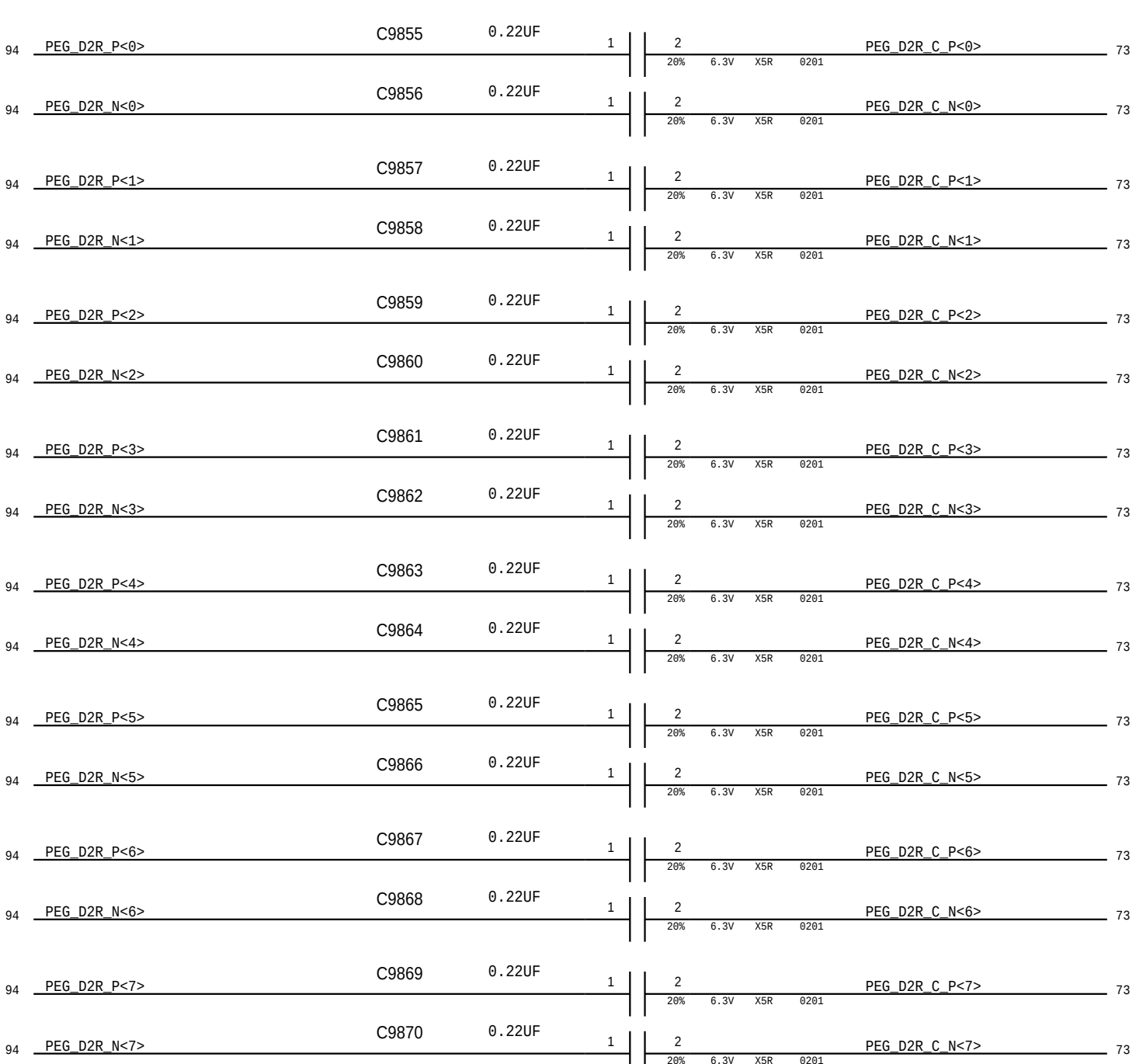
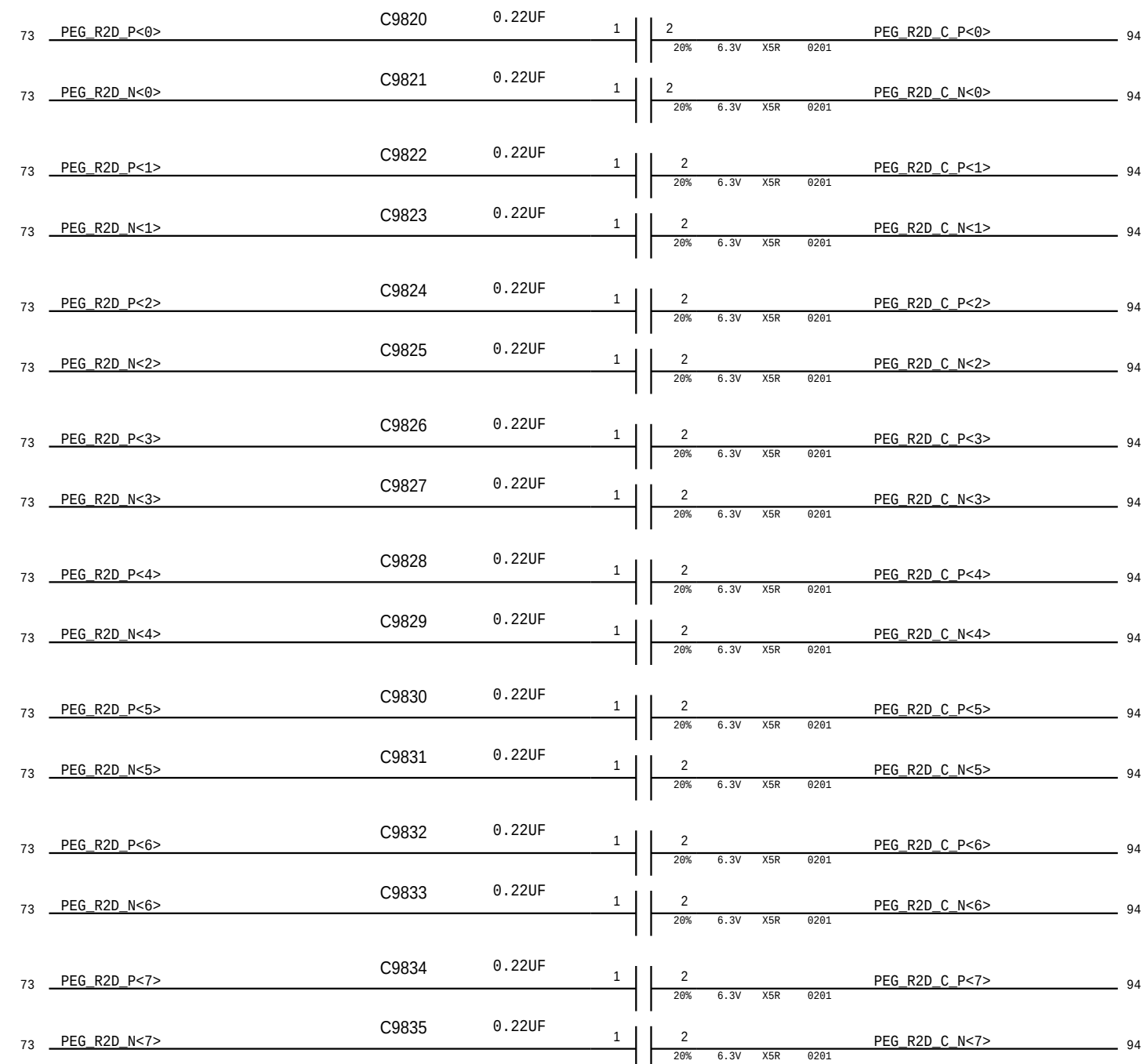
D

C

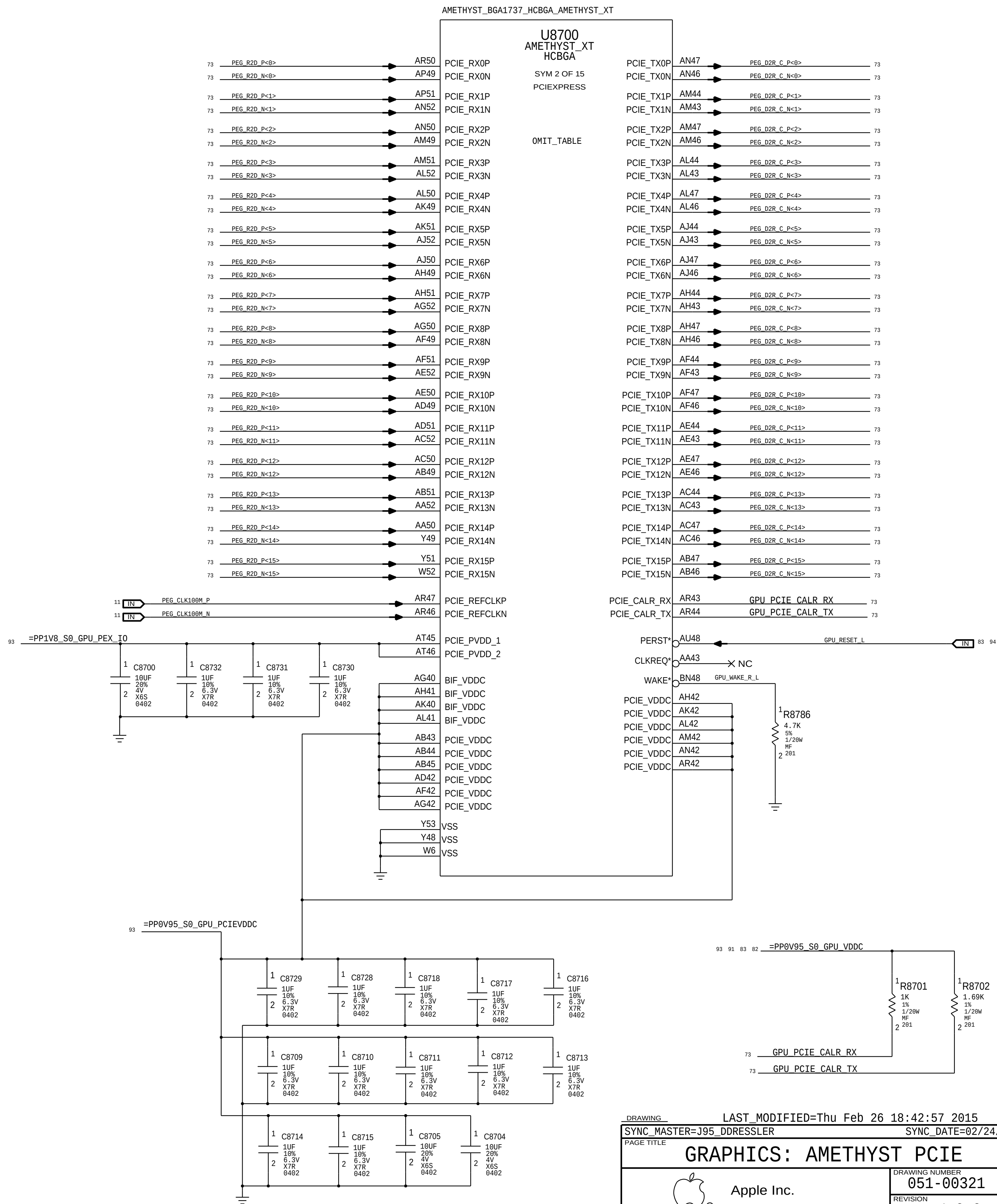
B

A

POLARITY SWAPS INTENDED ON LANES,SEE NOTES AT DIFF PAIRS.
ALL LANES ARE ALSO REVERSED, SEE ALIASES ON CSA 102

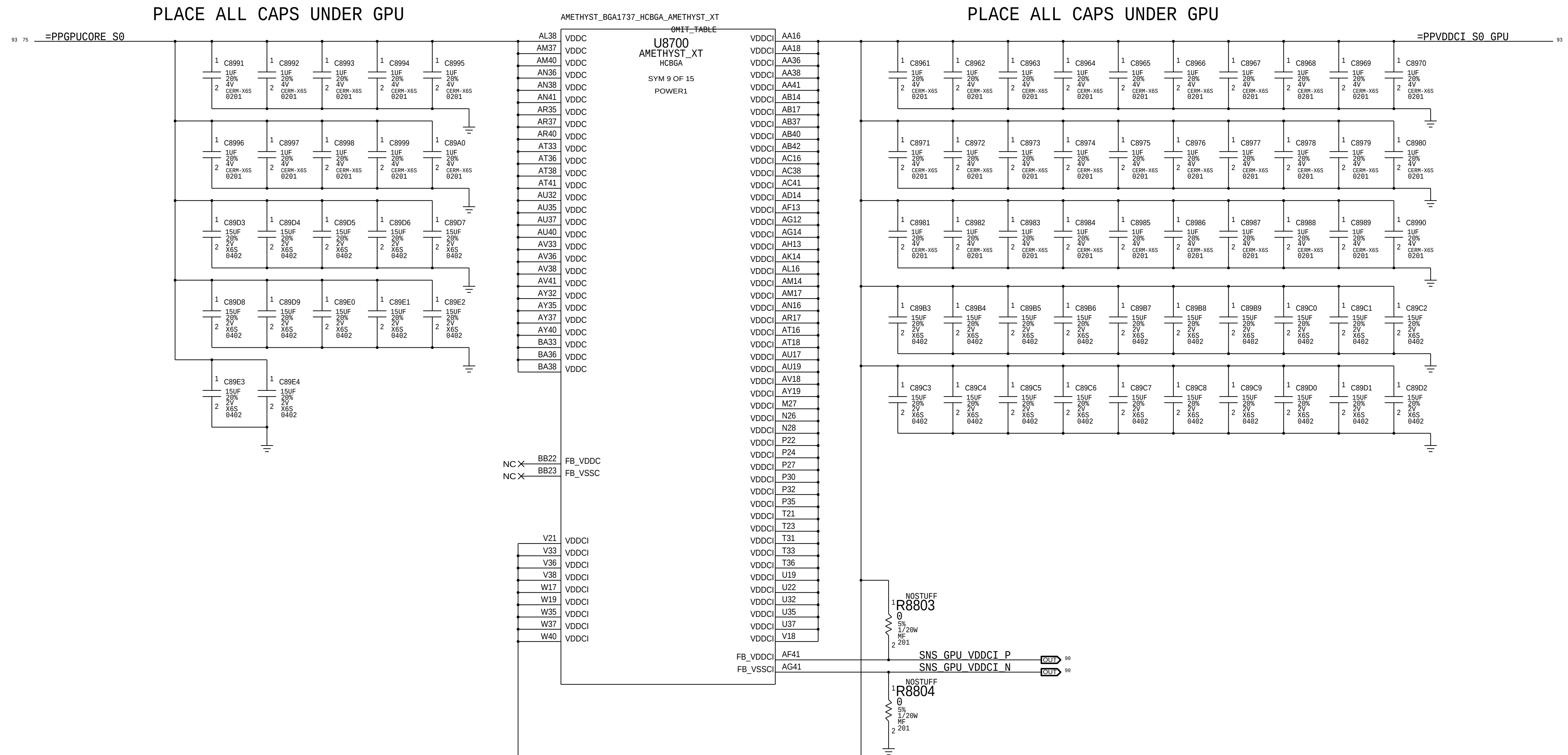


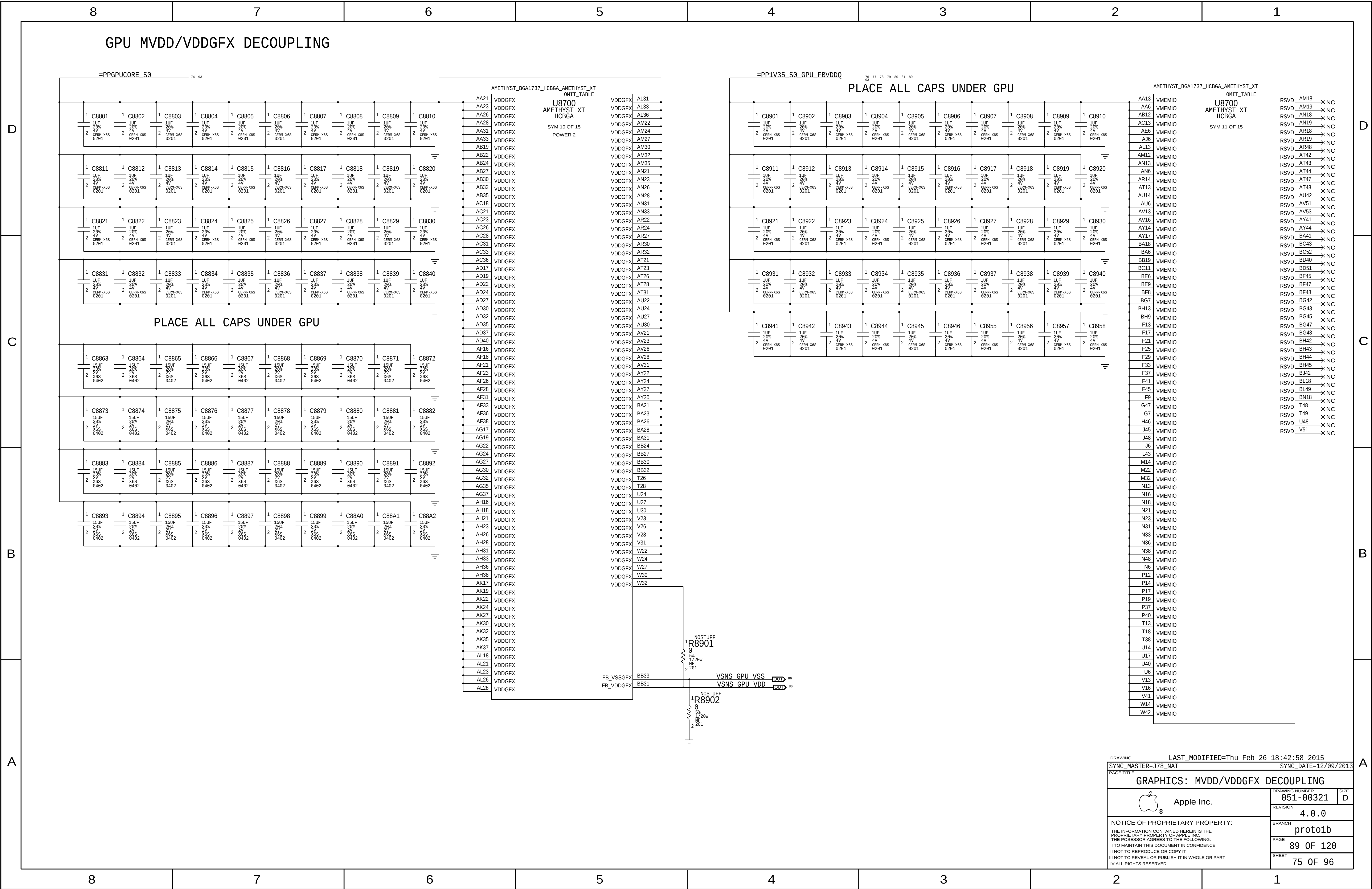
NOTE:
TONGA DOES NOT GENERATE PCIE CLKREQ.

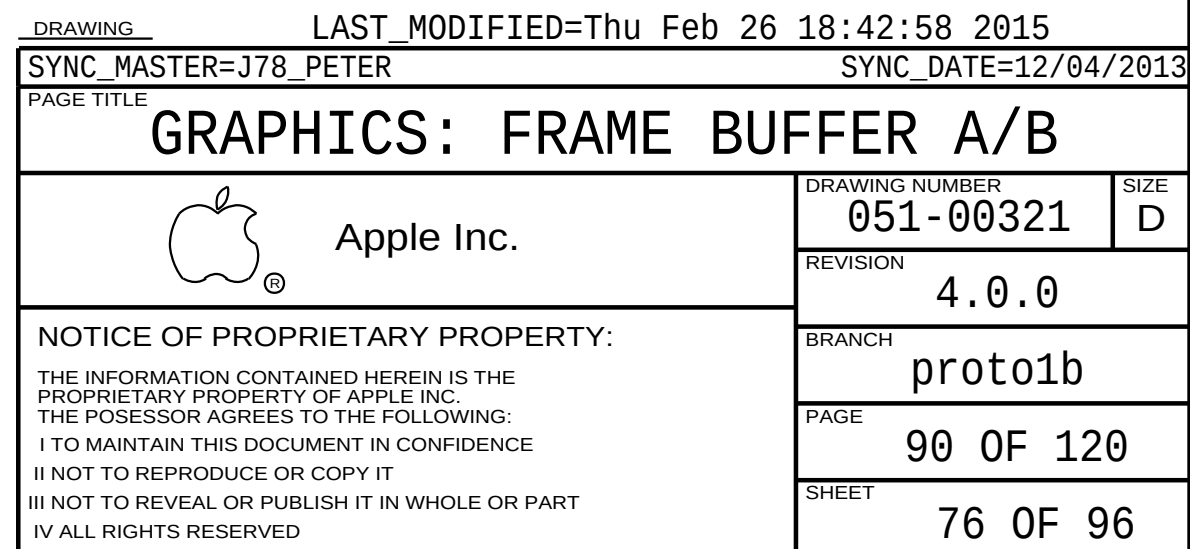


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GRAPHICS: AMETHYST PCIE			
 Apple Inc.		DRAWING NUMBER	051-00321
		REVISION	4.0.0
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		PAGE	87 OF 120
		SHEET	73 OF 96

GPU VDDC/VDDCI DECOUPLING







D

C

B

A

D

C

B

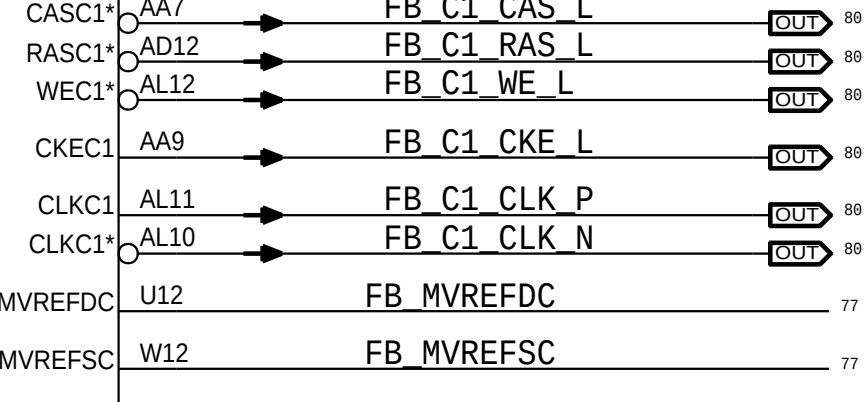
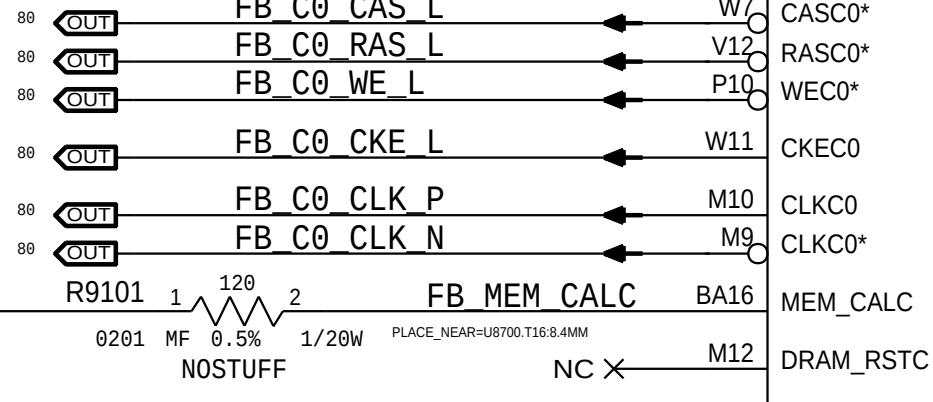
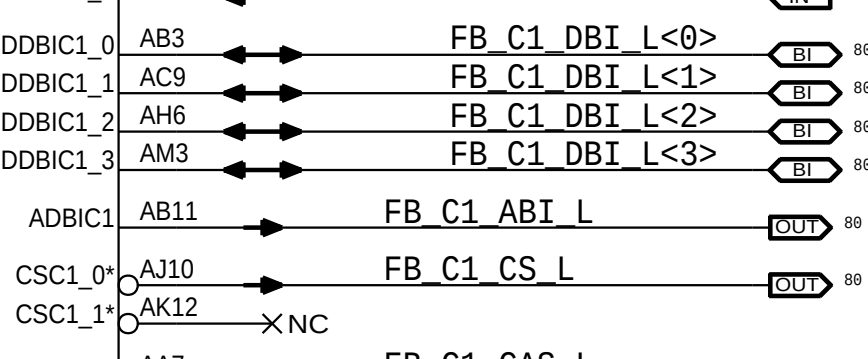
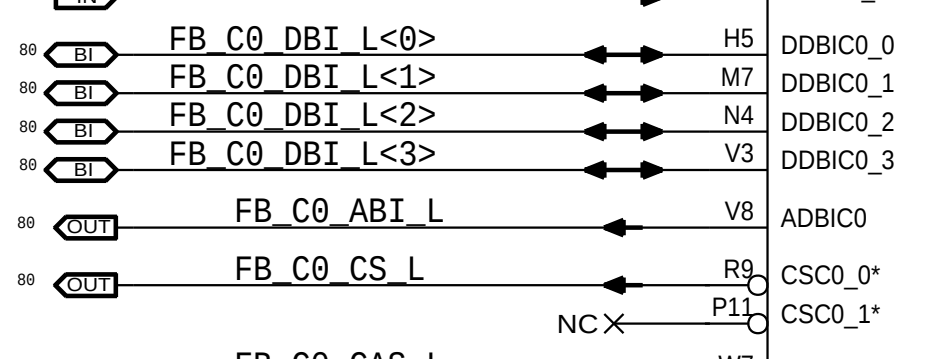
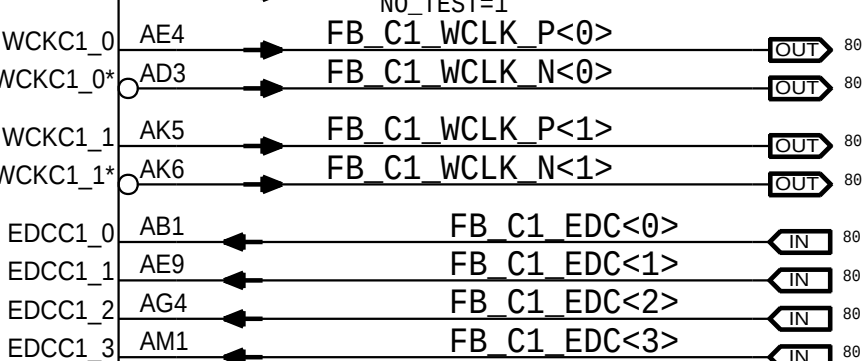
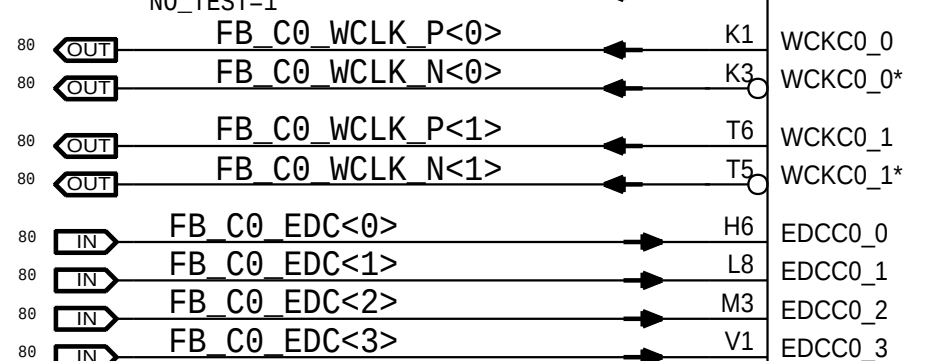
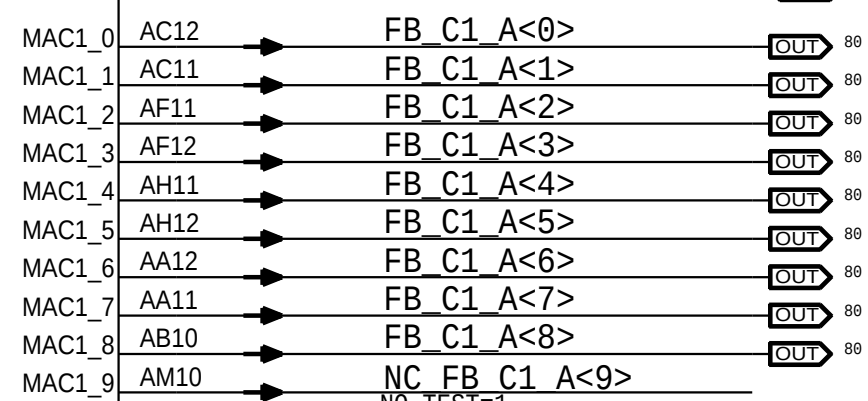
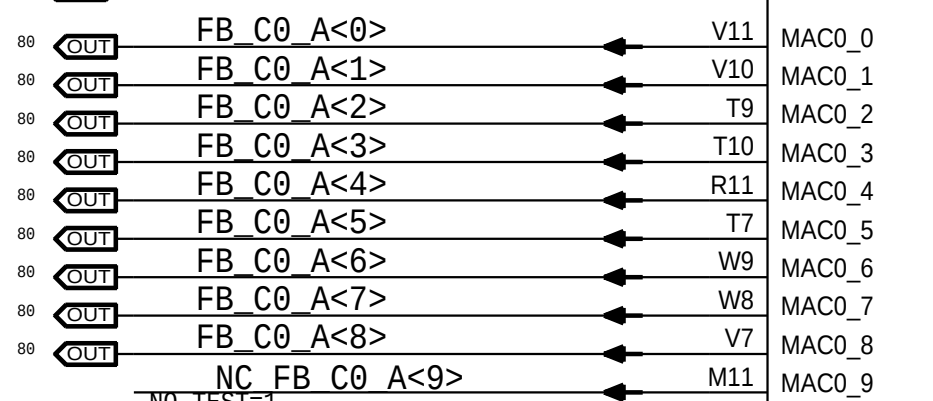
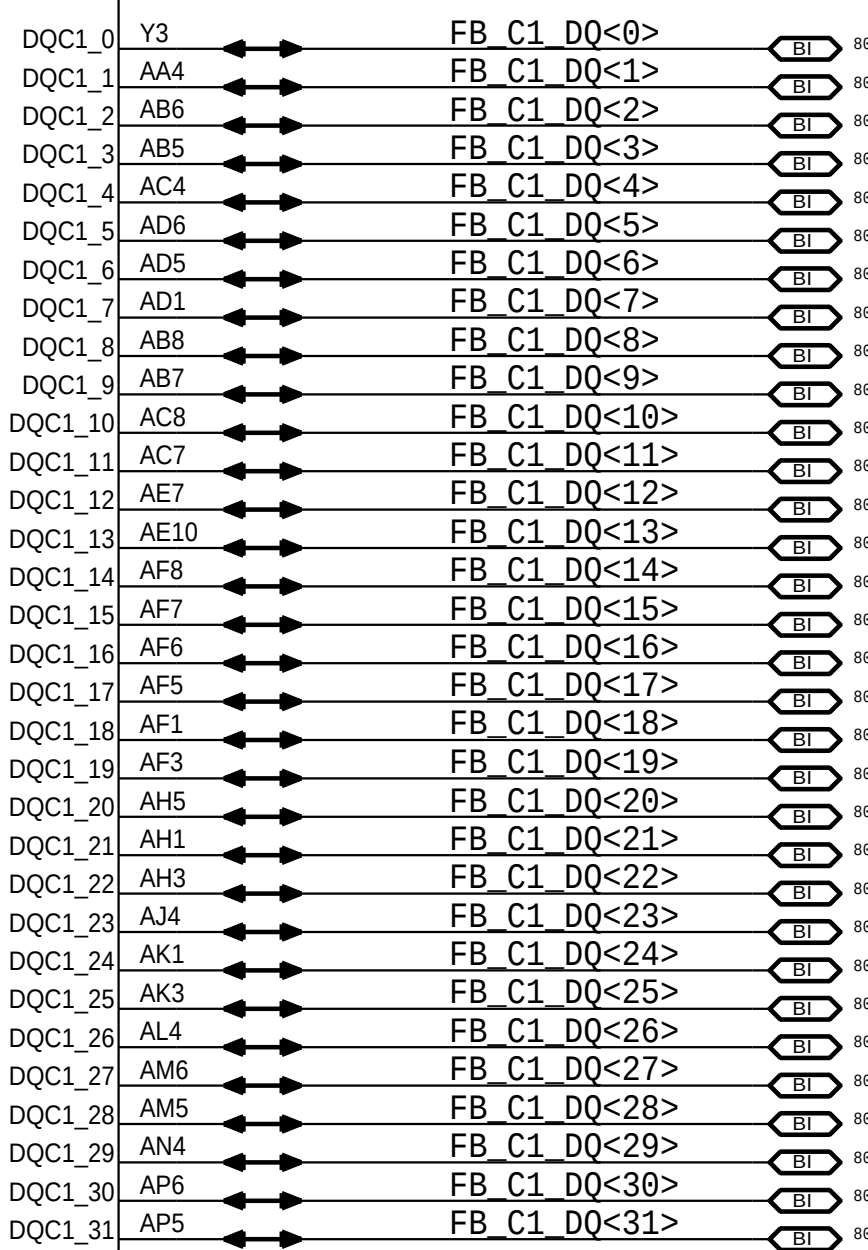
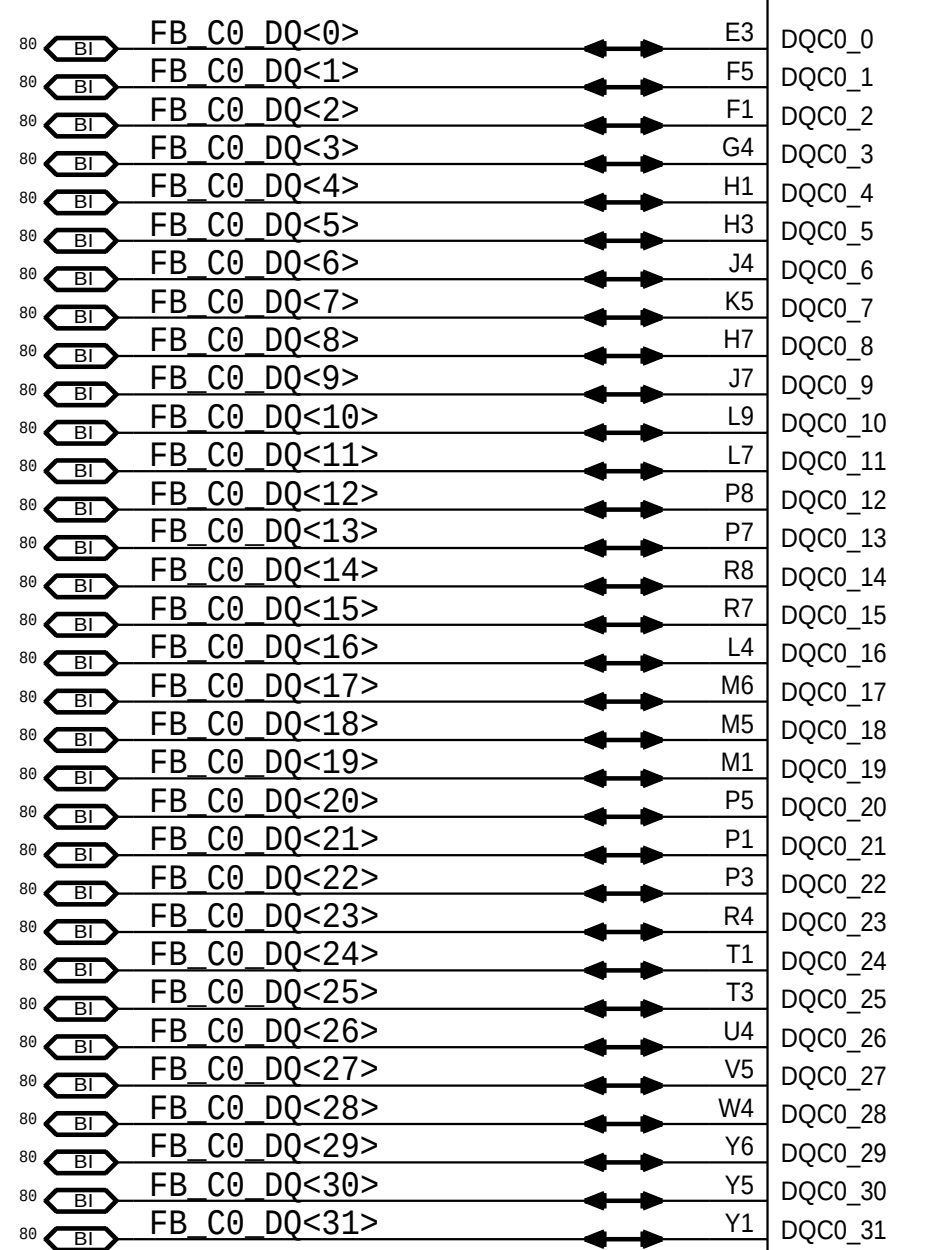
A

AMETHYST_BGA1737_HCBGA_AMETHYST_XT

OMIT_TABLE

U8700
AMETHYST_XT
HCBGA

SYM 5 OF 15
MEMORY INTERFACE
BANK C

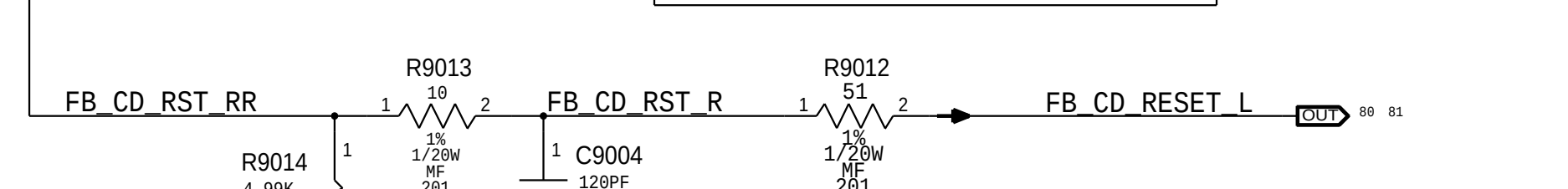
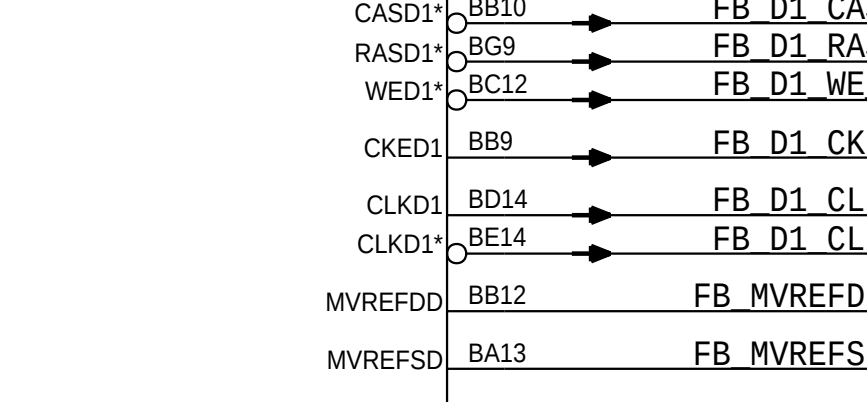
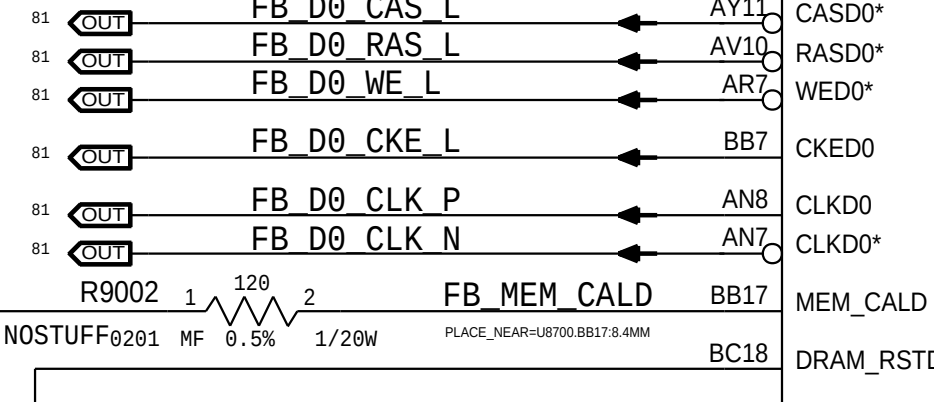
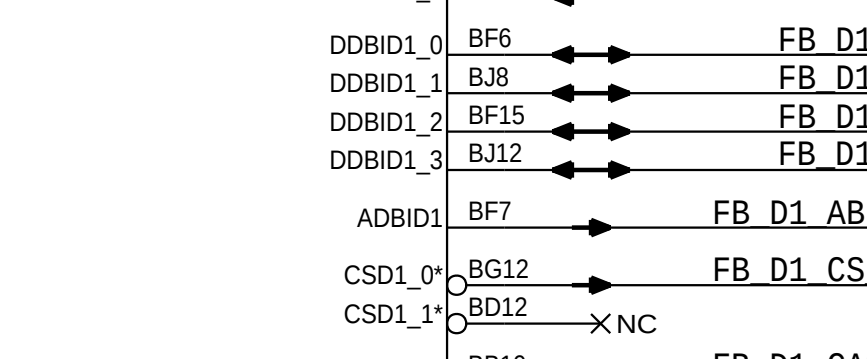
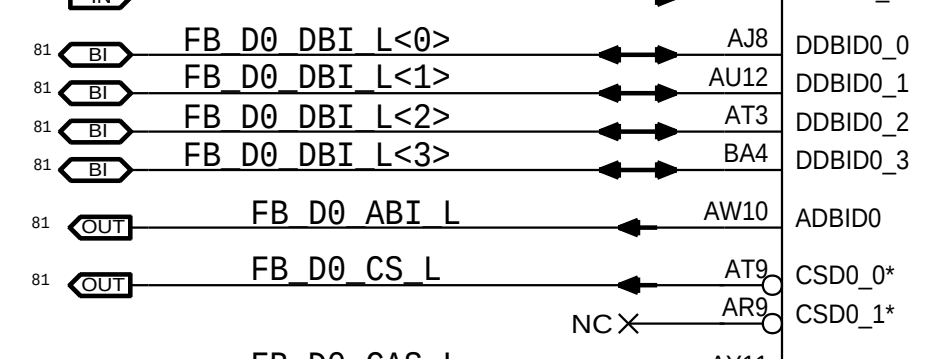
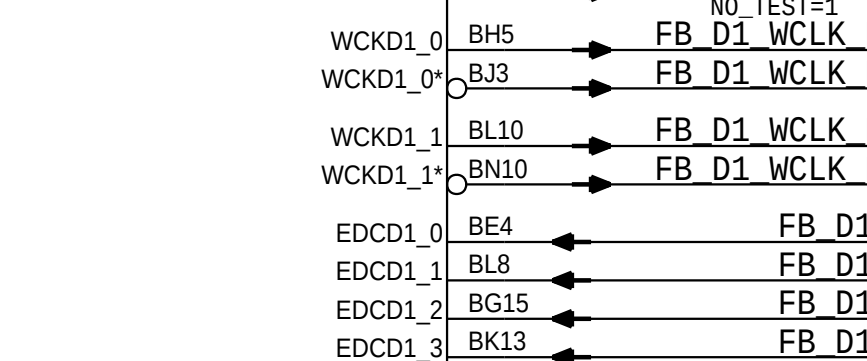
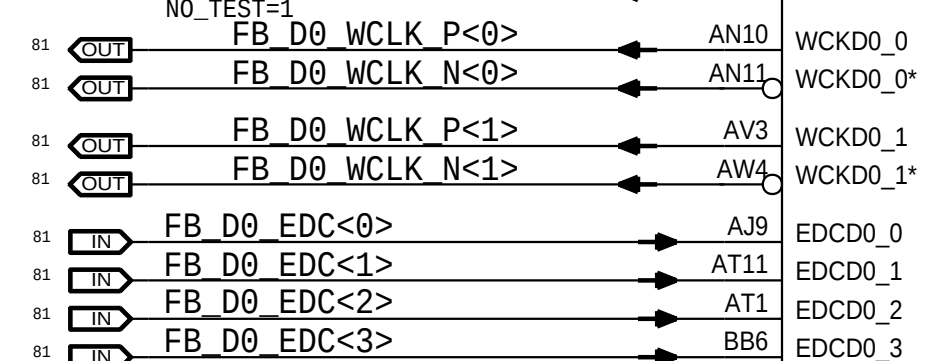
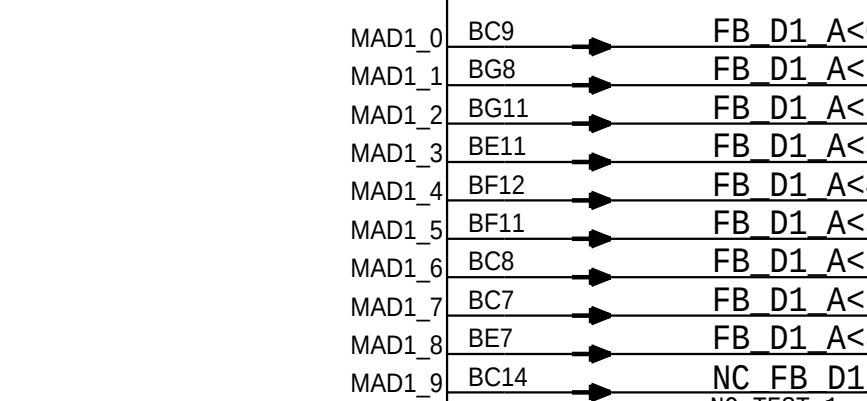
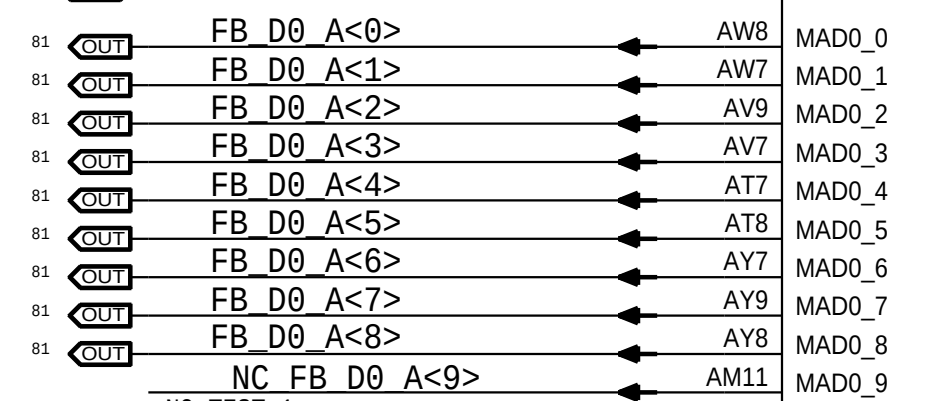
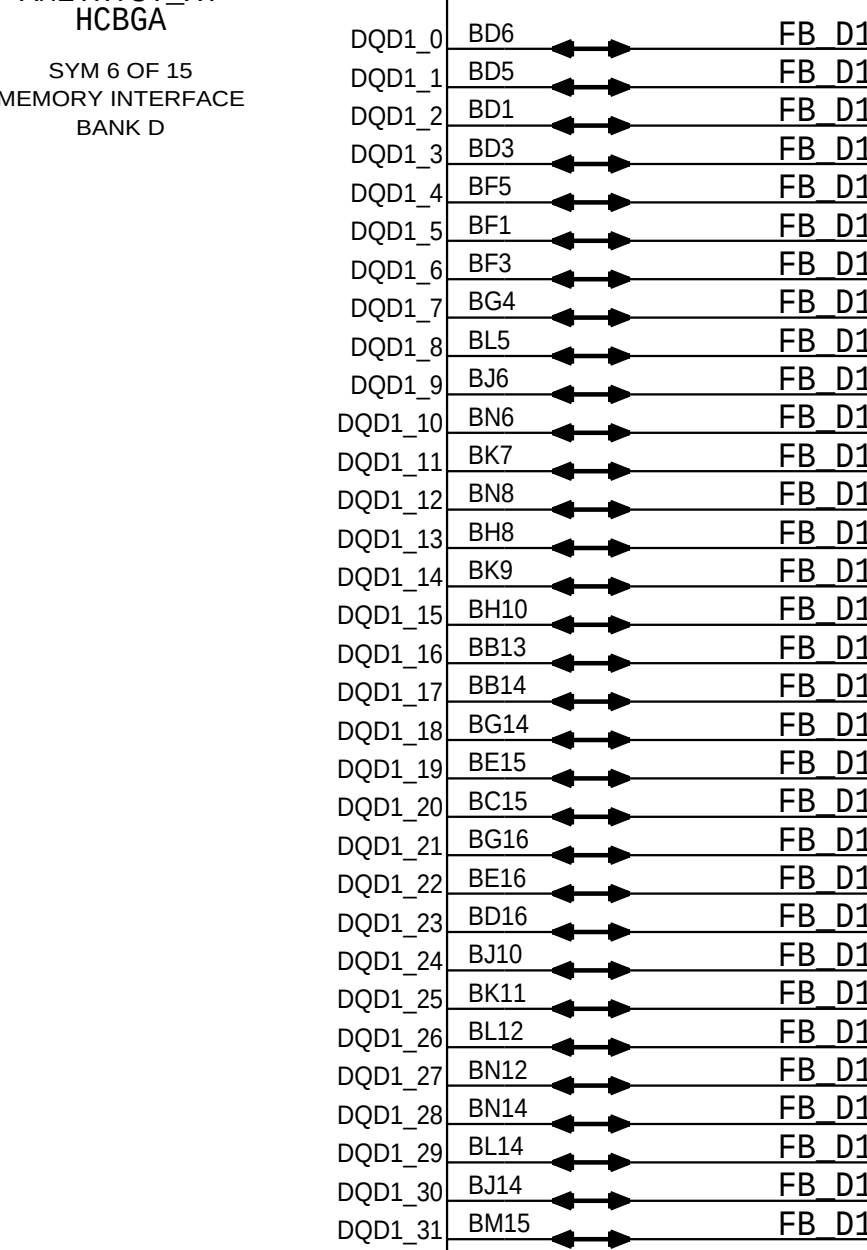
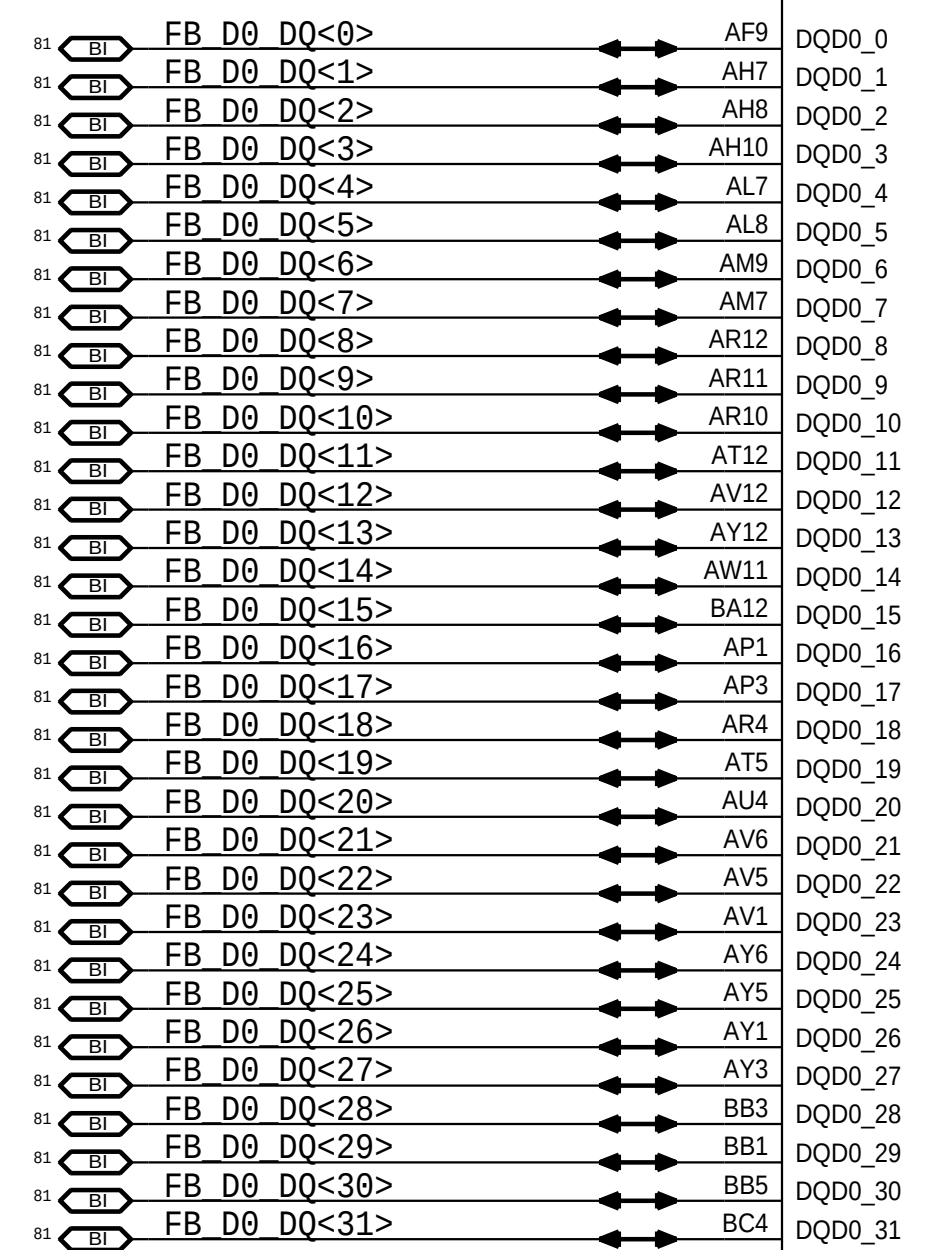


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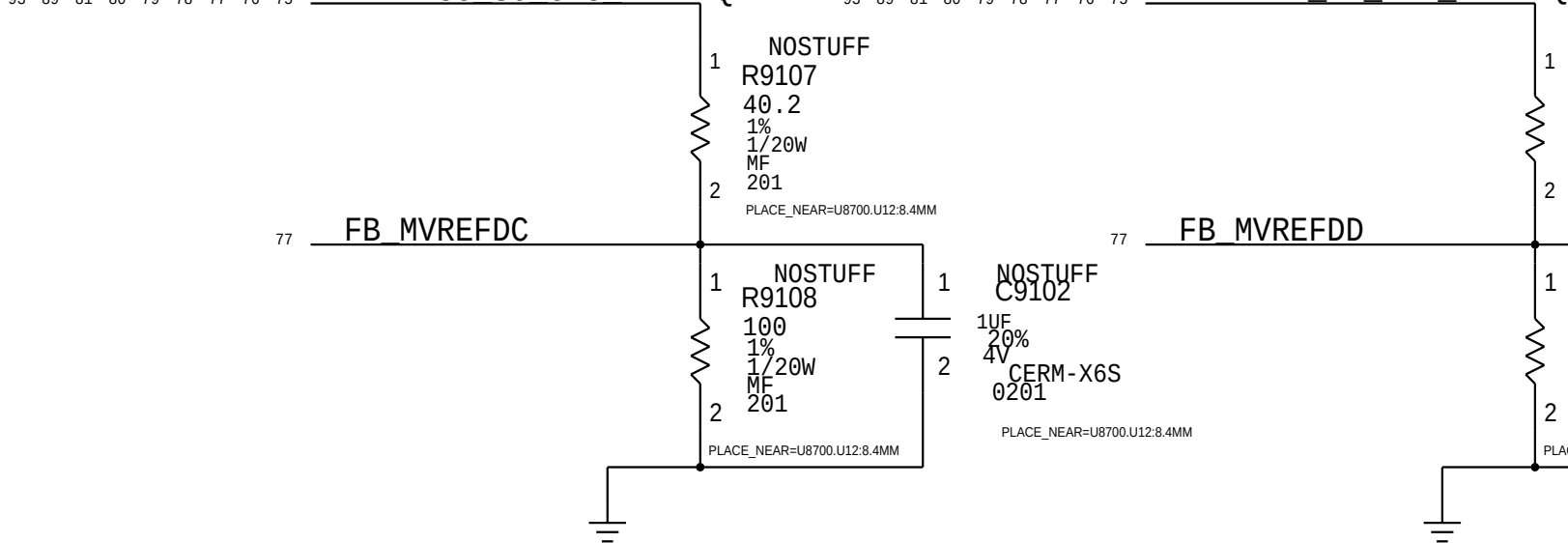
OMIT_TABLE

U8700
AMETHYST_XT
HCBGA

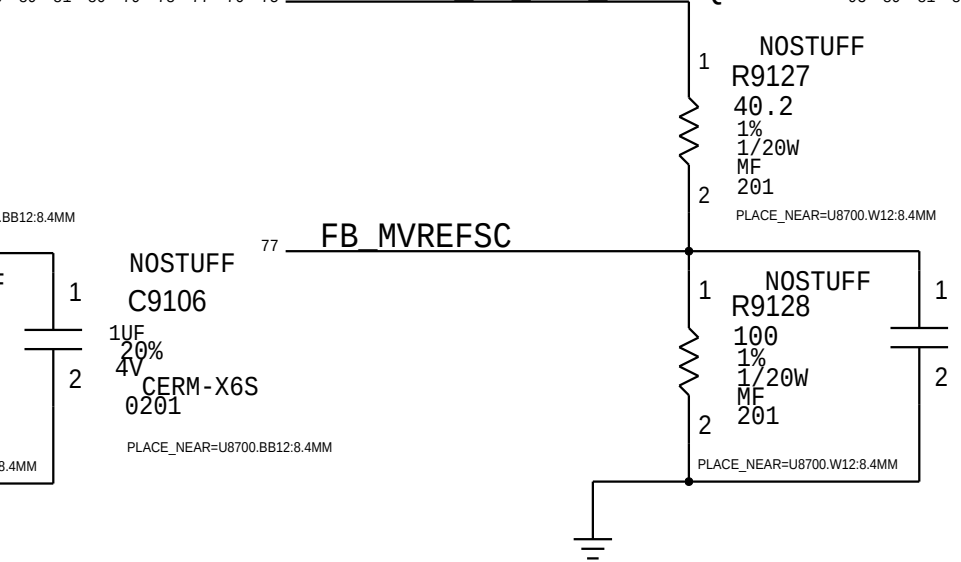
SYM 6 OF 15
MEMORY INTERFACE
BANK D



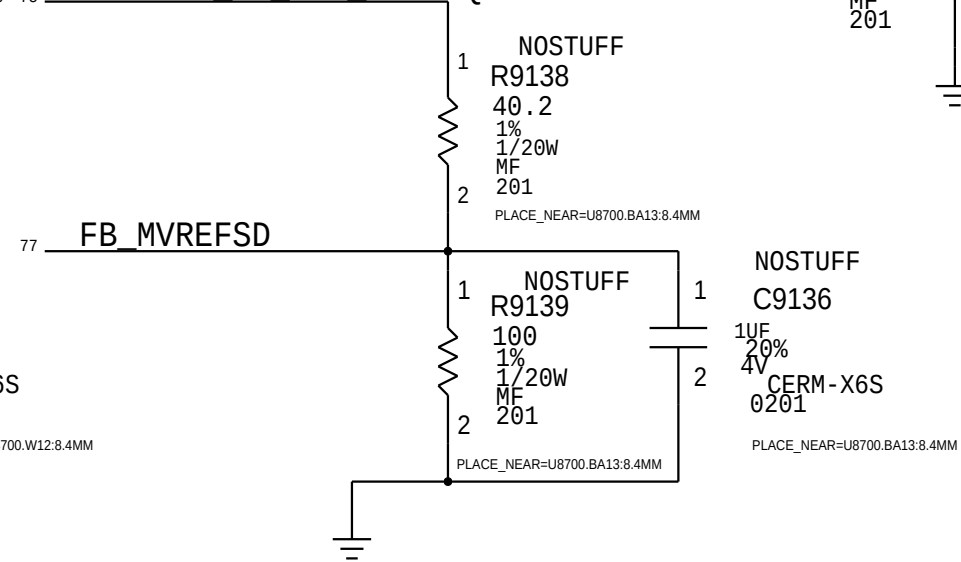
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=PP1V35_S0_GPU_FBVDDQ



=PP1V35_S0_GPU_FBVDDQ



=PP1V35_S0_GPU_FBVDDQ



DRAWING

LAST_MODIFIED=Thu Feb 26 18:42:58 2015

SYNC_MASTER=J78_PETER

SYNC_DATE=12/04/2013

PAGE TITLE

GRAPHICS: FRAME BUFFER C/D

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DRAWING NUMBER

051-00321

SIZE

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REVISION

4.0.0

BRANCH

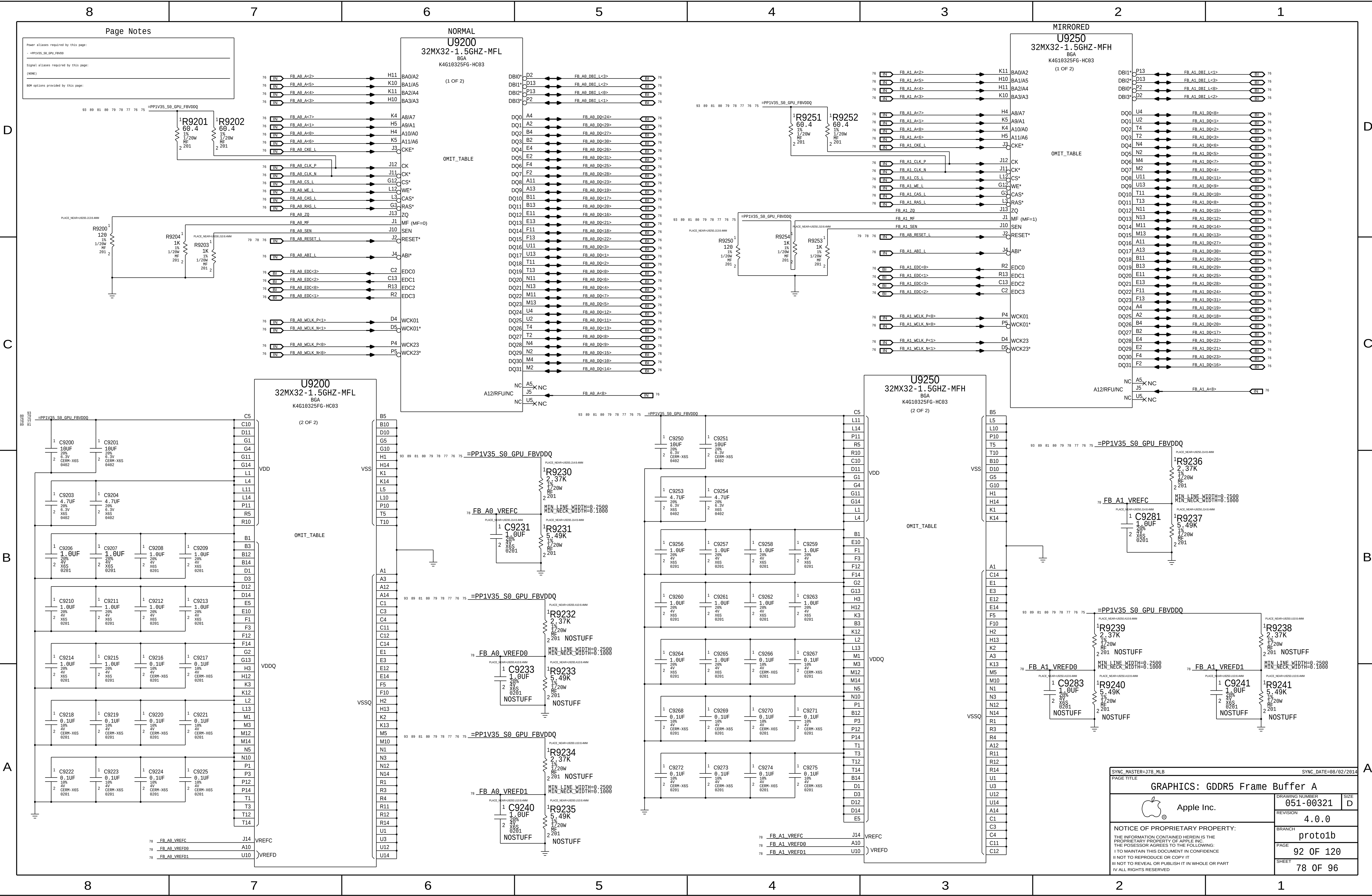
proto1b

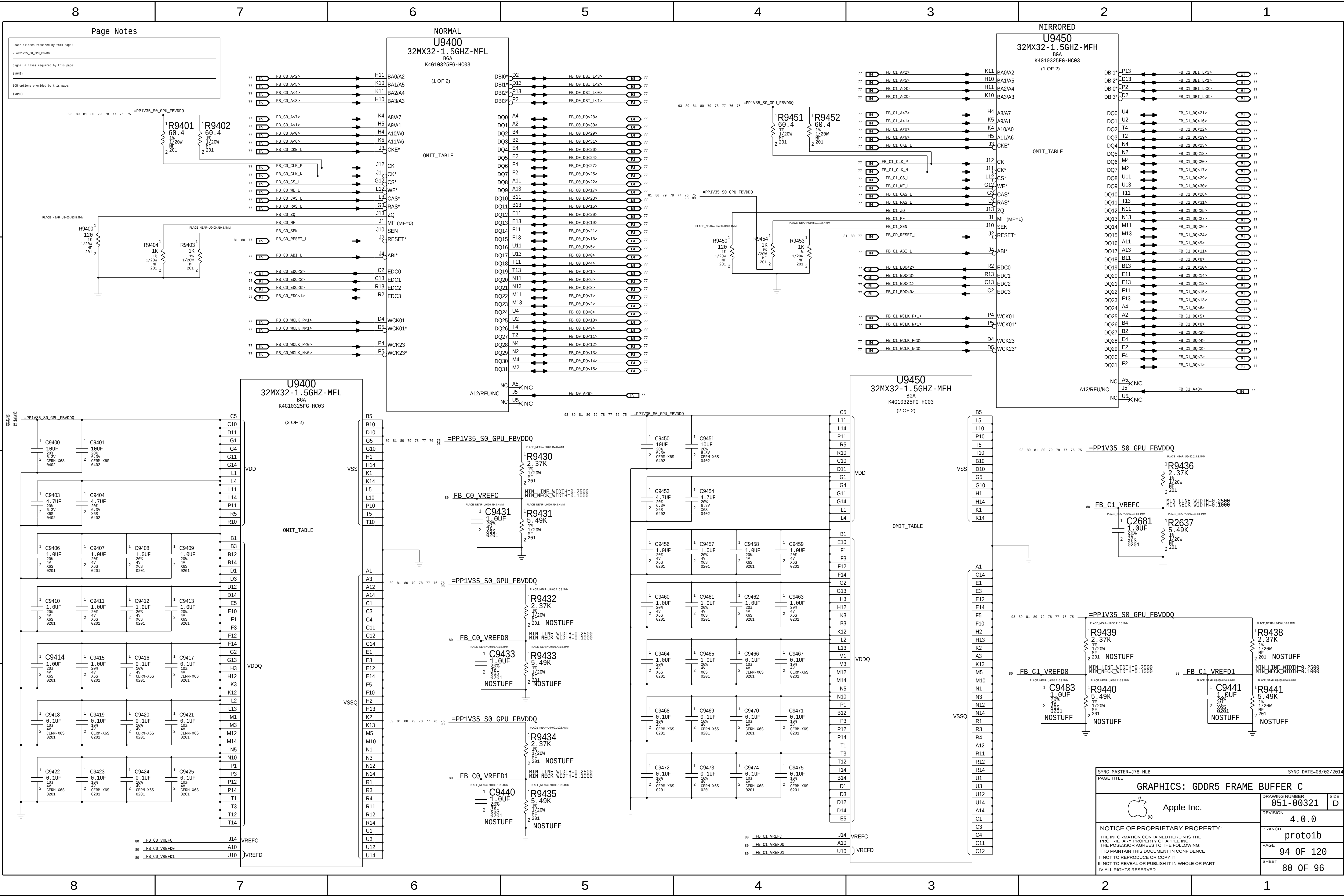
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Page Notes

Power aliases required by this page:
- PP1V35_S0_GPU_FBVDDQ

Signal aliases required by this page:
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BOM options provided by this page:
(NONE)

D

C

B

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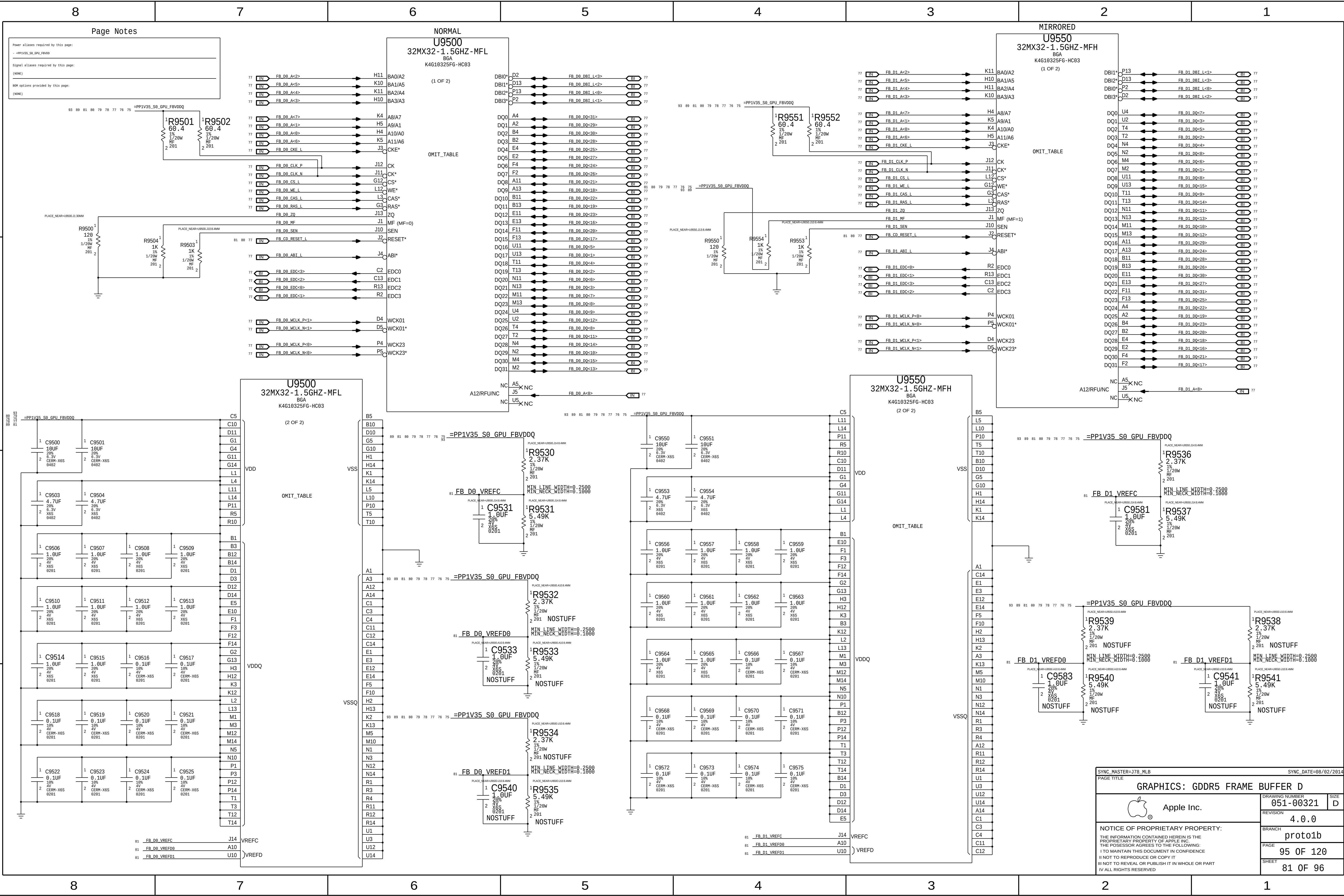
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		4.0.0	proto1b
		PAGE	SHEET
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Page Notes

Power aliases required by this page:

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BOM options provided by this page:

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D

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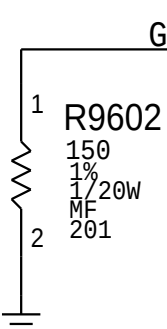
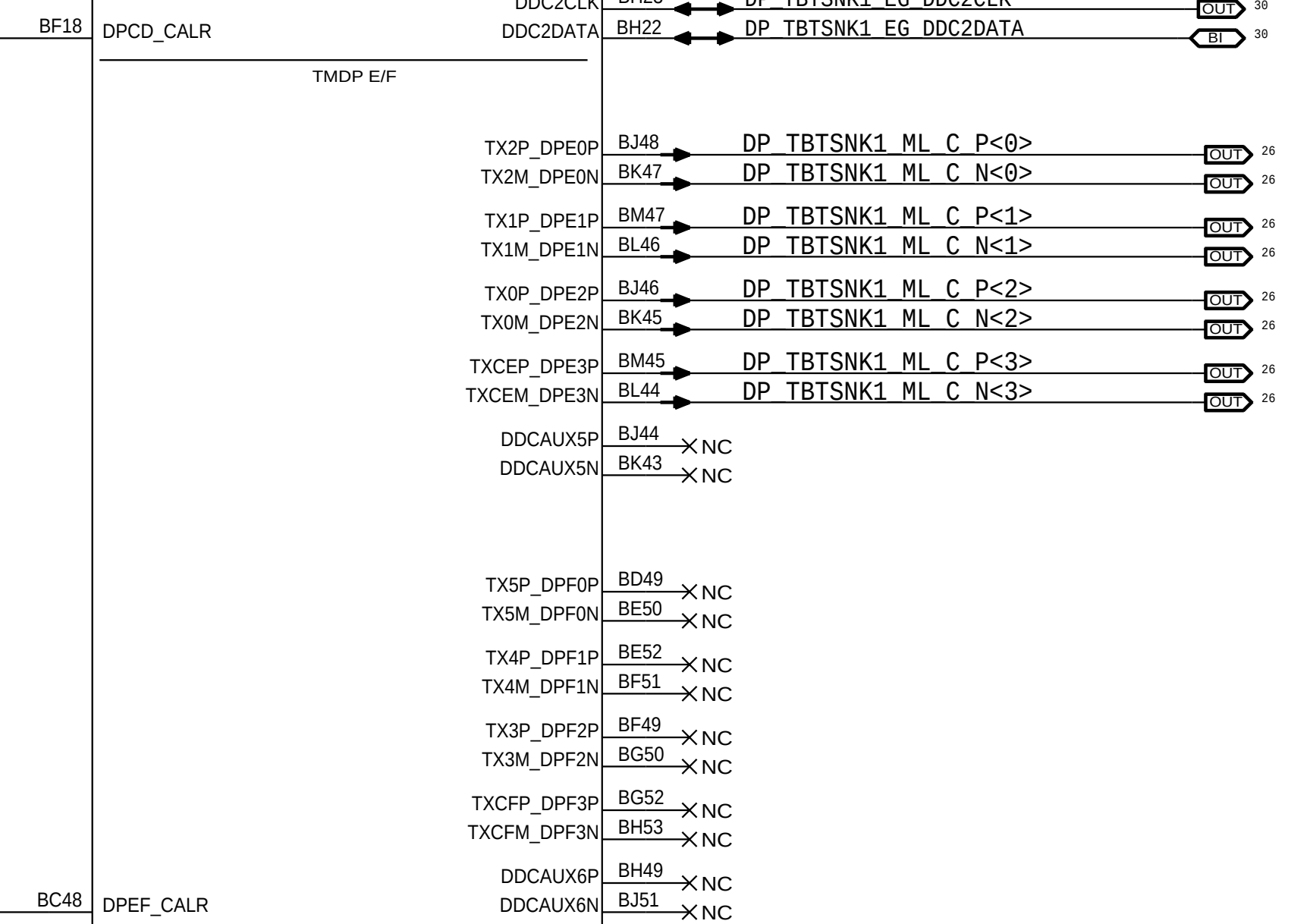
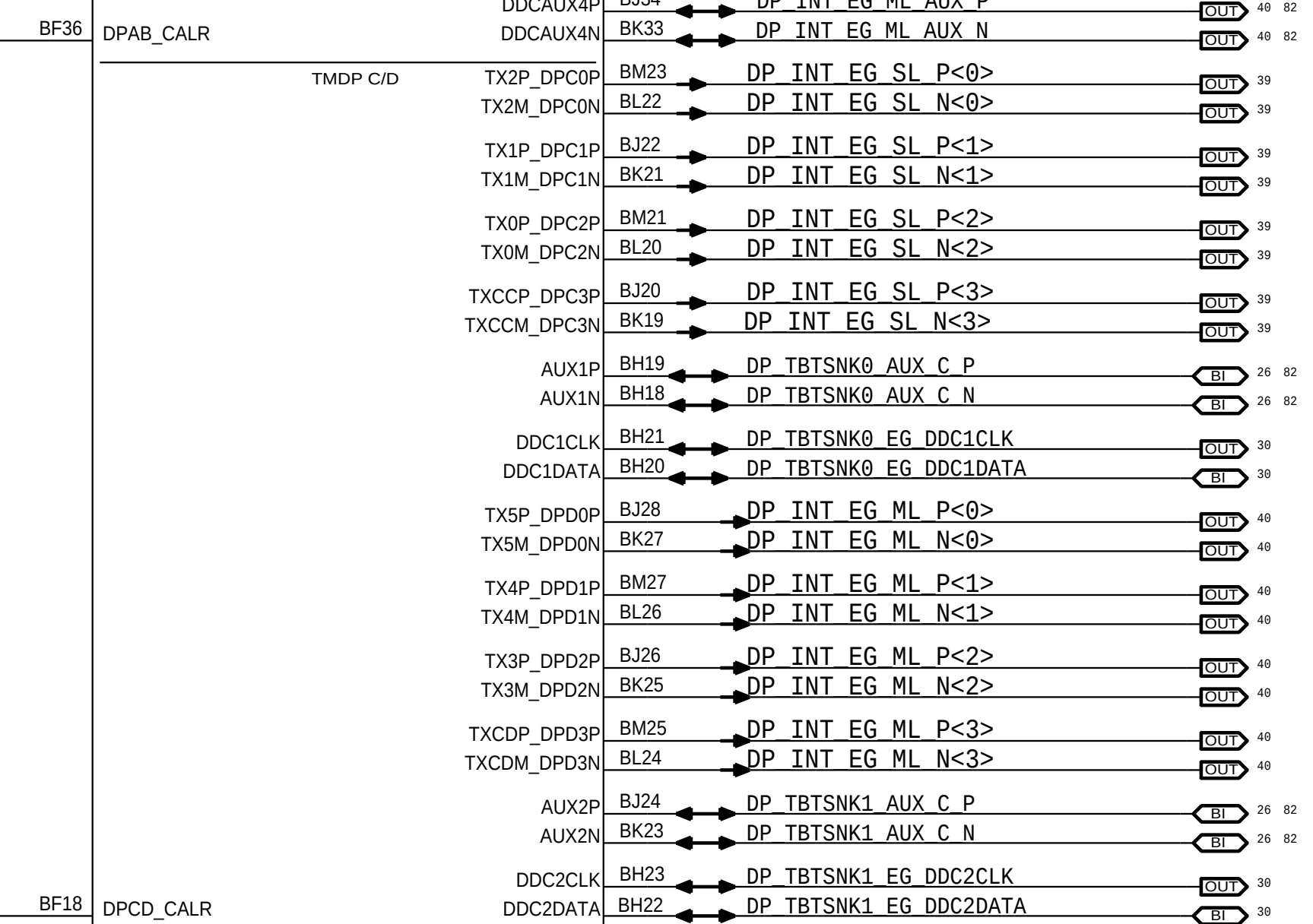
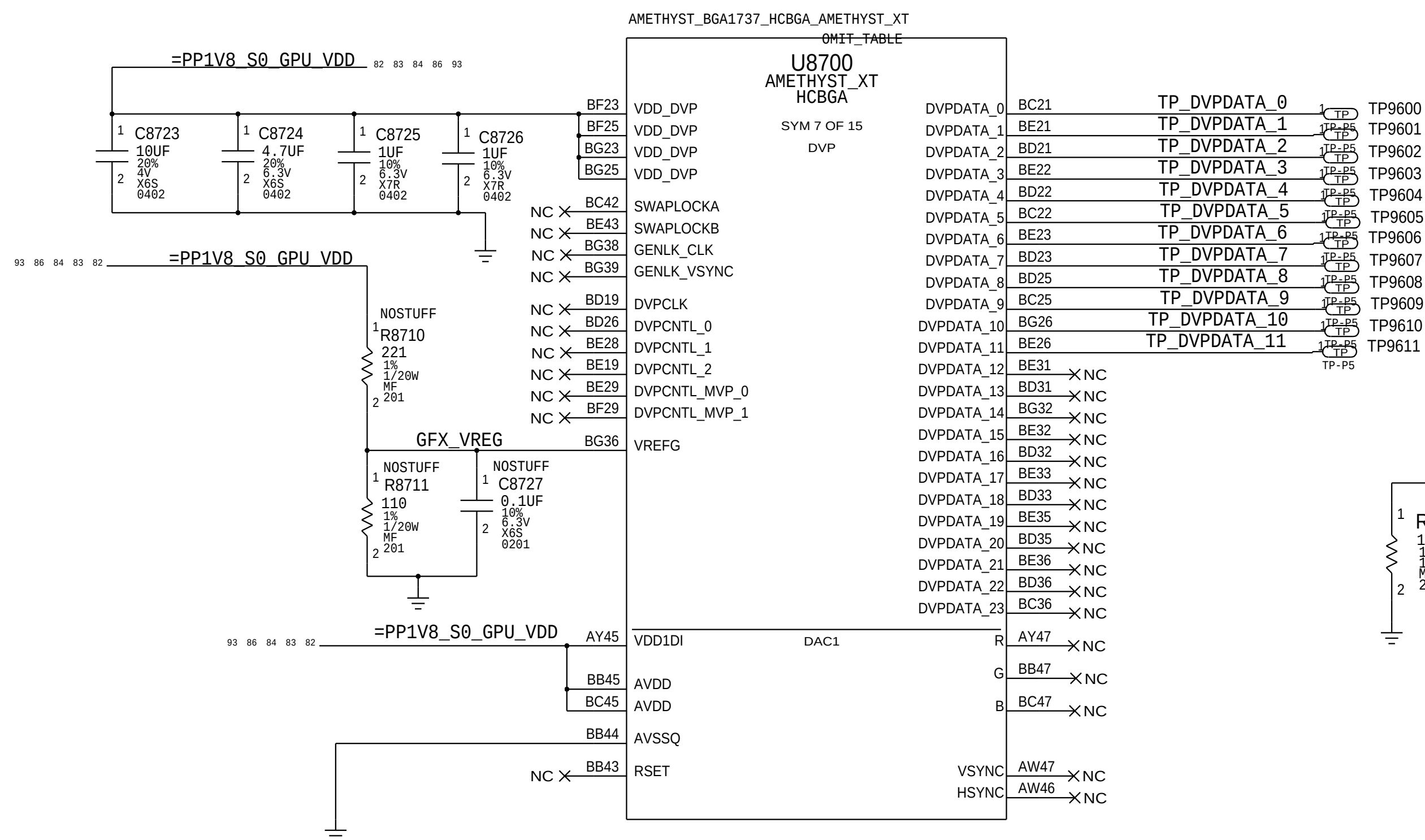
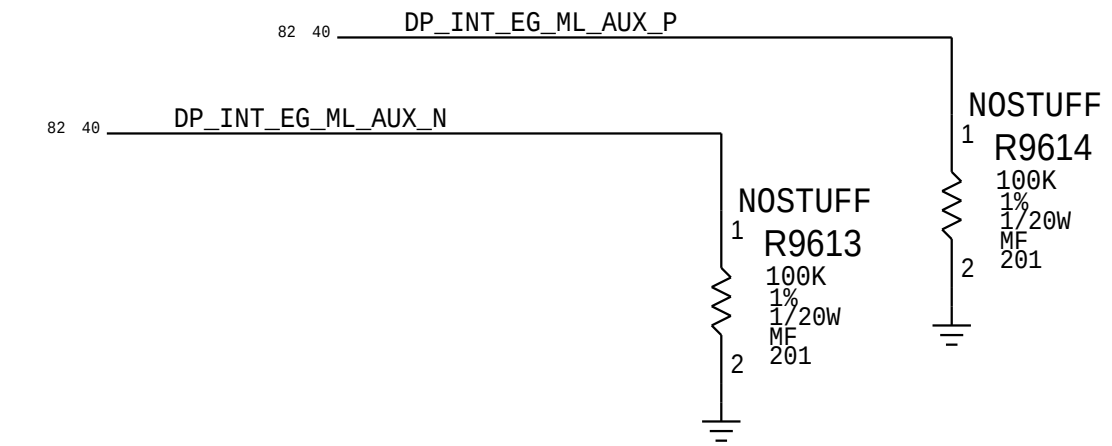
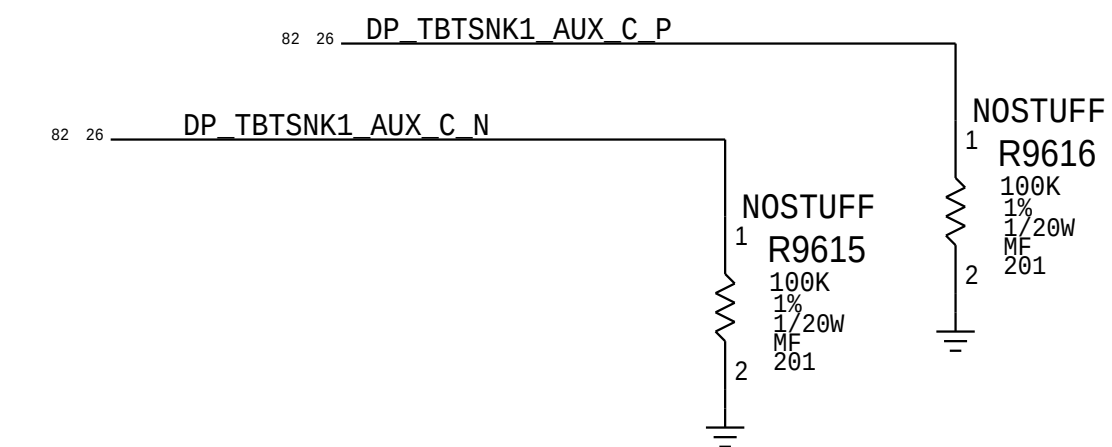
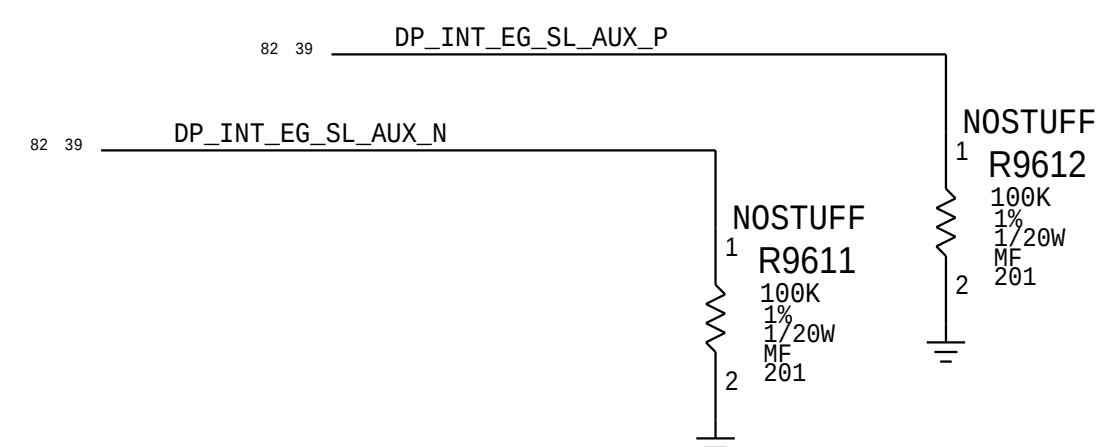
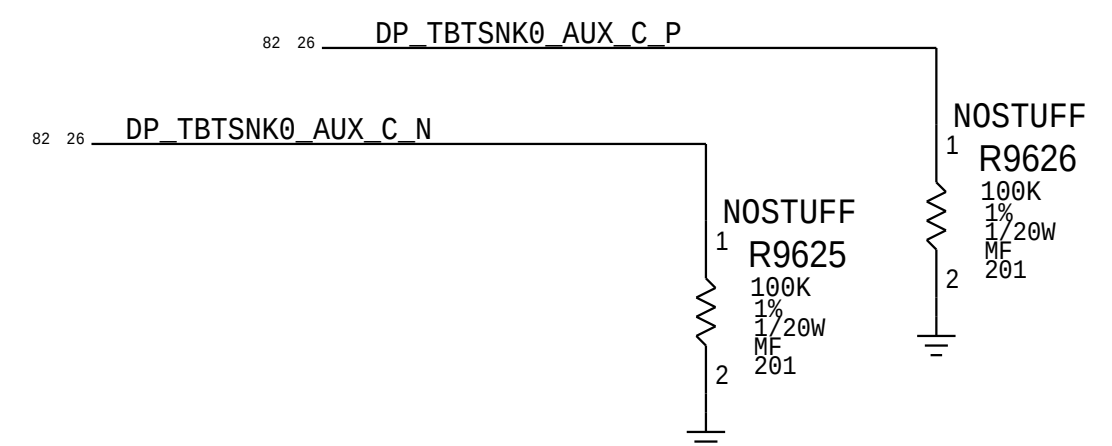
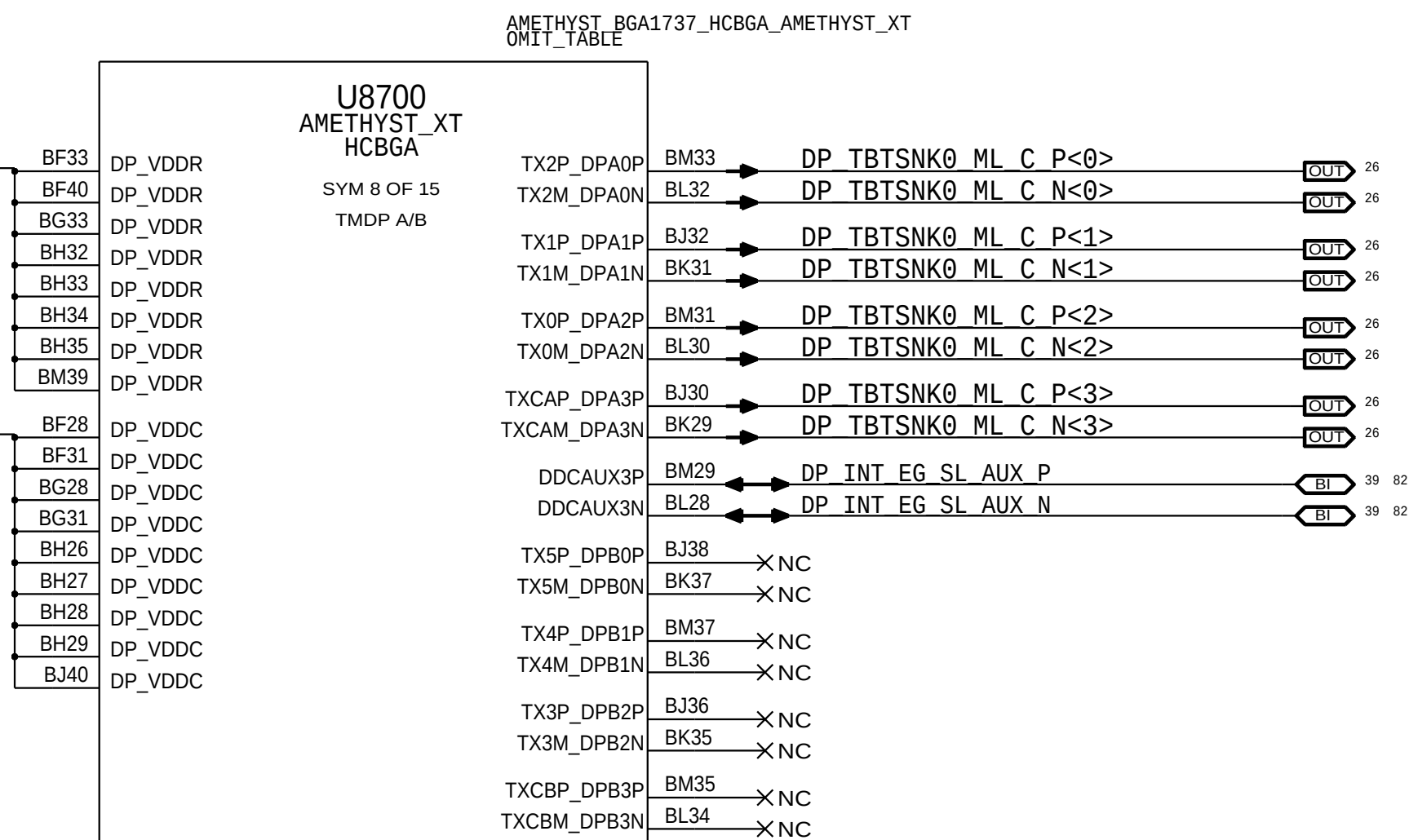
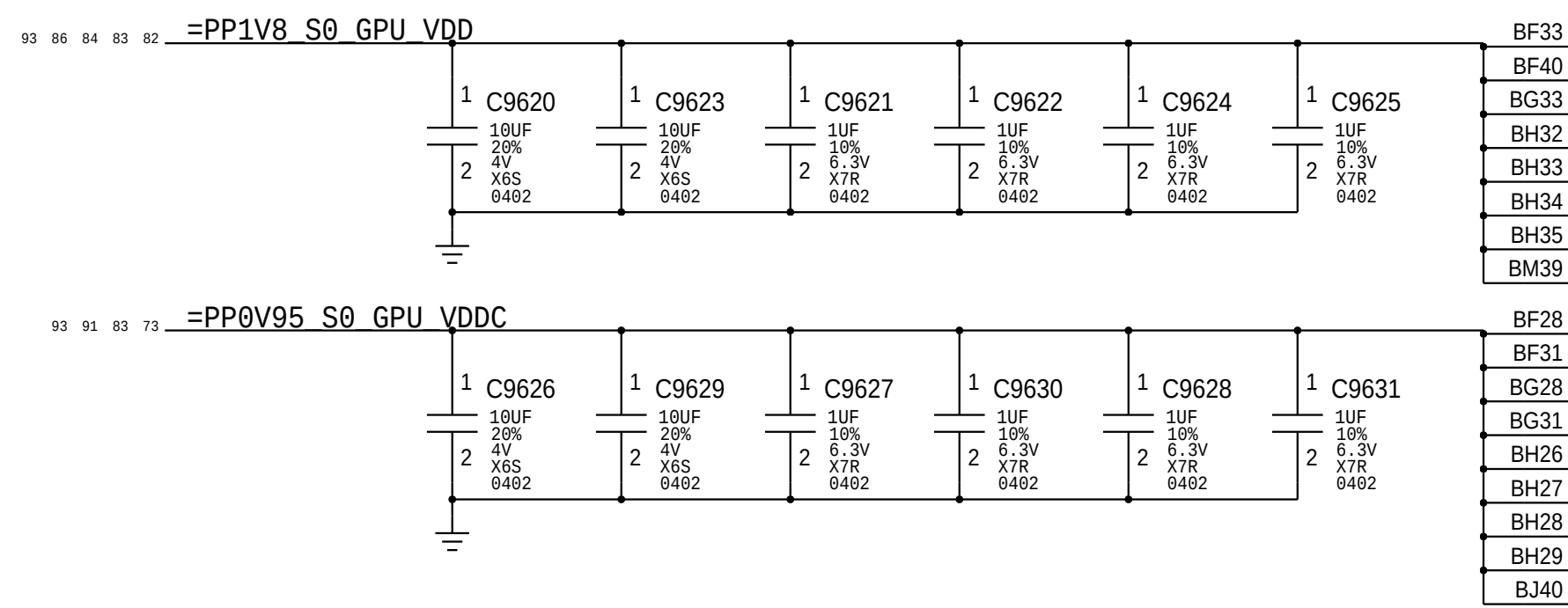
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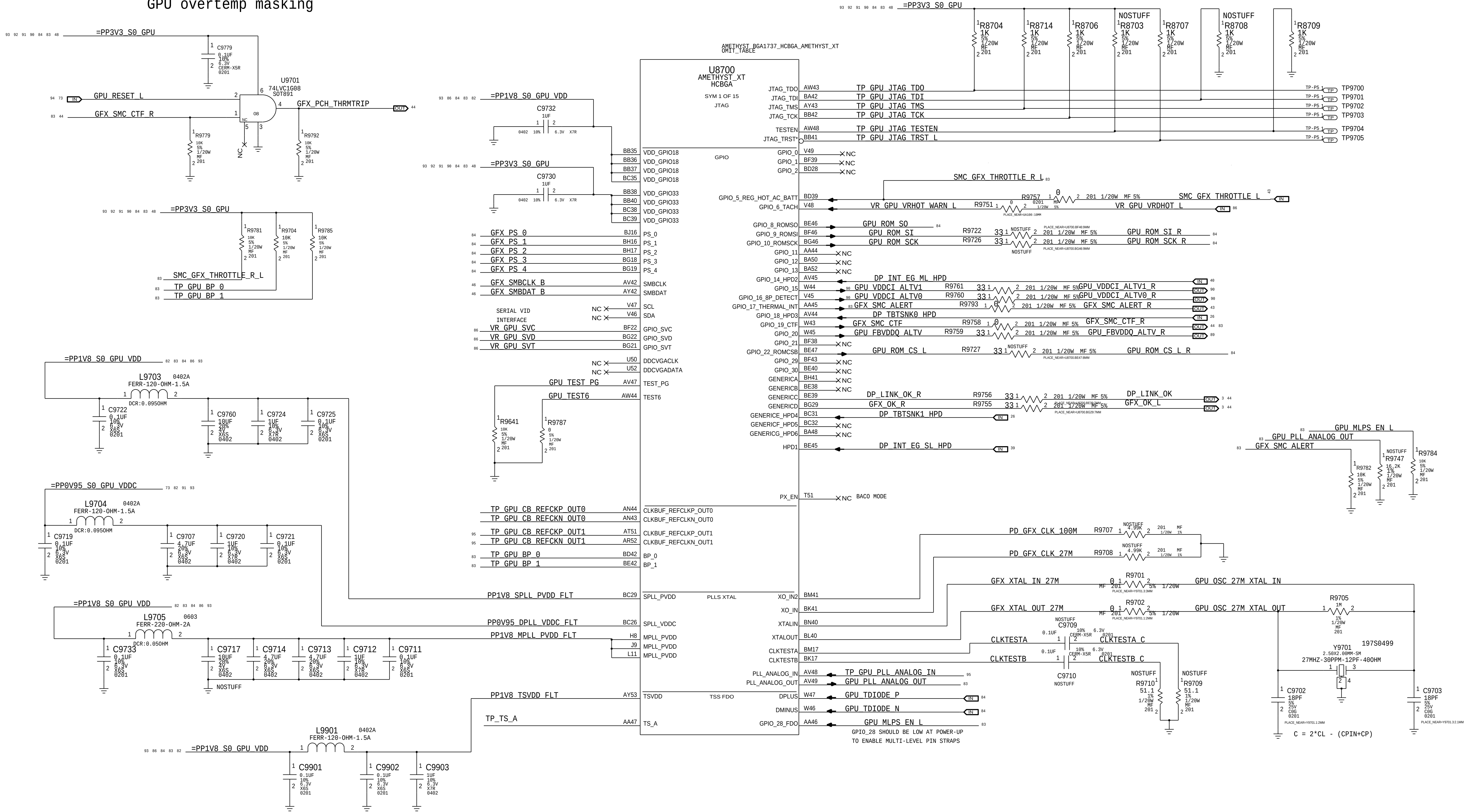
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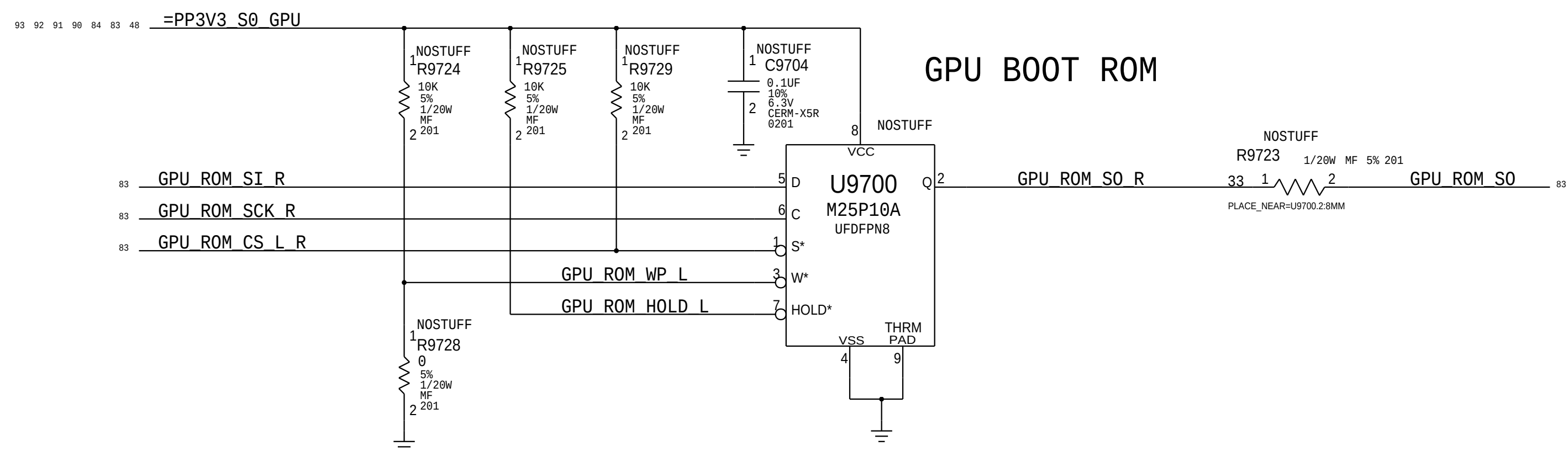
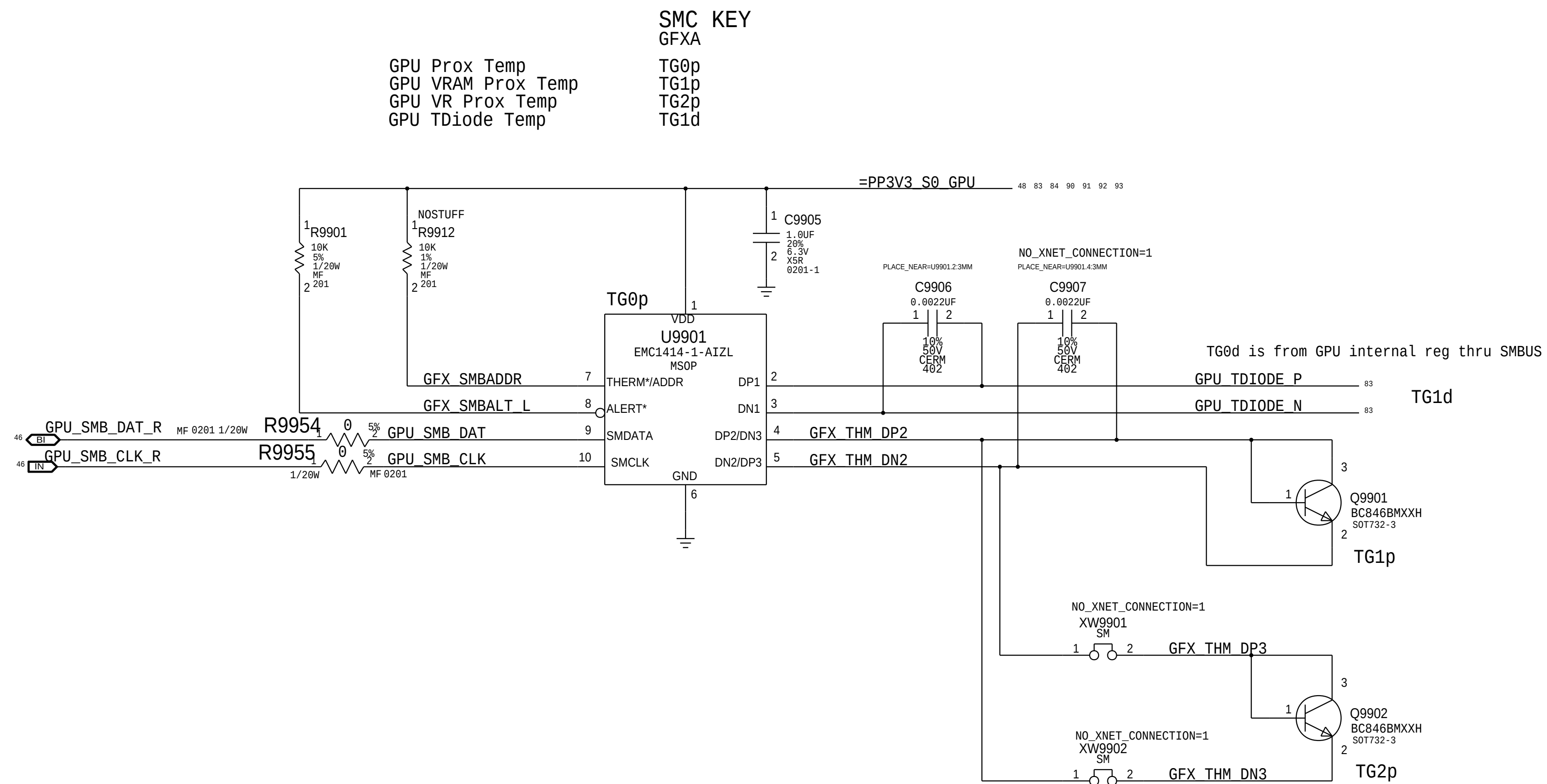
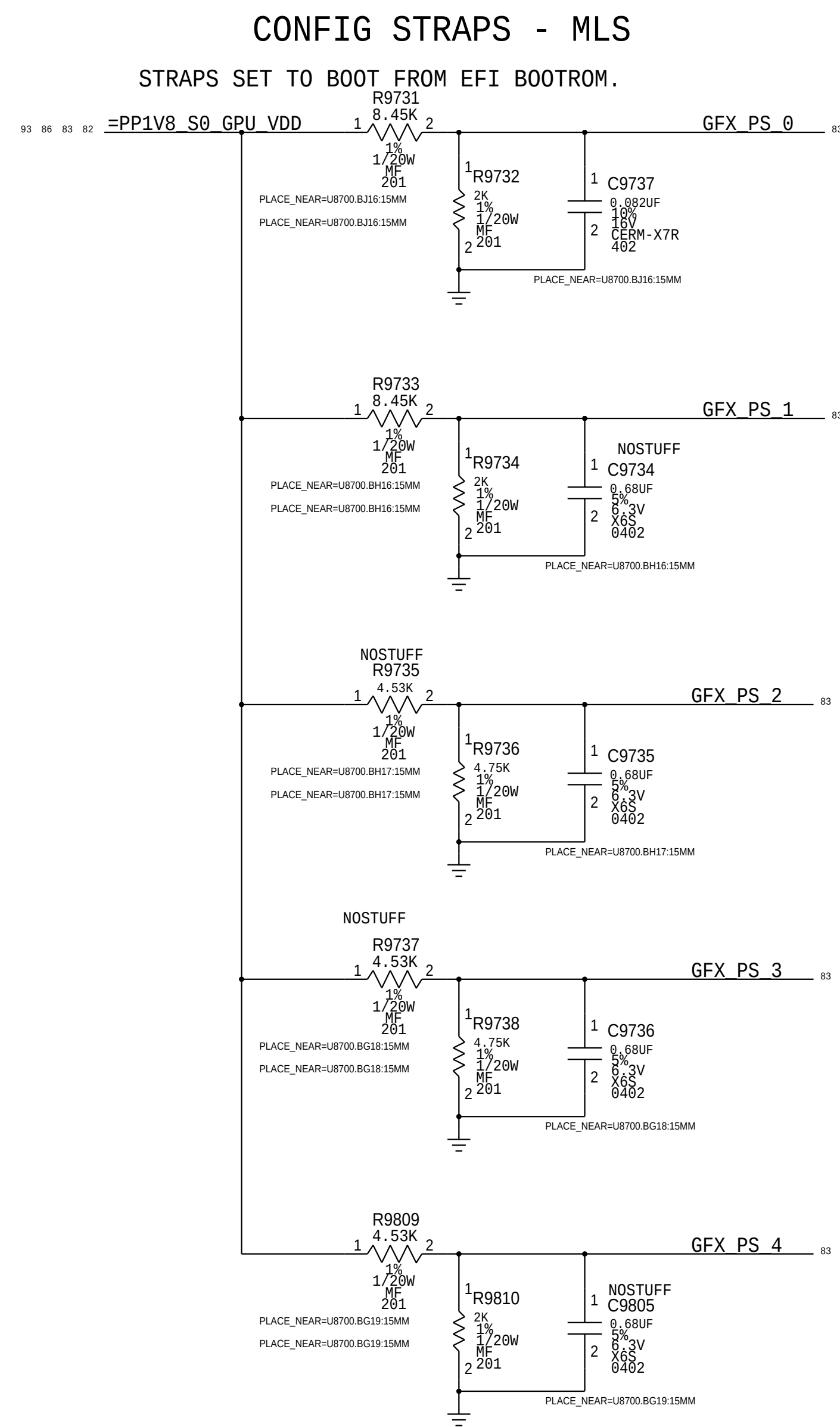
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051-00321		D	
REVISION		4.0.0	
BRANCH		proto1b	
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GPU overtemp masking





Page Notes

Power aliases required by this page:

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- +PP3V3_50_GPU
- +PP1V05_S8_EP2_PEX_PLLVDD

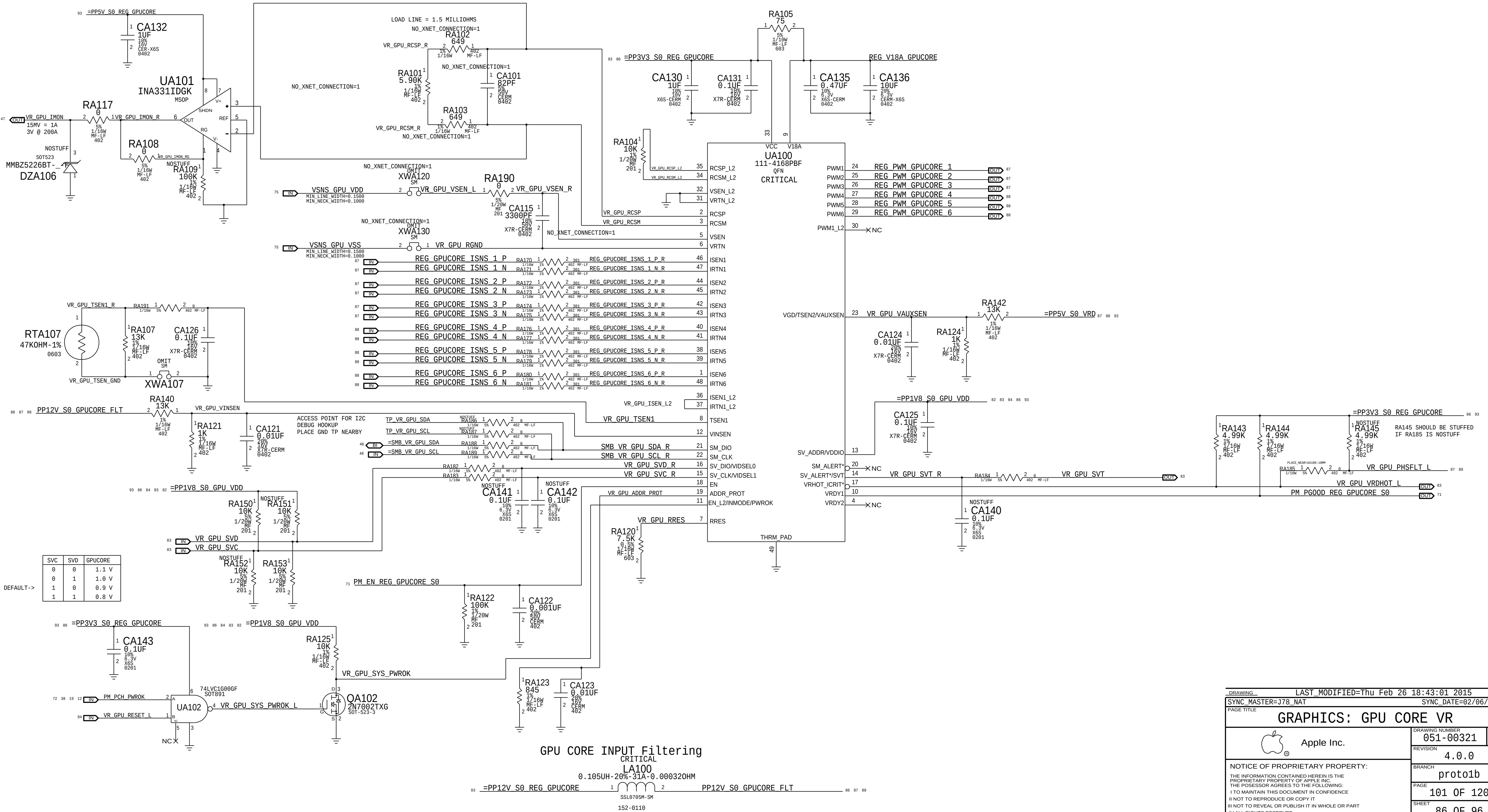
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SDM options provided by this page:

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GRAPHICS: GPU STRAPS, SENSORS, ROM			
 Apple Inc.	DRAWING NUMBER 051-00321		SIZE D
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O/P= PPGPUCORE_S0_REG

GPUCORE
VOUT = VCORE
PEAK = 195A
AVG = 154A



DRAWING		LAST MODIFIED=Thu Feb 26 18:43:01 2015	
SYNC_MASTER=J78_NAT		SYNC_DATE=02/06/2014	
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GRAPHICS: GPU CORE VR			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00321		D
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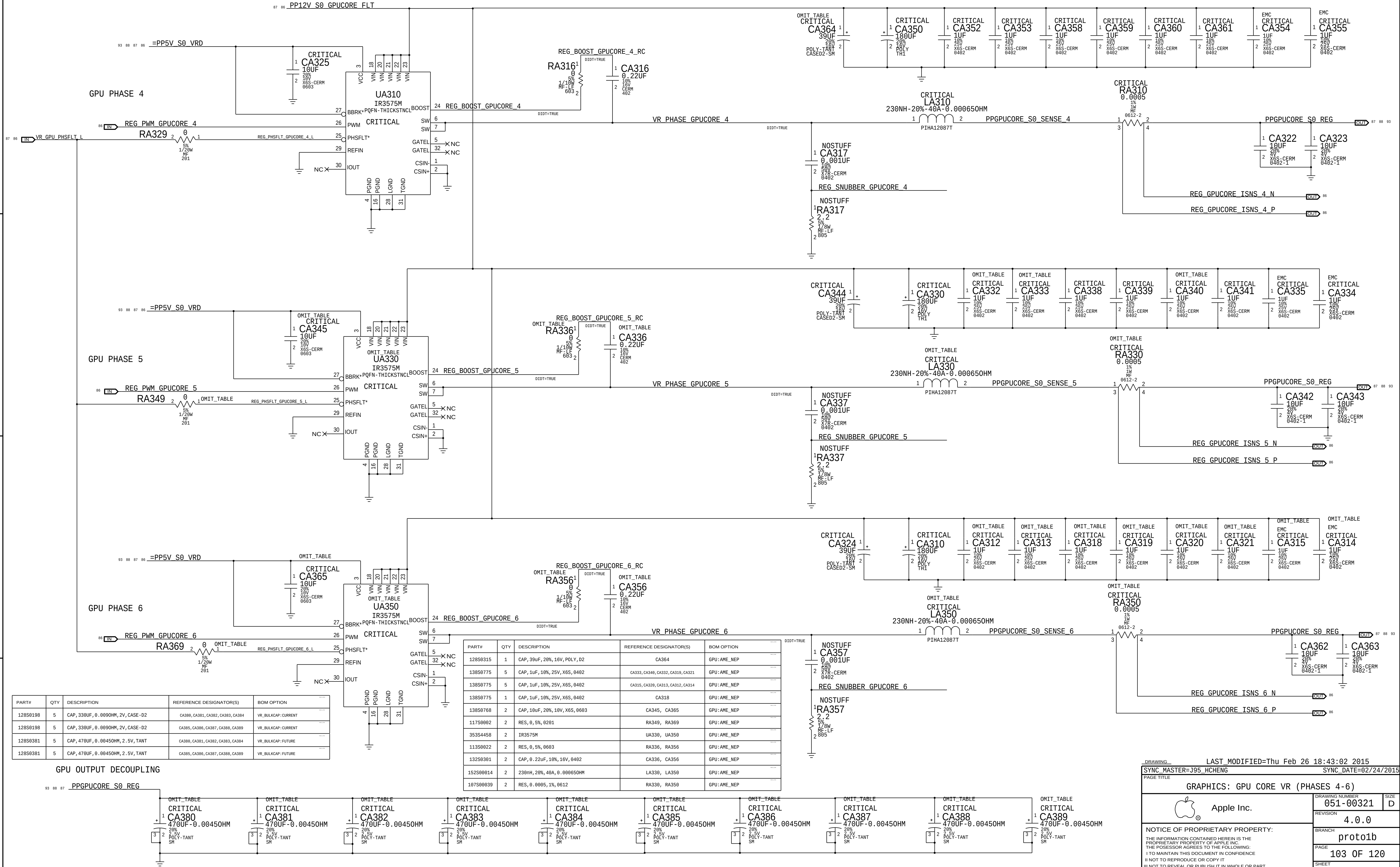
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	1	NO STUFF	CA224	GPU:EMERALD
138S9775	5	CAP, 1uF, 10%, 25V, X6S, 0402	CA254, CA228, CA255, CA239, CA261	GPU:AME_NEP
	5	NO STUFF	CA254, CA228, CA255, CA239, CA261	GPU:EMERALD

DRAWING		LAST_MODIFIED=Thu Feb 26 18:43:02 2015	
SYNC_MASTER=J95_HARPER		SYNC_DATE=02/12/2015	
PAGE TITLE			
GRAPHICS: GPU CORE VR (PHASES 1-3)			
 Apple Inc. ®	DRAWING NUMBER		051-00321
	REVISION		4.0.0
	BRANCH		proto1b
	PAGE		102 OF 120
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S8198	5	CAP, 330UF, 0.0090HM, 2V, CASE-D2	CA389, CA381, CA382, CA383, CA384	VR_BULKCAP: CURRENT
128S8198	5	CAP, 330UF, 0.0090HM, 2V, CASE-D2	CA385, CA386, CA387, CA388, CA389	VR_BULKCAP: CURRENT
128S8381	5	CAP, 470UF, 0.00450HM, 2.5V, TANT	CA389, CA381, CA382, CA383, CA384	VR_BULKCAP: FUTURE
128S8381	5	CAP, 470UF, 0.00450HM, 2.5V, TANT	CA385, CA386, CA387, CA388, CA389	VR_BULKCAP: FUTURE

GPU OUTPUT DECOUPLING

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S8315	1	CAP, 39uF, 20%, 16V, POLY, D2	CA364	GPU:AME_NEP
138S8775	5	CAP, 1uF, 10%, 25V, X6S, 0402	CA333, CA349, CA332, CA319, CA321	GPU:AME_NEP
138S8775	5	CAP, 1uF, 10%, 25V, X6S, 0402	CA315, CA329, CA313, CA312, CA314	GPU:AME_NEP
138S8775	1	CAP, 1uF, 10%, 25V, X6S, 0402	CA318	GPU:AME_NEP
138S8768	2	CAP, 10uF, 20%, 10V, X6S, 0603	CA345, CA365	GPU:AME_NEP
117S8802	2	RES, 0, 5%, 0201	RA349, RA369	GPU:AME_NEP
113S8822	2	RES, 0, 5%, 0603	RA336, RA356	GPU:AME_NEP
132S89301	2	CAP, 0.22uF, 10%, 16V, 0402	CA336, CA356	GPU:AME_NEP
152S88014	2	230nH, 20%, 40A, 0.000650HM	LA330, LA350	GPU:AME_NEP
107S88839	2	RES, 0.0805, 1%, 0612	RA330, RA350	GPU:AME_NEP



GRAPHICS: GPU CORE VR (PHASES 4-6)

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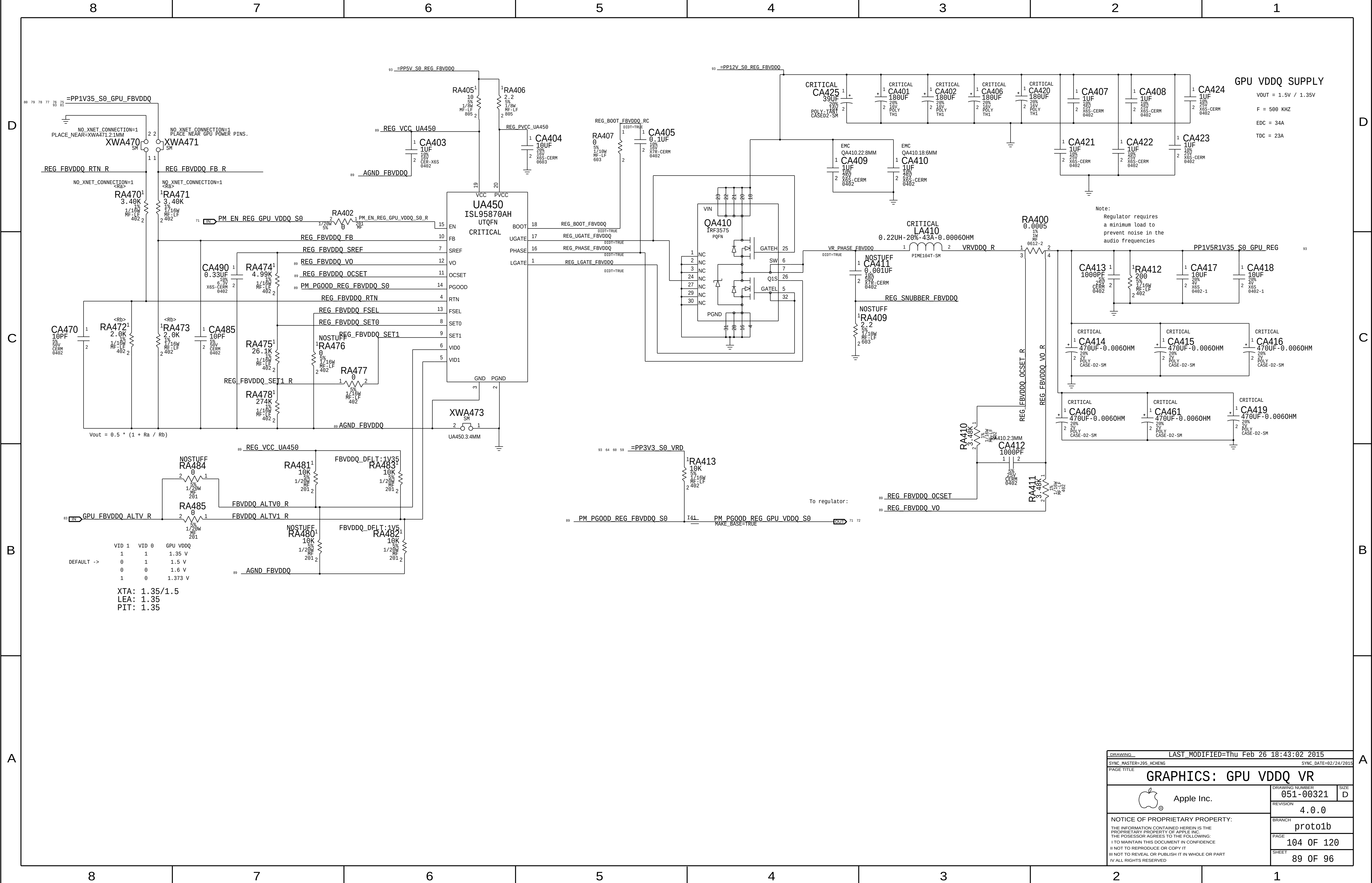
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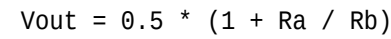
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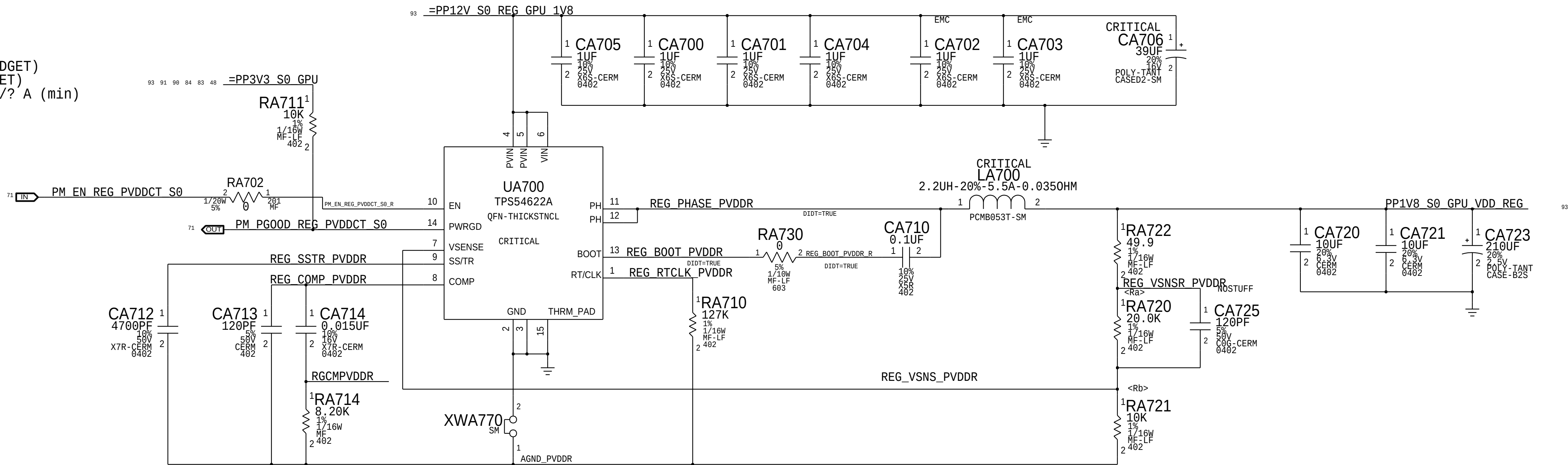
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Max peak current: 4.5 A
Switching freq: 300 kHz



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
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128S9381	2	CAP, 470UF, 0.00450HM, 2.5V, SM	CA640, CA641	VR_BULKCAP:FUTURE

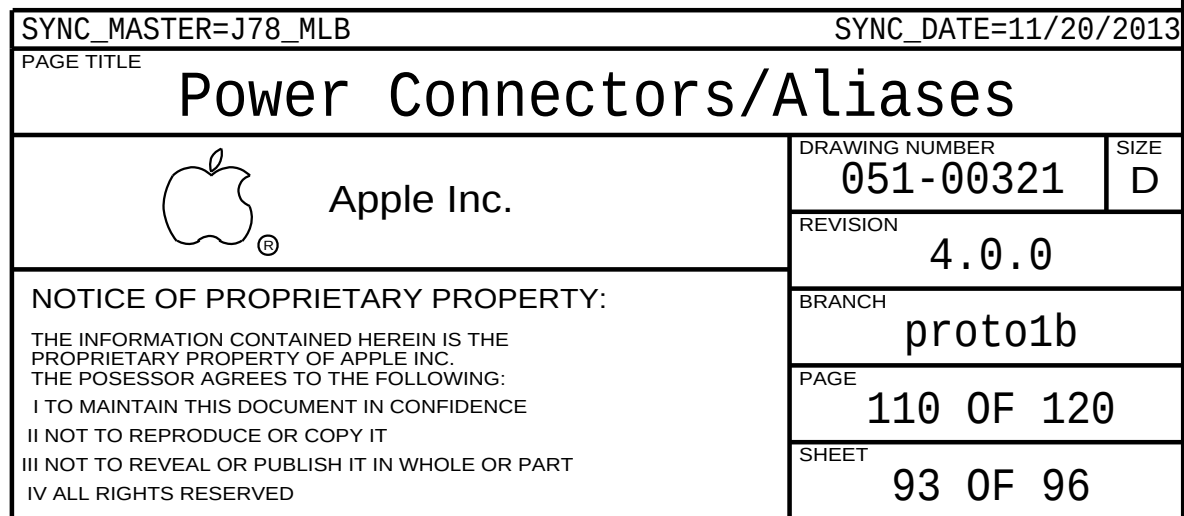
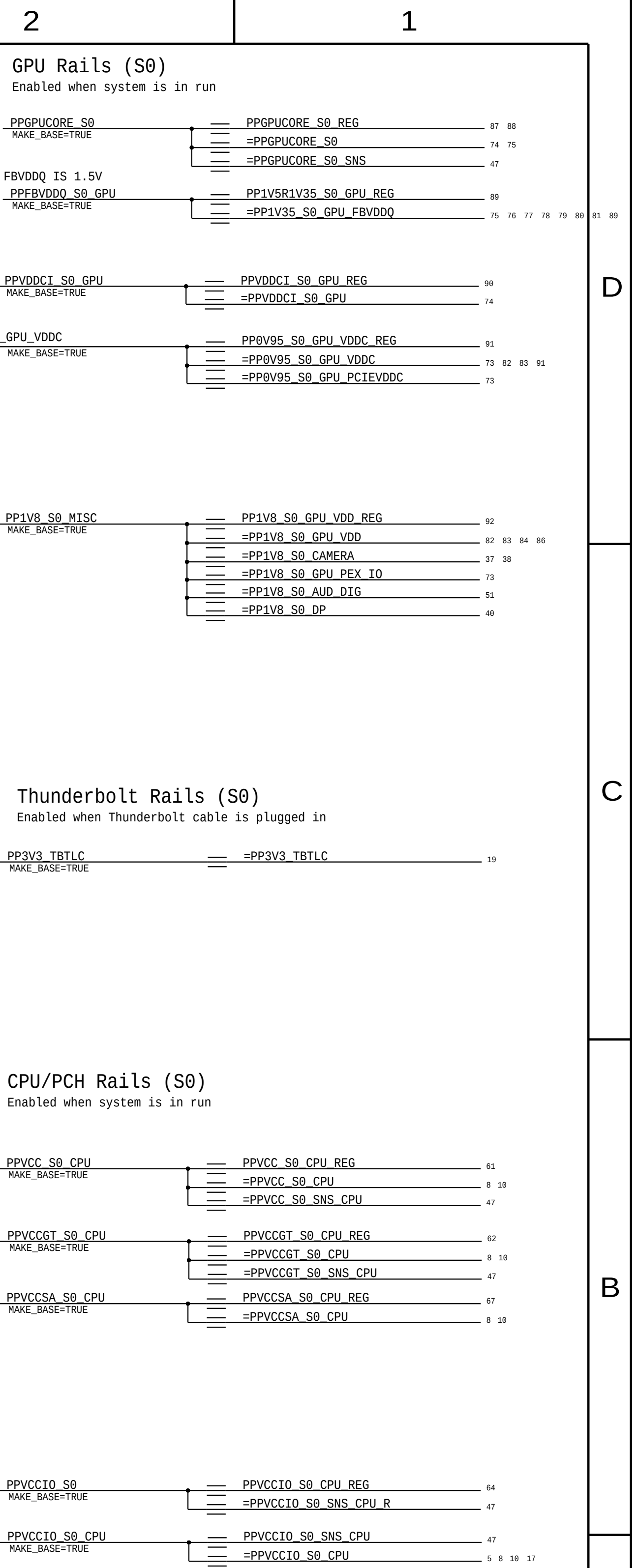
1.8V S0 Regulator

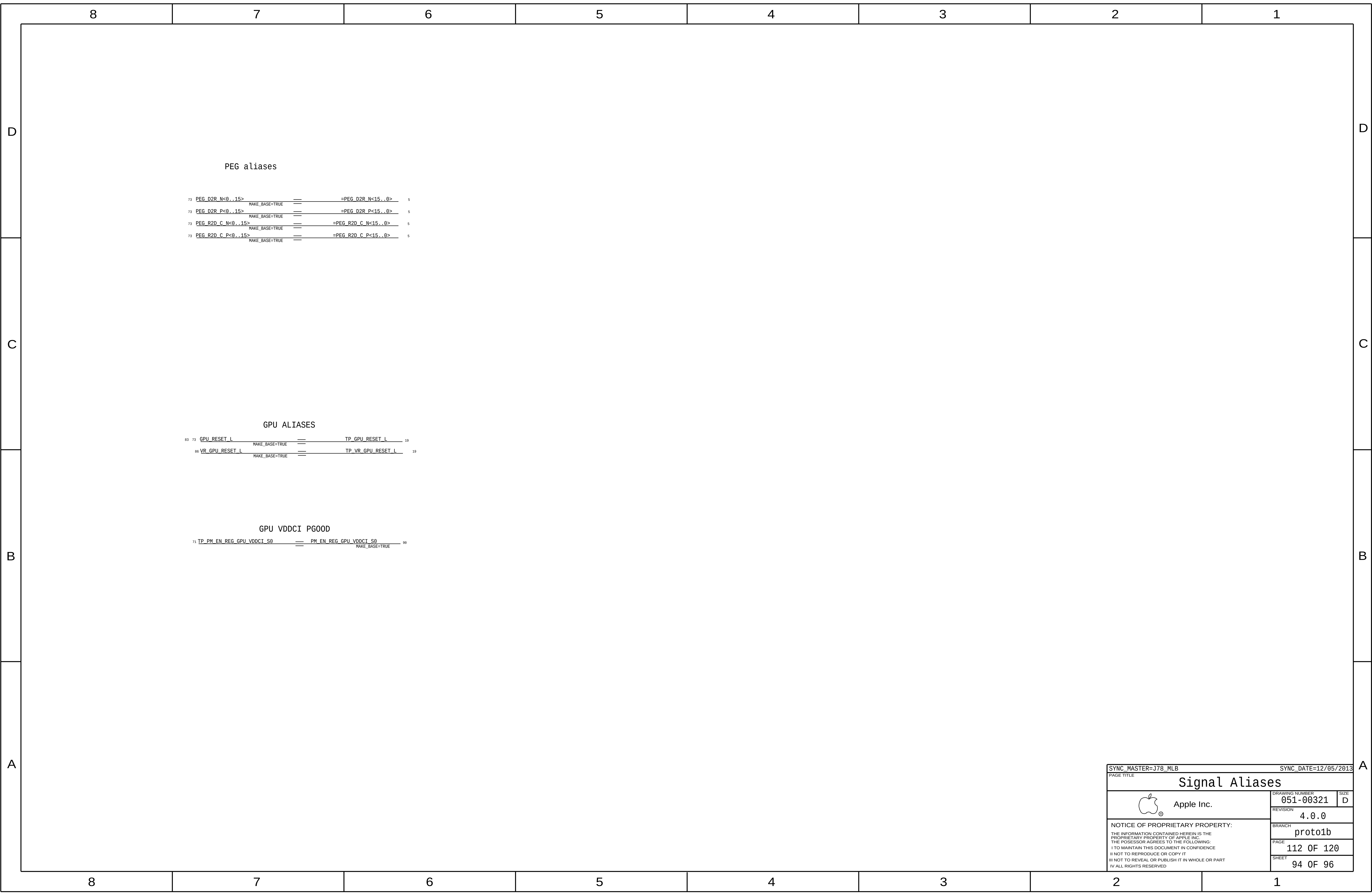
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Max peak current: 3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz

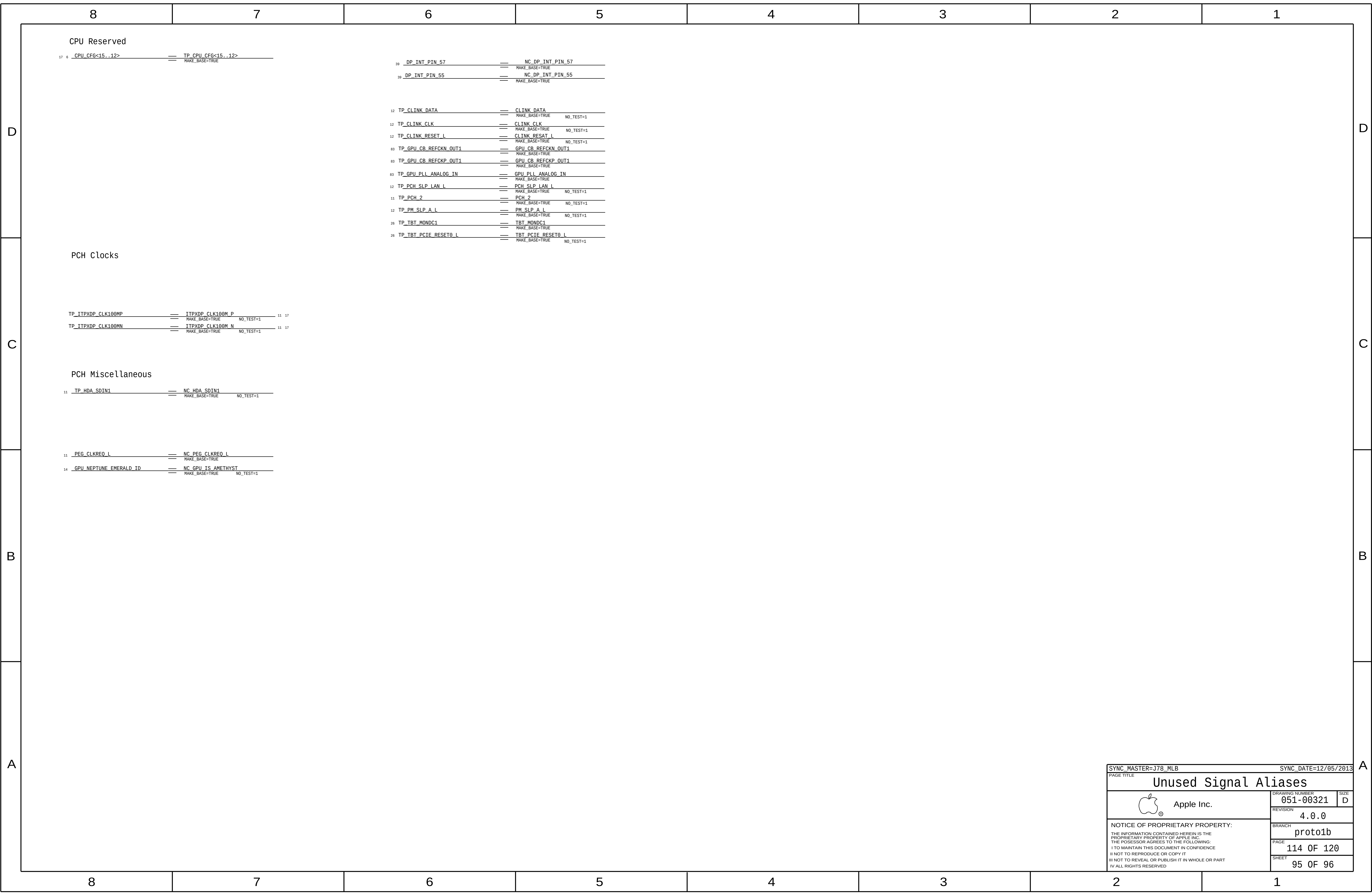


$V_{OUT} = 0.6 * (1 + R_A / R_B)$

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SYNC_MASTER=J95_HARPER		REVISION: 4.0.0		BRANCH: proto1b	
PAGE TITLE: GRAPHICS: GPU 1V8 VR		PAGE: 107 OF 120		SHEET: 92 OF 96	
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SYNC_MASTER=J78_MLB

SYNC_DATE=12/05/2013

PAGE TITLE

Unused Signal Aliases



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